

**ACADEMIC REGULATIONS
AND
COURSE STRUCTURE**
CHOICE BASED CREDIT SYSTEM
MLR18

AERONAUTICAL ENGINEERING

for

Bachelor of Technology (B.Tech)

For the batches admitted 2018-19

B. Tech. - Regular Four Year Degree Program
(For batches admitted from the academic year 2018 - 19)

&

B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2019 - 20)



MLRInstitute of Technology

(Autonomous)

Laxman Reddy Avenue, Dundigal (V), Quthbullapur (M),
Hyderabad – 500043, Telangana State
www.mlrit.ac.in, Email: director@mlrinstitutions.ac.in

FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

INDEX

S.No	Details	Page No
1.	Regulations	01
2.	Course Structure	17
3.	I B. Tech I Sem Syllabus	25
4.	I B. Tech II Sem Syllabus	41
5.	II B. Tech I Sem Syllabus	60
6.	II B. Tech II Sem Syllabus	79
7.	III B. Tech I Sem Syllabus	100
8.	III B. Tech II Sem Syllabus	116
9.	IV B. Tech I Sem Syllabus	143
10.	IV B. Tech II Sem Syllabus	169
11.	Open Elective Syllabus	186

B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2018 - 19)
&
B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2019 - 20)

For pursuing four year under graduate Bachelor Degree Programme of study in Engineering (B.Tech) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2018-19 onwards. Any reference to “Institute” or “College” in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

1. ADMISSION

1.1. Admission into first year of four year B. Tech. degree Programmes of study in Engineering

1.1.1. Eligibility:

A candidate seeking admission into the first year of four year B. Tech. degree Programmes should have:

(i) Passed either Intermediate Public Examination (I.P.E) conducted by the Board of Intermediate Education, Telangana, with Mathematics, Physics and Chemistry as optional subjects or any equivalent examination recognized by Board of Intermediate Education, Telangana or a Diploma in Engineering in the relevant branch conducted by the Board of Technical Education, Telangana or equivalent Diploma recognized by Board of Technical Education for admission as per guidelines defined by the Regulatory bodies of Telangana State Council for Higher Education (TSCHE) and AICTE.

(ii) Secured a rank in the EAMCET examination conducted by TSCHE for allotment of a seat by the Convener, EAMCET, for admission.

1.1.2. Admission Procedure:

Admissions are made into the first year of four year B. Tech. Degree Programmes as per the stipulations of the TSCHE.

- (a) Category A seats are filled by the Convener, TSEAMCET.
- (b) Category B seats are filled by the Management.

1.2. Admission into the second year of four year B. Tech. degree Program in Engineering

1.2.1 Eligibility:

A candidate seeking admission under lateral entry into the II year I Semester B. Tech. degree Programmes should have passed the qualifying exam (B.Sc. Mathematics or Diploma in concerned course) and based on the rank secured by the candidate at Engineering Common Entrance Test ECET (FDH) in accordance with the instructions received from the Convener, ECET and Government of Telangana.

1.2.2 Admission Procedure:

Admissions are made into the II year of four year B. Tech. degree Programmes through Convener, ECET (FDH) against the sanctioned strength in each Programmes of study as lateral entry students.

2. PROGRAMMES OFFERED

MLR Institute of Technology, an autonomous college affiliated to JNTUH, offers the following B. Tech. Programmess of study leading to the award of B. Tech. degree under the autonomous scheme.

- 1) B.Tech. - Aeronautical Engineering
- 2) B.Tech. - Computer Science and Engineering
- 3) B.Tech. - Electronics and Communication Engineering
- 4) B.Tech - Electrical & Electronics Engineering
- 5) B.Tech. - Information Technology
- 6) B.Tech. - Mechanical Engineering

3. DURATION OF THE PROGRAMMES

3.1 Normal Duration

- 3.1.1 B. Tech. degree programme extends over a period of four academic years leading to the Degree of Bachelor of Technology (B.Tech.) of the Jawaharlal Nehru Technological University Hyderabad.
- 3.1.2 For students admitted under lateral entry scheme, B. Tech. degree programme extends over a period of three academic years leading to the Degree of Bachelor of Technology (B. Tech.) of the Jawaharlal Nehru Technological University Hyderabad.

3.2 Maximum Duration

- 3.2.1 The maximum period within which a student must complete a full-time academic programme is 8 years for B. Tech. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.2 For students admitted under lateral entry scheme in B. Tech. degree programme, the maximum period within which a student must complete a full-time academic programme is 6 years. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.3 The period is reckoned from the academic year in which the student is admitted first time into the degree Programme.

4. AWARD OF B.Tech. DEGREE

A student will be declared eligible for the award of the B.Tech. degree if he/she fulfils the following academic regulations:

- 4.1 The candidate shall pursue a course of study for not less than four academic years and not more than eight years.
- 4.2 The candidate shall register for 160 credits and secure 160 credits.
- 4.3 The degree will be conferred and awarded by Jawaharlal Nehru Technological University Hyderabad on the recommendations of the Chairman, Academic Council.

5. PROGRAMME STRUCTURE

- 5.1 UGC/AICTE specified Definitions/ Descriptions are adopted appropriately for various terms and abbreviations used in these Academic Regulations/ Norms, which are listed below.

Semester Scheme:

Each UGP is of 4 Academic Years (8 Semesters), each year divided into two Semesters of 22 weeks (≥ 90 working days), each Semester having - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)'. Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) as denoted by UGC, and Curriculum/Course Structure as suggested by AICTE are followed.

5.1.2 The B.Tech. Programme of MLR Institute of Technology are of Semester pattern, with 8 Semesters constituting 4 Academic Years, each Academic Year having TWO Semesters (First/Odd and Second/Even Semesters). Each Semester shall be of 22 Weeks duration (inclusive of Examinations), with a minimum of 90 Instructional Days per Semester.

5.1.3 Credit Courses:

a) All Courses are to be registered by a student in a Semester to earn Credits. Credits shall be assigned to each Subject/ Course in a L: T: P: C (Lecture Periods: Tutorial Periods: Practical Periods : Credits) Structure, based on the following general pattern ..

- One Credit - for One hour/Week/Semester for Theory/Lecture(L)/Tutorial(T) Courses; and
- One Credit - for Two hours/Week/Semester for Laboratory/Practical (P) Courses, Mini Project...
- Mandatory Courses will not carry any Credits.

5.1.4 **Course Classification:**

All Courses offered for the UGP are broadly classified as:

- Basic Science Courses (BSC) include Mathematics, Physics, Chemistry, Biology etc
- **Engineering Science Courses (ESC)** courses include Materials, Workshop, Basics of Electrical/Electronics/ Mechanical/Computer Science & Engineering, Engineering Graphics, Instrumentation, Engineering Mechanics, Instrumentation etc.
- **Humanities and Social Science including Management Courses (HSMC)** courses include English, Communication skills, Management etc.
- Professional Core Courses(PCC) are core courses relevant to the chosen specialization/branch.
- **Professional Elective Courses (PEC)** are courses relevant to the chosen specialization/ branch offered as electives.
- **Open Elective Courses (OEC)** courses from other technical and/or emerging subject areas offered in the College by the Departments of Engineering, Science and Humanities.
- **Mandatory Course:** Course work on peripheral subjects in a programme, wherein familiarity considered mandatory. To be included as non-Credit, Mandatory Courses, with only a pass in each required to qualify for the award of degree from the concerned institution.
- **Project Work** and/or internship in industry or elsewhere, seminar.

5.1.5 **Course Nomenclature:**

The Curriculum Nomenclature or Course-Structure Grouping for the each of the UGP E&T (B.Tech. Degree Programme), is as listed below (along with AICTE specified Range of Total Credits).

S. No.	Broad Course Classification	Course Group/ Category	Course Description	Range of Credits
1)	BSC,ESC & HSMC	BSC – Basic Sciences Courses	Includes - Mathematics, Physics and Chemistry Subjects	25*
2)		ESC - Engineering Sciences Courses	Includes fundamental engineering subjects.	24*
3)		HSMC – Humanities and Social Sciences including Management	Includes subjects related to Humanities, Social Sciences and Management.	12*
4)	PCC	PCC – Professional Core Courses	Includes core subjects related to the Parent Discipline/ Department/ Branch of Engg.	48*
5)	PEC	PEC– Professional Elective Courses	Includes Elective subjects related to the Parent Discipline / Department / Branch of Engg.	18*
6)	OEC	OEC – Open Elective Courses	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the Parent Discipline/ Department / Branch of Engg.	18*
7)	PWC	Project Work	Major Project.	15*
8)		Industrial Training/ Mini-Project	Industrial Training/ Internship/ Mini-Project.	
9)		Seminar	Seminar / Colloquium based on core contents related to Parent Discipline/ Department/ Branch of Engg.	
10)	MC	Mandatory Courses	Mandatory Courses (non-credit)	-
Total Credits for UGP (B. Tech.)Programme				160 (100%)

- Minor variations as per AICTE guidelines

6. COURSE REGISTRATION

- 6.1 A 'Faculty Advisor or Counsellor' shall be assigned to each student, who advises him/her about the UGP, its Course Structure and Curriculum, Choice/Option for Subjects/Courses, based on his/her competence, progress, pre-requisites and interest.
- 6.2 Academic Section of the College invites 'Registration Forms' from students prior (before the beginning of the Semester), ensuring 'DATE and TIME Stamping'. The Registration Requests for any 'CURRENT SEMESTER' shall be completed BEFORE the commencement of SEEs (Semester End Examinations) of the 'PRECEDING SEMESTER'
- 6.3 A Student can apply for Registration, which includes approval from his faculty advisor, and then should be submitted to the College Academic Section through the Head of Department

(a copy of the same being retained with Head of Department, Faculty Advisor and the Student).

- 6.4 A student may be permitted to register for his/her course of CHOICE with a Total of prescribed credits per Semester (permitted deviation being $\pm 12\%$), based on his PROGRESS and SGPA/CGPA, and completion of the 'PRE-REQUISITES' as indicated for various courses in the Department Course Structure and Syllabus contents.
- 6.5 Choice for 'additional Courses' must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Counsellor.
- 6.6 If the Student submits ambiguous choices or multiple options or erroneous (incorrect) entries during Registration for the Course(s) under a given/specified Course Group/ Category as listed in the Course Structure, only the first mentioned Course in that Category will be taken into consideration.
- 6.7 Dropping of Courses or changing of options may be permitted, ONLY AFTER obtaining prior approval from the Faculty Advisor, 'within 15 Days of Time' from the commencement of that Semester. Course Options exercised through Registration are final and CAN NOT be changed, and CAN NOT be inter-changed; further, alternate choices will also not be considered. However, if the Course that has already been listed for Registration (by the Head of Department) in a Semester could not be offered due to any unforeseen or unexpected reasons, then the Student shall be allowed to have alternate choice - either for a new Subject (subject to offering of such a Subject), or for another existing Subject (subject to availability of seats), which may be considered. Such alternate arrangements will be made by Head of the Department, with due notification and time-framed schedule, within the FIRST WEEK from the commencement of Class-work for that Semester.

7. COURSES TO BE OFFERED

- 7.1 A typical section (or class) strength for each Semester shall be 60.
- 7.2 courses may be offered to the Students, only if minimum of 20 students ($1/3^{\text{rd}}$ of the section strength) opt for it.
- 7.2 More than ONE TEACHER may offer the SAME SUBJECT (Lab/Practical's may be included with the corresponding Theory Subject in the same Semester) in any Semester. However, selection choice for students will be based on - 'CGPA Basis Criterion' (i.e., the first focus shall be on early Registration in that Semester, and the second focus, if needed, will be on CGPA of the student).
- 7.3 If more entries for Registration of a Subject come into picture, then the concerned Head of the Department shall take necessary decision, whether to offer such a Subject/Course for TWO (or multiple) SECTIONS or NOT.
- 7.4 OPEN ELECTIVES will be offered by a department to the students of other departments.

8. ATTENDANCE REQUIREMENTS

- a. A student will be eligible to appear for the End Semester Examinations, if he acquires a minimum of 75% of attendance in aggregate of all the Subjects/Courses (excluding Mandatory or Non-Credit Courses) for that Semester.
- b. Condoning of shortage of attendance in aggregate up to 10% (65% and above, and below 75%) in each Semester may be granted by the College Academic Committee on genuine and valid grounds, based on the student's representation with supporting evidence by following the govt. rules in vogue.
- c. A stipulated fee shall be payable towards condoning of shortage of attendance.
- d. Shortage of Attendance below 65% in aggregate shall in No case be condoned.
- e. A student shall not be promoted to the next Semester unless he/she satisfies the attendance requirements of the current Semester. The student may seek readmission for the Semester when

offered next. He / She shall not be allowed to register for the subjects of the Semester while he/she is in detention. A student detained due to shortage of attendance, will have to repeat that Semester when offered next.

- f. Students whose attendance is less than 75% are not entitled to get the scholarship / fee reimbursement in any case as per the TS Govt. Rules in force.

9. ACADEMIC REQUIREMENTS FOR PROMOTION / COMPLETION OF REGULAR B.TECH PROGRAMME COURSE STUDY.

- 9.1 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to each Course, if he secures not less than 35% marks in the End Semester Examination, and a minimum of 40% of marks in the sum Total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of Letter Grades, this implies securing P Grade or above in that Course.
- 9.2 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to – Internship, if he/she secures not less than 40% of the Total marks to be awarded for each. The student would be treated as failed, if he/she - (i) does not submit a report on his/her Internship, or does not make a presentation of the same before the Evaluation Committee as per the schedule, or (ii) does not present the Seminar as required in the IV year I/II Semester, or (iii) secures less than 40% of marks in Internship evaluations. He/She may reappear once for each of the above evaluations, when they are scheduled again; if he/she fails in such 'one reappearance' evaluation also, he/she has to reappear for the same in the next subsequent Semester, as per the schedule.
- 9.3 A Student will not be promoted from I Year to II Year, unless he/she fulfils the Attendance requirements
- 9.4 A Student will not be promoted from II Year to III Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to II Year II Semester from all the relevant regular and supplementary examinations.
- 9.5 A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to III Year II Semester, from all the regular and supplementary examinations.
- 9.6 After securing the necessary 160 Credits as specified for the successful completion of the entire UGP, resulting in 160 Credits for UGP performance evaluation, i.e., the performance of the Student in these 160 Credits shall alone be taken into account for the calculation of the final CGPA.
If a Student registers for some more 'extra courses' (in the parent Department or other Departments/Branches of Engg.) other than those listed courses Totalling to 160 Credits as specified in the Course Structure of his/her Department, the performances in those 'extra courses' (although evaluated and graded using the same procedure as that of the required 160 Credits) will not be taken into account while calculating the SGPA and CGPA. For such 'extra courses' registered, % marks and Letter Grade alone will be indicated in the Grade Card, as a performance measure, subject to completion of the Attendance and Academic Requirements as stated in items 8 and 9.1-9.6.
- 9.7 Students who fail to earn minimum of 160 Credits as per the Course Structure, and as indicated above, within 8 Academic Years from the Date of Commencement of their I Year shall forfeit their seats in B.Tech Programme and their admissions shall stand cancelled.
When a Student is detained due to shortage of attendance/lack of credits in any Semester, he may be re-admitted into that Semester, as and when offered. However the regulations at the time of admissions hold good.

10. EVALUATION - DISTRIBUTION AND WEIGHTAGE OF MARKS

- 10.1 The performance of a student in each Semester shall be evaluated Course-wise (irrespective of Credits assigned) with a maximum of 100 marks for Theory. The B.Tech Project Work (Major Project) will be evaluated for 100 marks in Phase-I and 200 Marks in Phase-II.
- 10.2 For all Theory Courses as mentioned above, the distribution shall be 30 marks for CIE, and 70 marks for the SEE.
- 10.3 a) For Theory Subjects (inclusive of Minor Courses), during the Semester, there shall be 2 mid-term examinations for 30 marks each. Each mid-term examination consists of one subjective paper for 25 marks, and assignment for 5 marks for each subject.

Question paper contains 2 Parts (Part-A and Part-B.) The distribution of marks for PART-A and PART-B will be 10 marks & 15 marks respectively for UG programme.

Pattern of the question paper is as follows:

PART-A

Consists of **one compulsory question** with five sub questions each carrying two mark. For the I-Mid examinations the sub question would be from first 2 ½ units and for the II-Mid examination the sub question would be from the remaining 2 ½ units.

PART-B

Consists of five questions (out of which students have to answer three questions) carrying five marks each. Each question there will be an “either” “or” choice (that means there will be two questions from each unit and the student should answer any one question). The questions can consist of sub questions also.

- b) The first mid-term examination shall be conducted for the first 50% of the syllabus, and the second mid-term examination shall be conducted for the remaining 50% of the syllabus.
- c) First Assignment should be submitted before the commencement of the first mid-term examinations, and the Second Assignment should be submitted before the commencement of the second mid-term examinations. The assignments shall be specified/given by the concerned subject teacher.
- d) If any candidate is absent for the MID term examination or those who want to improve their internal marks in any subject can opt for Computer Based Test (CBT) as and when offered. The CBT is a 45 minutes duration ONLINE exam consisting of 30 objective questions from the entire syllabus of the subject. Best marks is consider as a final marks from the average of two mid examinations or CBT examination marks. The CBT can be taken after the payment of prescribed fee. There is no Internal Improvement for the courses Machine Drawing, Production Drawing, Engineering Drawing, Engineering Graphics and practical courses.
- 10.4 For Practical Courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned for Lab/Practical End Semester Examination (SEE). Out of the 30 marks for internals, day-to-day work in the laboratory shall be evaluated for 15 marks; and for the remaining 15 marks - two internal practical tests (each of 15 marks) shall be conducted by the concerned laboratory teacher and the average of the two tests is taken into account. The SEE for Practical's shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.
- 10.5 For the Subjects having Design and/or Drawing, (such as Engineering Graphics, Engineering Drawing, Machine Drawing, Production Drawing Practice, and Estimation), the distribution shall be 30 marks for CIE (10 marks for day-to-day work and 20 marks for internal tests) and 70 marks for SEE. There shall be two internal tests in a semester and the average of the two shall be considered for the award of marks for internal tests.

- 10.6 **Open Elective Course:** Students can choose One Open Elective Course (OEC-I) during V Semester, one (OEC-II) during VI Semester, one (OEC-III) in VII Semester, and one (OEC-IV) in VIII Semester from the list of Open Elective Courses given. However, Students cannot opt for an Open Elective Courses offered by their own (parent) Department, if it is already listed under any category of the Subjects offered by parent Department in any Semester.
- 10.7 There shall be an internship to be taken up during the vacation after II year II Semester (IV Semester) examination. However, the internship and its report shall be evaluated in III Year I Semester (VI Semester) SEE. The internship shall be submitted in a report form and presented before the committee. It shall be evaluated for 100 marks. The committee consists of head of the department, and a senior faculty member(s)(minimum Two) of the department. There shall be no internal marks for internship.
- 10.8 There shall be a Comprehensive Viva in VI Semester and will be conducted SEE by through a test or a committee consisting of One External Examiner, Head of the Department and two Senior faculty members of the Department. The comprehensive viva is intended to assess the student's understanding of the subjects he/she studied during the B.Tech course of study. The Comprehensive Viva is evaluated for 100 marks by the test or committee. There shall be no CIE for Comprehensive Viva.
- 10.9 Each Student shall start the Project Work Phase-I during the IV B.Tech I Semester(VII Semester), as per the instructions of the Project Guide/Project Supervisor assigned by the Head of Department. Total 100 marks allotted for the Project Work Stage-I. 40% of marks shall be evaluated Project Guide/Project supervisor CIE (Continuous Internal Evaluation) based on the reports submitted and conduct presentations. Remaining 60% of marks shall be evaluated by committee comprising of the Head of the Department, project supervisor and senior faculty member from concerned department based on Viva/Seminar Presentation. He/She must secure the 40% of the marks from CIE. For Project work Phase-II in IV Year II Sem. There is an internal marks of 50, the evaluation should be done by the supervisor. The There is an external marks of 150 and the same evaluated by the external examiner appointed by the Chief Controller of Examinations and he secures a minimum of 35% of marks in the Semester End Examination and a minimum aggregate of 40% of the total marks in the Semester End Examination and Continuous Internal Evaluation taken together.
- 10.10. **End Semester examination:**
- Question paper contains 2 Parts (Part-A and Part-B) having the questions distributed equally among all units.
 - The distribution of marks for PART-A and PART-B will be 20 marks & 50 marks respectively for UG programme. Pattern of the question paper is as follows:
- PART-A**
- Consists of one question which are compulsory. The question consists of ten sub-questions one from each unit and carry 2 marks each.
- PART-B**
- Consists of 5 questions carrying 10 marks each. Each of these questions is from one unit and may contain sub questions. Each question there will be an "either" "or" choice (that means there will be two questions from each unit and the student should answer any one question).
- 10.11 For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 35% marks in the SEE, and a minimum of 40% of marks in the sum Total of the CIE and SEE taken together in such a course, then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses '**Satisfactory**' or '**Unsatisfactory**' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

- 10.12. For social innovation and Engineering Exploration courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned End Semester Examination (SEE). Internal evaluations done through presentations in CIE-1 & CIE-2. The SEE for shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.

11. AWARD OF DEGREE

After a student has satisfied the requirement prescribed for the completion of the Programme and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes Shown in Table.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA $\geq$ 8.00
First Class	$\geq$ 6.50 to $<$ 8.00 CGPA
Second Class	$\geq$ 5.50 to $<$ 6.50 CGPA
Pass Class	$\geq$ 5.00 to $<$ 5.50 CGPA
FAIL	CGPA $<$ 5

a) Improvement of Grades and Completion of the Course

- Candidates who have passed in a theory course in a Semester are allowed to appear for improvement of Grade in the next immediate supplementary examination for a maximum of three subjects only. Candidates will not be allowed to improve grade in the Laboratory, Seminars, Internships and Project Work.
- Improved grade will not be counted for the award of prizes/medals and Rank. However the previous grade will be considered for the award of prizes/medals and rank in case of toppers.
- If the candidate does not show improvement in the grade, his/her previous grade will be taken into consideration.

12. LETTER GRADE AND GRADE POINT

- 12.1 Marks will be awarded to indicate the performance of each student in each Theory Subject, or Lab/Practical's, or Seminar, or Project, or Internship*/Mini-Project, Minor Course etc., based on the %marks obtained in CIE+SEE (Continuous Internal Evaluation + Semester End Examination, both taken together), and a corresponding Letter Grade shall be given.
* For internship only SEE will be taken into consideration. There is no CIE for Internship.

- 12.2 As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades (UGC Guidelines) and corresponding percentage of marks shall be followed...

% of Marks Secured (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
90% and above ($\geq$ 90% , $\leq$ 100%)	O (Outstanding)	10
Below 90% but not less than 80% ($\geq$ 80% , $<$ 90%)	A ⁺ (Excellent)	9
Below 80% but not less than 70% ($\geq$ 70% , $<$ 80%)	A (Very Good)	8
Below 70% but not less than 60% ($\geq$ 60% , $<$ 70%)	B ⁺ (Good)	7
Below 60% but not less than 50% ($\geq$ 50% , $<$ 60%)	B (above Average)	6
Below 50% but not less than 40%	C	5

($\geq 40\%$, $< 50\%$)	(Average)	
Below 40% ($< 40\%$)	F (FAIL)	0
Absent	AB	0

- 12.3 A student obtaining F Grade in any Subject shall be considered 'failed' and will be required to reappear as 'Supplementary Candidate' in the End Semester Examination (SEE), as and when offered. In such cases, his Internal Marks (CIE Marks) in those Subject(s) will remain same as those he obtained earlier.
- 12.4 A Letter Grade does not imply any specific % of Marks.
- 12.5 In general, a student shall not be permitted to repeat any Subject/Course (s) only for the sake of 'Grade Improvement' or 'SGPA/CGPA Improvement'. However, he has to repeat all the Subjects/Courses pertaining to that Semester, when he is detained.
- 12.6 A student earns Grade Point (GP) in each Subject/Course, on the basis of the Letter Grade obtained by him in that Subject/Course (excluding Mandatory non-credit Courses). Then the corresponding 'Credit Points' (CP) are computed by multiplying the Grade Point with Credits for that particular Subject/Course.

Credit Points (CP) = Grade Point (GP) x Credits For a Course

- 12.7 The Student passes the Subject/Course only when he gets $GP \geq 4$ (P Grade or above).
- 12.8 The Semester Grade Point Average (SGPA) is calculated by dividing the Sum of Credit Points (ΣCP) secured from ALL Subjects/Courses registered in a Semester, by the Total Number of Credits registered during that Semester. SGPA is rounded off to TWO Decimal Places. SGPA is thus computed as

$SGPA = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\}$ For each Semester,

where 'i' is the Subject indicator index (takes into account all Subjects in a Semester), 'N' is the no. of Subjects 'REGISTERED' for the Semester (as specifically required and listed under the Course Structure of the parent Department), C_i is the no. of Credits allotted to that ix Subject, and G_i represents the Grade Points (GP) corresponding to the Letter Grade awarded for that i Subject.

Illustration of Computation of SGPA

Course	Credit	Grade Letter	Grade Point	Credit Point (Credit x Grade)
Course1	3	A	8	$3 \times 8 = 24$
Course2	4	B+	7	$4 \times 7 = 28$
Course3	3	B	6	$3 \times 6 = 18$
Course4	3	O	10	$3 \times 10 = 30$
Course5	3	C	5	$3 \times 5 = 15$
Course6	4	B	6	$4 \times 6 = 24$

Thus, **$SGPA = 139/20 = 6.95$**

- 12.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student over all Semesters considered for registration. The CGPA is the ratio of the Total Credit Points secured by a student in ALL registered Courses in ALL Semesters, and the Total Number of Credits registered in ALL the Semesters. CGPA is rounded off to TWO Decimal Places. CGPA is thus computed from the I Year Second Semester onwards, at the end of each Semester, as per the formula

$CGPA = \{\sum_{j=1}^M C_j G_j\} / \{\sum_{j=1}^M C_j\}$... for all S Semesters registered

(i.e., up to and inclusive of S Semesters, $S \geq 2$),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all

Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the j th Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that j th Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

For CGPA Computation

Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6
Credits : 20 SGPA : 6.9	Credits : 22 SGPA : 7.8	Credits : 25 SGPA : 5.6	Credits : 26 SGPA : 6.0	Credits : 26 SGPA : 6.3	Credits : 25 SGPA : 8.0

$$\text{Thus, CGPA} = \frac{20 \times 6.9 + 22 \times 7.8 + 25 \times 5.6 + 26 \times 6.0 + 26 \times 6.3 + 25 \times 8.0}{144} = 6.73$$

- 12.10 For Merit Ranking or Comparison Purposes or any other listing, ONLY the 'ROUNDED OFF' values of the CGPAs will be used.
- 12.11 For Calculations listed in Item 12.6–12.10, performance in failed Subjects/Courses (securing F Grade) will also be taken into account, and the Credits of such Subjects/Courses will also be included in the multiplications and summations. However, Mandatory Courses will not be taken into consideration.
- 12.12 Conversion formula for the conversion of GPA into indicative percentage is
 $[\text{CGPA Earned} - 0.50] \times 10 = \% \text{ of marks scored}$

13. DECLARATION OF RESULTS

Computation of SGPA and CGPA are done using the procedure listed in 12.6– 2.10.

No SGPA/CGPA is declared, if a candidate is failed in any one of the courses of a given Semester.

14. WITH HOLDING OF RESULTS

If the student has not paid fees to College at any stage, or has pending dues against his name due to any reason what so ever, or if any case of indiscipline is pending against him, the result of the student may be withheld, and he will not be allowed to go into the next higher Semester. The Award or issue of the Degree may also be withheld in such cases.

15. REVALUATION

Students shall be permitted for revaluation after the declaration of end Semester examination results within due dates by paying prescribed fee. After revaluation if there is any betterment in the grade, then improved grade will be considered. Otherwise old grade shall be retained.

16. SUPPLEMENTARY EXAMINATIONS

- 16.1 Supplementary examinations for the odd Semester shall be conducted with the regular examinations of even Semester and vice versa, for those who appeared and failed or absent in regular examinations. Such candidates writing supplementary examinations may have to write more than one examination per day.

16.2 ADVANCED SUPPLEMENTARY EXAMINATION

Advanced supplementary examinations will be conducted for IV year II Semester after announcement of regular results.

17. TRANSCRIPTS

After successful completion of prerequisite credits for the award of degree a Transcript containing performance of all academic years will be issued as a final record. Duplicate transcripts will also be issued if required after the payment of requisite fee and also as per norms in vogue.

18. RULES OF DISCIPLINE

- 18.1 Any attempt by any student to influence the teachers, Examiners, faculty and staff of controller of Examination for undue favours in the exams, and bribing them either for marks or attendance will be treated as malpractice cases and the student can be debarred from the college.
- 18.2 When the student absents himself, he is treated as to have appeared and obtained zero marks in that course(s) and grading is done accordingly.
- 18.3 When the performance of the student in any subject(s) is cancelled as a punishment for indiscipline, he is awarded zero marks in that subject(s).
- 18.4 When the student's answer book is confiscated for any kind of attempted or suspected malpractice the decision of the Examiner is final.

19. MALPRACTICE PREVENTION COMMITTEE

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convenor
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

20. TRANSITORY REGULATIONS

Student who has discontinued for any reason, or has been detained for want of attendance or lack of required credits as specified, or who has failed after having undergone the Degree Programme, may be considered eligible for readmission to the same Subjects/Courses (or equivalent Subjects/Courses, as the case may be), and same Professional Electives/Open Electives (or from set/category of Electives or equivalents suggested, as the case may be) as and when they are offered (within the time-frame of 8 years from the Date of Commencement of his I Year I Semester).

21. AMENDMENTS TO REGULATIONS

The Academic Council of MLR Institute of Technology reserves the right to revise, amend, or change the regulations, scheme of examinations, and / or syllabi or any other policy relevant to the needs of the society or industrial requirements etc., without prior notice.

22. STUDENT TRANSFERS

There shall be no Branch transfers after the completion of Admission Process. Transfer of students from other colleges or universities are permitted subjected to the rules and regulations of TSCHE (TE Department) and JNTUH in vogue.

23. GRADUATION DAY

The College shall have its own Annual Graduation Day for the award of Degrees issued by the College/University.

24. AWARD OF MEDALS

Institute will award Medals to the outstanding students who complete the entire course in the first attempt within the stipulated time.

25. SCOPE

- i) Where the words “he”, “him”, “his”, occur in the write-up of regulations, they include “she”, “her”.
- ii) Where the words “Subject” or “Subjects”, occur in these regulations, they also imply “Course” or “Courses”.
- iii) The Academic Regulations should be read as a whole, for the purpose of any interpretation.
- iv) In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of the Academic Council is final.

Academic Regulations for B. Tech. (Lateral Entry Scheme)

(Effective for the students getting admitted into II year from the Academic Year 2018-2019 on wards)

- 1. The Students have to acquire 120 credits from II to IV year of B.Tech Programme (Regular) for the award of the degree.
- 2. Students, who fail to fulfil the requirement for the award of the degree in 6 consecutive academic years from the year of admission, shall forfeit their seat.
- 3. The same attendance regulations are to be adopted as that of B. Tech. (Regular)
- 4. **Promotion Rule:**
- 5. A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total of 50% Credits up to III Year II Semester, from all the regular and supplementary examinations.
- 6. **Award of Class:**
After the student has satisfied the requirements prescribed for the completion of the programme and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

The marks obtained for 120 credits will be considered for the calculation of CGPA and award of class shall be shown separately.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA $\geq$ 8.00
First Class	$\geq$ 6.50 to $<$ 8.00 CGPA
Second Class	$\geq$ 5.50 to $<$ 6.50 CGPA
Pass Class	$\geq$ 5.00 to $<$ 5.50 CGPA
FAIL	CGPA $<$ 5

7. All other regulations as applicable for B. Tech. Four-year degree programme (Regular) will hold good for B.Tech (Lateral Entry Scheme).

MALPRACTICES RULES- DISCIPLINARY ACTION FOR IMPROPER CONDUCT IN EXAMINATIONS

S. No	Nature of Malpractices / Improper Conduct	Punishment
1 (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the Principal.

3	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practical's and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.
4	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.

6	Refuses to obey the orders of the Addl. Controller of examinations / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the addl. Controller of examinations or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the addl. Controller of examinations, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat.

9	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year.
11	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that Semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the principal for further action to award suitable punishment.	

COURSE STRUCTURE

Department of Aeronautical Engineering

I B.Tech.- I-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS01	Calculus and Applications	BSC	3	1	-	4	30	70	100
A4BS09	Engineering Physics	BSC	3	1	-	4	30	70	100
A4ES01	Engineering Mechanics	ESC	3	1	-	4	30	70	100
A4ES02	Engineering Graphics and Design	ESC	1	-	4	3	30	70	100
A4BS13	Engineering Physics Lab	BSC	-	-	3	1.5	30	70	100
A4AE64	Workshop and Manufacturing Practices	ESC	1	-	4	3	30	70	100
A4HS03	Social Innovation	HS	-	-	2	1	30	70	100
Total			11	3	13	20.5	210	490	700
Mandatory Course (Non-Credit)									
A4MC04	Technical Seminar-I	MC	-	-	2	-	30	70	100

I B.Tech.- II-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS03	Integral Calculus and Transforms	BSC	3	1	-	4	30	70	100
A4BS11	Chemistry	BSC	4	-	-	4	30	70	100
A4CS01	Programming for Problem Solving	ESC	3	-	-	3	30	70	100
A4HS01	English	HS	2	-	-	2	30	70	100
A4CS02	Programming for Problem Solving Lab	ESC	-	-	4	2	30	70	100
A4BS12	Chemistry Laboratory	BSC	-	-	3	1.5	30	70	100
A4HS02	English Language and Communication Skills Lab	HS	-	-	2	1	30	70	100
A4HS04	Engineering Exploration	HS	-	-	2	1	30	70	100
Total			12	1	11	18.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC05	Technical Seminar-II	MC	-	-	2	-	30	70	100

II B.Tech.- I-Semester

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4AE01	Introduction to Aerospace Engineering	PCC	3	-	-	3	30	70	100
A4ME01	Mechanics of Solids	PCC	3	-	-	3	30	70	100
A4AE02	Mechanics of Fluids	PCC	3	-	-	3	30	70	100
A4ME04	Thermodynamics	ESC	3	-	-	3	30	70	100
A4EE64	Basic Electrical and Electronics Engineering	ESC	3	-	-	3	30	70	100
A4AE03	Mechanics of solids and Fluids Lab	PCC	-	-	3	1.5	30	70	100
A4EE65	Basic Electrical and Electronics Engineering Lab	ESC	-	-	3	1.5	30	70	100
A4AE04	Aircraft Engineering Drawing Lab	ESC	-	-	3	1.5	30	70	100
Total			15	-	9	19.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC01	Environmental Studies	MC	2	-	-	-	30	70	100

II B.Tech.- II-Semester

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS06	Numerical methods and probability statistics	BSC	3	-	-	3	30	70	100
A4AE06	Aircraft Production Technology	PCC	3	-	-	3	30	70	100
A4AE07	Aerodynamics-I	PCC	3	-	-	3	30	70	100
A4AE08	Aerospace Vehicle Structures - I	PCC	3	-	-	3	30	70	100
A4AE09	Flight Mechanics-I	PCC	3	-	-	3	30	70	100
A4AE10	Aerodynamics Lab	PCC	-	-	3	1.5	30	70	100
A4AE11	Aircraft Production Technology Lab	ESC	-	-	3	1.5	30	70	100
A4AE12	Aircraft Interior design Lab	PCC	-	-	3	1.5	30	70	100
Total			15	-	9	19.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC02	Gender Sensitization	MC	-	-	2	-	30	70	100

III B.Tech.- I-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4AE13	Aerospace Vehicle Structures-II	PCC	3	-	-	3	30	70	100
A4AE14	Aerodynamics- II	PCC	3	-	-	3	30	70	100
A4AE15	Flight mechanics - II	PCC	3	-	-	3	30	70	100
A4AE16	Aerospace Propulsion - I	PCC	3	-	-	3	30	70	100
	OPEN ELECTIVE - I	OEC	3	-	-	3	30	70	100
A4AE20	Propulsion and Flight Simulation Lab	PCC	-	-	3	1.5	30	70	100
A4AE21	Aerospace Vehicle Structures Lab	PCC	-	-	3	1.5	30	70	100
A4AE67	Aircraft Dynamics and Control Systems Lab	PCC	-	-	2	1	30	70	100
A4AE22	Summer Internship	PWC	-	-	-	2	--	100	100
Total			15	-	8	21	240	660	900
Mandatory Course (Non-Credit)									
A4MC06	Essence of Indian Traditional Knowledge	MC	2	-	-	-	30	70	100

III B.Tech.- II-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4AE23	Aero Materials	BSC	3	-	-	3	30	70	100
A4AE68	Aerospace Propulsion - II	PCC	3	-	-	3	30	70	100
	OPEN ELECTIVE-II	OEC	3	-	-	3	30	70	100
	PROFESSIONAL ELECTIVE - I	PEC	3	-	-	3	30	70	100
	PROFESSIONALELECTIVE - II	PEC	3	-	-	3	30	70	100
A4AE36	Aero Materials Lab	BSC	-	-	3	1.5	30	70	100
A4AE37	Computational Structural Analysis Lab	PCC	-	-	3	1.5	30	70	100
A4HS07	Advanced English Communication Skills Lab	HS	-	-	3	1.5	30	70	100
A4AE38	Comprehensive viva	PWC	-	-	-	1	--	100	100
Total			15	-	9	20.5	240	660	900
Mandatory Course (Non-Credit)									
A4MC03	Constitution of India	MC	2	-	-	-	30	70	100

IV B.Tech.- I-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4AE24	Flight Vehicle Design	PCC	3	-	-	3	30	70	100
A4AE39	Heat Transfer & Computational Fluid Dynamics	PCC	3	-	-	3	30	70	100
	OPEN ELECTIVE-III	OEC	3	-	-	3	30	70	100
	PROFESSIONAL ELECTIVE - III	PEC	3	-	-	3	30	70	100
	PROFESSIONAL ELECTIVE - IV	PEC	3	-	-	3	30	70	100
A4ME24	Heat Transfer Lab	PCC	-	-	3	1.5	30	70	100
A4AE51	Computational Fluid Dynamics Lab	PCC	-	-	3	1.5	30	70	100
A4AE52	Flight Vehicle Design Lab	PCC	-	-	3	1.5	30	70	100
A4AE53	Project Phase-I	PWC	-	-	8	4	100	--	100
Total			15	-	15	23.5	340	560	900
IV B.Tech.- II-Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
	Open Elective-IV	HS	3	-	-	3	30	70	100
	PROFESSIONAL ELECTIVE – V	PEC	3	-	-	3	30	70	100
	PROFESSIONAL ELECTIVE – VI	PEC	3	-	-	3	30	70	100
A4AE66	Project Phase-II	PWC	-	-	16	8	50	150	200
Total			9		16	17	140	360	500

PROFESSIONAL ELECTIVES			
PE-I		PE-II	
A4AE28	Additive Manufacturing	A4AE32	Finite Element Method
A4AE29	CAD/CAM	A4AE33	Turbo Machines
A4AE30	Space Technology	A4AE34	Geometric Dimensioning and Tolerances (GD&T)
A4AE31	Flight scheduling and operations	A4AE35	Optimization techniques
PE-III		PE-IV	
A4AE43	Mechanical Vibrations and Structural Dynamics	A4AE47	Aero Elasticity
A4AE44	Fatigue and Fracture Mechanics	A4AE48	Mechanisms and Machine Design
A4AE45	Rockets and Missiles	A4AE49	Avionics and Flight control Systems
A4AE46	Numerical techniques	A4AE50	Material characterization
PE-V		PE-VI	
A4AE55	Advanced Computational Fluid Dynamics	A4ME38	Product Life Cycle Management
A4AE56	Helicopter Engineering	A4AE60	Airport Management
A4AE57	Propellant Technology	A4AE61	Experimental Stress Analysis
A4AE58	Aero engine repair and maintenance	A4AE62	Fundamental of combustion

OPEN ELECTIVE COURSES

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1	A4AE17	Fabrication Processes	Aeronautical Engineering
2	A4AE18	Fundamentals of Avionics	
3	A4AE19	Introduction to jets and rockets	
4	A4CS43	Fundamentals of Databases	Computer Science and Engineering
5	A4CS44	Introduction to C++	
6	A4CS45	Core Java Programming	
7	A4EC59	Logic Design	Electronics & Communication Engineering
8	A4EC60	Principles of Communications	
9	A4EC61	Measurements and Instruments	
10	A4EE52	Electrical Wiring and Safety Measures	Electrical & Electronics Engineering
11	A4EE53	Electrical Materials	
12	A4EE54	New trends Electrical Energy	
13	A4IT11	Fundamentals of Data Structures	Information Technology
14	A4IT12	Software Engineering Principles	
15	A4IT13	Computer Organization and Operating Systems	
16	A4ME60	Elements Of Mechanical Engineering	Mechanical Engineering
17	A4ME61	Fundamentals Of Engineering Materials	
18	A4ME62	Fundamentals Of Theory Of Machines	
19	A4HS10	Business Economics and Financial Analysis	HS
20	A4HS11	Basics of Entrepreneurship	
21	A4HS12	Human Values and Professional Ethics	

OPEN ELECTIVE COURSE-II			
S. No.	Course Code	Course Name	Offering Department
1	A4AE25	Introduction to aircraft Industry	Aeronautical Engineering
2	A4AE26	Non destructive testing Methods	
3	A4AE27	Fundamentals of Finite element method	
4	A4CS46	Introduction to Computer Networks	Computer Science and Engineering
5	A4CS47	Introduction to Linux Programming	
6	A4CS48	Fundamentals of Artificial Intelligence	
7	A4EC62	Fundamentals of Integrated Circuits	Electronics & Communication Engineering
8	A4EC63	Signal Transmission through linear systems	
9	A4EC64	Fundamentals of VLSI Design	
10	A4EE55	Power Plant Engineering	Electrical & Electronics Engineering
11	A4EE56	Analysis of Linear Systems	
12	A4EE57	Neural Networks and Fuzzy Logic	
13	A4IT14	Basics of Python Programming	Information Technology
14	A4IT15	Human Computer Interaction	
15	A4IT16	Software Testing Fundamentals	
16	A4ME63	Fundamentals of Mechatronics	Mechanical Engineering
17	A4ME64	Basics Of Thermodynamics	
18	A4ME65	Fundamentals of Smart Materials	
19	A4HS13	Advanced Entrepreneurship	HS

OPEN ELECTIVE COURSE-III			
S. No.	Course Code	Course Name	Offering Department
1	A4AE40	Guidance and control of aerospace vehicles	Aeronautical Engineering
2	A4AE41	Wind tunnel Techniques	
3	A4AE42	Introduction to Aerospace Technology	
4	A4CS49	Soft Computing	Computer Science and Engineering
5	A4CS50	Introduction to Cloud Computing	
6	A4CS51	Web Programming	
7	A4EC65	Introduction of Microprocessors and Microcontrollers	Electronics & Communication Engineering
8	A4EC66	Fundamentals of Image processing	
9	A4EC67	TV Engineering	
10	A4EE58	Electrical Engineering Materials	Electrical & Electronics Engineering
11	A4EE59	Non-Conventional Power Generation	
12	A4EE60	Solar Energy and Applications	
13	A4IT17	E- Commerce	Information Technology
14	A4IT18	Cyber Forensics	
15	A4IT19	Discrete Structures	
16	A4ME66	Fundamentals of Operation Research	Mechanical Engineering
17	A4ME67	Fundamentals of Energy Sources and Applications	
18	A4ME68	Basics of Robotics	

19	A4HS14	Indian Ethos & Business Ethics	HS
OPEN ELECTIVE-IV			
S. No.	Course Code	Course Name	Offering Department
1	A4HS15	Management Science	HS
2	A4HS16	Intellectual Property Rights	
3	A4HS17	Number Theory	
4	A4HS18	Physics and Technology of Thin films	
5	A4HS19	Polymer chemistry	
7	A4CS52	Introduction to Network Security	Computer Science and Engineering
8	A4CS53	Data Analytics	
9	A4CS54	.NET & C# Programming	
10	A4EC68	Introduction to signal Processing	Electronics & Communication Engineering
11	A4EC69	Introduction to mobile communication	
12	A4EC70	Basic Embedded systems Design	
13	A4EE61	Instrumentation and Control	Electrical & Electronics Engineering
14	A4EE62	Energy Audit and Management Systems	
15	A4EE63	Energy Storage Systems	
16	A4IT20	Introduction to Mobile Application Development	Information Technology
17	A4IT21	Distributed Databases	
18	A4IT22	Big Data	
19	A4ME69	Introduction to Material Handling	Mechanical Engineering
20	A4ME70	Non Conventional Energy Sources	
21	A4ME71	Aspects of Heat Transfer	

I B.TECH I SEMESTER SYLLABUS

CALCULUS AND APPLICATIONS

I Semester:

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4BS01	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES:

To learn

1. Concept of Rank of a matrix, Consistency and solving system of linear equations.
2. The concept of differential equations and solve them using appropriate methods.
3. Usage of the appropriate test to find the convergence and divergence of the given series.
4. Evaluation of length, areas & volumes of different curves of revolution.
5. The partial derivatives of several variable functions.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Solve the system of linear equations using rank of the matrices.
2. Identify the different types of differential equations and solve them using appropriate methods.
3. Apply the appropriate test to find the convergence and divergence of the given series.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT-I	THEORY OF MATRICES	Classes: 10
Finding rank of a matrix by reducing to Echelon form, Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix, Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices.		
UNIT-II	ORDINARY DIFFERENTIAL EQUATIONS	Classes: 10
Introduction- Exact and reducible to Exact differential equations -Equations not of first degree: Equations solvable for p-equations, solvable for y- equations and solvable for x and Clairaut's type. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters.		
UNIT-III	SEQUENCES AND SERIES	Classes: 08
Basic definitions of Sequences and series – Convergence and divergence –Comparison Test- Ratio Test – Raabe's Test-Integral Test – Cauchy's nth root Test –Absolute and Conditional convergence – Power Series.		
UNIT-IV	BETA GAMMA FUNCTIONS AND APPLICATIONS OF DEFINITE INTEGRALS	Classes: 10
Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions. Application of definite integrals: Lengths, evaluate surface areas and volumes of revolution of curves (only in Cartesian co-ordinates).		

UNIT-V	CALCULUS OF SEVERAL VARIABLES	Classes: 10
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative –Chain rule, Jacobians -functional dependence & independence.Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints)		
Text Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
1.G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2.N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3.E.L.Ince, Ordinary differential Equations,Dover publications,1958.		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
1. https://www.e-booksdirectory.com/details.php?ebook=10166 2. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

ENGINEERING PHYSICS

I Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Understand the Optical phenomenon of light and gain knowledge in diffraction, interference of light. 2. Gain the knowledge of carrier concentration and recombination process of semiconductor materials. 3. Learn the basic principles of laser and optical fiber. 4. Know the origin of Electrical and Magnetic properties of various materials 5. Understand the acceleration as a function of displacement by using SHM and damped oscillations.								
COURSEOUTCOMES: The student will able to: 1. Analyze the Optical phenomenon of light and estimate their applications in different situations 2. Evaluate the mobility of charge carrier concentration of a given semiconductor material. 3. Justify how the graded index optical fiber is more efficient than step index optical fiber in fiber optic communication system. 4. Classify various magnetic, dielectric materials and can apply knowledge in various fields 5. Describe the behaviour of waves at interfaces (impedance) and their behaviour in dissipative media (damping)								
UNIT-I	WAVE OPTICS						Classes: 08	
Huygen's principle, Superposition of waves and interference of light by wave front splitting and amplitude splitting, Young's double slit experiment, Newton's rings, Michelson's interferometer, Mach-Zehnder interferometer, Fraunhofer diffraction from a single slit and circular aperture, Diffraction grating: Grating spectrum and resolving power.								
UNIT-II	QUANTUM MECHANICS						Classes: 06	
Introduction to quantum physics: Black body radiation, Planck's law, photoelectric effect Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.								
UNIT-III	LASER & OPTICAL FIBER						Classes: 10	
Laser: Characteristics of Laser beams, Energy levels in atoms, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: gas lasers -He-Ne, solid-state lasers -ruby, Neodymium, Applications of Lasers. Fiber Optic Communication: Structure of Optical fibers, Basic principle of fibers, Acceptance angle and Numerical aperture, Types of Optical Fibers-Step Index and Graded Index fibers; Modes of fibers-SMSI, MMSI, MMGI., Optical fiber Communication System with block diagram. Applications of fibers, fiber optic sensors – Basic principle, Intrinsic, Extrinsic sensors. Working of Pressure and Temperature Sensors.								

UNIT-IV	DIELECTRIC AND MAGNETIC PROPERTIES OF MATERIALS	Classes: 08
Dielectrics: Introduction, Types of polarizations (Electronic, Ionic and Orientation Polarizations) and calculation of their polarizabilities, Internal fields in a solid, Clausius-Mossotti relation. Magnetism: Introduction, Bohr magneton, classification of dia, para and ferro magnetic materials on the basis of magnetic moment, Hysteresis curve based on domain theory, Soft and hard magnetic materials, Properties of anti-ferro and ferri magnetic materials.		
UNIT-V	INTRODUCTION TO OSCILLATIONS	Classes: 07
Harmonic oscillators: Mechanical and electrical simple harmonic oscillators, Damped harmonic oscillator – heavy, critical and light damping, energy decay in a damped harmonic oscillator, quality factor. Non-dispersive transverse and longitudinal waves in one dimension Transverse wave on a string, the wave equation on a string, Harmonic waves, reflection and transmission of waves at a boundary, impedance matching, standing waves and their eigenfrequencies, longitudinal waves and the wave equation for them, acoustics waves and speed of sound, standing sound waves.		
Text Books:		
1. A textbook of Engineering Physics, Dr. M.N. Avadhanulu, Dr. P.G.Kshirsagar – S.Chand 2. Haliday and Resnick, Physics-Wiley 3. P K Palanisamy, Engineering Physics, Sitech Publications, 2008. 4. D. J. Griffiths, "Quantum mechanics", Pearson Education, 2014.		
Reference Books:		
1. Richard Robinett, Quantum mechanics. 2. O.Svelto, Principles of lasers. 3. Bottaccini M.R, E.E. Merill, Instruments and Measurements, Bell and Howell 4. I. G. Main, "Vibrations and waves in physics", Cambridge University Press, 1993. 5. Introduction to Solid State Physics by Charles Kittel, Wiley student edition 6. Ajoy Ghatak, "Optics", Mc Graw-Hill Education, 2012		
Web References:		
1. https://www.edx.org/course?search_query=semiconductor+physics 2. https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x 3. https://www.edx.org/course/physics-electronic-polymers-pep-purdue-nano600		
E-Text Books:		
1. https://www.researchgate.net/publication/274876688_Vibrations_and_Waves_in_Physics. 2. http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010_Fall/classnotes/NanoB_week14.pdf 3. https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf 4. http://www.hailienene.com/resources/nano-technology.pdf 5. https://books.google.co.in/books?id=erjymgEACAAJ&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. http://nptel.ac.in/noc/individual_course.php?id=noc16-ch06 2. http://nptel.ac.in/courses/104104085/2 (Lasers and its applications)		

ENGINEERING MECHANICS

I Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ES01	ESC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Resolve various force systems and calculate reactions of various supports on rigid bodies in equilibrium 2. Analyze the frictional forces of rigid bodies on rough horizontal and inclined planes. 3. Evaluate Geometric properties of composite areas and solids. 4. Analyze rigid bodies in motion, work-energy problems and their relationship to engineering applications 5. Develop mathematical model of dynamic systems subjected to simple harmonic motion.								
COURSE OUTCOMES: At the end of the course the student should be able to: 1. Evaluate magnitude, direction and position of resultant of different system of forces. 2. Analyze free body diagrams of rigid bodies which are in equilibrium 3. Calculate the centroid and centre of gravity of composite areas and solids. 4. Evaluate moment of inertia of plane areas and composite bodies. 5. Analyze the frictional forces of rigid bodies on rough horizontal and inclined planes.								
UNIT-I	INTRODUCTION TO ENGINEERING MECHANICS & EQUILIBRIUM OF PLANAR SYSTEM OF FORCES						Classes: 15	
Introduction to engineering mechanics: Basic concepts, Classification of a force system, Parallelogram law, Resultant of coplanar concurrent force system, Moment of a force, its applications, Varignon's theorem, Resultant of coplanar non-concurrent force systems, Couples. Equilibrium of Planar System of forces: Equilibrium, Conditions of equilibrium, Lami's theorem, free body diagrams, Types of supports, their reactions, Analysis of beams.								
UNIT-II	ANALYSIS OF PERFECT FRAMES & FRICTION						Classes: 12	
Analysis of perfect frames: Types of frames, Method of joints and Method of Sections Friction: Introduction, types of friction, laws of friction, coefficient of friction, angle of friction, angle of repose, analysis of bodies on rough horizontal, inclined planes, wedge friction, ladder friction and screw friction.								
UNIT-III	CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA						Classes: 12	
Centroid & Centre of gravity: Introduction, centroids & centre of gravity of simple figures (from first principles), centroid of composite sections, centre of gravity of composite solids, theorems of pappus & guldinus. Moment of Inertia: Area moment of inertia of plane sections (from first principles), Parallel axis theorem, Perpendicular axis theorem, Moment of inertia of standard sections and composite sections. Mass moment of inertia of circular disc, cylinder and sphere.								
UNIT-IV	KINEMATICS & KINETICS						Classes: 11	
Kinematics of a particle: Rectilinear motion, curvilinear motion, direct and oblique impact. Kinetics of rigid bodies: Analysis of connecting bodies, kinetics of rigid body rotation, D' Alembert's principle, work-energy theorem, impulse-momentum principle, their applications								

UNIT-V	VIRTUAL WORK & MECHANICAL VIBRATIONS	Classes: 10
Virtual work: Concept of virtual work, Principle of virtual work, its applications Mechanical Vibrations: Basic terminology, free and forced vibrations, resonance, simple harmonic motion, simple, compound and torsional pendulums.		
Text Books:		
1.Reddy Vijay Kumar and K. Suresh Kumar, Singer's Engineering Mechanics 2.Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company 3.Bansal R.K, A Text book of Engineering Mechanics, Laxmi Publications		
Reference Books:		
1.Engineering Mechanics/Irving Shames/Prentice Hall 2.Engineering Mechanics/N.H Dubey/Tata Mc Graw Hill 3.Engineering Mechanics/D. P Sharma/ Pearson 4.Engineering Mechanics /M. V Seshagiri Rao & D Rama Durgaiah / University Press 5.Tayal A.K, Engineering Mechanics, Umesh Publications 6. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3 rd Edition, 2014.		
Web References:		
1. http://nptel.ac.in/courses/112103109/ 2. http://www.iitg.ac.in/rkbc/me101/Presentation/L01-03.pdf 3. https://www.udemy.com/myengineeringmechanics/		
E-Text Books:		
1. https://www.pdfdrive.net/engineering-mechanics-e15629836.html 2. https://www.pdfdrive.net/engineering-mechanics-dynamics-si-version-e4362556.html 3. https://www.pdfdrive.net/engineering-mechanics-statics-3rd-ed-e4229691.html		
MOOC Course		
1. https://www.coursera.org/learn/engineering-mechanics-statics 2. https://www.edx.org/course/engineering-mechanics		

ENGINEERING GRAPHICS & DESIGN**I Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ES02	Engineering science courses	L	T	P	C	CIE	SEE	Total
		1	0	4	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Create awareness and emphasize the need for Engineering Drawing in various branches of engineering.
2. Enable the student with various concepts of dimensioning, conventions and standards related to engineering drawings.
3. Follow the basic drawing standards and conventions.
4. Develop skills in three-dimensional visualization of engineering component.

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Sketch the various curves used in engineering and their applications
2. Apply the knowledge of quadrant system and say to which quadrant and angle of project the object belongs.
3. Evaluate the given object position and draw the projections of objects
4. Analyze the given sectioned objects like in sheet metal applications.
5. Develop the new drawings for the industry requirements

UNIT-I	INTRODUCTION	Classes: 7
Introduction to Engineering Drawing covering: Principles of Engineering Graphics and their significance, usage of drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute.		
UNIT-II	DRAWING OF PROJECTIONS OR VIEWS: ORTHOGRAPHIC PROJECTION IN FIRST ANGLE PROJECTION ONLY	Classes: 10
Principles of orthographic projections – conventions – first and third angle projections. Projections of points- Projection of lines inclined to both the planes. PROJECTIONS OF PLANES: Projections of regular planes, inclined to both planes.		
UNIT-III	INTRODUCTION TO COMPUTER AIDED DRAFTING	Classes: 8
INTRODUCTION TO COMPUTER AIDED DRAFTING: Generation of points, lines, curves, polygons, simple solids, dimensioning. LAYERS: Concept of layers, working with layers, creating, display, locking, unlocking, and delete commands, Coordinate system in AutoCAD, UCS, WCS, MCS PROJECTION OF SOLIDS-Solids inclined to both planes(Auxiliary plane method) DEVELOPMENT OF SURFACES OF SOLIDS: Theory of development, development of lateral surface along with base		

UNIT-IV	ISOMETRIC DRAWINGS	Classes: 05
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections; Drawing Isometric circles, Dimensioning Isometric Objects; Conversion of Isometric view to Orthographic views and Orthographic to isometric views		
UNIT-V	3D MODELING	Classes: 04
Types of 3D models, 3D Coordinate Systems, basic commands in 3D, PEDIT command. CREATING SOLID MODELS: creating pre-defined Solid Primitives, Dynamic UCS, methods of creating solids by - Extrude, Revolve, Swept, Loft, & Presspull, in 3Dcreating solid models, Dynamic UCS. MODIFYING 3D OBJECTS: Fillet, Chamfer, Rotate, Mirror, Array, and Slicing solid Models. EDITING 3D OBJECTS: SOLVIEW, SOLDRAW, SOLPROF,		
Text Books:		
1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House 2. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education 3. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication 4. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers . 5. D.M. Kulkarni,A.P.Rastogi,A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications, 2013		
Reference Books:		
1. Johle (2009), Engineering Drawing, Tata Mc Graw Hill, New Delhi, India. 2. Trymbaka Murthy (2007), Computer Aided Engineering Drawing, I.K. International Publishers, New Delhi. 3. Sham Tickoo, D. saravanan, “AutoCAD 2010 for engineers and designers” Dreamtech Press, 2010 2. 4. Sham Tickoo “ AutoCAD 2011: A Problem solving approach” Autodesk Press, USA.		
Web References:		
1. nptel.ac.in/courses/112103019/ 2. web.iitd.ac.in/~achawla/public_html/201/lectures/sp46.pdf		
E-Text Books:		
https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing_A_Textbook		
https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing .		

ENGINEERING PHYSICS LABORATORY

I Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS13	BSC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100
COURSEOBJECTIVES:								
The course should enable the students to:								
1. To provide an experimental foundation for the theoretical concepts introduced in the lectures. 2. To teach how to make careful experimental observations and how to think about and draw conclusions from such data. 3. To help students understand the role of direct observation in physics and to distinguish between inferences based on theory and the outcomes of experiments. 4. To introduce the concepts and techniques which have a wide application in experimental science but have not been introduced in the standard courses. 5. To teach how to write a technical report which communicates scientific information in a clear and concise manner.								
COURSEOUTCOMES:								
The student will be able to:								
1. To make careful experimental observations and draw conclusions from such data. 2. To distinguish between inferences based on theory and the outcomes of experiments. 3. To write a technical report which communicates scientific information in a clear and concise manner.								
LIST OF EXPERIMENTS								
Week-1	DISPERSIVE POWER OF PRISM-SPECTROMETER							
a. Find the dispersive power of a material of prism using Spectrometer for angle of minimum deviation. b. Determine the angular rotation for a given wavelength for a given prism. What is your analysis? c. Using minimum deviation method how do you find the dispersion of a given prism. Analyze the results. d. Let the angle of the prism is 50 degree and refractive index of the prism is 1.33. Find the angle of minimum deviation. e. Define refractive index and minimum deviation. Evaluate the dispersive power of prism for blue and green colour.								
Week-2	SOLAR CELL							

- a. Study the V-I characteristics of a solar cell for the voltage interval of 0.4 V when the given electric bulb is at a distance of 10cm. Will there be any change if the distance is doubled?
- b. Study the P-V characteristics of a solar cell for the voltage interval of 0.3 V for four different distances of the given electric bulb. Discuss your observations.
- c. Study the P-I characteristics of a solar cell for the given electric bulb for the distances 30 cm. What is expected if the distance is doubled?

Week-3	OPTICAL FIBER
To determine the numerical aperture and acceptance angle of an optical fiber.	
Week-4	NEWTON'S RINGS
To determine the radius of curvature of a given convex lens by using Newton's Rings apparatus.	
Week-5	ENERGY GAP OF PN JUNCTION DIODE
a. Using PN junction diode, determine the energy gap of a semiconductor by applying 2V of reverse bias by increasing the temperature from 30-70 degree centigrade. Analyze the obtained results. b. Evaluate the energy gap of Si PN junction diode by applying 1.5V of reverse bias for values of temperature 70,60,50,40,30 centigrade. c. Evaluate the energy gap of a semiconductor by applying 0.5V of reverse bias for temperatures at 75, 65, 55,45,35 centigrade.	
Week-6	STEWART & GEE'S EXPERIMENT
a. Study the magnetic field along the axis of a current carrying coil of 500 turns for a constant current for the distances 0, 10,15 cm respectively. Analyze the graph obtained. b. Study the magnetic field along the axis of a current carrying coil of 50 turns by applying current 100mA for the distances 0,12,25 cm. Evaluate the result when the no. of turns are doubled.	
Week-7	LCR CIRCUIT
a. To determine the Resonance frequency and Quality factor of a LCR Circuit b. Study the Resonance frequency of an electrical circuit by varying values of inductance, Capacitance and resistors and analyze the results.	
Week-8	MALUS LAW
To study Polarization of light and verification of Maluslaw .	
Week-9	RC CIRCUIT
a. Analyze the time constant of R-C circuit by varying Resistance and Capacitance values in a electrical circuit. b. Determine the time constant of a given RC circuit by plotting a graph between Charging current versus time.	

Week-10	TORSIONAL OSCILLATOR
a. Making use of a torsional oscillator of 300gms circular disc, determine the rigidity modulus of given steel wire for 65 cm & 55 cm lengths. Analyze the results obtained. b. Determine the rigidity modulus of the given copper wire for 50,40,30 lengths by using torsional oscillator of 400gms circular disc. Analyze the results obtained. c. Give your analysis of $L&T^2$ behavior of a torsional oscillator. You may pick your own values for the analysis.	
Week-11	LASER - DIFFRACTION GRATING
a. Using a diffraction grating element of 2500 LPI determine the wavelength of LASER source for first and second order diffraction when the distance between the screen and grating is 50cm. What is your analysis? b. Determine the wavelength of a LASER source for first three orders of diffraction by maintaining a distance of 30 cms between grating material and the screen. Use diffraction grating element of 15000 LPI. What differences do you observe for the three orders.	
Week-12	MELDE'S EXPERIMENT
a. Determine the longitudinal frequency of tuning fork by using 100cm length of the thread by varying masses of 5gms, 10gms. Discuss your findings. b. Findout the transverse frequency of tuning fork by using 80cm length of the thread and by varying masses of 10gms and 40gms. Analyze the results obtained.	
Reference Books:	
1. "Optics, Principles and Applications" by K K Sharma. 2. "Principles of Optics" by M Born and E Wolf. 3. "Semiconductor Physics and Devices: Basic Principles" by Donald A Neamen. 4. "Oscillations and Waves" by Satya Prakash and Vinay Dua. 5. "Waves and Oscillations" by N Subrahmanyam and Brij Lal.	
Web References:	
1. http://www.arxiv.org/pdf/1510.00032 2. http://www.nptel.ac.in/courses/122103010/ 3. http://www.researchgate.net/.../276417736_Video_Presentations_in_Engineering-Ph... 4. http://www.wileyindia.com/engineering-physics-theory-and-practical.html	
Note: Students can perform any 8 experiments.	

WORKSHOP AND MANUFACTURING PRACTICES

I Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE64	Foundation	L	T	P	C	CIE	SEE	Total
		1	-	4	3	30	70	100

COURSE OBJECTIVES:

Student will

1. Get the hands on experience on various trades.
2. Perform various machining operations.
3. Capable to make useful products using one or more operations.
4. Learn various manufacturing processes.

COURSE OUTCOMES:

Student will be able to:

1. Fabricate components with their own hands
2. Get practical knowledge of the dimensional accuracies and tolerances possible with different manufacturing processes
3. Assemble different components
4. Produce small devices of their interest

VIDEO LECTURES

Week-1	Carpentry
Week-2	Fitting operations & Power Tools
Week-3	Casting
Week-4	Forming
Week-5	Machining
Week-6	Joining – Bolted & Riveted Joints
Week-7	Arc welding & Gas welding
Week-8	Brazing & Soldering
Week-9	Advanced Manufacturing methods
Week-10	CNC Machining
Week-11	Additive manufacturing
Week-12	Electrical & Electronics
Week-13	Plastic moulding, Glass cutting
Week-14	Metal casting

LIST OF EXPERIMENTS		
WEEKS	BASIC TRADES	BASIC MANUFACTURING
	Fitting	Machine Shop
Week 1	Filing Four Sides of Work piece	Facing & Step Turning on Lathe
Week 2	L Fit	Milling and Drilling
	Carpentry	Black Smithy
Week 3	Half Lap Joint	Convert round rod to S-hook
Week 4	Dove Tail Joint	Convert round rod to Chisel
	Tin Smithy	Casting
Week 5	Tin Smithy- Prepare a Rectangular Tray	Preparation of Mould Cavity for Multi Piece Pattern
Week 6	Prepare A Square Tin	Casting of Simple pattern
	Electrical	Welding Shop
Week 7	House Wiring Parallel and Series Connection	Lap/Butt joint Using Arc Welding
Week 8	House Wiring Two Way Switch	Lap/Butt joint Using Gas Welding
	Electronics	Plastic Moulding & Glass Cutting
Week 9	Soldering Parallel Connection	Injection moulding of Simple Components
Week 10	Soldering Series Connection	Glass Cutting
Week 11	Revision/Practice	Revision/Practice
Text Books:		
1. Workshop Manual by P. Kannaiah and K. L. Narayana. 2. Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II.		
Reference Books:		
1. HajraChoudhury S.K., HajraChoudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. H. S. Bawa, "Workshop Practice", Tata McGraw-Hill Publishing Company Limited, New Delhi, (2007). 3. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4 th edition, 2002.		
E-Text Books:		
1. https://blogpuneet.files.wordpress.com/2013/07/introduction-to-basic-manufacturing-processes-and-workshop-technology.pdf 2. https://soaneemrana.org/onewebmedia/Manufacturing%20Processes%20By%20H.N.%20Gupta.pdf		
MOOC Course:		
1. https://www.class-central.com/course/edx-fundamentals-of-manufacturing-processes-7224		

SOCIAL INNOVATION								
I Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS03	HS	L	T	P	C	CIE	SEE	Total
		2	-	-	1	30	70	100
COURSE OBJECTIVES:								
The course should enable the students to:								
1. Understand social innovation concepts and approaches.								
2. Understand the community problems, social and economical change.								
3. Identify new and unaddressed social needs.								
4. Analysis of social innovation disclosures in different sectors.								
5. Design innovative solutions with Social impact through application of new models of leadership, collective intelligence and creativity techniques.								
COURSE OUTCOMES:								
1. Increase familiarity with a range of perspectives on social innovation including public sector, philanthropic, civil society, and social service innovation, corporate social innovation, social finance and social entrepreneurship. Understanding of professional and ethical responsibility								
2. Understand how to create a safe container with groups that allows for trust, risk-taking, creativity and new forms of collective action								
3. Engage complex problems in new ways, including observing and mapping systems with greater clarity, diagnosing problems, experimenting and iterating								
4. Situate your work and that of others in a systemic context and see new possibilities for integrative solutions								
5. Know how, when and why to apply social innovation approaches and tools, including systems thinking, design and innovation methods, financial tools, and collaborative processes.								
UNIT-I	INTRODUCTION TO SOCIAL INNOVATION						Classes: 06	
Core definitions, core elements and common features of social innovation, a topology of social innovations, history of social innovation, social and economic change, Swachh Bharat, Unnat Bharat Abhiyan, National Service Scheme (NSS).								
UNIT-II	INTERACTION AND ENGAGEMENT WITH SOCIETY						Classes: 06	
Engage with community, interact with them to understand the community problems, Understanding social and economical change – individuals, organizations and movements.								
UNIT-III	PROCESS OF SOCIAL INNOVATION						Classes: 06	
Understanding the pain/need, description and problem definition, social and economic constraints for affordable and appropriate technology.								
UNIT-IV	SOCIAL INNOVATION ACROSS FOUR SECTORS IN INDIA AND GLOBAL SCENARIO						Classes: 06	

The four sectors – the non-profit sector, public sector, the private sector, the informal sector, links between and cross sectors.		
UNIT-V	SOCIAL INNOVATION – CASE STUDIES	Classes: 08
Designing and implementing social innovations, report writing and documentation, presentation of the case studies with a focus on impact and vision on society.		
Text Books:		
1. The Power of Social Innovation: How Civic Entrepreneurs Ignite Community Networks for Good 1st Edition by Stephen Goldsmith, Michael R. Bloomberg, Gigi Georges, Tim Glynn Burke. 2. The Open Book of Social Innovation: Ways to Design, Develop and Grow Social Innovation Paperback – March, 2010 by Robin Murray, Julia Caulier-Grice, Geoff Mulgan.		
Reference Books:		
1. Social innovator series: ways to design, develop and grow social innovation, the open book of social innovation by robin murrayjuliecaulier-gricegeoffmulgan. 2. The International Handbook on Social Innovation: Collective Action, Social Learning and Transdisciplinary Research Paperback by Frank Moulaert , Diana MacCallum. 3. Guide to Social Innovation by Johannes HAHN and Laszlo ANDOR		
Web References:		
1. http://s3platform.jrc.ec.europa.eu/documents/20182/84453/Guide_to_Social_Innovation.pdf 2. https://www.si-drive.eu/wp-content/uploads/2016/12/SI-DRIVE-CA-short-2016-11-30-Druckversion.pdf		
E-Text Books:		
1. https://epdf.tips/the-power-of-social-innovation-how-civic-entrepreneurs-ignite-community-networks.html 2. https://youngfoundation.org/wp-content/uploads/2012/10/The-Open-Book-of-Social-Innovationg.pdf 3. http://www.idmais.org/desislab/wp-content/media/social.pdf		
MOOC Course		
1. https://iversity.org/en/courses/social-innovation-101-en 2. https://www.class-central.com/tag/social%20innovation 3. https://www.mooc-list.com/tags/social-innovation		

I B.TECH II SEMESTER SYLLABUS

INTEGRAL CALCULUS AND TRANSFORMS

II Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS03	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	30	70	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			

COURSE OBJECTIVES:

To learn

1. Evaluation of the multiple integrals.
2. Finding the integrals in two and three dimensional space over given curve and surface.
3. Concept and application of Laplace transforms.
4. Fourier series for periodic functions.
5. Fourier transform and inverse transform of common functions.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate multiple integrals.
2. Verify vector integral theorems.
3. Solve the differential equations using Laplace transform techniques.
4. Find the Fourier series of the given functions.
5. Find the Fourier transforms of the given functions.

UNIT-I	MULTIPLE INTEGRALS	Classes: 10
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double, centre of mass and gravity.		
UNIT-II	VECTOR CALCULUS	Classes: 11
Scalar and vector point functions - Gradient, divergence, curl and their related properties - Solenoidal and irrotational vector point functions - Scalar potential function - Laplacian operator - Line integral - work done - surface integrals - volume integral - Vector integral theorems - Green's theorem in a plane - Stoke's theorem - Gauss divergence theorem (all theorem statements and their verification).		
UNIT-III	LAPLACE TRANSFORM S	Classes: 12
Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.		
UNIT-IV	FOURIER SERIES	Classes:10
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Even Odd periodic continuation-Half range Fourier sine and cosine expansions.		

UNIT-V	FOURIER TRANSFORMS	Classes: 09
Fourier integral theorem (statement)-Fourier sine and cosine integrals –Fourier transforms –Fourier sine and cosine transforms-properties- Inverse transforms-Finite Fourier transforms.		
Text Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9 th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
1. G.B.Thomas, calculus and analytical geometry,9 th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.		
Web References:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1. https://www.e-booksdirectory.com/details.php?ebook=10166 2. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

CHEMISTRY

II Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS11	BS	L	T	P	C	CIE	SEE	Total
		5	1	-	3	30	70	100
COURSE OBJECTIVES: Student will be able to: 1. Bring adaptability to the concepts of chemistry and to acquire the required skills to become a perfect engineer. 2. To impart the basic knowledge of atomic, molecular and electronic modifications which makes the student to understand the technology based on them. 3. To acquire the knowledge of electrochemistry, corrosion and water treatment which are essential for the Engineers and in industry. 4. To acquire the skills pertaining to spectroscopy and to apply them for medical and other fields. 5. To impart the knowledge of stereochemistry and synthetic aspects useful for understanding reaction pathways.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. The knowledge of atomic, molecular and electronic changes, band theory related to conductivity. 2. The required principles and concepts of electrochemistry, corrosion And in understanding the problem of water and its treatments. 3. The required skills to get clear concepts on basic spectroscopy and application to medical and other fields. 4. The knowledge of configurational and conformational analysis of molecules and reaction mechanisms.								
UNIT-I	MOLECULAR STRUCTURE AND THEORIES OF BONDING						Classes: 10	
Atomic and Molecular orbitals. Linear Combination of Atomic Orbitals (LCAO), molecular orbitals of diatomic molecules, molecular orbital energy level diagrams of N ₂ , O ₂ and F ₂ molecules. π molecular orbitals of butadiene and benzene. Crystal Field Theory (CFT): Salient Features of CFT – Crystal Field Splitting of transition metal ion d- orbitals in Tetrahedral, Octahedral and square planar geometries. Band structure of solids and effect of doping on conductance.								
UNIT-II	WATER AND ITS TREATMENT						Classes: 12	
Introduction – hardness of water – Causes of hardness - Types of hardness: temporary and permanent – expression and units of hardness – Estimation of hardness of water by complexometric method. Potable water and its specifications. Steps involved in treatment of water – Disinfection of water by chlorination and ozonization. Boiler feed water and its treatment – Calgon conditioning, Phosphate conditioning and Colloidal conditioning. External treatment of water – Ion exchange process. Desalination of water – Reverse osmosis. Numerical problems.								
UNIT-III	ELECTROCHEMISTRY AND CORROSION						Classes: 12	
Electro chemical cells – electrode potential, standard electrode potential, types of electrodes – calomel, Quinhydrone and glass electrode. Nernst equation Determination of pH of a solution by using quinhydrone and glass electrode. Electrochemical series and its applications. Numerical problems. Potentiometric titrations. Batteries – Primary (Lithium cell) and secondary batteries (Lead – acid storage battery).								

Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion – mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection– Sacrificial anode and impressed current cathodic methods. Surface coatings – metallic coatings –Hot dipping , metal cladding and electro plating(copper plating)..		
UNIT-IV	STEREOCHEMISTRY, REACTION MECHANISM AND SYNTHESIS OF DRUG MOLECULES	Classes: 10
Introduction to representation of 3-dimensional structures, Structural and stereoisomers, configurations, symmetry and chirality. Enantiomers, diastereomers, optical activity and Absolute configuration. Conformation a analysis of n- butane. Substitution reactions: Nucleophilic substitution reactions: Mechanism of SN1, SN2 reactions. Electrophilic and nucleophilic addition reactions: Addition of HBr to propene. Markownikoff and anti Markownikoff's additions. Grignard additions on carbonyl compounds. Elimination reactions: Dehydro halogenation of alkylhalides. Saytzeff rule. Oxidation reactions: Oxidation of alcohols using KMnO ₄ and chromic acid. Reduction reactions: reduction of carbonyl compounds using LiAlH ₄ & NaBH ₄ . Hydroboration of olefins. Structure, synthesis and pharmaceutical applications of Aspirin.		
UNIT-V	SPECTROSCOPIC TECHNIQUES AND APPLICATIONS	Classes:8
Principles of spectroscopy, selection rules and applications of electronic spectroscopy, vibrational and rotational spectroscopy. Basic concepts of Nuclear magnetic resonance Spectroscopy, chemical shift. Introduction to Magnetic resonance imaging. Fluorescence and its applications in medicine.		
Text Books:		
1. Engineering Chemistry 14th edition by P.C.Jain & M.Jain; Dhanpat Rai Publishing Company (P) Ltd., New Delhi. 2. Fundamentals of Molecular Spectroscopy 5th edition by C.N. Banwell Mc.Graw-Hills book company.		
Reference Books:		
1. Organic Chemistry 7th Edition, Robert Thornton Morrison, Robert Neilson Boyd, Saibal Kanti Bhattacharjee Pearson Education Singapore Pvt.Ltd. 2. Engineering Chemistry (NPTEL Web-book), 11th edition by B.L. Tembe, Kamaluddin and M.S. Krishnan. 3. Physical Chemistry 11th edition by P.W. Atkins OUP Oxford.		
Web References:		
1. https://www.scribd.com/document/23180395/Engineering-Chemistry-Unit-I-Water-Treatment 2. https://chem.libretexts.org/Core/Inorganic_Chemistry/Descriptive_Chemistry/Periodic_Trends_of_Elemental_Properties/Periodic_Properties_of_the_Elements 3. https://www.khanacademy.org/science/biology/chemistry--of-life/chemical-bonds-and-reactions/v/intermolecular-forces-and-molecular-bonds 4. https://study.com/academy/lesson/the-relationship-between-free-energy-and-the-equilibrium-constant.html		
E-Text Books:		
1. http://www.freebookcentre.net/Chemistry/Chemistry-Books-Online.html 2. https://www.sdu.dk/en/om_sdu/institutter_centre/fysik_kemi_og_farmaci		
MOOC Course		
1. http://nptel.ac.in/courses/122101001/34 2. https://ocw.mit.edu/courses/chemistry/		

PROGRAMMING FOR PROBLEM SOLVING

II Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS01	Foundation	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To impart basic knowledge about simple algorithms for arithmetic and logical problems. 2. To understand how to write a program, syntax and logical errors. 3. To enable them how to implement conditional branching, iteration and recursion. 4. To understand how to decompose a problem into functions and synthesize a complete program. 5. To enable them to use arrays, pointers, strings and structures in solving problems. 6. To understand how to solve problems related to matrices, Searching and sorting. 7. To make them to understand the use files to perform read and write operations.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Formulate simple algorithms for arithmetic and logical problems. 2. Test and execute the programs and correct syntax and logical errors. 3. Implement conditional branching, iteration and recursion. 4. Decompose a problem into functions and synthesize a complete program. 5. Use arrays, pointers, strings and structures to formulate algorithms and programs. 6. Apply programming to solve problems related to matrices, Searching and sorting.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction to Programming: Computer system, components of a computer system, computing environments, computer languages, creating and running programs, Algorithms, flowcharts. Introduction to C language: History of C, basic structure of C programs, process of compiling and running a C program, C tokens, keywords, identifiers, constants, strings, special symbols, variables, data types, I/O statements.								
UNIT-II	OPERATORS, EXPRESSIONS AND CONTROL STRUCTURES						Classes: 15	
Operators and expressions: Operators, arithmetic, relational and logical, assignment operators, increment and decrement operators, bitwise and conditional operators, special operators, operator precedence and associativity, evaluation of expressions, type conversions in expressions. Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, jump statements, break, continue, goto statements.								
UNIT-III	ARRAYS AND FUNCTIONS						Classes: 17	
Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays, Basic Algorithms: Searching, Basic Sorting Algorithms- Bubble sort, Insertion sort and Selection sort. Functions: User defined and built-in Functions, storage classes, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, Quick sort or Merge sort.								
UNIT-IV	STRINGS AND POINTERS						Classes: 10	

Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, string handling functions. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, functions returning pointers, Dynamic memory allocation.		
UNIT-V	STRUCTURES AND FILE HANDLING	Classes: 10
Structures and unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, structures and functions, self referential structures, unions, typedef, enumerations. File handling: command line arguments, File modes, basic file operations read, write and append, example programs		
Text Books:		
1. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3 rd edition, 2017. 2. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 6 th Edition, 2012.		
Reference Books:		
1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2 nd Edition, 1988. 2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2 nd Edition, 2003. 3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4 th Edition, 2014. 4. R. S. Bichkar, "Programming with C", Universities Press, 2 nd Edition, 2012. 5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2 nd Edition, 2006. 6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4 th Edition, 2014. 7. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3 rd Edition, 2014.		
Web References:		
1. https://www.bfoit.org/itp/Programming.html 2. https://www.khanacademy.org/computing/computer-programming 3. https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0 4. https://www.edx.org/course/introduction-computer-science-harvardx-cs50x		
E-Text Books:		
1. http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm 2. http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/ 3. http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf		
MOOC Course		
1. https://onlinecourses.nptel.ac.in/noc18_cs33/preview 2. https://www.alison.com/courses/Introduction-to-Programming-in-c 3. http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm		

ENGLISH**II Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS01	Foundation	L	T	P	C	CIE	SEE	Total
		2	-	-	2	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Improve language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations

COURSE OUTCOMES:

Students should be able to:

1. Use English Language effectively in spoken and written forms.
2. Comprehend the given texts and respond appropriately.
3. Communicate confidently in various contexts and different cultures.
4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.
5. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

UNIT-I	THE RAMAN EFFECT	Classes: 12
Chapter entitled 'The Raman Effect' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- The concept of Word Formation, Root Words and their use in English G- Types of Sentences, Parts of Speech, Nouns, Pronouns, Adjectives R- Reading and its importance W- Sentence Structures, Punctuation, Writing Introductions and Conclusions		
UNIT-II	LETTER WRITING	Classes: 12
Letter Writing. V- Synonyms and Antonyms, Standard Abbreviations G- Verbs, Modal Auxiliaries, Adverbs, Prepositions, Conjunctions R- Improving Comprehension Skills, Reading and Comparing Two Articles W- Writing Paragraphs, Letter Writing- Letters of Request, Apology and Complaint- Letter of Application with Resume.		
UNIT-III	BLUE JEANS	Classes: 12

Chapter entitled 'Blue Jeans' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- Prefixes and Suffixes, Idioms and Phrasal verbs G- Articles, Tenses and its forms R- Sub skills of Reading- Skimming and Scanning W- Essay writing and Describing Objects, Places and Events		
UNIT-IV	WHAT SHOULD YOU BE EATING	Classes: 12
Chapter entitled 'What Should You Be Eating' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- One word Substitutes, Words often confused G- Active and passive Voice Subject Verb Agreement (Concord) R- Reading Comprehension- Intensive and Extensive W- Technical Report Writing, E-mail writing, Picture Essay		
UNIT-V	HOW A CHINESE BILLIONAIRE BUILT HER FORTUNE	Classes: 12
Chapter entitled 'How a Chinese Billionaire Built Her Fortune' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- Technical Vocabulary and their Usage, Misplaced Modifiers, Redundancies G- Direct and Indirect Speech, Degrees of Comparison, Common Errors in English R- Effective Reading and Exercises for Practice W- Memo, Précis and Resume Writing		
Text Books:		
1. Sudarshan, N. P. and Savitha, C. (2018). English for Engineers. Cambridge University Press. 2. Wren & Martin. (2017). High School English Grammar and Composition Book. S Chand Publishing.		
Reference Books:		
1. Murphy, R. (2015). Essential Grammar in Use. Cambridge University Press. 2. Wood, F.T. (2007). Remedial English Grammar. Macmillan. 3. Swan, M. (2016). Practical English Usage. Oxford University Press. 4. Kumar, S and Lata, P. (2018). Communication Skills. Oxford University Press. 5. Zinsser, William. (2001). On Writing Well. Harper Resource Book. 6. Hamp-Lyons, L. (2006). Study Writing. Cambridge University Press. 7. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.		
Web References:		
1. https://www.oxfordlineenglish.com 2. https://www.bbclearningenglish.com 3. https://www.learnenglish.britishcouncil.org 4. https://www.fluentu.com/english		
E-Text Books:		
1. https://www.uop.edu.io/download/research/members/oxford_guide_to_english_grammar.pdf 2. http://www.espressoenglish.net/wp-content/uploads/2012/07/Free-Grammar-Ebook-Level-2.pdf 3. https://update24hour.com/2018/03/wren-and-martin-english-grammar-pdf		
MOOC Course		
1. http://www.cambridgeenglish.org/learning-english/free-resources/mooc/ 2. https://www.britishcouncil.org/tr/en/english/mooc 3. https://mooc.com		

PROGRAMMING FOR PROBLEM SOLVING LABORATORY

II Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS02	Foundation	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100
COURSE OBJECTIVES: 1. To understand how to formulate the algorithms for simple problems 2. To be able to translate given algorithms to a working and correct program 3. To make them understand how to correct syntax errors as reported by the compilers 4. To be able to identify and correct logical errors encountered at run time 5. To understand how to write iterative as well as recursive programs 6. To enable them to represent data in arrays, strings and structures 7. To impart the knowledge of declare pointers of different types and their usage. 8. To understand how to create, read and write to and from simple text files.								
COURSE OUTCOMES: At the end of the course, student will be able to 1. Formulate the algorithms for simple problems 2. Translate given algorithms to a working and correct program 3. Correct syntax errors as reported by the compilers 4. Identify and correct logical errors encountered at run time 5. Write iterative as well as recursive programs 6. Represent data in arrays, strings and structures and manipulate them through a program 7. Declare pointers of different types and use them in defining self-referential structures. Create, read and write to and from simple text files.								
LIST OF EXPERIMENTS								
Week-1	INTRODUCTION TO LINUX COMMANDS							
a. Basic Linux commands b. Write a C program to use printf() and scanf() functions c. Write C programs to implement basic arithmetic operations – sum, average, product, difference, quotient and remainder of given numbers etc.								
Week-2	OPERATORS AND EVALUATION OF EXPRESSIONS							
a. Write a C program to check whether a number is even or odd using ternary operator. b. Write a C program to perform the addition of two numbers without using +operator. c. Write a C program to evaluate the arithmetic expression ((a + b / c * d - e) * (f - g)). Read the values a, b, c, d, e, f, g from the standard input device. d. Write a C program to find the sum of individual digits of a 3 digit number. e. Write a C program to read the values of x and y and print the results of the following expressions in one line: i (x + y) / (x -y) ii (x + y)(x - y)								
Week-3	CONDITIONAL STATEMENTS							

a. Write a C program to find largest and smallest of given numbers. b. Write a C program to find roots of a quadratic equation. c. Write a C program which takes two integer operands and one operator from the user(+, -, *, /, % use switch)	
Week-4	LOOPING STATEMENTS
a. Write a C program to find Sum of individual digits of given integer b. Write a C program to generate first n terms of Fibonacci series c. Write a C program to generate prime numbers between 1 and n	
Week-5	LOOPING STATEMENTS
a. Write a C Program to find the Sum of Series $SUM = 1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$ b. Write a C program to generate Pascal's triangle. c. Write a C program to generate pyramid of numbers. 1 1 3 1 1 3 5 3 1	
Week-6	ARRAYS
a. Write a C Program to implement following sorting methods - Bubble sort - Selection sort - Insertion sort b. Write a C program to find largest and smallest number in a list of integers	
Week-7	ARRAYS
a. Write a C program - To add two matrices - To multiply two matrices b. Write a C program to find Transpose of a given matrix	
Week-8	FUNCTIONS
a. Write a C program to find the factorial of a given integer using functions b. Write a C program to find GCD of given integers using functions c. Write a C Program to find the power of a given number using functions	
Week-9	RECURSION
a. Write a C Program to find binary equivalent of a given decimal number using recursive functions. b. Write a C Program to print Fibonacci sequence using recursive functions. c. Write a C Program to find LCM of 3 given numbers using recursive functions	
Week-10	STRINGS

a. Write a C program using functions to a. Insert a sub string into a given main string from a given position b. Delete n characters from a given position in a string b. Write a C program to determine if given string is palindrome or not	
Week-11	POINTERS AND STRUCTURES
a. Write a C program to print 2-D array using pointers b. Write a C program to allocate memory dynamically using memory allocation functions (malloc, calloc, realloc, free) c. Write a C Program using functions to a. Reading a complex number b. Writing a complex number c. Add two complex numbers d. Multiply two complex numbers Note: represent complex number using structure.	
Week-12	FILES
a. Write a C program to copy one file to other b. Write a C program to copy one file to other c. Write a C Program to merge two files into a third file	
Text Books:	
Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13 th Edition, 2012. Oualline Steve, "Practical C Programming", O'Reilly Media, 3 rd Edition, 1997.	
Reference Books:	
1. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2 nd Edition, 2015. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3 rd Edition, 2004. 4. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1 st Edition, 1994.	
Web References:	
http://www.sanfoundry.com/c-programming-examples http://www.geeksforgeeks.org/c http://www.cprogramming.com/tutorial/c http://www.cs.princeton.edu	

CHEMISTRY LABORATORY

II Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks	
A4BS12	BS	L	T	P	C	CIE	SEE
		-	-	3	2	30	70

COURSE OBJECTIVES:

The course should enable the students to:

1. Estimation of hardness and chloride content in water to check its suitability for drinking purpose.
2. To determine the rate constant of reactions from concentrations as a function of time.
3. The measurement of physical properties like adsorption and viscosity.
4. To synthesize the drug molecules and check the purity of organic molecules by thin layer chromatographic (TLC) technique.

COURSE OUTCOMES:

The course should enable the students to:

1. Determination of parameters like hardness and chloride content in water.
2. Estimation of rate constant of a reaction from concentration – time relationships.
3. Determination of physical properties like adsorption and viscosity.
4. Calculation of R_f values of some organic molecules by TLC technique

LIST OF EXPERIMENTS

Week-1	Determination of total hardness of water by complexometric method using edta
Week-2	Estimation of an hcl by conductometric titrations
Week-3	Estimation of acetic acid by conductometric titrations
Week-4	Estimation of hcl by potentiometric titrations
Week-5	Estimation of acetic acid by potentiometric titrations
Week-6	Determination of rate constant of acid catalysed hydrolysis of methyl acetate
Week-7	Synthesis of aspirin
Week-8	Thin layer chromatography calculation of r_f values. eg ortho and para nitro phenols
Week-9	Verification of freundlich adsorption isotherm-adsorption of acetic acid on charcoal
Week-10	Determination of viscosity of castor oil and ground nut oil by using ostwald's viscometer
Week-11	Determination of surface tension of a give liquid using stalagmometer

Week-12	Synthesis of thiokol rubber
Week-13	Determination of chloride content of water using argentometric method
Week-14	Determination of rate constant of acid catalysed hydrolysis of methyl acetate
Reference Books:	
1. Senior practical physical chemistry, B.D. Khosla, A. Gulati and V. Garg (R. Chand & Co., Delhi). 2. An introduction to practical chemistry, K.K. Sharma and D. S. Sharma (Vikas publishing, N. Delhi). 3. Vogel's text book of practical organic chemistry 5th edition. 4. Text book on Experiments and calculations in Engineering chemistry – S.S. Dara.	
Web References:	
1. http://amrita.olabs.edu.in/?sub=73&brch=8&sim=153&cnt=2 2. https://chem.libretexts.org/Textbook_Maps/Analytical_Chemistry_Textbook_Maps/Map%3A_Analytical_Chemistry_2.0_(Harvey)/11_Electrochemical_Methods/11.2%3A_Potentiometric_Methods 3. http://fch.upol.cz/skripta/fcc_and_zvem_english/FCH/Adsorption%20of%20oxalic%20acid%20on%20activated%20charcoal.htm 4. https://www.askiitians.com/iit-jee-chemistry/physical-chemistry/electrolytic-conductance-molar-conductance-and-specific-conductance.aspx 5. https://owlcation.com/stem/tlc-thin-layer-chromatography-Principle-Procedure	

ENGLISH LANGUAGE COMMUNICATION SKILLS LABORATORY

II Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS02	HS	L	T	P	C	CIE	SEE	Total
		0	0	2	1	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Facilitate computer-assisted multi-media instruction enabling individualized and independent language learning. 2. Enhance English language skills, communication skills and to practice soft skills. 3. Improve fluency and pronunciation intelligibility by providing an opportunity for practice in speaking. 4. Train students in different interview and public speaking skills such as JAM, debate, role play, group discussion etc. 5. Instill confidence and make them competent enough to express fluently and neutralize their mother tongue influence.								
COURSE OUTCOMES: The course should enable the students to: 1. Better perception of nuances of English language through audio- visual experience 2. Neutralization of accent for intelligibility 3. Take part in group activities 4. Speaking skills with clarity and confidence which in turn enhances their employability								
LIST OF ACTIVITIES								
Week-1	GENERAL INTRODUCTIONS AND FORMAL GREETINGS							
a. Introductions and greetings in formal and informal situations b. Worksheets to extract information c. Questionnaires to enquire about the expressions used during formal introductions d. Ice Breaking activity by preparing and asking five questions each e. Creation of dialogues using greetings, leave- taking and introductions								
Week-2	JAM- JUST A MINUTE							
a. Strengthen the ability to analyze a topic and logical organization of thoughts. b. Logically starting with introductory sentence, points of discussion and closing sentence. c. Practicing to speak within one minute d. Activity based on JAM on a familiar topic e. Planning and executing JAM considering the parameters								
Week-3	PHONETICS							
a. Speech sounds and their prominence in pronunciation b. Understanding and practicing word stress c. Neutralizing the accent and practicing the right intonation d. Knowing the differences between different accents e. Increase fluency with the help of Phonetics								

Week-4	LISTENING SKILLS
a. Developing good listening skills for effective communication b. Enhancing listening skills through audio tracks and oral conversation c. Empathizing others point of view while they speak d. Incorporating non verbal communications while listening e. Improving overall performance listening to the audio tracks	
Week-5	SITUATIONAL DIALOGUES AND GIVING DIRECTIONS
a. Creating dialogues in any given situations b. Framing and choosing appropriate words to frame the dialogues in any situation c. Guiding and giving directions using appropriate expressions d. Activities on how to make polite requests, offers, rejections etc e. Practicing to speak confidently in different situations	
Week-6	ROLE PLAY
a. Understanding a Role play and its procedure b. Planning and Executing a Role Play accordingly c. Practicing to get into the role and perform within stipulated time d. Activities based on Role Play with different situations e. Performing a Role Play considering the parameters	
Week-7	GROUP DISCUSSIONS
a. Understanding a Group Discussion (GD) and its procedure b. Following the rules of a GD c. Planning and Executing a GD within the stipulated time d. Activities based on GD e. Performing a GD considering the parameters	
Week-8	DEBATE
a. Understanding the procedure of a Debate b. Planning and executing a Debate following its rules c. Strengthen the ability to analyze a topic and logical organization of thoughts. d. Logically arranging the arguments e. Performing a Debate considering the parameters	
Week-9	TELEPHONIC ETIQUETTES
c. Understanding basic Telephonic Etiquettes d. The approach one needs to follow while making and answering a call e. Making a formal telephonic conversation f. Activities based on modulating voice and tone g. Interpersonal skills required to overcome rude and hostile behavior	

Week-10	PRESENTATION SKILLS
a. Planning a Presentation b. Enhancing skills required for making effective presentations c. Usage of different tools that help us to give effective presentations d. Executing a presentation effectively e. Activities based on presentations	
Week-11	ORAL PRESETATIONS AND EXTEMPORE
a. Planning an oral presentation or an Extempore b. Preparing good PPT c. Using appropriate body language in public speaking domain d. Planning and Executing oral presentation e. Activities based on oral presentations and extempore	
Week-12	INTERVIEW SKILLS
a. Preparing to succeed in Interviews b. Preparing a strong Resume for interviews c. Practicing different techniques to overcome nervousness in interviews d. Using appropriate body language in interviews e. Activities based on Interviews skills	
Week-13	INFORMATION TRANSFER
a. Extracting Information Transfer from different kinds of representation b. Reading and decoding the information given in various types c. Representing the information in charts or graphs in a written document d. Developing writing skills from these aspects e. Activity on transferring given data into graphs or charts for presentation skills	
Reference Books:	
1. E. Suresh Kumar. A Handbook for English Language Laboratories (with CD) Revised Edition 2. Normal Whitby. Business Benchmarch. Cambridge University Press(with CD) 2 nd Edition 3. Liz Hamp-Lyons and Ben Heasly. Study Writing. Cambridge University Press. 2006. 4. Sanjay Kumar and Pushp Lata. Communication Skills. Oxford University Press. 2011. 5. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press 6. Raman Sharma, Technical Communications, Oxford Publication, London, 2004. 7. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004. 8. T. Balasubramanian, A Text book of English Phonetics for Indian Students, Macmillan, 2008. 9. Edgar Thorpe, Winning at Interviews, Pearson Education, 2006. 10. J. Sethi et al, A Practical Course in English Pronunciation (with CD), Prentice Hall of India, 2005. 11. Hari Mohan Prasad, How to Prepare for Group Discussions and Interviews, Tata McGraw Hill, 2006	
Web References:	
https://www.britishcouncil.org https://www.bbc.co.uk https://www.grammarly.com https://www.fluentu.com https://www.cambridgeenglish.org/exams-and-tests/business-preliminary https://www.cambridgeenglish.org/exams-and-tests/business-vantage	

ENGINEERING EXPLORATION

II Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS04	HS	L	T	P	C	CIE	SEE	Total
		2	-	-	1	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the Engineering attributes and Ethics.
2. Identify the community problem and its stakeholder.
3. Examine required specifications and gap in existing and required product.
4. Build sustaining interactions among people that create social value by transforming ideas into tangible products, services, or initiatives.
5. Develop skills to work collaboratively, reports and progress updates throughout the lifecycle of the project.

COURSE OUTCOMES:

1. Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change with in communities.
2. To demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
3. Students will contribute to the welfare of society and environment by taking sound professional and responsible technical decisions
4. To develop technically competent graduates capable of providing engineering solutions to real life problems in regard of a community
5. Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change with in communities.

UNIT-I	INTRODUCTION TO ENGINEERING AND ENGINEERING EXPLORATION	Classes: 06
Engineering Projects in Community Service, Design Thinking Process-Empathize, Define, Ideate, Prototype, Test.		
Engineering Ethics: Introduction to ethics, moral values, significance of professional ethics, code of conduct for engineers, identify ethical dilemmas in different tasks of engineering, applying moral theories and codes of conduct for resolution of ethical dilemmas.		
UNIT-II	PROBLEM IDENTIFICATION	Classes: 06
Authentic need in the community or society. Identify a real user or stake holder, Interaction with Stakeholders, Viewpoints, Interviewing, Scenario.		
UNIT-III	SPECIFICATION DEVELOPMENT	Classes: 06
Clear and measurable requirements, criteria for success, Identifying relevant benchmarks, identifying the gap between the available and required products, requirements documentation.		

UNIT-IV	CONCEPTUAL DESIGN	Classes: 06
Ideation-generated multiple ideas, evaluation of ideas, systems model, Architectural Design, prototype development, testing – real/simulated users, feedback.		
UNIT-V	PROJECT MANAGEMENT	Classes: 08
Importance of team work, importance of project life cycle, project management, tools, various tools used in electronics documentation, importance of communication, usage of communication media.		
Text Books:		
1. Software Engineering: A Practitioner's Approach, Roger S. Pressman, 7th Edition, Mc Graw Hill Education (India) Pvt. Ltd. 2. Software Engineering, Sommerville Ian, 7th Edition, Pearson Education. 3. EPICS Design Process https://sharepoint.ecn.purdue.edu/epics/teams/Public%20Documents/EPICS_Design_Process.pdf 4. Examples of good practice in Special Needs Education & Community Based Programs, UNESCO PRESS. 5. Project Management, GRY r. Heerkens, McGraw-Hill		
Web References:		
1. http://www.purdue.edu/epics 2. http://epics.ieee.org/ 3. https://www.uninettunouniversity.net/en/epics.aspx		
E-Text Books:		
1. http://www.uoitc.edu.iq/images/documents/informatics-institute/exam_materials/Software%20Engineering%20(9th%20Edition)%20by%20Ian%20Sommerville.pdf 2. https://engineering.purdue.edu/EPICS/k12/resources/1.6%20Teacher%20Toolbox%20EPICS%20High%20Design%20Process%20and%20Cycle.pdf 3. https://launchschool.com/books/agile_planning/read/epics_and_stories http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf		
MOOC Course		
1. https://www.mooc-list.com/tags/design-thinking 2. https://www.class-central.com/tag/design%20thinking		

II B.TECH III SEMESTER SYLLABUS

INTRODUCTION TO AEROSPACE ENGINEERING**III Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE01	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Describe the History of aeronautical engineering.
2. Apply the Basic aerodynamics
3. Examine the Structures and materials
4. Explain the Elements of airplane performance
5. Explain the Space flight (Astronautics) dynamics
6. Develop the design from the given specifications

COURSE OUTCOMES:

Students should able

1. Compare the atmosphere conditions of different altitudes for spacecraft system
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in a satellite mission
5. Design lighter than air vehicle using Catia software

UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT	Classes: 12
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments		
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION	Classes: 13
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.		
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY	Classes: 12
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes.		

UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION	Classes: 13
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity.		
UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION	Classes: 13
Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.		
Text Books:		
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082-		
Reference Books:		
1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2 nd Edition, 1988. 2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2 nd Edition, 2003. 3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4 th Edition, 2014. 4. R. S. Bichkar, "Programming with C", Universities Press, 2 nd Edition, 2012. 5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2 nd Edition, 2006. 6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4 th Edition, 2014. 7. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3 rd Edition, 2014.		
Web References:		
https://www.grc.nasa.gov/www/k-12/airplane/index.html		
E-TextBooks:		
https://www.researchgate.net/publication/259576416_Fundamentals_of_Aerospace_Engineering http://www.adl.gatech.edu/extrovert/Ebooks/ebook_Intro.pdf		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_ae07/		

MECHANICS OF SOLIDS

III Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME01	PC	L	T	P	C	CIA	SEE	Total
		4	1	-	4	30	70	100

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Apply the concept of stress and strain to analyse and design structural members
2. Develop the shear force and bending moment diagrams for different beams subjected to various loads.
3. Determine the bending stress and develop the shear stress distribution across various beam sections.
4. Determine the principle stresses and deflection of beams.
5. Design the circular shafts and analyze the thin cylinders.

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the concept of stress and strain to analyze and design structural members
2. Develop the shear force and bending moment diagrams for different beams subjected to various loads.
3. Determine the bending stress and develop the shear stress distribution across various beam sections.
4. Determine the principal stresses and deflection of beams.
5. Design the circular shafts and analyze the thin cylinders.

UNIT-I	SIMPLE STRESSES AND STRAINS	Classes: 12
SIMPLE STRESSES AND STRAINS: Elasticity and plasticity, Types of stresses and strains, Hooke's law stress, strain diagram for mild steel, Working stress, Factor of safety, Lateral strain, Poisson's ratio and volumetric strain, Elastic module and the relationship between them, Bars of varying section, composite bars, Temperature stresses. Strain energy, Resilience - Gradual, Sudden, Impact and Shock loadings.		
UNIT-II	SHEAR FORCE AND BENDING MOMENT	Classes: 11
SHEAR FORCE AND BENDING MOMENT: Definition of beam, Types of beams, Concept of shear force and bending moment, Relation between Shear Force and Bending Moment. and rate of loading at a section of a beam. Shear Force and Bending Moment diagrams for cantilever simply supported and overhanging beams subjected to point loads, U.D.L., uniformly varying loads and combination of these loads, Point of contra flexure.		
UNIT-III	FLEXURAL STRESSES & SHEAR STRESSES	Classes: 12
FLEXURAL STRESSES: Theory of simple bending, Assumptions, Derivation of bending equation: $M/I = f/y = E/R$ Neutral axis, Determination bending stresses, section modulus of rectangular and circular sections (Solid and Hollow), I, T, Angle and Channel sections, Design of simple beam sections.		
SHEAR STRESSES: Derivation of formula, Shear stress distribution across various beams sections like rectangular, circular, I, T, angle sections.		

UNIT-IV	PRINCIPAL STRESSES AND STRAINS & DEFLECTION OF BEAMS	Classes: 13
PRINCIPAL STRESSES AND STRAINS: Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses - Principle stresses and strains - Analytical and graphical solutions. DEFLECTION OF BEAMS: Bending into a circular arc slope, deflection and radius of curvature, Differential equation for the elastic line of a beam, Double integration and Macaulay's methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads		
UNIT-V	TORSION OF CIRCULAR SHAFTS & THIN CYLINDERS	Classes: 12
TORSION OF CIRCULAR SHAFTS: Theory of pure torsion - derivation of Torsion equations: $T/J = q/r = N/L$ - Assumptions made in the theory of pure torsion - Torsional moment of resistance - Polar section modulus - Power transmitted by shafts THIN CYLINDERS: Thin seamless cylindrical shells, Derivation of formula for longitudinal and circumferential stresses, hoop, longitudinal and volumetric strains, changes in dia, and volume of thin cylinders, Riveted boiler shells, Thin spherical shells.		
Text Books:		
1. Ramamrutham. S (2012), Strength of materials, 17th edition, Dhanpat Rai Publications, Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company New Delhi, India. 2. Dr.Bansal R.K(2007), Strength of materials, 10th edition, Laxmi Publications, Hyderabad 3. Timoshenko. S (2004), Strength of materials, 3rd edition, CBS Publishers, New Delhi, India.		
Reference Books:		
1. Ryder G. H (2007), Strength of materials, 3rd edition, Macmillan, New Delhi, India. 2. Bhavikathi S. S (2008), Strength of materials, 3rd edition, Vikas Publishing House, New Delhi, India		
Web References:		
1. http://nptel.ac.in/courses/112107146/ 2. http://nptel.ac.in/courses/112107147/		
E-Text Books:		
http://www.engineering108.com/Data/Engineering/Mechanical/SM/Strength_Of_Materials_parts_landII-Timoshenko.pdf		
MOOC Course		
https://www.class-central.com/tag/solid%20mechanics		

MECHANICS OF FLUIDS

III Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE02	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The objectives of the course are to enable the student:

- 1.To understand the basic principles of fluid mechanics
- 2.To identify various types of flows
- 3.To understand concepts of flow measuring and calculate the energy losses
- 4.To understand type of flow such as laminar and turbulent flows
- 5.To understand the difference between compressible and incompressible flow

COURSE OUTCOMES:

Student is able to

1. Determine the fluid properties for different types of fluids
2. Evaluate the flow depends upon it's flow property
3. Choose the appropriate notch for the real-time applications
4. Estimate the type of flow such as laminar and turbulent flows
5. Choose the type of flows such as compressible or in compressible to determine the effect of compressibility on flow properties

UNIT-I	FLUID PROPERTIES AND FLUID STATICS	Classes: 12
Density, Specific weight, Specific gravity, viscosity, Vapour pressure,compressibility, Surface tension Pressure at a point, Pascal's law, pressure variation with temperature, density and altitude. Hydro static law, Piezometer, Simple and differential manometers, pressure gauges, total pressure and center of pressure plane, vertical and inclined surfaces. Buoyancy and stability of floating bodies.		
UNIT-II	FLUID KINEMATICS, FLUID DYNAMICS	Classes: 12
Stream line, path line, streak line, stream tube, classification of flows, steady, unsteady,uniform, non-uniform, laminar, turbulent, rotational, irrotational flows, one, two and three dimensional flows, Continuity equation in 3D flow, stream function, velocity potential function. Surface and Body forces, Euler's and Bernoulli's equation derivation, Navier stokes equation(explanation only), Momentum equation - applications, Vortex Free and Forced. Forced vortex with free surface.		
UNIT-III	SIMILITUDE AND FLOW MEASUREMENT, REYNOLDS EXPERIMENT	Classes: 12
Flow through venturimeter and orifice meter, flow through notches and weirs Viscometers hot weir anenometers, pitot tube flow through nozzles, Characteristics of real fluids. Darcy's equation, Minor losses - pipes in series, pipes in parallel, total energy line and hydraulic gradient line, numerical problems.		

UNIT-IV	BOUNDARY LAYER CONCEPTS	Classes: 12
Definition, thicknesses, characteristics along thin plate, laminar and turbulent layers(No Derivation) boundary layer in transition, separation of boundary layer submerged objects drag and lift.		
UNIT-V	FLOW OF COMPRESSIBLE FLUID	Classes: 12
Introduction, Thermodynamic relations, basic equations of compressible flow, Velocity of sound wave in a fluid for isothermal and adiabatic process, Mach number and its applications, Mac angle propagation of pressure waves and stagnation properties.		
Text Books:		
1. P. N. Modi, S. M. Seth (2011), Hydraulics and fluid mechanics including hydraulic machines, 18th revised edition Standard Book House, India. 2. Yumus A. Cengel, John M. Cimbala (2010), Fluid Mechanics (SI Units), 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.		
Reference Books:		
1. R. K. Bansal (2011), A Textbook of Fluid Mechanics and Hydraulic Machines, 10th edition, Laxmi Publications, New Delhi, India. 2. Frank M. White (2011), Fluid Mechanics, 7th edition, Tata McGraw Hill, New Delhi, India. 3. John F. Dauglas (2005), Fluid Mechanics, 5th edition, Pearson Education Limited, New Delhi, India. 4. Mechanics of Fluids by pijush K Kundu, B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.		
Web References:		
http://mechanicsoffluids.blogspot.com/2008/11/fluid-mechanics.html http://www.engineerblogs.net/course-categories/physics/fluid-mechanics/		
E-Text Books:		
http://varunkamboj.typepad.com/files/engineering-fluid-mechanics-1.pdf		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_me47/announcements?force=true		

THERMODYNAMICS

III Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME04	PC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

OBJECTIVES:

The course should enable the students to:

1. Analyze laws of thermodynamics.
2. Analyze changes in entropy in various thermodynamic processes.
3. Analyze the thermodynamic properties of pure substances undergoing various thermodynamic processes.
4. Evaluate the thermodynamic properties of gases and mixture of gases.
5. Realize the working of different power cycles.

COURSE OUTCOMES:

Students should able to

1. Apply first law of thermodynamics to real time applications
2. Evaluate the performance parameters of energy transfers
3. Evaluate the properties of gases during phase changes
4. Compare the process parameters in mixture of gases
5. Distinguish the processes of various power cycles

UNIT-I	INTRODUCTION	Classes: 13
--------	--------------	-------------

Basic Concepts: Macroscopic and Microscopic viewpoints, Quasi static Process, various flow and non flow process, energy in State and in Transition, Work and Heat, Point and Path function. Zeroth Law of Thermodynamics, First law of Thermodynamics, PMM-I, Corollaries, First law applied to a Process, applied to a flow system.

UNIT-II	LIMITATIONS OF FIRST LAW	Classes: 13
---------	--------------------------	-------------

LIMITATIONS OF THE FIRST LAW: Thermal Reservoir, Heat Engine, Heat pump Refrigerator. Parameters of performance, Second Law of Thermodynamics-Kelvin Planck and Clausius Statements and their Equivalence / Corollaries, PMM of Second kind, Carnot's principle, Carnot cycle and its specialties, Clausius Inequality, Entropy, Principle of Entropy Increase, Elementary Treatment of the Third Law of Thermodynamics.

UNIT-III	PURE SUBSTANCES	Classes: 11
----------	-----------------	-------------

PURE SUBSTANCES: P-V-T surfaces, T-S and h-s diagrams, Mollier Charts, Phase transformations, Triple point at critical state properties during change of phase, Dryness Fraction. Various thermodynamic processes and energy transfer, Steam calorimetry.

UNIT-IV	GAS LAWS AND MIXTURES OF PERFECT GASES	Classes: 10
Gas laws: Equation of State, specific and Universal Gas constants, Deviations from perfect Gas Model, Vander Waals Equation of State. Mixtures of perfect gases: Mole fraction, Volume fraction, Mass fraction, Gravimetric and volumetric analysis, Dalton's law of partial pressure, Avogadro's law of additive volumes, Equivalent gas const and Molecular internal energy, enthalpy, Sp. heats and entropy of mixture of perfect gases and vapour.		
UNIT-V	POWER CYCLES	Classes: 13
POWER CYCLES: Otto, Diesel, Dual Combustion cycles, Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis, comparison of Cycles.		
Text Books:		
1. Engineering Thermodynamics, P.K. Nag, 6th Edition, Mc Graw Hill Education. 2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8th Edition, Mc Graw Hill Companies. 3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7th Edition, Wiley.		
Reference Books:		
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi. 3. Applied Thermodynamics, Onkar Singh, 3rd Edition, New Age, India		
Web References:		
1. http://nptel.ac.in/courses/112104113/ 2. http://nptel.ac.in/courses/112105123/ 3. https://www.udemy.com/fundamentals-of-engineering-thermodynamics/		
E-Text Books:		
1. https://www.pdfdrive.net/engineering-thermodynamics-e18770201.html 2. https://www.pdfdrive.net/fundamentals-of-thermodynamics-e18762062.html 3. https://www.pdfdrive.net/thermodynamics-e15057935.html		
MOOC Course		
1. https://onlinecourses.nptel.ac.in/ 2. https://www.coursera.org/learn/thermodynamics-intro 3. https://swayam.gov.in/courses/4642-july-2018-engineering-thermodynamics		

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

III Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE64	ESC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Describe basic fundamentals of Electric Circuits, their components and the mathematical tools used to represent and analyze Electrical circuits.
2. Develop fundamentals, including Ohm's law, Kirchoff's laws and be able to solve for currents, voltages and power in complex circuits.
3. Demonstrate to write and solve loop current and node voltage equations for arbitrary DC, AC networks including resistors, capacitors, inductors, dependent and independent sources.
4. Extrapolate the concept of magnetic circuit, Faraday's laws and analyze the series and parallel magnetic circuits.
5. Summarize various two port network parameters and their relations and develop the design and analysis of basic DC and AC circuits with network topologies.

COURSE OUTCOMES:

Student should able to

- 1 Understand basic electrical concepts, including electric charge, current, electrical potential, electrical Power and energy.
- 2 Distinguish the relationship of voltage and current in resistors, capacitors, inductors, and mutual Inductors.
- 3 Differentiate circuits with ideal, independent, and controlled voltage and current sources and able to apply Kirchhoff's voltage and current laws to the analysis of electric circuits.
- 4 Apply concepts of electric network topology, nodes, branches, and loops to solve circuit problems, including the use of computer simulation.
- 5 Emphasize on basic laws and techniques to develop a working knowledge of the methods of analysis used in electric circuits.
- 6 Interpret to solve series and parallel magnetic circuits
- 7 Design various two port network parameters and relations between mutual and self inductances.

UNIT-I	ELECTRICAL CIRCUITS	Classes: 12
Basic definitions-Ohm's Law – Kirchhoff's Laws – simple problems. types of elements, types of sources, resistive networks, inductive networks, capacitive networks, series & parallel circuits, star to delta and delta to star transformations.		
UNIT-II	DC MACHINES	Classes: 12
Principle of Operation of DC Motor, types of DC motor, Losses and Torque equation, DC Generator construction and working Principle, EMF Equation, working principle and Construction of transformer.		
UNIT-III	AC MACHINES & INSTRUMENTS	Classes: 12
Principle of operation of 3-phase induction motor, slip and torque Equation, principle of operation of 3-phase Alternator. Basic principle of indicating instruments, permanent magnet moving coil and moving iron instruments.		

UNIT-IV	SEMICONDUCTOR DEVICES	Classes: 12
Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers – filters capacitor, inductor-Voltage Regulation.		
UNIT-V	TRANSISTOR	Classes: 12
Bipolar Junction Transistor – CB, CE, CC Configurations and Characteristics – Transistor Amplifier.		
Text Books:		
1. Basic Electrical Engineering by <i>M.S.Naidu and S.Kamakshaiah</i> TMH 2. Electronic Devices and circuits by <i>J.Millman, C.C.Halkias and Satyabrata Jit</i> 2ed.,		
Reference Books:		
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, “Basic Electrical, Electronics and Computer Engineering”, Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press (2005). 3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, (2002). 3. B. A. Forouzan, R. F. Gillberg, “C Programming and Data Structures”, Cengage Learning, India, 3 rd Edition, 2014.		
Web References:		
1. https://www.electronics-tutorials.ws/diode/diode_3.html 2. https://www.youtube.com/watch?v=i8HG-n1e724 3. jntuh-elsdm.in/		
E-Text Books:		
S K Bhattacharya, Basic Electrical and Electronics Engineering, 2011, Pearson, Kindle Edition		
MOOC Course		
https://nptel.ac.in/courses/108108076/ https://nptel.ac.in/courses/108102097/		

MECHANICS OF SOLIDS AND FLUIDS LAB**III Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE03	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OUTCOMES:

The students should be able to:

1. Recognise the concept of one and two dimensional simple and compound stress and strain
2. Determine the torsion in circular members
3. Estimate impact load on materials under different loading conditions.
4. Determine stiffness of the linear springs.
5. Understand the properties of fluid along with pressure measurement techniques
6. Analyse flows through pipes and open channels.
7. Recognise the concept of laminar and turbulent flow

COURSE OBJECTIVES:

1. Analyze the behavior of the solid bodies subjected to various types of loading
2. Apply knowledge of materials and structural elements to the analysis of simple structures
3. Evaluate the problem identification, formulation and solution using a range of analytical methods.
4. Analyze and interpret laboratory data relating to behavior of structures and the materials
5. Develop the expectation and capacity to undertake lifelong learning.
6. Determine the coefficient of discharge and coefficient of contraction for loss of head in a sudden contraction, orifice, venturimeter, small orifice, external mouthpiece by variable head method.
7. Determine coefficient of discharge for flow through different notches
8. Justify the Bernoulli equation by calculating the total head and also calculate the impact force on different types of vanes

LIST OF EXPERIMENTS**MOS LAB**

1. Direct tension test
2. Bending test on Simple supported beam
3. Bending test on Cantilever beam
4. Torsion test
5. Brinell's hardness test
6. Test on springs
7. Impact test

MOF LAB

- 1 Calibration of Venturi meter
- 2 Calibration of Orifice meter
- 3 Determination of Coefficient of discharge for a small orifice by a constant head method.
- 4 Determination of Coefficient of discharge for an external mouthpiece by variable head method.
- 5 Calibration of contracted Triangular Notch
- 6 Determination of Coefficient of loss of head in a sudden contraction and friction factor.
- 7 Verification of Bernoulli's equation.

Note: Total 10 experiments should be done. At least 4 from each lab.

Reference Books:

1. R. K. Bansal (2011), A Textbook of Fluid Mechanics and Hydraulic Machines, 10th edition, Laxmi Publications, New Delhi, India.
2. Ramamrutham. S (2012), Strength of materials, 17th edition, Dhanpat Rai Publications,
3. Timoshenko and D.H. Young, Engineering Mechanics, Mc Graw Hill Book Company New Delhi, India

Web References:

<https://nptel.ac.in/courses/105104160/>

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB**III Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE65	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

1. The theoretical concepts of KVL and KCL, Diode, Transistor are verified experimentally
2. The performance of A.C and D.C machines are studied practically
3. The efficiency and regulation of Transformer are determined experimentally
4. The fundamentals of A.C. and D.C supply are studied practically.
5. The characteristics of P-N junction diode, Zener diode, transistors and rectifiers.
6. Working principles of CRO.

COURSE OUTCOMES

Student should able to

- 1 To analyze basic concepts to electric circuits
- 2 To apply electrical fundamentals to real time applications.
- 3 To apply electronics components to electronics circuits.
- 4 To create circuits containing basic electrical elements.
- 5 To apply electrical and electronics engineering concepts to real time applications

LIST OF EXPERIMENTS**SECTION A: ELECTRICAL ENGINEERING:**

1. Verification of KCL and KVL.
2. Magnetization characteristics of D.C. Shunt generator.
3. Speed control of DC motor.
4. Swinburne's Test on DC shunt machine.
5. Brake test on DC shunt motor.
6. OC and SC tests on Single-phase transformer.
7. Brake test on 3-phase Induction motor.
8. Regulation by an alternator by synchronous impedance method.

SECTION B: ELECTRONICS ENGINEERING:

1. PN Junction Diode Characteristics (Forward bias, Reverse bias)
2. Transistor CE Characteristics (Input and Output)
3. Study of CRO.
4. Class A Power Amplifier
5. Zener Diode Characteristics
6. Transistor CE Characteristics
7. Rectifier without Filters (Full wave & Half wave)
8. Rectifier with Filters (Full wave & half wave).

Note: Total 12 experiments are to be conducted.

(Six experiments from PART-A, Six experiments from PART-B)

Reference Books:

1. Basic Electrical Engineering by *M.S.Naidu and S.Kamakshaiah* TMH
2. Electronic Devices and circuits by *J.Millman, C.C.Halkias and Satyabrata Jit* 2ed.,

Web References:

<https://nptel.ac.in/courses/108108076/>

<https://nptel.ac.in/courses/108102097/>

AIRCRAFT ENGINEERING DRAWING LAB

III Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE04	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OUTCOMES:

1 Understand the importance of drawing and design process and phases involved in the design process.

2 Ability to draw various individual components, sub-assemblies and main assemblies in drawing lab.

3 Ability to draw various orthographic and isometric projections in drawing sheets.

4 Ability to develop and understand Basic Concepts of aircraft

5 Ability to draw various orthographic and isometric projections of an aircraft components by using auto-cad software.

COURSE OBJECTIVES:

Students Should able to

1. Draw various machine components in drawing lab.

2 Draw various individual components, sub-assemblies and main assemblies in drawing lab.

3. Discuss the importance of design process and studying the different phases of designing process involved in the design.

4. Understand the Integrated product development and principles of baseline design cost

5. Understand the design of aircraft concepts.

6. Design various aircraft components by using Auto-Cad software

LIST OF EXPERIMENTS

UNIT – I

Machine Drawing conventions Need for Drawings conventions – Introduction to ISI - conventions.

1 Conventional representation of material, common machine elements and parts such as screws, nuts, bolts, keys, gears and welding.

2 Types of sections – Selection planes and drawing of section and auxiliary sectional views. Parts not usually sectioned.

3 Methods of dimensioning, general rules for sizes and placement of dimensions for holes, centers, curved and Tapered features

UNIT - II

Drawing of Machine Elements and simple parts. Section of views, additional views for the following machine elements and parts with every drawing proportions.

1 Popular forms of screw threads, bolts, set screws and bolted joints.

2 Keys, cottered joint and knuckle joint

3 Riveted joints for plates

4 Shaft couplings - spigot and socket joint

UNIT - III

Introduction to Autocad –Advantages, Features and merits over manual drawing.

- 1 Object snap commands – Function keys.
- 2 Practice on Draw commands
- 3 Practice on Modify commands
- 4 Practice on View and other commands
- 5 Practice on Simple excersies

Reference Books:

K. L. Narayana, P. Kannaiah, Venkata Reddy, Machine Drawing, New Age publication.

Web References:

<https://www.slideshare.net/umeshchikhale/assembly-and-details-machine-drawing-pdf>

Note: 40% Course Work should be done on Drawing Board & 60% Course Work should be done by computer

ENVIRONMENTAL STUDIES**III Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC01	MC	L	T	P	C	CIA	SEE	Total
		2	-	-	-	30	70	100

COURSE OBJECTIVES:

1. Determine the Natural resources on which the structure of development is raised for sustainability of the society through equitable maintenance of natural resources.
2. Illustrate about biodiversity that raises an appreciation and deeper understanding of species, ecosystems and also the interconnectedness of the living world and thereby avoids the mismanagement, misuse and destruction of biodiversity.
3. Summarize a methodology for identification, assessment and quantification of global environmental issues in order to create awareness about the international conventions for mitigating global environmental problems.
4. Sustainable development that aims to meet raising human needs of the present and future generations through preserving the environment.
5. Outline green environmental issue provides an opportunity to overcome the current global environmental issues by implementing modern techniques like CDM, green building, green computing etc.

COURSE OUTCOMES:

Upon successful completion of the course, the student should be familiar with and be able to:

1. Classify ecosystem and construct various ecological pyramids.
2. Distinguish between resources of renewable and non renewable origin and justify the need for conservation of natural resources
3. Identify the various families of biodiversity.
3. Classify pollution and also to identify various natural and anthropogenic sources of pollution.
5. Support eco-friendly projects and recommend environmental protection measures.

UNIT-I	ECOSYSTEM	Classes:9
Introduction, definition: Scope and Importance of environmental studies. Classification, structure and function of anecosystem, Food chains, food webs and ecological pyramids. Biogeochemical cycles, Bioaccumulation, Biomagnifications.		
UNIT-II	NATURAL RESOURCES	Classes: 9
Classification of Resources, water resources: useand over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Forest resources, Energy resources: renewable and non renewable energy sources.		
UNIT-III	BIODIVERSITY	Classes:9

Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife: co-existence and conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation.		
UNIT-IV	ENVIRONMENTAL POLLUTION	Classes:9
Environmental Pollution: Air Pollution: types of pollutants. Water pollution: Sources and types of pollution, Soil Pollution: Sources and types, Impacts of modern agriculture, degradation of soil. Noise Pollution: Sources and Health hazards, standards, composition and characteristics of e-Waste and its management. Global Environmental Problems: Ozone depletion and Ozone depleting substances (ODS).		
UNIT-V	ENVIRONMENTAL POLICY & LEGISLATION	Classes: 9
International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Towards Sustainable Future: Concept of Sustainable Development, Environmental Education, urbanization, Urban Sprawl. Concept of Green Building.		
Text Books:		
1. Erach Bharucha (2005)., Textbook of Environmental Studies for Undergraduate Courses, Hyderabad, Universities Press.		
Reference Books:		
1. Anubha Kaushik(2006)., Perspectives in Environmental Science, 3 rd Edition, New Delhi, New age international. 2. Textbook of Environmental Studies by OVK Reddy		
Web References:		
https://www.tutorialspoint.com/environmental_studies/environmental_studies_environment.htm		
E-Text Books:		
https://bookboon.com/en/environmental-science-ebooks		
MOOC Course		
https://swayam.gov.in/course/141-enviromental-studies-i		

II B.TECH IV SEMESTER SYLLABUS

NUMERICAL METHODS AND PROBABILITY STATISTICS

IV Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS06	BSC	L	T	P	C	CIA	SEE	Total
		3	0	--	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The concepts of finite differences, operators and relations between them.
2. Evaluation of integrals by using numerical methods.
3. The concepts of discrete and continuous random variables, probability distribution and density function.
4. The concept of correlation and regression to find covariance.
5. Evaluation of the given data for appropriate test of hypothesis and finding the variance.

COURSE OUTCOMES:

student should able to

1. Find Interpolating polynomial for the given tabular data.
2. Solve the first order ordinary differential equations using numerical techniques.
3. Evaluate the discrete and continuous random variables, mathematical expectation of mean and variance.
4. Apply the concepts of correlation and regression to find covariance and sampling distribution of mean and variance.
5. Evaluate the given data for appropriate test of hypothesis.

UNIT-I	INTERPOLATION&CURVE FITTING	Classes: 10
INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Differences of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae and Stirling's formula. Interpolation with unequal intervals – Lagrange's interpolation. CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$, $y = a x^b$, $y = a b^x$ by the method of least squares.		
UNIT-II	NUMERICAL TECHNIQUES	Classes: 08
ROOT FINDING TECHNIQUES: Bisection method-Iteration method and Newton Raphson method. NUMERICAL INTEGRATION: Newton-cotes quadrature formula - Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor's series method – Picard's method - Euler's - modified Euler's Method – Runge-Kutta method		
UNIT-III	SINGLE RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS	Classes:08
Basic definitions of probability, Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution- mathematical expectation, Mean, Variance. Binomial, Poisson & Normal distributions and their Properties.		

UNIT-IV	CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS	Classes:07
Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression-Regression Coefficient, The lines of regression. Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation.		
UNIT-V	TESTING OF HYPOTHESIS	Classes:09
Testing of hypothesis: Null hypothesis, Alternate hypothesis, type I & type II errors – critical region, confidence interval, Level of significance. One sided test, two sided test, Large sample tests: (i) Test of Equality of means of two samples equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion & difference between two sample proportions. SMALL SAMPLE TESTS: Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, it's properties, Chi-square test of goodness of fit.		
Text Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9 th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010. 3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning		
Reference Books:		
1. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand 2. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press		
Web References:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
https://www.e-booksdirectory.com/details.php?ebook=10166 https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOC Course		
https://nptel.ac.in/courses/111107062/ https://nptel.ac.in/courses/122102009/		

AIRCRAFT PRODUCTION TECHNOLOGY**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE06	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Emphasis is given to process capabilities and limits, tooling considerations, materials requirements and constraints, economics of production and design producibility
2. Introduction to Manufacturing Technology is a broad exploratory course that introduces students to the manufacturing industry
3. Though hands-on activities students will learn how manufactures use technology to change raw materials into finished products
4. The courses also introduce students to the principles and practices of modern aircraft manufacturing. Specific manufacturing processes are addressed as they relate to the production of major aircraft components made of conventional metals as well as fibre reinforced plastic composites.

COURSE OUTCOMES:

Students should able to

1. Compare between primary shaping and joining processes
2. Analyze the working of various material removing techniques by conventional machining processes
3. Compare the working of various material removing techniques by unconventional machining processes
4. Analyze the various heat treatment technique used in aerospace industries
5. Determine the methods used to identify the manufacturing defects

UNIT-I	CASTING & WELDING:	Classes: 12
General principles of various Casting Processes - Sand casting, die-casting, centrifugal casting, investment casting and shell molding types. Principles and equipment used in arc welding, as welding, resistance welding, Laser welding , Electron Beam welding, Soldering and brazing techniques.		
UNIT-II	MACHINING AND FORMING	Classes: 12
General Principles (with schematic diagram only) of working and types-lathe, shaper, milling machines, grinding, drilling m/c, CNC machining. Sheet metal operations-shearing, punching, super plastic forming and diffusion bonding. Bending, Automation in bend forming and different operations in bending like stretch forming, spinning, drawing etc.		
UNIT-III	UNCONVENTIONAL MACHINING	Classes: 12
Principles of working and applications of abrasive jet machining, ultrasonic machining, electric discharge machining, electro chemical machining, laser beam, electron beam and plasma arc machining.		

UNIT-IV	HEAT TREATMENT AND SURFACE FINISHING	Classes: 12
Heat treatment of Aluminum alloys, titanium alloys, steels, case hardening, Initial stresses and the stress alleviation procedures. Corrosion prevention, protective treatment for aluminum alloys, steels, anodizing of titanium alloys, organic coating, and thermal spray coatings.		
UNIT-V	JIGS & FIXTURES, NDT and Other Inspection Techniques	Classes: 12
Jigs, fixtures, stages of assembly, types and equipment for riveted joints, bolted joints (only). Aircraft Tooling Concepts. Dye Penetrant Test, X - ray, magnetic particle and ultrasonic testing. Acoustic holography.		
Text Books:		
1. Introduction to Physical Metallurgy, Sidney H. Avenner. 2. Kalpakjian serope (2011), Manufacturing Engineering and Technology, 5 th edition, Pearson Education, New Delhi, India. 3. Manufacturing Technology, P.N. Rao, TMH, R.K. Jain		
Reference Books:		
1. Material Science and Metallurgy, kodgire. 2. Science of Engineering Materials, Agarwal 3. Materials Science and Engineering, William and collister. 4. Manufacturing Technology, Kalpak Jain, Pearson education 5. Production Technology, R.K. Jain 6. Process and materials of manufacturing –Lindberg, PE 7. Keshu S. C, Ganapathy K. K (2012), Air craft production techniques, E-book, Interline Publishing House, Bangalore. 8. P. C. Sharma (2011), Manufacturing Technology - I & II, 1 st edition, S. Chand & Company Ltd. New Delhi		
Web References:		
https://www.researchgate.net/publication/252435685_Aircraft_Production_Technology		
E-Text Books:		
Douglas F Horne, Aircraft Production Technology		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_me51/		

AERODYNAMICS-I**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE07	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

- 1 Understand the theoretical concepts underlying the development of lift, drag, and movement forces on aeronautical vehicle
- 2 Be able to visualize the flow around objects and physically understand the concepts of path lines, streamlines and vorticity
- 3 Understand the concept of superposition of elementary flows for linear incompressible flow
- 4 Analyze the characteristics of airfoil geometries and planform shapes to assist in determining aircraft performance
- 5 Be equipped to evaluate new lift-enhancement or drag-reduction devices and appreciate the directions and promise of upcoming developments in aerodynamic technology
- 6 Determine basic aerodynamic characteristics of propeller

COURSE OUTCOMES

Upon successful completion of this course, the student will be able to

1. Determine the flow similarities over different bodies like airfoil, cylinder and sphere
2. Analyze the combination of elementary flows.
3. Determine aerodynamics characteristics for symmetric and cambered airfoil section in an incompressible flow condition
4. Compare the flow over an arbitrary bodies over cylinder and sphere with respect to non-lifting and lifting flows
5. Explain the flow visualization techniques over symmetrical and cambered airfoil by using PIV and hot wire anemometers.

UNIT-I	BASICS	Classes: 12
Flow regimes, definition of incompressible and compressible flows, isentropic relations, Flow Similarity, Dimensional Analysis, Wing and airfoil geometry, aerodynamic force and moments, estimation of lift, drag and pitching moment from the pressure distribution, aerodynamic centre, centre of pressure, types of drag, characteristics of symmetric and cambered airfoils.		
UNIT-II	ELEMENTARY FLOWS & BASIC ELEMENTARY FLOWS	Classes: 12
ELEMENTARY FLOWS: Scalar and vector fields, Velocity Potential and Stream function for 2-D incompressible flow, Incompressible flow condition, governing equation for irrotational incompressible flow Laplace's equations, boundary conditions. BASIC ELEMENTARY FLOWS: uniform flow, source flow, doublet flow and vortex flow, superimposition of elementary flows, Combination of uniform flow with a Source and Sink, Doublet, non-lifting and lifting flow over a circular cylinder. Kutta - Joukowski theorem and the generation of lift, numerical source panel method, comparison with real flow over circular cylinder, D'Alembert's paradox.		
UNIT-III	COMPLEX TRANSFORMATION & THIN AIRFOIL THEORY	Classes: 12
COMPLEX TRANSFORMATION: The vortex filament and vortex sheet. Kelvin circulation theorem, The Complex potential function, Blasius theorem, principles of conformal transformation, the Joukowski transformation of a circle into flat plate, airfoils and ellipses, modified Joukowski transformation THIN AIRFOIL THEORY: The vortex sheet, Kutta condition and Kelvin's circulation theorem. Classical		

thin airfoil theory: symmetric and cambered airfoil, vortex panel numerical method, experimental characteristics of airfoils and comparison with theoretical results.		
UNIT-IV	FINITE WING THEORY	Classes: 12
Downwash, induced drag, Biot-Savart's law and Helmholtz's theorem. Prandtl's classical lifting line theory and fundamental equations. Elliptic and general lift distribution over finite unswept wings, effect of aspect ratio, flow past swept and delta wings. Drag polar, correlation of C_L distribution over other aspect ratios, lifting surface theory and ground effect.		
UNIT-V	AERODYNAMIC CHARACTERISTICS OF AIRFOILS AND WINGS	Classes: 12
Introduction to airfoils- nomenclature, Types of Airfoils- NACA Series and their applications, Laminar flow airfoils, Low Reynolds number airfoils, Subsonic compressible flow past airfoils; Critical Mach number, drag divergence Mach number, supercritical airfoils, effect of sweep, area rule. Introduction to Wind tunnels- approaches for solution of Fluid Dynamic problems velocity and pressure measurement, flow visualization techniques- PIV, Hot wire Anemometers, Smoke Testing		
Text Books:		
1. Anderson J .D.(2011), Fundamental of Aerodynamics, 5th edition, McGraw-Hill International Edition, New York 2. E. L. Houghton, P.W. Carpenter (2010), Aerodynamics for Engineering Students, 5th edition, Elsevier, New York.		
Reference Books:		
1. L. J. Clancy, Aerodynamics, 1/e, Shroff Publications, 2006 2. J. J. Bertin and R. Cummings, Aerodynamics for Engineers, 6/e, Pearson, 2013. 3. B. W. McCormick, Aerodynamics, Aeronautics and Flight Mechanics, 2/e, John Wiley and Sons, 1995 4. C. Y. Chow and A. M. Kuethe, Foundations of Aerodynamics, 5/e, Wiley India, 2009.		
Web References:		
https://www.nasa.gov/audience/forstudents/k-4/stories/nasa-knows/what-is-aerodynamics-k4.html		
E-Text Books:		
Arnab Roy, A First Course on Aerodynamics		
MOOC Course		
https://nptel.ac.in/courses/101105059/ https://www.edx.org/course/introduction-to-aerodynamics		

AEROSPACE VEHICLE STRUCTURES – I**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE08	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

Upon the completion of course students should be able to understand.

1. Understanding the basic concepts of stresses and strains on the 3D elastic body
2. Apply the degree of redundancy (DOR) for statically determinate and indeterminate beam structure.
3. Evaluate the Equation of equilibrium for a curved beam stresses
4. Analyze and differentiate the behaviour in members subjected to combined loading and used to study the theories of failure effects.
5. Demonstrate the knowledge of the critical loading conditions of the columns.
6. Summarize the direct application of concepts of shear flow for a closed thin walled section beam to aircraft structures

COURSE OUTCOMES:

This course uses lectures, assignments and home works to the students. The teaching methods include regular class work, Problem solving, technical quiz and seminars to enable the students:

1. Develop the mathematical model for a physical structure subjected to external loads applied on it.
2. Calculate the degree of redundancy for statically determinate and indeterminate beam structure
3. Determine the curved deflection due to forces and moments on an aircraft fuselage structure
4. Illustrate the stresses and strains in members subjected to combined loading and apply the theories of failure for static loading
5. Estimate the buckling load equations for different boundary conditions of the columns
6. Analyze the shear flow distribution in an aircraft structures like wings & fuselage

UNIT-I	INTRODUCTION TO THEORY OF ELASTICITY	Classes: 12
Introduction To Aircraft Structural Components, Equilibrium and Compatibility conditions for elastic solids, 2D elasticity equations for plane stress, plane strain and generalized plane strain causes Airy's stress function. Simple problems in plane stress/plane strain stresses and strains on arbitrary planes and transformations. Concept of principle planes, stress and Strains, Construction of Mohr's circle		
UNIT-II	REDUNDANT STRUCTURES	Classes: 12
Analysis of statically determinate structure and Indeterminate structure, order of redundancy, Introduction to redundant analysis, use of free body diagrams to explain compatibility and redundant analysis principles, Singularity method for uniform beams with various boundary and support conditions (props, hinges and fixities) subjected to distributed / discrete loads (including moment) moment area method analysis of various types of beams, three moment equation (Clapeyron theorem) of a continuous beams		
UNIT-III	BEAMS WITH ELASTIC SUPPORTS AND INITIAL CURVATURE, THEORIES OF FAILURE	Classes: 12
Direct solution of beams on elastic foundation, deflection of beams, Equation of equilibrium for curved beam and deflection of typical curved beams (Bulkhead sections on Fuselage) Introduction – Various theories of failure - Maximum Principal Stress Theory, Maximum Principal Strain Theory, Strain Energy and Shear Strain Energy Theory (Von Mises Theory) and fracture modes		

UNIT-IV	STABILITY OF COLUMNS	Classes: 12
Stability of structural systems, Models of instability of columns. Euler's formula for critical loads of column, end conditions of a column, Slenderness ratio, Effect of boundary conditions on mode shapes and critical loads. Column with initial curvature, effect of eccentricity Long, Medium and short column ranges. Rankine and Johnson's formulae. Effect of intermediate supports. Concept of beam column.		
UNIT-V	ENERGY PRINCIPLES AND METHODS, SHEAR FLOW IN CLOSED SECTIONS	Classes: 12
Introduction to energy principle and methods. Principle of virtual Displacements and principle of Virtual Force Castigliano's theorems, maxwell's reciprocal theorem and unit load method.		
Bredt- Batho formula. Single and multi-cell closed box structures. Semi monocoque and monocoque structures. Approximate method for box beams. Shear flow in single and multicell monocoque and semi monocoque box beams subject to torsion.		
Text Books:		
1. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5th edition, Elsevier, New York. 2. Strength of Materials by R.K Rajput S.chand publishers.		
Reference Books:		
1. Irving Herman Shames, Clive L. Dym (2003), Energy and finite element methods structural analysis, McGraw Hill, New Delhi, India. 2. B. C. Punmia (2011), Theory of Structures, 13th edition, Laxmi Publications Ltd, Hyderabad. 3. Donaldson B. K (2008), Analysis of Aircraft Structures an introduction to Aeronautical Structures Analysis, 2nd edition, Cambridge University Press, USA.		
Web References:		
https://cosmolearning.org/courses/introduction-aerospace-structures/ http://www.triumphgroup.com/business-units/aerospace-structures/		
E-Text Books:		
Aircraft Structures (Dover Books on Aeronautical Engineering) Aircraft Structures - Federal Aviation Administration		
MOOC Course		
https://www.edx.org/course/introduction-to-aerospace-structures-and-materials		

FLIGHT MECHANICS – I

IVSemester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE09	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

Upon the completion of course students should understand

1. Demonstrate competence in evaluating lift and drag of airplanes, and installed thrust of their engines
2. To familiarize with the concepts of Flight performance
3. To understand the parameters effecting the performance
4. To familiarize with the concept of Stability and control of Aircraft
5. To familiarize with the concept of dynamic stability of Aircraft
6. Ability to understand and apply the governing equations related to longitudinal and lateral static stability and control of an aircraft
7. Effects of the aerodynamic design, center of gravity location and moments of inertia on static and dynamic stability and control of an aircraft

COURSE OUTCOMES:

Students should be able to:

1. Compare and contrast between the performance parameters for civil transport and military aircrafts
2. Calculate the range and endurance of a fixed-wing aircraft with either a jet or a propeller-driven propulsion system in straight and level flight and analyze the various types of cruise techniques
3. Evaluate the performance of an aircraft during its climb and descend
4. Evaluate the factors effecting the take-off and landing performance of the aircraft
5. Apply flight mechanics analytical concepts and aircraft performance data to the preliminary design of a new aircraft to meet defined performance requirements

UNIT-I	INTRODUCTION TO AIRCRAFT PERFORMANCE	Classes: 12
The role and design mission of an aircraft specification of the performance requirements and mission profile. Importance of performance analysis, estimation and measurements .Scheduled performance and operational performance of aircraft. The international standard Atmosphere. Off - design atmosphere. Measurements of air data. Air data computers.		
UNIT-II	THE FORCE SYSTEM OF THE AIRCRAFT, CRUISE PERFORMANCE	Classes: 12
The aircraft force system. The lift force, side force the drag force. Total airplane drag- drag estimation - drag reduction methods. The propulsive forces the thrust producing engine, power and specific fuel consumption with altitude and flight speed. The minimum drag speed, minimum power speed. Aerodynamic relationships for a parabolic drag polar.		
The maximum & minimum speeds in level flight -Range and Endurance of aircraft with thrust producing engines and with power producing engines. Cruise techniques: constant angle of attack, constant Mach number, constant Mach number methods, comparison of performance. The effect alternative fuel flow laws, the effect of weight, altitude and temperature on cruise performance. Cruise performance of aircraft with mixed power plants.		
UNIT-III	CLIMB AND DESCENT PEROFRMANCE, AIRCRAFT MANEUVER PERFORMANCE	Classes: 12
Importance of climb and descent performance- safety considerations. Climb and descent techniques, generalized performance analysis for thrust producing, power producing and mixed power plants, maximum climb gradient, climb rate. Energy height and specific excess power, energy methods for optimal climbs, minimum time climbs, minimum fuel climbs, Measurements of climb performance. Descent performance in aircraft operations. Effect of wind on climb and descent performance.		

The general equations of accelerated motion of aircraft maneuver envelope, significance. Longitudinal aircraft maneuver, the pull – up maneuver, pull down maneuver-Lateral maneuvers, turn performance, instantaneous turns - specific excess power, the energy turns, aircraft maneuver performance.		
UNIT-IV	TAKE-OFF AND LANDING PERFORMANCE PLANNING, AIRCRAFT PERFORMANCE MEASUREMENTS AND DATA HANDLING	Classes: 12
Flight safety criteria. Performance classification civil. Flight plans, performance planning and fuel planning Estimation of take off distances. The effect on the take – off distance, of weight wind runway conditions, ground effect. Take off performance safety factors. Estimation of landing distances, the discontinued landing baulked landing air safety procedures and requirements on performance. Fuel planning fuel requirements trip fuel, fuel reserves, tankering. Purpose of performance measurements in flight. Flight testing Principle performance variables weight, altitude and ambient temperature (WAT). Parametric performance data analysis. Dimensional analysis. Measurements data reduction. The equivalent weight method. Correction to cruise climb take – off and landing performance for weight and temperature.		
UNIT-V	THE APPLICATION OF PERFORMANCE DATA	Classes: 12
The performance summary and fleet selection-the block performance, payload – range diagram. Route analysis and optimization. Operational analysis procedure. Operational performance data for flight planning, take off field performance runway correction chart, aircraft datum performance (WAT) chart, determination of the maximum takeoff weight		
Text Books:		
1. Eshelby M. E (2000), <i>Aircraft performance: Theory and Practice</i>, AIAA Education Series, USA. 2. Brandt S. A. et al (2004), <i>Introduction to Aeronautics: A Design perspective</i>, 2nd edition, AIAA Education Series, USA. 3. Anderson J. D. (2011), <i>Aircraft Performance and Design</i>, international edition, McGraw Hill, New Delhi.		
Reference Books:		
1. Dole C. E. (2010), <i>Flight Theory and Aerodynamics; a practical Guide for Operational Safety</i>, Wiley India Ltd, New Delhi, India. 2. Mc Cormic B. W. (2010), <i>Aerodynamics, Aeronautics and Flight Mechanics</i>, 2nd edition, Wiley India Ltd. India 3. Raymer D. P (2006), <i>Aircraft Design: A Conceptual Approach</i>, 4th Edition, AIAA Education Series, USA 4. Yethout (2003), <i>Introduction to Aircraft Flight Mechanics</i>, AIAA Education Series, USA		
Web References:		
https://nptel.ac.in/courses/101106041/		
E-Text Books:		
https://soaneemrana.org/onewebmedia/MECHANICS%20OF%20FLIGHT%20BY%20A.C%20KERMODE.pdf		
MOOC Course		
https://nptel.ac.in/courses/101104062/		

AERODYNAMICS LAB

IV Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE10	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the student:

1. To understand and study the operation of subsonic wind tunnel.
2. To study experimentally the pressure distribution of circular, symmetric and cambered airfoil
3. To study Flow visualization studies in low speed flow over airfoil with different angle of incidence and over cylinder
4. To determine the drag coefficient for a Cylinder.
5. They can analyse the performance of axial and centrifugal compressors.

COURSE OUTCOMES:

The students should be able to:

1. Evaluate the pressure distribution over different aerodynamic shapes
2. Apply the flow visualization techniques over different aerodynamic bodies
3. Determine the efficiency of axial and centrifugal compressor
4. Determine the efficiency of blower
5. Predict the aerodynamic forces

LIST OF EXPERIMENTS

1. Calibration of subsonic wind tunnel.
2. Pressure distribution over smooth and rough cylinder.
3. Pressure distribution over symmetric airfoils.
4. Pressure distribution over cambered airfoils
5. Flow visualization studies in low speed flows over sphere.
6. Flow over a flat plate at different angles of incidence.
7. Flow visualization studies in low speed flows over cylinders.
8. Flow visualization studies in low speed flows over airfoil with different angle of incidence.
9. Fluid flow studies using a blower.
10. Drags of different bodies.
11. Performance of Axial flow compressor.
12. Performance of Centrifugal flow compressor.

Note: Ten experiments should be performed.

Reference Books:

Anderson J .D.(2011), Fundamental of Aerodynamics, 5th edition, McGraw-Hill International Edition, New York

Web References:

<https://www.nasa.gov/audience/forstudents/k-4/stories/nasa-knows/what-is-aerodynamics-k4.html>

AIRCRAFT PRODUCTION TECHNOLOGY LAB**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE11	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the student:

1. To understand and study the various manufacturing process in aircraft industries.
2. To understand and perform various operation on conventional machines.
3. To understand and perform various operation on moulding process.
4. To understand and perform various welding processes used in industry.

COURSE OUTCOMES:

The students should be able to:

1. Manufacturing of a products by using various joining techniques
2. Analyse micro structure of ferrous and non ferrous material
3. Operate conventional machines to remove material with single point cutting tools like lathe, shaping, planning and slotting.
4. Operate conventional machines to remove material with multi point cutting tools like milling, drilling, etc
5. Perform surface finishing using surface grinding machines

LIST OF EXPERIMENTS

1. Lathe operations (Step turning, taper turning, grooving, knurling, threading)
2. Milling
3. Drilling and tapping
4. Shaping (making U grooves)
5. Slotting (making slots on pulleys)
6. Surface grinding
7. CNC lathe
 - I. Facing
 - II. Step turning
 - III. Taper turning
 - IV. Multiple turning
8. CNC milling
 - I. Plain milling
 - II. Step milling
 - III. End milling
9. Blow molding process
10. Mechanical press working
11. Spot welding, soldering & brazing
12. TIG welding

Note: Ten experiments should be performed.

Reference Books:

Manufacturing Engineering and Technology by Kalpakjian.

Web References:

https://www.researchgate.net/publication/252435685_Aircraft_Production_Technology

AIRCRAFT INTERIOR DESIGN LAB**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE12	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

1. To design various Aircraft interior components in lab.
2. To design various individual components, sub-assemblies and main assemblies in lab.
3. Discuss the importance of design process and studying the different phases of designing process involved in the design.
4. Understand the design of aircraft concepts.
5. To design various aircraft components by using Catia software

COURSE OUTCOMES:

1. Understand the importance of drawing and design process and phases involved in the design process
2. Design various individual components, sub-assemblies and main assemblies in design lab
3. Design various orthographic and isometric projections in drawing sheets
4. Develop the basic concepts of aircraft interiors
5. Design and develop aircraft interior components

LIST OF EXPERIMENTS

- 1 Aircraft Seating Design Arrangements (General Class & Business Class) (2D)
- 2 Aircraft Single Aisle Design(3D & 2D)
- 3 Aircraft Twin Aisle Design(3D & 2D)
- 4 Aircraft Seat Design (3D- Business Class)
- 5 Aircraft Seat Design (3D- First Class)
- 6 Design of Laptop Tray Sizing for Seating in Civil Aircrafts.
- 7 Design of Window for Civil Aircraft
- 8 Design of Wind Shield for Fighter Aircraft(Select any one airplane)
- 9 Design of Wind shield for Passenger Aircraft(Select any one Airplane)
- 10 Design of Arm rest cap height from surface in Civil Aircraft
- 11 Design of Joystick(Spacing) in Fighter Aircraft

Note: Ten experiments should be performed.

Reference Books:

Basics in Catia V5 with Simulation by Ranjan Chikesh

Web References:

<https://www.youtube.com/channel/UCKgl0VNNTToH-xje5XlwBAhg>

GENDER SENSATIZATION**IV Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC02	MC	L	T	P	C	CIE	SEE	Total
		1	--	--	Nil	30	70	100
Contact Classes: 16	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 4216			
COURSE OBJECTIVES: 1. To provide a critical perspective on the socialization of men and women. 2. To introduce students to information about some key biological aspects of genders. To expose the students to debates on the politics and economics of work. 3. To help students reflect critically on gender violence. 4. To expose students to more egalitarian interactions between men and women.								
COURSE OUTCOMES: 1. Develop a better understanding of important issues related to gender in contemporary India. 2. Sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender 3. Attain a finer grasp of how gender discrimination works in our society and how to counter it 4. Men and Women students and professionals will be better equipped to work and live together as equals. 5. Develop a sense of appreciation of women in all walks of life.								
UNIT-I	UNDERSTANDING GENDER						Classes: 03	
Gender: Why Should We Study It? (Towards a World of Equals: Unit -1) Socialization: Making Women, Making Men (Towards a World of Equals: Unit -2)Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.								
UNIT-II	GENDER AND BIOLOGY						Classes: 03	
Missing Women: Sex Selection and Its Consequences (Towards a World of Equals: Unit - 4)Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (Towards a World of Equals: Unit -10)Two or Many? Struggles with Discrimination.								
UNIT-III	GENDER AND LABOUR						Classes:03	
Housework: the Invisible Labour (Towards a World of Equals: Unit -3) “My Mother doesn’t Work.” “Share the Load.” Women’s Work: Its Politics and Economics (Towards a World of Equals: Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.								
UNIT-IV	ISSUES OF VIOLENCE						Classes:04	
Sexual Harassment: Say No! (Towards a World of Equals: Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading:“Chupulu”. Domestic Violence: Speaking Out (Towards a World of Equals: Unit -8) Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading:New Forums for Justice.								

Thinking about Sexual Violence (Towards a World of Equals: Unit -11) Blaming the Victim-"I Fought for my Life..." - Additional Reading: The Caste Face of Violence.		
UNIT-V	GENDER: CO - EXISTENCE	Classes:03
Just Relationships: Being Together as Equals (Towards a World of Equals: Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers. Additional Reading: Rosa Parks-The Brave Heart.		
Text Books:		
All the five Units in the Textbook, " Towards a World of Equals: A Bilingual Textbook on Gender " written by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad , Telangana State in the year 2015 .		
Reference Books:		
1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012 2. Abdulali Sohaila. "I Fought For My Life...and Won." Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/		
Web references:		
1. http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/		
E-Text Books:		
Abdulali Sohaila. " <i>I Fought For My Life...and Won.</i> " Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/		

III B.TECH V SEMESTER SYLLABUS

AEROSPACE VEHICLE STRUCTURES – II**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE13	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To analyse the Aerospace structures under major loading conditions
2. To conduct stress analysis on aircraft components

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Illustrate the tension field and axial flow diagrams
2. Explain the failure stresses in plates and stiffened panels.
3. Demonstrate a shear loaded thin walled beams- general stress, strain and displacement relationships- direct stress and shear flow system- shear centre, twist and warping
4. Distinguish between buckling of thin plates and deflections due to bending
5. Develop wing spars and box beams- tapered wing spar, open and closed section beam

UNIT-I	THIN PLATE THEORY, STRUCTURAL INSTABILITY	Classes: 12
Analysis of thin rectangular plates subject to bending, twisting, distributed transverse load, combined bending and in-plane loading-thin plates having small initial curvature, energy methods of analysis.		
UNIT-II	BUCKLING OF THIN PLATES AND BENDING OF THIN WALLED BEAMS	Classes: 12
Buckling of thin plates-elastic, inelastic, experimental determination of critical load for a flat plate, local instability, Tension field beams- complete diagonal tension, incomplete diagonal tension. Unsymmetrical bending- resolution of bending moments, direct stress distribution, position of neutral axis. Deflections due to bending-approximations for thin walled sections, temperature effects.		
UNIT-III	SHEAR AND TORSION OF THIN WALLED BEAMS	Classes: 12
Shear loaded thin Walled beams-general stress, strain and displacement relationships, direct stress, shear centre, twist and warping. Torsion of beams of closed section-displacements associated with Bredt-Batho shear flow. Torsion of open section beams, warping of cross section conditions for zero warping. Bending, shear, torsion of combined open and closed section beams		
UNIT-IV	STRUCTURAL IDEALIZATION OF THIN WALLED BEAMS, TORSION OF THIN WALLED BEAMS	Classes: 12
Structural idealization-principal assumptions, idealization of panel, effect on the analysis of thin Walled beams Under bending, shear, and torsion loading-application to determining deflection.		
UNIT-V	STRESS ANALYSIS OF AIRCRAFT COMPONENTS- WING, FUSELAGE	Classes: 12

Wing spars and box beams-tapered wing spar, open and closed section beams, beams having variable Stringer areas. Wings-Three-boom shell in bending, torsion, shear, tapered wings, deflections, cut-outs in wings. Bending, shear, torsion, cut-outs in fuselages, fuselage frames and wing ribs-principles of stiffener/ web construction, wing ribs.

Text Books:

1. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5th edition, Elsevier, New York.
2. David J. Perry (2011), Aircraft Structures, 2nd Edition, McGraw- Hill, New Delhi.
3. E F Bruhn (1973), Analysis and Design of Flight Vehicle Structures, Tri-state Offset Company, USA

Reference Books:

1. Irving Herman Shames, Clive L. Dym (2003), Energy and finite element methods structural analysis, McGraw-Hill, New Delhi.
2. B. C. Punmia (2011), Theory of Structures, 13th edition, Laxmi Publication, Hyderabad.
3. Donaldson B. K. (2008), Analysis of Aircraft Structures An introduction to Aeronautical Structures Analysis, 2 Edition, Cambridge University Press, USA.
4. S Ramamrutham & R Narayan (2018), Theory of Structures, Dhanpat Rai Publishing Company, India.

Web References:

<https://cosmolearning.org/courses/introduction-aerospace-structures/>

<http://www.triumphgroup.com/business-units/aerospace-structures/>

E-Text Books:

1. Aircraft Structures (Dover Books on Aeronautical Engineering)
2. Aircraft Structures - Federal Aviation Administration

MOOC Course

1. <https://www.edx.org/course/introduction-to-aerospace-structures-and-materials>
2. <https://nptel.ac.in/courses/112101095/>

AERODYNAMICS-II**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE14	PCC	L	T	P	C	CIA	SEE	Total
		3		-	3	30	70	100

COURSE OBJECTIVES

Students undergoing this course are expected:

- To familiarize the features of inviscid compressible flows, including shock and expansion waves and the governing differential equation of motion of steady compressible flows
- To familiarize the estimation of the lift and drag for basic aerodynamic shapes in compressible in viscous flows and the importance of compressibility effects in wind tunnel design

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Formulate and predict the aerodynamic characteristics of a body in supersonic flows
2. Apply perturbation theories to evaluate the changes in performance of a body due to disturbances
3. Design a nozzle using Method of Characteristics and evaluate flow inside nozzle.
4. Evaluate the aerodynamic characteristics of supersonic airfoils.
5. Calculate flow variable across shock and expansion waves using normal shocks tables, oblique shock charts, Prandtl-Meyer function tables as a preparation toward design of high speeds aircrafts and high speed wind tunnel

UNIT-I	ONE DIMENSIONAL FLOWS	Classes: 12
Compressibility, Review of Fundamentals: Concepts from Fluid Mechanics, Basic Thermodynamic Relations. Velocity of sound. Mach number, flow regimes. Governing equations for inviscid compressible flow-Continuity, Momentum and Energy equations in Integral and Differential form. Stagnation conditions..		
UNIT-II	SHOCKS AND EXPANSION WAVES	Classes: 14
Basic equations, relations across a normal shock, calculation of normal shock wave properties, measurements of airspeed in subsonic and supersonic flows. Entropy rise across normal shock and its relation to pressure rise, Numerical exercises with normal shock tables. Oblique shock relations. Supersonic Mach number relations strong and weak shock solutions / Shock flow over a wedge polar. Regular reflection from a solid boundary. Intersections of shock wave. Expansion waves. Prandtl – Meyer Expansion.		
UNIT-III	FLOW THROUGH NOZZLES AND DUCTS	Classes: 10
Flow Through a nozzle: Convergent Nozzles, CD Nozzles, Exit Pressure variation vs Stagnation pressure variation. Choked flow conditions. Normal shock. Under and over expansion conditions. Flow through diffusers, wave reflections from a free boundary. Description of supersonic wind tunnels and rocket engine. Hugoniot Equation, Flow with Friction, Friction choking, variation of flow properties, tables and charts for Fanno flow, Flow with heat addition, Thermal choking, variation of flow properties, tables and charts for Rayleigh flow.		
UNIT-IV	PERTURBATION THEORIES AND LINEARIZATION	Classes: 12
Introduction, Velocity potential equation, subsonic small perturbation equation, Prandtl-Glauert compressibility corrections, Critical Mach number, Drag divergence Mach number, Area rule, Supercritical airfoil. Linearized supersonic flow, Linearized supersonic flow over airfoil and wings. Shock Expansion theory.		

Detached shock. Axi-symmetrical flows, flow past slender bodies of revolution, conical flows, Numerical integration procedure		
UNIT-V	FLOW MEASUREMENTS AND MODEL TESTING	Classes: 10
Non dimensional parameters and numbers Similarity of flows, Model testing in wind tunnels. Pressure, Velocity measurements, Force measurements-Wind tunnel balances, Scale effects and corrections, wall interferences, induced drag and other computations/corrections. Experimental Methods, Shock Tube, Supersonic Wind tunnel, Flow visualization, Supersonic Probes., Methods of characteristics. Design of nozzles, External flow around bodies, Experimental characteristics of airfoils in compressible flow.		
Text Books:		
1. A.H.Shapiro, The Dynamics And Thermodynamics Of Compressible Fluid, Volume -1, 2. H.W.Liepmann and Roshko, Elements of Gas Dynamics, Dover Publications 3. Rathakrishnan E.E. (2010), Gas Dynamics, 3rd Edition, Prentice Hall of India, New Delhi.		
Reference Books:		
1. Anderson J .D. (2011), Fundamental of Aerodynamics, 5th edition, McGraw-Hill, New Delhi. 2. Hodge B. K, Koenig K (1995), Compressible Fluid Dynamics with Computer Application, 1st edition, Prentice Hall, New York. 3. JJ. Bertin, Aerodynamics for Engineers, 5th edition, Pearson.		
Web References:		
https://nptel.ac.in/courses/112103021/ https://nptel.ac.in/courses/101106044/		
E-Text Books:		
https://open.umn.edu/opentextbooks/textbooks/fundamentals-of-compressible-flow-mechanics		
MOOC Course		
1. https://www.classcentral.com/course/nptel-fundamentals-of-gas-dynamics-5293 2. https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-120-compressible-flow-spring-2003/		

FLIGHT MECHANICS – II**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE15	PCC	L	T	P	C	CIA	SEE	Total
		4	1	-	4	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Describe the contribution of various components to the static longitudinal stability of the aircraft
2. Evaluate the effect of centre of gravity on the static longitudinal stability of the aircraft
3. Ability to distinguish between stick fixed and stick free stability conditions
4. Discuss the aerodynamic stability and control derivatives of the aircraft
5. Explain the longitudinal and lateral motions of the aircraft

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Calculate airplane performance for steady and accelerated flights
2. Analysis the aircraft trim analysis. importance of the neutral point, maneuver point etc.
3. Distinguish between stick fixed and stick free stability conditions
4. Illustrate how an aircraft can maintain static and dynamic stability
5. Assess if a particular aircraft configuration is statically stable, and suggest design changes needed to affect the static stability of an aircraft

UNIT-I	STATIC LONGITUDINAL STABILITY AND CONTROL	Classes: 15
Introduction Static and dynamic stability. Need for controlled flight - Equilibrium, stability, control, trim-definitions- examples Need for stability in airplanes. Purpose of controls, inherently and marginally stable airplanes. Tailless aircraft and aircraft with fore planes Basic equilibrium equation, Stability criterion, Contribution of wing and tail and elevator to pitching moments. Effect of fuselage and nacelles, Effects of center of gravity location, Power effects Stabilizer setting and center of gravity location, Elevator power, Elevator to trim. Trim gradients. Control fixed static stability.		
UNIT-II	STATIC LONGITUDINAL STABILITY - CONTROL FREE, MANEUVER STABILITY	Classes: 12
Effects of releasing the elevator. Hinge moment coefficients, Control forces to trim. Control free neutral point - Trim tabs. Aerodynamic balancing of control surfaces. Means of augmentation of control. Contribution of pitch damping to pitching moment of flight vehicle, Effect on trim and stability. Control deflections and control forces for trim in symmetric manoeuvres and coordinated turns. Control deflection and force gradients. Control fixed and control free manoeuvre stability. Manoeuvre points. Manoeuvre margins.		
UNIT-III	INTRODUCTION- EQUATIONS OF MOTION	Classes: 14
Degree of freedom of a system, Need for stability in airplanes. Purpose of controls, inherently and marginally stable airplanes. Tailless aircraft and aircraft with fore planes. Equations of motion of a rigid body. Inertial forces and moments. Equations of motion of flight vehicles. Aerodynamic forces and moments.		

UNIT-IV	AERODYNAMIC STABILITY DERIVATIVES	Classes: 12
Aerodynamic stability and control derivatives. Relation to geometry, flight configuration. Effects of power, compressibility and flexibility. Derivatives of side force, rolling and yawing moments with respect to the angle of sideslip, rate of sideslip.		
UNIT-V	STATIC LATERAL AND DIRECTIONAL STABILITY AND CONTROL,DYNAMIC STABILITY AND RESPONSE TO CONTROL	Classes: 14
Dihedral effect,Coupling between rolling and yawing moment, Adverse yaw, Aileron power, Aileron reversal. Weather cocking effects, Rudder power. Lateral and directional stability- definition. Control surface deflections in steady sideslips, rolls and turns one engine inoperative conditions, Rudder lock,Solutions to the stability quadratic of the linearised equations of motion. The principal modes. Phugoid, Short Period Dutch Roll and Spiral modes, Further approximations. Restricted degrees of motion. Solutions. Response to controls. Auto rotation and spin recovery, pilot techniques.		
Text Books:		
1. Perkins C. D, Robert Hage E (2003), Airplane Performance, Stability and Control, Wiley Toppan, USA. 2. Nelson R. C (2007), Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York. 3. T. R. Yechout, S. L. Morns (2003), Introduction to Aircraft Flight Mechanics, AIAA Publishers, USA.		
Reference Books:		
1. Houghton E. L, Carruthers N. B. (2010), Aerodynamics for Engineering Students, 5 th edition, Elsevier, USA. 2. Mc. Cormic B. W. (2010), Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA.		
Web References:		
https://nptel.ac.in/courses/101106043/6 https://nptel.ac.in/courses/101106042/		
E-Text Books:		
T. R. Yechout, S. L. Morns (2003), Introduction to Aircraft Flight Mechanics, AIAA Publishers, USA		
MOOC Course		
https://swayam.gov.in/nd1_noc19_ae10 https://www.mooc-list.com/course/abc-flight-mechanics-how-do-planes-fly-futurelearn		

AEROSPACE PROPULSION – I**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE16	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course is intended to build up necessary background for understanding the basics of propulsion.

1. To understand the application of various experimental fluid mechanics correlations in propulsion.
2. To understand the basic concepts of compressible fluid flow.

COURSE OUTCOMES:

1. Analyze principles of various engines and thermodynamic cycles
2. Analyze the characteristics and performance of various types of jet engines
3. Examine the various types of inlets and principle of axial flow compressors
4. Evaluate the performance of combustion chamber
5. Analyze the characteristics of turbine and nozzles

UNIT-I	FUNDAMENTALS OF GAS TURBINE	Classes: 12
Engines, Types Introduction to IC Engines Types of Reciprocating Engine (SI, CI), Piston Engine Thermodynamic Cycles, Piston Rings, Crank Casing, Rotation Mechanisms. Airbreathing Engines, types Working of Gas Turbine Engine, Cycle Representation Fundamentals Of Gas Turbine: Engines Thermodynamic Cycle of Gas Turbine Engine, Industrial Gas turbines, Types Advantages, Disadvantages Application to Industry Based, Aerospace Thrust, TSFC, SFC, Mass Flow rate Numerical Problems on Gas Turbine Basic Cycle, CI Engine, SI Engine		
UNIT-II	AIR BREATHING ENGINES FOR AIRPLANES	Classes: 12
Characteristics of turboprop, turbojet, Turbofan, Performance parameters characteristics Thrust Single Two-Shaft Turbojet & Turbo-prop, Turbofan Engines. Engine Design Operations. Engine - Design Point Operations. Principles, Aerothermodynamic Process involved in Turbo & Ram Air engines and Pulsejets in Aircraft propulsion. Ideal Cycle Analysis for Turbojet Engine, Turbofan Engine, Turboprop Engine, Real Analysis for Turbojet Engine, Turbofan Engine, Turbo Prop Engine. Numerical Problems		
UNIT-III	INLETS & COMPRESSORS	Classes: 12
Subsonic Inlets: Flow Characteristics, Governing Equations of Inlets, Stall in Subsonic inlets, Relation between minimum area ratio and external deceleration ratio. Diffuser performance. Supersonic Inlets, Inlet Guide Vanes, Characteristics of IGV's Axial Flow Compressors: Elementary theory of axial flow compressor, Velocity triangles, Degree of reaction. Flow through an Axial flow Compressor stage; Losses in axial flow compressor stage; Compressor blade design. Losses and Blade Single and Multi-stage Axial compressor characteristics HPC, LPC Design.		
UNIT-IV	COMBUSTION CHAMBER, BURNERS AND PERFORMANCE	Classes: 12
Classification of combustion process, important factors affecting combustion chamber design, Combustion chamber performance. Combustor liner materials and thermal barrier coatings. Thermo-chemistry of combustion, Chemical Equilibrium, Premixed Flame and flame stabilization, Jet Diffusion flame and droplet combustion, Flame Propagation, Chemical composition of Fuels in Aviation, BioFuels,		

Advanced Combustion Chemistry (Numerical problems on fuel compositions) Burners, Types of Burners, Materials for Combustion Chamber, Thermal Conduction, Convection, Heating & Cooling methodologies Performance Sensitivity, Flame Tube Cooling, Flame Stabilization. Use of Flame Holders, Afterburners Ignition methods for GT afterburners

UNIT-V**TURBINES & NOZZLES****Classes: 12**

Axial Flow Turbines: Introduction; Turbine stage; Turbine Blade 2-D (cascade) analysis Work Done; Degree of Reaction; Losses and Efficiency; Flow Passage; Subsonic, transonic and supersonic turbines, Multi-staging of Turbine, Turbine Cooling; Turbine Blade design & Profiles, Airfoil Data and Profile construction. Radial Turbine: Introduction; Thermodynamics and Aerodynamics of radial turbines; Radial Turbine Characteristics; Losses and efficiency; Design of radial turbine Numerical Problems
Nozzles: Nozzle Types, Nozzle Throat Conditions. Nozzle Efficiency, Losses In Nozzles. Nozzle Expansion Process Ejector And Variable Area Nozzles, Thrust Reversal thrust Vectoring, Noise reduction methods.

Text Books:

1. Ahmed F EL Sayed 2008 —Aircraft Propulsion and Gas Turbine EnginesII, CRC Press LLC
2. Ronald D. Flack (2010), Fundamentals of Jet Propulsion with Application, Cambridge University Press, USA.
3. H. I. H. Saravanamuttoo, Cohen H. Rogers (2009), Gas Turbine Theory, 6th edition, Pearson Education, New Delhi, India.
4. V. Ganesan (2010), Gas Turbines, Tata McGraw-Hill, New Delhi, India.

Reference Books:

1. Oates G. C (1986), AeroThermodynamics of Aircraft Engine Components, AIAA Education Series, USA.
2. Rolls- Royce (2005), Jet Engine, 6th edition, Rolls - Royce Ltd, USA.
3. S. Burack (2011), Gas Turbines and Jet and Rocket Propulsion, Goldberg Press, New York.
4. Saeed Farokhi (2009), Aircraft Propulsion, 2nd edition, John Wiley, USA.

Web References:

<https://nptel.ac.in/downloads/101101001/>

E-Text Books:

Aircraft Propulsion

https://www.sil.si.edu/smithsoniancontributions/AnnalsofFlight/pdf_lo/SAOF-0001.4.pdf

MOOC Course

<https://nptel.ac.in/courses/101106033/>

PROPULSION AND FLIGHT SIMULATION LAB**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE20	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To understand the physical behaviour of fluid, like, viscosity, density and, incompressible and compressible flow.
2. To learn fundamental calculations in heat transfer applicable to propulsion.
3. To provide real time experience on handling aircraft instruments and systems.

COURSE OUTCOMES

At the end of the course the students are able to:

1. Analyze the working principle, components and their functions of 2-s, 4-s piston engine and jet engines.
2. Calculate the dynamic and static engine balancing and analyze the calorific value of major fuels.
3. Understand the application of various experimental fluid mechanics correlations in engineering calculations especially in heat transfer area.
4. Analyse the performance of an aircraft under various operating conditions.
5. Improve the aspect of stability and control of an aircraft.
6. Analyse the static and dynamic response of aircraft for both voluntary and involuntary changes in flight conditions.

PROPULSION LAB

Exp.- I	Study of piston engine (Valve Timing)	Classes: 3
Exp.-II	Study of piston engine (Port Timing Diagram)	Classes: 3
Exp -III	Stripping of a piston engine, visual inspection and reasoning for common troubles and trouble shooting	Classes: 3
Exp -IV	Performance of piston engine 2-stroke	Classes: 3
Exp -V	Performance of piston engine 4-stroke	Classes: 3
Exp-VI	Heat Balance Test on piston engine	Classes: 3
Exp-VII	Engine Balancing	Classes: 3
Exp-VIII	Characterization of Aviation fuels (SOLID) (BOMB CALORIMETER)	Classes: 3
Exp- IX	Study of an aircraft piston engine.(Includes study of assembly of sub systems, various components, their functions and operating Principles)	Classes: 3
Exp- X	Study of an aircraft jet engine (Includes study of assembly of sub systems, various components, their functions and operating principles)	Classes: 3

FLIGHT SIMULATION LAB		
Exp.- I	Effect of speed on glide performance	Classes: 3
Exp.-II	Calculation of CL and CD using the Speed-Power Method	Classes: 3
Exp -III	Effect of Velocity on Climb Rate	Classes: 3
Exp -IV	Effect of Altitude on Range & Endurance of the Aircraft	Classes: 3
Exp -V	Turn Performance	Classes: 3
Exp-VI	Effect of Flaps and Weight on Take off Performance	Classes: 3
Exp-VII	Effect of Flaps and Weight on Landing Performance	Classes: 3
Exp-VIII	Longitudinal Stability Modes	Classes: 3
Exp.- IX	Lateral Stability Modes Stefan Boltzmann Apparatus	Classes: 3

Note: Twelve experiments should be performed by taking at least four experiments from each lab.

EQUIPMENT NEEDED	
I	2-S piston engine cut section
II	4-S piston engine cut section
III	Heat Engine Test Rig.
IV	Balancing test Rig
V	Calorimeter apparatus
VI	Piston Engine
VII	Aircraft jet engine
VIII	Flight Simulation setup

Reference Books:

1. H. Cohen, G. F. C. Rogers and H. I. H. Saravanamuttoo, "Gas Turbine Theory", Longman, 2006.
2. . C. Oates, "Aero thermodynamics of Aircraft Engine Components", AIAA Education Series, New York, 2005.
3. M. L. Mathur and R. P. Sharma, "Gas Turbine, Jet and Rocket Propulsion", Standard Publishers & Distributors, Delhi, 2007.
4. Courtland D. Perkins, Robert E. Hage, "Airplane Performance stability and Control", Wiley India Pvt Ltd, 2011.
5. Robert Nelson, "Flight Stability and Automatic Control", 2nd Edition, McGraw Hill Education (India) Private Limited, 2007.
6. Bernard Etkin, Lloyd Duff Reid, "Dynamics of Flight Stability and Control", 3rd Edition, Wiley India Pvt Ltd, 2010.
7. A. W. Babister, "Aircraft Dynamic Stability and Response", Pergamon Press, 2013.

Web References:

<http://nptel.ac.in/courses/101101002/>
<http://nptel.ac.in/courses/101104061/>
<http://nptel.ac.in/courses/101104062/>

AEROSPACE VEHICLE STRUCTURES LAB**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE21	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Install strain gauges correctly on a structure and apply appropriate electronics to measure strain with the gauges.
2. Experimentally determine the load-displacement behaviour of common aerospace thin-walled structures
3. Identify design features of aerospace structures, and calculate load factors and margins of safety;
4. Analyze the behaviour of thin-walled beams subjected to combined loads, including bending, torsion, and shear;
5. Analyze the stability of structural elements and determine critical buckling loads

COURSE OUTCOMES

At the end of the course the students are able to:

1. Classify types of shear loads acting on beam
2. Estimate crippling load acting on short and long column.
3. Determine surface flaws as well as internal flaws using ndt techniques
4. Analyze the various strain pressure using thin walled pressure vessel
5. Determine the failures on riveted plates using utm

LIST OF EXPERIMENTS

1. Tensile testing using universal Testing Machine
2. Bending tests: Deflection of beams for various end conditions. Verification of Maxwell's and Castiglianos theorems
3. Compression tests on long and short columns, Critical buckling loads.
4. Test on riveted joints.
5. Test using NDT inspection method.
6. Thin walled cylinders subjected to internal pressure.
7. Shear centre of an open sections beam
8. Shear center of a closed section beam
9. Vibrations testing
 - i) Free longitudinal vibrations
 - ii) Forced vibrations
 - iii) Torsional vibrations

Text Books:

1. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5th edition, Elsevier, New York.
2. David J. Perry (2011), Aircraft Structures, 2nd Edition, McGraw- Hill, New Delhi.

Reference Books:

1. Irving Herman Shames, Clive L.Dym(2003), Energy and finite element methods structural analysis, McGraw-Hill, New Delhi.
2. B. C. Punmia (2011), Theory of Structures, 13th edition, Laxmi Publication, Hyderabad.
3. Donaldson B. K.(200nd8), Analysis of Aircraft Structures An introduction to Aeronautical Structures Analysis, 2 Edition, Cambridge University Press, USA.

Web References:

<https://cosmolearning.org/courses/introduction-aerospace-structures/>

<http://www.triumphgroup.com/business-units/aerospace-structures/>

E-Text Books:

1. Aircraft Structures (Dover Books on Aeronautical Engineering)
2. Aircraft Structures - Federal Aviation Administration

MOOC Course

<https://www.edx.org/course/introduction-to-aerospace-structures-and-materials>

EQUIPMENT NEEDED:

1. UTM – 20 / 40 Tons.
2. Shear center Test rig
3. NDT Equipment
 - a. Ultrasonic apparatus
 - b. Magnetic Particle test rig
 - c. Dye penetration test
4. Vibration test rig.
5. Deflection test rig
6. Column test rig

AIRCRAFT DYNAMICS AND CONTROL SYSTEMS LAB**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE67	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

To develop a Matlab code to analyze the behavior of the governing equations of aircraft dynamics and control system.

COURSE OUTCOMES:

After successful completion of Course, The student must be able to

1. Enhance ability to apply knowledge of mathematics, science and engineering.
2. Apply knowledge to design of a system, process or component by developing a mathematical model of autonomous model
3. Improve ability to identify, formulate and solve engineering problems.
4. Improve ability to use modern engineering tools by providing practice in using Matlab and computer programming language to solve realistic engineering problems.
5. Generate MATLAB code to analyze the behaviour of the governing equations.
6. Write a MATLAB code generate grid over an profiles.

LIST OF EXPERIMENTS

1. Array and Matrix operation in Matlab program
2. Working with files-Scripts and function
3. Plotting and output
4. Errors in Numerical computation
5. Generation of higher order differential formulae
6. Generation of partial differential formulae
7. Generation of Numerical integration formulae
8. Generation of Newton Raphson-Lorenz equation
9. Basics of linear algebra.
10. Generation of Range kutta second order equation

Reference Books:

1. Essential MATLAB for Engineers and Scientists by Brian D. Hahn and Daniel T. Valentine

Web References:

1. https://nptel.ac.in/noc/individual_course.php?id=noc16
2. <https://www.youtube.com/watch?v=qFeHWZ-xkL0>

ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE**V Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC06	MC	L	T	P	C	CIA	SEE	Total
		2	-	-	-	30	70	100

COURSE OBJECTIVE

To facilitate the students with the concepts of Indian traditional knowledge and to make them understand the Importance of roots of knowledge system.

COURSE OUTCOMES

After completion of the course, students will be able to:

1. Upon completion of the course, the students are expected to:
2. Understand the concept of Traditional knowledge and its importance
3. Know the need and importance of protecting traditional knowledge.
4. Know the various enactments related to the protection of traditional knowledge.
5. Understand the concepts of Intellectual property to protect the traditional knowledge.

UNIT-I	INTRODUCTION TO TRADITIONAL KNOWLEDGE	CLASSES: 12
Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, the physical and social contexts in which traditional knowledge develop, the historical impact of social change on traditional knowledge systems. Indigenous Knowledge (IK), characteristics, traditional knowledge vis-à-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge vis-à-vis formal knowledge		
UNIT-II	PROTECTION OF TRADITIONAL KNOWLEDGE	CLASSES: 12
Need for protecting traditional knowledge Significance of TK Protection, value of TK in global economy, Role of Government to harness TK.		
UNIT-III	LEGAL FRAME WORK AND TK: [6 PERIODS] A	CLASSES: 12
The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001 (PPVFR Act); B: The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016. Geographical indicators act 2003.		
UNIT-IV	TRADITIONAL KNOWLEDGE AND INTELLECTUAL PROPERTY	CLASSES: 12
Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Certain non IPR mechanisms of traditional knowledge protection, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge, global legal FORA for increasing protection of Indian Traditional Knowledge.		
UNIT-V	TRADITIONAL KNOWLEDGE IN DIFFERENT SECTORS	CLASSES: 12
Traditional knowledge and engineering, Traditional medicine system, TK and biotechnology, TK in agriculture, Traditional societies depend on it for their food and healthcare needs, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of TK. 139		

Text Books:

1. Traditional Knowledge System in India, by Amit Jha, 2009.
2. Traditional Knowledge System and Technology in India by Basanta Kumar Mohanta and Vipin Kumar Singh, Pratibha Prakashan 2012.

Reference Books:

1. Traditional Knowledge System in India by Amit Jha Atlantic publishers, 2002
2. "Knowledge Traditions and Practices of India" Kapil Kapoor¹, Michel Danino²

Web References:

<https://www.youtube.com/watch?v=LZP1StpYEPM>

MOOC Course

<http://nptel.ac.in/courses/121106003>

III B.TECH VI SEMESTER SYLLABUS

AERO MATERIALS**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE23	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To give a thorough treatment of the classification and properties of materials and how they can be analysed, with emphasis on physical understanding
2. To provide the necessary knowledge and experience to enable the student to perform independent analyses

COURSE OUTCOMES

Upon successful completion of this course, the student are able to

1. Exemplify the properties and applications of ferrous and non-ferrous alloys used in aircraft
2. Illustrate the major materials (fibres, resins, cores) and its properties used in composites
3. Determine the material properties of the composites using micro and macro mechanics approach
4. Construct laminated plates and sandwich structures with desired properties and predict its failure criteria
5. Illustrate major manufacturing techniques used to fabricate composites

UNIT-I	FERROUS AND NON-FERROUS ALLOYS	Classes: 12
Types of steels - SAE and MIL specifications - effect of alloying elements on properties of steels - heat treatment of aircraft steels, super alloys - Nickel, Iron and Cobalt based alloys. Aluminium alloys - heat treatment, strengthening mechanisms - precipitation hardening, dispersion strengthening - magnesium and titanium alloys - properties and applications.		
UNIT-II	COMPOSITE MATERIALS	Classes: 12
Definition – Classification of composite materials – Advantages, limitations and application of composite materials – Properties and classifications of reinforcements and matrices - Factors contribute to mechanical performance of composites - Generalized Hooke's law		
UNIT-III	METHODS OF ANALYSIS	Classes: 12
Micro mechanics - Mechanics of materials approach, elasticity approach to determine material properties - Macro mechanics - Stress-strain relations - Determination of material properties – Mechanical properties of composites: Tensile test, compression test, flexural test, shear test and inter-laminar shear strength		
UNIT-IV	LAMINATED PLATES AND SANDWICH CONSTRUCTION	Classes: 12
Governing differential equation for a general laminate, angle ply and cross ply laminates - Failure criteria for composites. Basic design concepts of sandwich construction - Materials used for sandwich construction - Failure modes of sandwich panels		
UNIT-V	MANUFACTURING TECHNIQUES	Classes: 12
Degree of Cure – Gel time test – Lay-up – Prepregs – Molding: Compression molding, bag molding, autoclave molding and resin transfer molding – Filament winding – Pultrusion – Centrifugal casting – Extrusion methods		

Text Books:

1. R. M. Jones, "Mechanics of Composite Materials", 2nd Edition, Taylor & Francis
2. Madhujit Mukhopadhyay, "Mechanics of Composite Materials and Structures", Orient Blackswan Pvt Ltd.
3. George F. Titterton, "Aircraft materials and processes", 5th Edition, Indian Reprint 2004

Reference Books:

1. Marc Andre Meyers and Krishna Kumar Chawla, "Mechanical behavior of materials", Prentice-Hall, Inc. USA, 2007.
2. T. H. Courtney, "Mechanical behavior of materials", McGraw-Hill, USA, 2011.
3. D. Agarwal, L. J. Broutman and K. Chandrashekhara, "Analysis and Performance of Fiber Composites", 3rd Edition, John Wiley & Sons.
4. R.F. Gibson, "Principles of composite material mechanics", 3rd Edition, CRC press.

Web References:

- <http://www.ae.iitkgp.ac.in/ebooks/chapter1.html>
<https://www.sciencedirect.com/topics/materials-science/structural-composite-materials>

E-Text Books:

1. Prof. R. Velmurugan, "e-Book on Composite Materials", IIT Madras.
2. Dr. K. V. Nagendra Gopal, "e-Book on Composite Structures", IIT Madras.

MOOC Course

1. <https://nptel.ac.in/courses/112104168/>
2. <https://nptel.ac.in/courses/101106038/>
3. <https://nptel.ac.in/courses/101104010/>

AEROSPACE PROPULSION - II**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE68	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The course is intended to build up necessary background for understanding the Aircraft and Rocket propulsion systems.

1. To learn the propulsion system performance information at various condition.
2. To understand the application of various propellant systems and their properties.

COURSE OUTCOMES

1. Analyze the basic concepts about planets and orbits
2. Determine rocket launching principle and concept of nozzles
3. Determine the selection criterion of various propellants.
4. Examine the configuration of propellants and thrust chamber performance.
5. Analyze the combustion instability in rockets and principles of electrical rockets.

UNIT-I	BACKGROUND AND REQUIREMENTS	Classes: 12
Introduction, Motion of bodies in space, parameters describing motion of bodies, frame of reference Impulse, force, universal law for gravitational force, motion in rotating frame of reference, pseudo-centrifugal force Orbits, orbit velocities, orbital period, geosynchronous and geostationary orbits, eccentricity and inclination, polar, sun-synchronous and other orbits Energy requirements for orbit, escape velocity, orbital and suborbital flight, state of weightlessness under free fall.		
UNIT-II	ROCKET PRINCIPLE	Classes: 12
ROCKET PRINCIPLE: Means of achieving orbit, Motion of a sled initially at rest Motion of giant squids, Rocket principle and Rocket equation Mass ratio of rocket, desirable parameters to achieve high velocities, propulsive efficiency Performance parameters of a rocket, staging and clustering, classification of rockets. NOZZLES : Rocket nozzles: Expansion of gases from high pressure chamber, efflux velocity, shape of nozzle Convergent divergent nozzle, choking, variation of parameters in nozzle Expansion ratio of nozzles, Performance loss in nozzles Under-expanded and over-expanded nozzles, flow separation, Contour nozzles, adapted nozzles and unconventional nozzles, mass flow rates and characteristic velocities Thrust developed by a rocket, thrust coefficient, vacuum and sea level specific impulse, efficiencies and thrust correction factor		
UNIT-III	CHEMICAL PROPELLANTS	Classes: 12
Chemical propellants: Choice from considerations of molecular mass, specific heats, specific heat ratios, temperature and pressure Choice of chemical propellants: heats of formation, moles and mixture ratio; choice of mixture ratio Calculation of heat of combustion, temperature, molecular mass and rocket performance parameters Solid propellants: Double base, composite, composite modified double base and nitramine propellants Liquid propellants; Energy content and classification, Earth storable and space storable propellants, hypergolic and other features, hybrid propellants Influence of dissociation on propellant performance, frozen and equilibrium analysis.		

UNIT-IV	SOLID PROPELLANT ROCKETS	Classes: 12
Solid propellant rockets: burn rate of double base and composite propellants, parameters influencing burn rates Choice of burn rates for stable operation Propellant grain configurations: design of solid propellant rocket Ignition of solid propellant rockets, ignition problems and solutions Characteristic burn times and action times of solid propellant rockets, variation of burn rates with rocket size, erosive burning, components of solid propellant rocket LIQUID PROPELLANT ROCKETS: Introduction to liquid propellant rockets, propellant feed systems, cycles of operation, gas generator, topping/staged combustion cycle, expander and other cycles, factors influencing choice of cycle Thrust chamber, injector types, combustion chamber Calculation of efficiency of liquid propellant rockets from non uniform distribution of propellants and incomplete vaporization, characteristic length of chamber Cooling of thrust chamber and nozzle Performance and choice of feed system cycle, Choice of parameters for liquid propellant rockets Turbo-pumps for liquid propellant rockets.		
UNIT-V	COMBUSTION INSTABILITY IN ROCKETS	Classes: 12
Combustion instability in rockets; illustration through examples, bulk and wave modes of instability Modeling of bulk mode of instability in solid and liquid propellant rockets. ELECTRICAL ROCKETS Electrical rockets: electrical and magnetic fields, electro-thermal, arc-jet, electrostatic and electromagnetic thrusters Gridded ion thrusters, neutralization and thrust limitations, Hall effect thrusters Choice of parameters for electrical thrusters, specific mass and optimum efflux velocity		
Text Books:		
1.Hill, P. G. and Peterson, C.R., Mechanics and thermodynamics of propulsion, 2nd ed., Reading, Massachusetts: Addison Wesley Publishing Company,, 1992. 2.Sutton, G. P. and Biblarj, O. Rocket propulsion elements, 7th Ed., New York: Wiley Intescience Pulications, 2001. 3.Mukunda, H. S., Understanding aerospace propulsion, Bangalore: Interline Publishing, 2004. 4.Ramamurthi, K., Rocket Propulsion, Macmillan (in press) 2009		
Reference Books:		
1.Timnat, Y. M., Advanced chemical rocket propulsion, London: Academic Press,,1987. 2.Shapiro, A. H., The dynamics and thermodynamics of compressible fluid flow, vol. 1, New York: John Wiley and Sons Inc., 1953. 3.E. Stulinger, Electric propulsion development, Progress in Astronautics and Aeronautics, vol. 9, Academic Press, New York, 1963. 4. M. Barrere, A. Jaumotte, B.J. Veubeke and J. Vanderkerckhove, Rocket Propulsion, Elsevier Publishing Company, Amsterdam, 1960		
Web References:		
https://nptel.ac.in/downloads/112106073/		
E-Text Books:		
Rocket Propulsion Elements https://pdfs.semanticscholar.org/6cce/fa84a00b396c5b50b66063772f4851392b8f.pdf		
MOOC Course		
https://nptel.ac.in/courses/112106073/ https://nptel.ac.in/courses/101104019/		

ADDITIVE MANUFACTURING**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE28	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with the theoretical background and engineering applications. 1. To know the principle methods, areas of usage, possibilities and limitations as well as environmental effects of the Additive Manufacturing technologies 2. To be familiar with the characteristics of major materials used in Additive Manufacturing.								
COURSE OUTCOMES: At the end of the course the students are able to: 1. Analyze the characteristics of major materials used in Additive Manufacturing. 2. Compare various digitization techniques 3. Compare major methods of additive manufacturing 4. Discuss the effects of the Additive Manufacturing technologies 5. Compare major tools used in additive manufacturing techniques.								
UNIT-I	INTRODUCTION						Classes: 10	
Overview – History – Need-Classification -Additive Manufacturing Technology in product development-Materials for Additive Manufacturing Technology – Tooling – Applications.								
UNIT-II	CAD AND REVERSE ENGINEERING						Classes:12	
Basic Concept – Digitization techniques – Model Reconstruction – Data Processing for Additive Manufacturing Technology: CAD model preparation – Part Orientation and support generation – Model slicing –Tool path generation – Software for Additive Manufacturing Technology: MIMICS, MAGICS.								
UNIT-III	LIQUID BASED AND SOLID BASED ADDITIVE MANUFACTURING SYSTEMS						Classes: 12	
Classification – Liquid based system – Stereolithography Apparatus (SLA)- Principle, process, advantages and applications – Solid based system –Fused Deposition Modeling – Principle, process, advantages and applications, Laminated Object Manufacturing								
UNIT-IV	POWDER BASED ADDITIVE MANUFACTURING SYSTEMS						Classes: 12	
Selective Laser Sintering – Principles of SLS process – Process, advantages and applications - Three Dimensional Printing – Principle, process, advantages and applications- Laser Engineered Net Shaping (LENS), Electron Beam Melting.								
UNIT-V	TOOLING						Classes: 8	
Classification of Soft tooling, Production tooling, Bridge tooling, direct and indirect tooling, Fabrication processes, Applications, Case studies on automotive, aerospace and electronics industries								

Text Books:

1. Chua C.K., Leong K.F., and Lim C.S., "Rapid prototyping: Principles and applications", Third Edition, World Scientific Publishers, 2010.
2. Gebhardt A., "Rapid prototyping", Hanser Gardener Publications, 2003.

Reference Books:

1. Liou L.W. and Liou F.W., "Rapid Prototyping and Engineering applications : A tool box for prototype development", CRC Press, 2007.
2. Kamrani A.K. and Nasr E.A., "Rapid Prototyping: Theory and practice", Springer, 2006.
3. Hilton P.D. and Jacobs P.F., "Rapid Tooling: Technologies and Industrial Applications", CRC press, 2000.

Web References:

<https://www.worldscientific.com/worldscibooks/10.1142/10200>

E-Text Books:

"Additive Manufacturing Technologies" by Ian Gibson
"Additive Manufacturing: Advanced Manufacturing Technology in 3d Print Deposit" by Sabrie Soloman

MOOC Course

<https://engineering.purdue.edu/ProEd/certifications/additive-manufacturing>

CAD/CAM**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE29	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To know the basic idea about the computers in industrial manufacturing
2. To gain knowledge about computer aided graphics, drafting and modelling

COURSE OUTCOMES

At the end of the course the students are able to:

1. Analyze the major working processes in product cycle
2. Compare the major commands used for solid modelling
3. Compare the difference between NC machines and CNC machines
4. Analyze the major inspection techniques used in manufacturing process
5. Compare the major types of manufacturing systems

UNIT-I	COMPUTER GRAPHICS	Classes: 10
Computers in Industrial Manufacturing, Product cycle, CAD / CAM Hardware, Basic structure, CPU, Memory types, input devices, display devices, hard copy devices, storage devices. Raster scan graphics coordinate system, database structure for graphics modeling, transformation of geometry, 3D transformations, mathematics of projections, clipping, hidden surfaceremoval.		
UNIT-II	GEOMETRIC MODELING, DRAFTING AND MODELING SYSTEMS	Classes:10
Requirements, geometric models, geometric construction models, curve representation methods, surface representation methods, modeling facilitiesdesired. Basic geometric commands, layers, display control commands, editing,dimensioning, solid modelling.		
UNIT-III	NUMERICAL CONTROL AND GROUP TECHNOLOGY	Classes: 12
NC, NC modes, NC elements, NC machine tools, structure of CNC machine tools, features of m machining center, turning center, CNC Part Programming: fundamentals, manual part programming methods, Computer Aided Part Programming. Part family, coding and classification, production flow analysis, advantages and limitations, ComputerAided Processes Planning, Retrieval type and Generative type.		
UNIT-IV	COMPUTER AIDED QUALITY CONTROL	Classes: 10
Terminology in quality control, the computer in QC, contact inspection methods, non contact inspection methods-optical, non contact inspection methods- non optical, computer aided testing, integration of CAQC with CAD/CAM.		
UNIT-V	COMPUTER INTEGRATED MANUFACTURING SYSTEMS	Classes: 10
Types of Manufacturing systems, Machine tools and relatedequipment, material handling systems, computer control systems, human labor in the manufacturing systems, CIMS benefits.		
Text Books:		
1. A. Zimmers, P. Groover (2010), CAD / CAM, 3rd edition, Prentice Hall of India, NewDelhi. 2. IbrahimZeid(2011),CAD/CAMTheoryandPractice,4th edition,TataMcGrawHilleducation(P) Ltd, New Delhi, India.		
Reference Books:		

1. P. Groover(2011), Automation, Production systems and Computer integrated Manufacturing, 3rd edition, Pearson Publications,India.
2. Radhakrishnan, Subramanian (2009),CAD / CAM / CIM, New Age International Pvt. Ltd, New Delhi,India.
3. Alavala, C. R (2012),CAD/CAM: Concepts and Applications, 1st edition, Prentice Hall of India, New Delhi, India.

Web References:

http://home.iitk.ac.in/~jrkumar/download/ME761A_Lecture%201%20Introduction.pdf

E-Text Books:

<http://www.engineeringbookspdf.com/cadcamcim/>

http://www.engineering108.com/pages/Mechanical_Engineering/CAD_CAM/Mechanical_engineering_ebook_free%20download.html

MOOC Course

<https://nptel.ac.in/courses/Webcourse-contents/IIT-Delhi/Computer%20Aided%20Design%20&%20ManufacturingI/index.htm>

SPACE TECHNOLOGY

VI Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE30	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with theoretical background and engineering applications. 1. To introduce the basic concepts of design, propulsion system considerations and trajectory estimation of rockets. 2. To study the basic concepts of orbital Mechanics with particular emphasis on interplanetary trajectories.								
COURSE OUTCOMES This course uses lectures, assignments and home works to the students. The teaching methods include regular class work, Problem solving, technical quiz, and seminars to enable the students: 1. Categorize space missions 2. Compare the working of solid and liquid propellant rockets 3. Identify suitable trajectory for the re-entry vehicle 4. Analyze the transfer method of a spacecraft from one orbit to another 5. Apply various mechanisms for attitude control of satellite and space crafts								
UNIT-I	INTRODUCTION						Classes: 8	
Space Mission, Types, Space Environment, Launch Vehicle Selection.								
UNIT-II	FUNDAMENTALS OF ROCKET PROPULSION						Classes: 10	
Introduction to rocket propulsion, fundamentals of solid propellant rockets, Fundamentals of liquid propellant rockets. Rocket equation.								
UNIT-III	ASCENT FLIGHT MECHANICS OF ROCKETS AND MISSILES						Classes: 15	
Two-dimensional trajectories of rockets and missiles. Multi-stage rockets, Vehicle sizing. Two stage Multi-stage Rockets, Trade-off Ratios-Single Stage to Orbit, Sounding Rocket-Aerospace Plane, Gravity Turn Trajectories-Impact point calculation, Injection conditions, and Flight dispersions. ATMOSPHERIC REENTRY: Introduction, Steep Ballistic Reentry, Ballistic Orbital Reentry, Skip Reentry, —Double- Dip Reentry, Aero-braking, Lifting Body Reentry.								
UNIT-IV	FUNDAMENTALS OF ORBITAL MECHANICS						Classes: 13	
Two-body motion-Circular, elliptic, hyperbolic, and parabolic orbits. Kepler laws of planetary motion, Basic Orbital Elements, Ground Trace. ORBITAL MANEUVERS: In-Plane Orbit changes, Hohmann Transfer, Bielliptical Transfer, Plane Changes, Combined Maneuvers, and Propulsion for Maneuvers.								
UNIT-V	SATELLITE ATTITUDE DYNAMICS						Classes: 12	
Torque free Axi-symmetric rigid body, Attitude Control for Spinning Spacecraft. Attitude Control for Non-spinning Spacecraft. The Yo-Yo Mechanism, Gravity, Gradient Satellite. Dual Spin Spacecraft, Attitude Determination. SPACECRAFT POWER AND COMMUNICATION SYSTEMS: Spacecraft Power, Telecommunications.								

Text Books:

1. W. E. Wiesel (2010), Spaceflight Dynamics, 3rd edition, McGraw-Hill, New Delhi.

Reference Books:

1. J. Sellers (2005), Understanding Space: An Introduction to Astronautics, 3rd edition, McGrawHill, New Delhi.
2. Francis J. Hale (1994), Introduction to Space Flight, 1st edition, Prentice-Hall, New York.
3. D. Brown Charles (1998), Spacecraft Mission Design, 2nd edition, AIAA Education Series, USA.
4. Meyer Rudolph X (1999), Elements of Space Technology for Aerospace Engineers, Academic Press New York

Web References:

<http://www.braeunig.us/space/orbmech.htm>

https://www.tutorialspoint.com/satellite_communication/satellite_communication_orbital_mechanics.htm

E-Text Books

<https://onlinelibrary.wiley.com/doi/book/10.1002/9780470742433>

MOOC Course

<https://nptel.ac.in/courses/101106046/19#>

FLIGHT SCHEDULING AND OPERATIONS**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE31	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To discuss airline network flows and to Understand mathematical formulation- decision variables, objective function, constraints, and methods of solution for airline scheduling.
2. Analyze aircraft routing and management of irregular operation.
3. Demonstrate the significance of flight scheduling.
4. Understand the importance fleet assignment and crew and manpower scheduling.
5. Demonstrate assignment and aircraft boarding strategy and to understand the common strategies for aircraft boarding.

COURSE OUTCOMES

1. Apply knowledge in understanding the complexity of airline planning, operations and dispatch.
2. Differentiate and analyze the problems in aircraft routing and management of irregular operations.
3. Analyze the route development and flight scheduling process and Apply the formulation of crew pairing problem
4. Discuss fleet assignment and crew and manpower scheduling.
5. Analyze the gate assignment and aircraft boarding strategy.

UNIT-I	NETWORK FLOWS AND INTEGER PROGRAMMING MODELS	Classes: 13
Complexity of airline planning, operations and dispatch- need for optimization- role of operations research and simulation. Networks- definitions, network flow models- shortest path problem, minimum cost flow problem, maximum flow problem, multi-commodity problem. Integer programming models- set covering/ partitioning problems, traveling salesman problem- mathematical formulation- decision variables, objective function, constraints, and methods of solution. Solution by simulation.		
UNIT-II	AIRCRAFT ROUTING AND MANAGEMENT OF IRREGULAR OPERATIONS	Classes: 12
Goal of aircraft routing- maintenance requirements, other constraints. Routing cycles, route generators. Mathematical models of routing- decision variables, objective functions, alternatives, constraints- flight coverage and aircraft available. Example problems and solutions. The problem statement, the time band approximation model- formulation of the problem- the scenarios- solution.		
UNIT-III	FLIGHT SCHEDULING	Classes: 11
Significance of flight scheduling. The route system of the airlines- point-to-point flights, hub and spoke flights. Schedule construction- operational feasibility, economic viability. Route development and flight scheduling process- load factor and frequency- case study.		
UNIT-IV	FLEET ASSIGNMENT AND CREW AND MANPOWER SCHEDULING	Classes: 13
Purpose of fleet assignment. Fleet types, fleet diversity, fleet availability- performance measures, Formulation of the fleet assignment problem- decision variables, objective function, constraints, solution. Scenario analysis, fleet assignment models. Crew scheduling process- significance. Development of crew pairing- pairing generators- mathematical formulation of crew pairing problem- methods of solution. Crew rostering- rostering practices. The crew rostering problem- formulation, solutions. Manpower scheduling- modeling, formulation of the problem, solutions.		

UNIT-V	GATE ASSIGNMENT AND AIRCRAFT BOARDING STRATEGY	Classes: 11
Gate assignment- significance- the problem- levels of handling-passenger flow, distance matrix- mathematical formulation, solution. Common strategies for aircraft boarding process, mathematical model, interferences, model description, aisle interferences.		
Text Books:		
1. Bazargan,M., 'AirlineOperationsandScheduling', 2 nd edn., AshgatePublishingLtd, 2010.		
Reference Books:		
1. Belobaba,P., Odoni,A., Barnhart,C. 'TheGlobalAirlineIndustry', Wiley, 2009. 2. Wu,Cheng-Lung, 'AirlineOperationsandDelayManagement', AshgatePublishingLtd, 2010. 3. Wensveen,J.G., 'AirTransportation:AManagementPerspective', 6 th edn., AshgatePublishingLtd, 2007. 4. Yu,G., "OperationsResearchinAirlinesIndustry", AcademicPublishers, 1998.		
Web References:		
https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-206j-airline-schedule-planning-spring-2003/lecture-notes/ www.airlinestechology.net		
E-Text Books:		
Airline Network Planning and Scheduling by Ahmed Abdelghany and Khaled Abdelghany, Kindle Edition.		
MOOC Course		
https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-206j-airline-schedule-planning-spring-2003/		

FINITE ELEMENT METHOD

VI Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE32	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The objectives of the course are to enable the student;

1. Introduction of Finite Element Method (FEM) which is one of the Numerical Methods with which solutions can be obtained for problems with complex geometries, material properties and boundary conditions.
2. Utility of FEM as Engineering solution tool to problems (both vector and scalar) involving various fields for Design Analysis and Optimization.
3. Development of Mathematical Model (Governed by Differential equations) for physical problems and concept of discretization of continuum.
4. Ability to understand, to improve or refine the approximate solution by spending more computational effort by using higher interpolation continuities unlike expensive experimental methods / exact solutions.

COURSE OUTCOMES:

1. Develop elastic equations, formulate and solve the axially loaded bar structures using FEM.
2. Apply finite element method to truss and beam analysis.
3. Implement finite element method to solve two dimensional problems and apply numerical integration to one and two dimensional problems.
4. Solve and analyze heat transfer problems using FEM.
5. Apply FEM to dynamic analysis of one dimensional bars and beams.

UNIT-I	INTRODUCTION TO FEM	Classes: 12
INTRODUCTION TO FEM: Basic concept, historical background, application of FEM, general description, comparison of FEM with other methods. Basic equations of elasticity, Stress, strain relations, Strain, Displacement relations. ONE DIMENSIONAL PROBLEM: Finite element modeling coordinates and shape functions. Potential Energy approach: Assembly of Global stiffness matrix and load vector. Finite element equations, Treatment of boundary conditions, Quadratic shape functions		
UNIT-II	ANALYSIS OF TRUSSES	Classes: 12
ANALYSIS OF TRUSSES: Stiffness Matrix for plane truss and space truss elements, stress calculations. ANALYSIS OF BEAMS: Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, deflection, stresses.		
UNIT-III	2-D PROBLEMS	Classes: 12
2-D PROBLEMS: CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems. FINITE ELEMENT MODELLING of Axisymmetric solids subjected to Axisymmetric loading with triangular elements. Two dimensional four noded isoparametric elements and numerical integration		

UNIT-IV	STEADY STATE HEAT TRANSFER ANALYSIS	Classes: 12
STEADY STATE HEAT TRANSFER ANALYSIS: one dimensional analysis of a fin and two dimensional analysis of thin plate.		
UNIT-V	DYNAMIC ANALYSIS	Classes: 12
DYNAMIC ANALYSIS: Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar and a beam		
Text Books:		
1. R. Tirupathi Chandrapatla (2011), Introduction to Finite Elements in Engineering, 4th edition, Pearson Education, India. 2. S. S. Rao (2012), The Finite Element Methods in Engineering, 5th edition, Elsevier, USA. 3. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1st edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India		
Reference Books:		
1. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India. 2. J. N. Reddy (2010), Anintroduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 3. Kenneth H. Huebner, Donald L. rDd ewhirst, Douglas E. Smith , Ted G. Byrom (2009), The Finite 4. Element Methodfor Engineers, 3 edition, John Wiley & sons (ASIA) Pvt. Ltd., New York.		
Web References:		
https://nptel.ac.in/courses/112104116/		
E-Text Books:		
http://web.mit.edu/kjb/www/Books/FEP_2nd_Edition_4th_Printing.pdf		
MOOC Course		
https://nptel.ac.in/courses/112104193/		

TURBO MACHINES

VI Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE33	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Overview of major types of turbo machinery used for energy transformation
2. Turbomachinery applications in power generation, transport, refrigeration and the built environment.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Explain the working principles of turbo machines and apply it to various types of machines
2. Determine the performance parameters in turbo machinery stages operating at design and offdesign conditions
3. Exemplify the importance of gas dynamics in turbo machinery.
3. Explain the working principle of turbines, compressor and nozzles and know their application range
4. Perform the preliminary design of turbo machines (pumps, compressors, turbines) on a 1-D basis
5. Determine the off-design behaviour of turbines and compressors and relate it to changes in the velocity triangles

UNIT-I	FUNDAMENTALS OF TURBO MACHINES	Classes: 12
FUNDAMENTALS OF TURBO MACHINES: Classifications, Applications, Thermodynamic analysis, Isentropic flow. Energy transfer. Efficiencies, Static and Stagnation conditions, Continuity equations, Euler's flow through variable cross sectional areas, Unsteady flow in turbo machines		
UNIT-II	STEAM NOZZLES	Classes: 12
STEAM NOZZLES: Convergent and Convergent-Divergent nozzles, Energy Balance, Effect of back pressure of analysis. Designs of nozzles. STEAM TURBINES: Impulse turbines, Compounding, Work done and Velocity triangle, Efficiencies Constant reactions, Blading, Design of blade passages, Angle and height, Secondary flow. Leakage losses, Thermodynamic analysis of steam turbines.		
UNIT-III	GAS DYNAMICS	Classes: 10
GAS DYNAMICS: Fundamental thermodynamic concepts, isentropic conditions, mach numbers and area, Velocity relations, Dynamic Pressure, Normal shock relation for perfect gas. Supersonic flow, oblique shock waves. Normal shock recoveries, Detached shocks, Aerofoil theory. CENTRIFUGAL COMPRESSOR: Types, Velocity triangles and efficiencies, Blade passage design, Diffuser and pressure recovery. Slip factor, Stanitz and Stodolas formula's, Effect of inlet mach numbers, Pre whirl, Performance		

UNIT-IV	AXIAL FLOW COMPRESSORS	Classes: 10
AXIAL FLOW COMPRESSORS: Flow Analysis, Work and velocity triangles, Efficiencies, Thermodynamic analysis. Stage pressure rise, Degree of reaction, Stage Loading, General design, Effect of velocity, Incidence, Performance CASCADE ANALYSIS: Geometrical and terminology. Blade forces, Efficiencies, Losses, Free end forced vortex, Vortex Blades.		
UNIT-V	AXIAL FLOW GAS TURBINES	Classes: 10
AXIAL FLOW GAS TURBINES: Work done, Velocity triangle and efficiencies, Thermodynamic flow analysis, Degree of reaction, Zweifel's relation, Design cascade analysis, Soderberg, Hawthorne, Ainley, Correlations, Secondary flow, Free vortex blade, Blade angles for variable, degree of reaction, Actuator disc, Theory, Stress in blades, Blade assembling, Material and cooling of blades, Performances, Matching of compressors and turbines, Off design performance ratios.		
Text Books:		
S Larri Dixon and Cesare Hall, Fluid Mechanics and Thermodynamics of Turbomachinery, 7 th edition, Butterworth-Heinemann, 2013.		
Reference Books:		
1. Nicholas Cumpsty, Compressor Aerodynamics, 2004, Kreiger Publications, USA. 2. Johnson I.A., Bullock R.O. NASA-SP-36, Axial Flow Compressors, 2002 (re-release), NTIS. 3. El-Wakil, M M; Powerplant Technology, 1984, McGraw-Hill Pub. 4. NASA-SP-290, Axial Flow turbines, 2002 (re-release), NTIS, USA. 5. J H Horlock, Axial flow compressors, Butterworths, 1958, UK. 6. J H Horlock, Axial Flow Turbines, Butterworths, 1965, UK. 7. B Lakshminarayana; Fluid Mechanics and Heat Transfer in turbomachineries, 1995, USA. 8. Ahmed F. El-Sayed; Aircraft Propulsion and Gas Turbine Engines; CRC press, 2008.		
Web References:		
https://nptel.ac.in/courses/112106200/		
E-Text Books:		
Principle of turbo machinery By R.K. Turton		
MOOC Course		
https://nptel.ac.in/downloads/101101058/		

GEOMETRIC DIMENSIONING AND TOLERANCES (GD&T)**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE34	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To impart through knowledge in technical graphics to communicate the technical information required for manufacture and assembly of machine components.
2. To familiar with industrial drafting process and through understanding of production drawings to make themselves fit in industries

COURSE OUTCOMES:

1. Use limits, fits and tolerances in real world problems.
2. Sketch the sectional views of simple elements.
3. Sketch and dimension the standard mechanical elements like bolt, nut, screw etc.
4. Sketch the assembly drawings and sectional views of automobile components from orthographic part drawings.
5. Sketch the detailed drawing of automobile components and able to prepare bill of materials and tolerance sheet. .

UNIT-I	LIMITS, FITS AND TOLERANCES	Classes: 12
Limit System- Tolerance, Limits, Deviation, Actual Deviation, Upper Deviation, Lower Deviation, Allowance, Basic Size, Design Size, Actual Size. Fits- Types, Tolerances of Form and Position- Form and Position Variation, Geometrical Tolerance, Tolerance Zone, Indicating Geometrical Tolerances. Indication of Surface Roughness, Standard Abbreviations and Symbols used in industries.		
UNIT-II	SECTIONAL VIEWS	Classes: 12
Sections- Hatching of Sections, Cutting Planes, Revolved or Removed Section, Sectional Views- Full Section, Half Sections and Auxiliary Sections- Conventional Representation-One-view, Two-view and three view Drawings.		
UNIT-III	INTRODUCTION TO MACHINE ELEMENT DRAWINGS	Classes: 12
Drawing standards and Designation of Bolts, nuts, screws, keys, pins, Rivets, Welded Joints- Principle of dimensioning - Dimensioning of Welds, Belt Driven Pulleys, Chain and Gears Drives.		
UNIT-IV	ASSEMBLY DRAWINGS AND SECTIONAL VIEWS	Classes: 12
Preparation of manual parts drawing and assembled sectional views from orthographic part drawings, Automobile components - stuffing box, Machine Tool Parts plummer block, Joints knuckle joints, Couplings Protected type flanged coupling, Bearings swivel bearing.		
UNIT-V	REAL PRODUCTS TO MACHINE DRAWING CONVERSION	Classes: 12
Preparation of manual parts drawing and assembled sectional views from real time products- Internal combustion engine parts, connecting rod, couplings - universal coupling, machine tool parts -tailstock, Automobile components screw jack, stuffing box - Commercial products - Preparation of Bill of materials and tolerance data sheet.		

Text Books:

1. Gary R Bertoline, Leonard O Nasman, Technical Graphics Communication (Irwin Graphics Series), 2nd edition, Richard D Irwin Publisher, 1997.
2. William P. Spence, Engineering Graphics, Printice - Hall Inc, Engle Wood Cliff, 1984.
3. S. Bogolyubov. A. Voinov., Engineering Drawing, Van Nostr and Reinhold Company, 1976.

Reference Books:

1. N.D. Bhatt, Machine Drawing, Charotar Publishing House Pvt. Ltd., 2014
2. P.S.Gill, A Textbook of Machine Drawing, Katson books, 2013
3. K.C. John, Textbook of Machine Drawing, PHI Learning Pvt. Ltd., 2009

Web References:

<https://www.machinedesignonline.com/>

<https://ocw.mit.edu/courses/mechanical-engineering/2-72-elements-of-mechanical-design-spring-2009/lecture-notes/>

E-Text Books:

Machine drawing, 3rd edition by K L Narayana, P kannaiiah and K Venkata Reddy.
A Textbook of Machine Drawing by R K Dhawan.
Principles and techniques for designing precision machines by Hale, Layton Carter.

MOOC Course

<https://ocw.mit.edu/courses/mechanical-engineering/2-72-elements-of-mechanical-design-spring-2009/>

<https://www.coursera.org/learn/machine-design1>

OPTIMIZATION TECHNIQUES**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE35	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To formulate design optimization problems for engineering applications
2. To provide knowledge on single variable unconstrained problems
3. To learn multi-objective unconstrained optimization problems
4. To introduce concepts of constrained non-linear optimization problems
5. To interpret non-traditional optimization techniques for engineering problems

COURSE OUTCOMES:

1. Formulate design optimization problem from real world applications.
2. Compute the solution for single variable unconstrained optimization problems.
3. Determine the solution for multivariable unconstrained optimization problems.
4. Find the solution for the constrained non-linear optimization problems.
5. Apply non-traditional optimization techniques to solve engineering problems.

UNIT-I	INTRODUCTION TO OPTIMIZATION TECHNIQUES	Classes: 12
Introduction to design optimization-Historical development, the design process, Conventional Vs Optimum design process - Statement of an optimization problem- Optimum design problem formulation - process steps, Problem formulation for engineering applications - Two-bar bracket, Design of coil springs - Classifications of optimization problems.		
UNIT-II	SINGLE VARIABLE NONLINEAR UNCONSTRAINED OPTIMIZATION ALGORITHMS	Classes: 12
Optimality criteria - Unimodal function - Eliminating methods - Exhaustive search, Dichotomous search, Interval halving method, Fibonacci search method, Golden section search method. Point estimation method (Powell's algorithm) - Gradient-based methods - Newton-Raphson method (Taylor's series expansion), Bisection method, Secant method, Cubic search method.		
UNIT-III	MULTI VARIABLE NONLINEAR UNCONSTRAINED OPTIMIZATION ALGORITHMS	Classes: 12
Optimality criteria - Unidirectional search - Direct search methods - Evolutionary optimization method, Random search methods, Simplex search method, Hooke-Jeeves pattern search method, Indirect search (gradient) methods- Cauchy's (steepest descent) method, Newton's method, Conjugate gradient method.		
UNIT-IV	CONSTRAINED NONLINEAR OPTIMIZATION ALGORITHMS AND SPECIALIZED PROGRAMMING	Classes: 12
Introduction, Characteristics - Indirect search methods - Transformation methods, Penalty function method, Method of multipliers - Sensitivity analysis - Kuhn-Tucker conditions, Theorems. Test problems on three-bar truss, welded beam design. Direct search minimization methods- Variable elimination method, Complex search method and Random search methods - Feasible direction method. Integer programming - Penalty function method, Branch and Bound method.		
UNIT-V	NONTRADITIONAL OPTIMIZATION TECHNIQUES	Classes: 12
Genetic Algorithms (GA)- principle, difference and similarities between GA and traditional methods, constrained optimization, GA operators, Real-coded and Advanced GAs - Simulated Annealing - Neural Network based Optimization.		

Text Books:

1. Singiresu S. Rao, Engineering Optimization: Theory and Practice, Fourth Edition, Wiley India Pvt Ltd, Delhi, 2009.
2. Kalyanmoy Deb, Optimization for Engineering Design- Algorithms and Examples, Second Edition, PHI Learning Pvt. Ltd., New Delhi, 2012.

Reference Books:

1. Jasbir Singh Arora, Introduction to Optimum design, Third Edition, Elsevier India Pvt.Ltd. New Delhi, 2011.
2. R.Saravanan, Manufacturing optimization through intelligent techniques, First Edition, Taylor & Francis Publications, CRC Press, New Delhi, 2006.
3. Prof. C. Balaji, Essentials of Thermal System Design and Optimization, Aue Books, New Delhi in India and CRC Press in the rest of the world.
4. Y.Jaluria, Design and optimization of thermal systems, Mc Graw Hill, 1998.
5. J.S.Arora, Introduction to optimum design, Mc Graw Hill, 1989.

Web References:

http://www.nptel.ac.in/courses/105108127/pdf/Module_1/M1L1.ppt
http://www.zfm.ethz.ch/e/v/opt/handouts/ETHZ_Lecture1.ppt
<https://www.resolvedanalytics.com/s/Optimization-Methods-in-Engineering-Design.pdf>
<http://www.ist.edu.pk/downloads/cacss/workshops/national/optimum-engineering-design-workshop/fundamental-engineering-optimization-methods-2e.pdf>

E-Text Books:

1. Optimization for Engineering Design: Algorithms and Examples, 2nd ed by Kalyanmoy Deb
2. Optimization Methods From Theory to Design Scientific and Technological Aspects in Mechanics by Cavazzuti & Marco

MOOC Course

<https://nptel.ac.in/downloads/105108127/>
<https://nptel.ac.in/courses/112106064/>

AERO MATERIALS LAB**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE36	BSC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

To impart the knowledge of major fabrication processes and mechanical characterization of composite materials

COURSE OUTCOMES:

Upon successful completion of this course, the student are able to

1. Fabricate polymer matrix composite material according to the requirement of mechanical properties.
2. Demonstrate testing methods to evaluate mechanical properties of the material.
3. Predict the size and location of the defects in the composite material using Non Destructive Testing method.
4. Explain the specimen preparation procedure for metallographic study.
5. Exemplify the impact of microstructure of the materials on its mechanical properties.

LIST OF EXPERIMENTS

1. Preparation of fiber reinforced polymer composites using hand-lay-up technique.
2. Evaluation of flexural properties of polymer based composites.
3. Determine the hardness of fiber reinforced polymers.
4. Impact test study of polymer based composite materials.
5. Identifying location of voids and flaws in polymer composites using Ultrasonic Machine.
6. Specimen Preparation For Metallographic Examination
7. Study of Microstructure of Non-Ferrous Metals
8. Study of Microstructure of Cast Irons
9. Study of Microstructure of Carbon Steels
10. Jominy End Quench Test
11. Age Hardening of Al-Cu Alloys

Note: Ten experiments should be performed.

Reference Books:

1. Tariq Siddiqui, Aircraft Materials and Analysis, McGraw-Hill Education, 2015.
2. Titterton, Aircraft Materials and Processes, Shroff, 1998.
3. Bruce Yu, Aircraft Composite Design Handbook for Training: Basic Knowledge of Aircraft Composite Material and Process, Amazon Digital Services, 2017.

Web References:

<https://nptel.ac.in/courses/112107146/11>

<https://nptel.ac.in/courses/101106038/>

<https://nptel.ac.in/courses/113107078/25>

COMPUTATIONAL STRUCTURAL ANALYSIS LAB**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE37	PCC	L	T	P	C	CIA	SEE	Total
			-	3	1.5	30	70	100

COURSE OBJECTIVES

The purpose of this subject is

1. To train the students for structural analysis using FEM based software packages.
2. To introduce the problems and modern calculation methods in stress analysis of aircraft structures, as well as their application to solving real problems.
3. To introduce the modern computational methods for stress analysis related to airframe structures.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Understand the basic features of ANSYS software tools.
2. Analyze the torsion and shear stress for a section beams
3. Analyze and design the various models in ANSYS.
4. Analyze and design of fuselage and wing box
5. Understand the different types design methodologies in CATIA

LIST OF THE EXPERIMENTS

1. Introduction to ANSYS.
2. Stress analysis using bar element.
3. Drawing SFD and BMD using beam element.
4. Finding the member force in truss structure
5. Structural analysis using 2D elements
6. Structural analysis using axi-symmetric elements
7. Structural analysis using solid elements
8. Thermal Conduction analysis of structures.
9. Thermal stress analysis of structures
10. Model analysis of structures.
11. Structural analysis of a column.
12. Aircraft Applications based structural problem solving using FEM Packages

Note: Ten experiments should be performed.

Software's Required

ANSYS & its equivalent

Reference Books

1. Tadeusz Stolarski, Y Nakasone and S Yoshimoto, Engineering Analysis with ANSYS Software, Butterworth-Heinemann, 2006.
2. Divya Zindani, Apurba Kumafr Roy and Kaushik Kumar, Working with ANSYS: A Tutorial Approach, I K International Publishing House Pvt. Ltd, 2017.

Web References:

<https://www.youtube.com/channel/UCCmK2-v3mgXmIOBxKYliN1g>

ADVANCED ENGLISH COMMUNICATION SKILLS LAB

VI Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
+A4HS07	HS	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

1. This Lab focuses on using computer-aided multimedia instruction for language development to meet the following targets:
2. To improve the students' fluency in English, through a well-developed vocabulary and enable them
3. To listen to English spoken at normal conversational speed by educated English speakers
4. Respond appropriately in different socio-cultural and professional contexts.

COURSE OUTCOMES:

1. Better perception of nuances of English language through audio- visual experience.
2. Neutralization of accent for intelligibility.
3. Participate in group activities.
4. Speaking skills with clarity and confidence which in turn enhances their employability

LIST OF EXPERIMENTS

The following course content is prescribed for the Advanced Communication Skills Lab:

FUNCTIONAL ENGLISH - starting a conversation – responding appropriately and relevantly – using the right body language – role play in different situations.

VOCABULARY BUILDING – synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, analogy, idioms and phrases.

READING COMPREHENSION – reading for facts, guessing meanings from context, scanning, skimming, inferring meaning, Critical reading.

WRITING SKILLS – structure and presentation of different types of writing – *Resume writing / e-correspondence/Technical report writing/Portfolio writing* – planning for writing – *research abilities/data collection/organizing data/tools/analysis* – improving one's writing.

GROUP DISCUSSION – dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and coherence.

PRESENTATIONSKILLS –Oral presentations (individual and group) through JAM sessions/seminars and written presentations through posters/projects/reports/PPTs/e-Mails/assignmentsetc.

INTERVIEW SKILLS – concept and process, pre-interview planning, opening strategies, answering strategies, interview through tele and video-conferencing.

MINIMUM REQUIREMENT:

The English Language Lab shall have two parts:

1. The Computer aided Language Lab for 60 students with 60 systems, one master console, LAN facility and English language software for self- study by learners.
2. The Communication Skills Lab with movable chairs and audio-visual aids with a P.A System,
T. V., a digital stereo –audio & video system and camcorder etc.

SYSTEM REQUIREMENT (HARDWARE COMPONENT):

Computer network with Lan with minimum 60 multimedia systems with the following

specifications:

1. P – IV Processor
2. Speed – 2.8 GHZ
3. RAM – 512 MB Minimum
4. Hard Disk – 80 GB
5. IHeadphones of High quality

SUGGESTED SOFTWARE:

The software consisting of the prescribed topics elaborated above should be procured and used.

SUGGESTED SOFTWARE:

1. Clarity Pronunciation Power – part II
2. Oxford Advanced Learner's Compass, 7th Edition
3. DELTA's key to the Next Generation TOEFL Test: Advanced Skill Practice.
4. Lingua TOEFL CBT Insider, by Dreamtech
5. TOEFL & GRE(KAPLAN, AARCO & BARRONS, USA, Cracking GRE by CLIFFS)
6. The following software from 'train2success.com' Preparing for being interviewed,
 1. Positive Thinking,
 2. Interviewing Skills,
 3. Telephone Skills,
 4. Time Management
 5. Team Building,
 6. Decision making

Reference Books:

1. **TECHNICAL COMMUNICATION** by Meenakshi Raman & Sangeeta Sharma, Oxford University Press 2009.
2. **ADVANCED COMMUNICATION SKILLS LABORATORY MANUAL** by Sudha Rani, D, Pearson Education 2011.
3. **ENGLISH LANGUAGE COMMUNICATION: A Reader cum Lab Manual** Dr A Ramakrishna Rao, Dr GNatanam & Prof SA Sankaranarayanan, Anuradha Publications, Chennai 2008.
4. **English Vocabulary** in Use series, Cambridge University Press 2008.
5. **MANAGEMENT SHAPERS SERIES**: by Universities Press(India) Pvt Ltd., Himayatnagar, Hyderabad 2008.
6. **COMMUNICATION SKILLS**: by Leena Sen, PHI Learning Pvt Ltd., New Delhi, 2009.
7. **Handbook for Technical Writing** by David A McMurrey & Joanne Buckely CENGAGE Learning 2008.
8. **JOB HUNTING** by Colm Downes, Cambridge University Press 2008.
9. **MASTER PUBLIC SPEAKING**: by Anne Nicholls, JAICO Publishing House, 2006.
10. **ENGLISH FOR TECHNICAL COMMUNICATION FOR ENGINEERING STUDENTS**:, Aysha Vishwamohan, Tata Mc Graw-Hil 2009.
11. Books on **TOEFL/GRE/GMAT/CAT/IELTS** by Barron's/DELTA/Cambridge University Press.
12. **INTERNATIONAL ENGLISH FOR CALL CENTRES** by Barry Tomalin and Suhashini Thomas, Macmillan Publishers, 2009.

CONSTITUTION OF INDIA**VI Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC03	MC	L	T	P	C	CIA	SEE	Total
		2	-	-	-	30	70	100

COURSE OBJECTIVES:

Students will be able to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

COURSE OUTCOMES:

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy ,its institutions and processes at the local,state and union levels.
5. Explain the functions and responsibilities of election commission of india and union public service commission.

UNIT-I	HISTORY OF MAKING OF THE INDIAN CONSTITUTION	Classes: 12
Introduction to the constitution of India, the making of the constitution and salient features of the constitution.		
UNIT-II	PHILOSOPHY OF THE INDIAN CONSTITUTION	Classes: 12
Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.		
UNIT-III	UNION GOVERNMENT	Classes: 12
Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.		
UNIT-IV	STATE GOVERNMENT	Classes: 12
State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)		
UNIT-V	ELECTION COMMISSION	Classes: 12
Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.		

Text Books:

1. M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional
2. Law, Prentice – Hall of India Pvt. Ltd. New Delhi
3. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Callenges to Civil Rights Guarantees in India, Oxford University Press, 2012
4. P.M.Bakshi, The constitution of India, Universal Law Publishing Co.,
5. The Constitution of India, 1950 (Bare Act), Government Publication.
6. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.

Reference Books:

1. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014.
2. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015.
3. Indian constitution at work, NCERT
4. SubashKashyap, Indian Constitution, National Book Trust
5. J.A. Siwach, Dynamics of Indian Government & Politics
6. D.C. Gupta, Indian Government and Politics
7. H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal LawPublication)
8. J.C. Johari, Indian Government and Politics Hans
9. J. Raj Indian Government and Politics

Web References:

1. www.hss.iitb.ac.in/en/lecture-details
2. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution

MOOC Course

1. nptel.ac.in/courses/109104074/8
2. nptel.ac.in/courses/109104045/
3. nptel.ac.in/courses/101104065/

IV B.TECH VII SEMESTER SYLLABUS

FLIGHT VEHICLE DESIGN**Semester: VII**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE24	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Discuss the importance of conceptual design process and studying the different phases of designing process involved in the design.
2. Understand the Integrated product development and principles of baseline design– stability & control, performance and constraint analysis
3. Understand the designing process of how a aircraft production company does.
4. Discuss the trade design and trade studies of the major types of aircrafts.

COURSE OUTCOMES:

At the end of the course the students are able:

1. Define the design process and compare the designing process involved in the design of different aircraft's
2. To distinguish the different types of aerodynamic geometries and determine the initial sizing weight estimations for fixed engine and rubber engine.
3. To apply the basic concepts of propulsion systems, fuel systems and landing gear system to determine the lift curve slope; maximum lift coefficient and complete drag build up.
4. To analyze the integrated product development, cost estimation, parametric analysis, optimization, refined sizing and principles of baseline design such as stability & control, performance and constraint analysis
5. To design an DC – 1, DC – 2, DC- 3 aircraft airfoil geometry by using different design techniques like lofting, conic lofting, flat warp lofting

UNIT-I	OVERVIEW OF THE DESIGN PROCESS, SIZING FROM A CONCEPTUAL SKETCH	Classes: 12
Phases of aircraft design. Aircraft conceptual design process, project brief / request for proposal, Problem definition, information retrieval, aircraft requirements, configuration options. Integrated Product development and aircraft design. The initial conceptual sketches, L / D estimation. Initial takeoff weight build-up, empty weight estimation, historical trends, fuel fraction estimation, mission profiles, mission segment weight fractions.		
UNIT-II	AIRFOIL AND GEOMETRY SELECTION, THRUST TO WEIGHT RATIO, WING LOADING INITIAL SIZING AND CONFIGURATION LAYOUT, CREW STATION, PASSENGERS AND PAYLOAD	Classes: 12
Airfoil selection, airfoil design, design lift coefficient, stall, airfoil thickness ratio and other airfoil considerations. Wing geometry and wing vertical location, wing tip shapes. Tail geometry and arrangements. Thrust to weight ratio, statistical estimation, thrust matching. Wing loading performance constraints. Selection of thrust-to-weight ratio and wing loading. Sizing with fixed engine and with rubber engine. Geometry sizing of fuselage, wing, tail, control surfaces. Development of configuration lay out from conceptual sketch. The inboard profile drawing, wetted area, volume distribution and fuel volume plots. Lofting- definition, significance and methods, flat wrap lofting. Special consideration in configuration lay out. Isobar tailoring, Sears-Haack volume distribution, structural load paths. Radar, IR, visual detectability, aural signature. Considerations of vulnerability, crashworthiness, producibility, maintainability. Fuselage design, crew station, passenger compartment, cargo provisions, weapons carriage, gun installation.		

UNIT-III	PROPULSION AND FUEL SYSTEM INTEGRATION, LANDING GEAR AND SUBSYSTEMS BASELINE DESIGN ANALYSIS- AERODYNAMICS & PROPULSION, STRUCTURES & WEIGHT AND BALANCE	Classes: 12
Propulsion selection, jet engine integration, engine dimensions, inlet geometry, inlet location, capture area calculation, boundary layer diverters, nozzle integration, engine cooling provisions, engine size estimation. Fuel system design and integration. Landing gear arrangements, guidelines for lay out. Shock absorbers—types, sizing, stroke determination, gear load factors. Gear retraction geometry. Aircraft subsystems, significance to configuration lay out. The baseline design layout and report of initial specifications. Estimation of lift curve slope, maximum lift coefficient, complete drag build up. Installed performance of an engine, installed thrust methodology, net propulsive force, part power operation. Aircraft loads, categories: maneuver, gust, inertial, power plant, landing gear loads. Limit loads, the V, n diagram. Air load distribution on lifting surfaces. Review of methods of structural analysis. Material selection. Weights and moments statistical group estimation method, centre of gravity excursion control.		
UNIT-IV	BASELINE DESIGN - STABILITY AND CONTROL, PERFORMANCE AND CONSTRAINT ANALYSIS COST ESTIMATION, PARAMETRIC ANALYSIS, OPTIMISATION, REFINED SIZING AND TRADE STUDIES	Classes: 12
Estimation of static pitch stability, velocity stability and trim. Estimation of stability and control derivatives. Static lateral, directional stability and trim. Estimation of aircraft dynamical characteristics, handling qualities. Cooper – Harper scale, relation to aircraft dynamic characteristics. Performance analysis and constraint analysis— steady level flight, minimum thrust required for level flight, range and loiter endurance. Steady climbing and descending flight, best angle and rate of climb, time to climb and fuel to climb. Level turning flight, instantaneous turn rate, sustained turn rate. Energy maneuverability methods of optimal climb trajectories and turns. The aircraft operating envelope. Take off analysis, Balanced field length. Landing analysis. Fighter performance measures of merit. Effects of wind on aircraft performance. Initial technical report of baseline design analysis and evaluation. Refined baseline design and report of specifications. Elements of lifecycle cost, cost estimating method, RDT&E and production costs, operation and maintenance costs, fuel and oil costs, crew salaries, maintenance expenses, depreciation. Cost measures of merit. Aircraft and airline economics, DOC and IOC, airline revenue, breakeven analysis, investment cost analysis. Parametric analysis and optimization. Refined conceptual sizing methods. Sizing matrix plot and carpet plot. Trade studies, design trades, requirement trades, growth sensitivities. Multivariable design optimization methods. Measures of merit. Determination of final baseline design configuration, preparation of type specification report		
UNIT-V	CASE STUDIES AND DESIGN OF UNIQUE AIRCRAFT CONCEPT	Classes: 12
Design of the DC – 1, DC – 2, DC- 3 aircraft, Boeing B-47 and 707, General Dynamics F-16, SR-71 Blackbird, Northrop Grumman B-2 Stealth Bomber. A survey of the Indian aircraft design effort. Design of VTOL aircraft, helicopters, hypersonic vehicles, delta and double delta wings, forward swept wings, uninhabited air vehicles.		
Text Books:		
1. Raymer, Daniel P. (2006), Aircraft Design: A Conceptual Approach, 4th edition, AIAA Educational Series, USA. 2. J. F. Marchman, L. R. Jenkinson (2003), Aircraft Design Projects for Engineering students, AIAA Publishers, USA. 3. Ajoy Kumar Kunda (2010), Aircraft Design, Cambridge University Press, UK.		
Reference Books:		
1. Torenbeek E. (1986), Synthesis of Subsonic Airplane Design, Delft University Press, New York. 2. Bruhn. E. H (1973), Analysis and Design of Flight Vehicles Structures, New Edition, Jacobs Publishing House, USA. 3. Scheler E. E, Dunn L.G (1963), Airplane Structural Analysis and Design, John Wiley & Sons, USA. 4. D. Howe (2005), Aircraft conceptual Design Synthesis, John Wiley & Sons Publishers, USA.		

Web References:

<https://nptel.ac.in/downloads/101106035/>

<https://nptel.ac.in/downloads/101104069/>

E-Text Books:

<https://www.abebooks.com/9781563478291/Aircraft-Design-Conceptual-Approach-Fourth-1563478293/plp>

MOOC Course

<https://nptel.ac.in/courses/101106035/>

<https://nptel.ac.in/courses/101104069/>

HEAT TRANSFER & COMPUTATIONAL FLUID DYNAMICS**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE39	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Identify the various techniques of solving heat transfer and CFD problems
2. Analyze the discretization techniques for governing equations.
3. Describe modelling of diffusion problems using finite volume method
4. Describe how the flow is modelled using structured and unstructured grids.
5. Analyse the multiphase modelling using various techniques of heat transfer and computational fluid dynamics.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Identify the various techniques of solving heat transfer and CFD problems
2. Analyze the discretization techniques for governing equations.
3. Describe modelling of diffusion problems using finite volume method
4. Describe how the flow is modelled using structured and unstructured grids.
5. Analyse the multiphase modelling using various techniques of heat transfer and computational fluid dynamics.

UNIT-I	HIGHER-ORDER NUMERICAL SCHEMES FOR HEAT, MASS, AND MOMENTUM TRANSFER IN FLUID FLOW	Classes: 12
Basics of heat transfer, fluid flow, mathematical description of fluid flow and heat transfer: conservation equations for mass, momentum, energy and chemical species, classification of partial differential equations, coordinate systems single-grid schemes, new numerical simulation strategy, novel multigrid numerical procedure application of NIMO Scheme to laminar flow, steady laminar flow in pipes, steady laminar flow over a fence, application of the NIMO higher-order scheme to the turbulent flow in pipes		
UNIT-II	DISCRETIZATION TECHNIQUES	Classes: 12
Discretization techniques using finite difference methods: Taylor-series and control volume formulations. Finite element discretization techniques. governing equations, non-dimensional form of fluid flow equations, non-dimensional form of turbulent flow equations, porous media flow: the generalized model equations, the finite element method, strong and weak forms, weighted residual approximation, the Galerkin, finite element, method, characteristic Galerkin scheme for convection-diffusion equation stability conditions, characteristic-based split scheme		
UNIT-III	MODELLING OF DIFFUSION PROBLEMS USING FINITE VOLUME METHOD	Classes: 12
One dimensional steady state diffusion problem, discretization technique, Solution methodology for linear and non-linear problems: Point-by-point iteration, TDMA, Two and three dimensional discretization, Discretization of unsteady diffusion problems: Explicit, Implicit and Crank-Nicolson's algorithm; stability of solutions, Modelling of Convection- Diffusion Problems: One dimensional convection-diffusion problem: Central difference scheme, Discretization based on analytical approach (exponential scheme), Hybrid and power law discretization techniques, Higher order schemes (QUICK algorithm).		

UNIT-IV	FLOW MODELLING	Classes: 10
Discretization of incompressible flow equations, Pressure based algorithm: SIMPLE, SIMPLER etc. Unstructured grids: Introduction to Finite Volume Method with unstructured grids, numerical methods for incompressible flow , numerical methods for compressible flow, governing equations for two-dimensional flow		
UNIT-V	MULTIPHASE MODELLING	Classes: 14
Modelling of multiphase problems: enthalpy method, volume of fluid (VOF) and Level Set Methods. Introduction to turbulence modelling: Large Eddy Simulation (LES), Present State of Direct Numerical Simulations , Instantaneous and Reynolds-Averaged Governing, Equations for Flow and Heat Transfer, Numerical Procedures of DNS , DNS using high-accuracy finite-difference method , DNS using spectral method , DNS of Turbulent heat Transfer in Channel Flow with Transverse-Rib Roughness: Finite-Difference Method, Heat transfer and skin friction coefficients, Velocity and thermal fields around the rib , Statistical characteristics of velocity field and turbulent structures, Statistical characteristics of thermal field and related turbulent structures		
Text Books:		
1. K. Muralidhar, T.Sundararajan, (2011), Computational Fluid Flow and Heat Transfer ,Second Edition, Narosa 2. P. S. Ghoshdastidar ,Computer Simulation of Flow and Heat Transfer (4th Edition, Tata McGraw-Hill), 1998. 3. Hirch C , Numerical Computation of Internal and External Flows , Elsevier 2007.		
Reference Books:		
1. S. V. Patankar ,Numerical Heat Transfer and Fluid Flow by (Hemisphere Series on Computational Methods in Mechanics and Thermal Science) 2. Zikanov.O., Essential Computational Fluid Dynamics Wiley 2010. 3. Chung T. J., Computational Fluid Dynamics, Cambridge University Press, 2003.		
Web References:		
http://home.iitk.ac.in/~gtm/turbulence/ui/TOC.htm		
E-Text Books:		
1. Computational Fluid Dynamics, John D Anderson, Jr, McGraw Hill Book Company. 2. Finite Elements in Engineering, T R Chandruputla and A D Belegundu, Prentice Hall of India.		
MOOC Course		
https://nptel.ac.in/courses/112104030/		

MECHANICAL VIBRATIONS AND STRUCTURAL DYNAMICS**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE43	PEC	L	T	P	C	CIA	SEE	Total
		4	1-	-		30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Identify the various elements of stiffened panels and types of mechanical load systems like transverse, longitudinal and torsional
2. Analyze one-degree-of-freedom system
3. Describe various terms like free vibrations, forced vibrations, damped vibrations and their engineering applications
4. To study the effect of time dependent forces on mechanical systems and to get the natural characteristics of system with more degree of freedom systems.
5. To study the aeroelastic effects of aircraft wing.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Analyze the problem and estimate the governing equation of motion
2. apply linear mathematical models of real-life engineering systems
3. Determine vibratory responses of single degree of freedom and multi-degree of freedom systems to the harmonic, periodic and non-periodic excitation
4. Develop the mathematical model of dynamic systems with a single degree of freedom,
5. Develop mathematical model of dynamic systems with multiple degrees of freedom

UNIT-I	SINGLE DEGREE OF FREEDOM SYSTEMS	Classes: 12
INTRODUCTION: Simple harmonic motion, terminology, Newton's Law, D'Alembert's Principle, Resonance, Introduction to mechanism of damping. Oscillations. Degrees of freedom. Various mechanisms of damping. Equivalent stiffness concept, vibration isolation SINGLE DEGREE OF FREEDOM SYSTEMS: Free vibrations – damped vibrations– forced vibrations, with and without damping – support excitation – transmissibility – vibration measuring instruments. Amplitude and Phase response diagrams. Applications of single degree of freedom systems		
UNIT-II	MULTI DEGREE OF FREEDOM SYSTEMS	Classes: 12
Two degrees of freedom systems - static and dynamic couplings - vibration absorber- Multi degree of freedom systems - principal co-ordinates - principal modes and orthogonal conditions – Eigen value problems - Hamilton's principle - Lagrangean equations and application. VIBRATION OF CONTINUOUS SYSTEMS Longitudinal vibration, lateral vibration, torsional vibration of shafts, dynamical equations of equilibrium of elastic bodies, natural frequencies and mode shapes determination.		
UNIT-III	DETERMINATION OF NATURAL FREQUENCIES AND MODE SHAPES	Classes: 12
Determining natural frequencies and mode shape. Natural Vibrations of solid continua. Determination of Eigen Values and Eigen modes – Matrix method. Kinetically consistent Load systems and determination of [K], [M], [C] and [I] matrices ROTATING SHAFTS: Natural frequency of rotating shafts Whirling of shafts. Dynamic balancing of rotating shafts. Dynamic dampers.		

UNIT-IV	APPROXIMATE METHODS FOR FREQUENCY	Classes: 12
Introduction to approximate methods for frequency analysis, Rayleigh Ritz method for vibration analysis. Diagonalization of stiffness, mass and damping matrices using orthogonality conditions. Approximate methods - Rayleigh's method - Dunkerley's method – Rayleigh-Ritz method, matrix iteration method.		
UNIT-V	INTRODUCTION TO THEORY OF AEROELASTIC STABILITY	Classes: 12
Aeroelastic and inertial coupling- aeroelastic problems. Collar's triangle. Static and dynamic aeroelastic phenomena. Aeroelatic instabilities and their prevention. Wing divergence, control reversal and wing flutter– buffeting, flutter speed. Aeroelastic tailoring. Elements of servo elasticity		
Text Books:		
1. R. W. Clough and Penzien (2010), Dynamics of Structures, 2 nd edition, McGraw Hill, New Delhi. 2. S. S. Rao (2011), Mechanical Vibrations, 5 th editions, Prentice Hall of India, New Delhi. 3. J. S. Rao, Gupta K. (2002), Theory and practice of Mechanical vibrations, Wiley Eastern Ltd, USA.		
Reference Books:		
1. Fug Y. C. (2008), An Introduction to Theory of Aeroelasticity, Dover Publications, US 2. Timoshenko S (2011), Vibration Problems in Engineering, 2 nd edition, Oxford city press, USA		
Web References:		
https://easyengineering.net/vibrationbooks/ https://easyengineering.net/mechanical-vibrations-book-v-p-singh-free-download/		
E-Text Books:		
1. Mechanical Vibrations By V.P. Singh 2. Mechanical vibrations graham kelly solutions pdf		
MOOC Course		
https://www.classcentral.com/course/nptel-introduction-to-mechanical-vibration-7929		

FATIGUE AND FRACTURE MECHANICS

VII Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE44	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The Main objective of this course is to

1

Understand the major concepts of fracture mechanisms and fracture modes associated with failure

2

Have a fundamental understanding of various regimes of fatigue crack growth and life estimation

3

Analyse the fatigue failure of aircraft components made up of metals and composites

4

Conduct case study to understand the impact of fracture mechanics in design and life prediction of aircraft components

COURSE OUTCOMES:

This course uses lectures, assignments and home works to the students. The teaching methods include regular class work, Problem solving, technical quiz, and seminars to enable the students:

1.

Predict material failure for the combination of applied stress

2.

Identify and exemplify the basic fatigue mechanisms and apply that knowledge to failure analysis

3.

Predict lifetimes for fatigue and environmentally assisted cracking

4.

Apply stress analysis to calculate the crack driving force in linear and nonlinear materials and formulate appropriate fracture criteria for stationary and growing cracks

5.

Evaluate well defined fracture mechanics problems for both linear and nonlinear materials subjected to both monotonic and cyclic loading

UNIT-I	FATIGUE OF STRUCTURES	Classes: 12
Introduction to fatigue, Endurance limits - Effect of mean stress, Goodman, Gerber and Soderberg relations and diagrams - Notches and stress concentrations - Neuber's stress concentration factors - Plastic stress concentration factors - Notched S.N. Curves.		
UNIT-II	PHYSICAL ASPECTS OF FATIGUE	Classes: 12
Phase in fatigue life - Crack initiation - Crack growth - Final fracture - Dislocations - Fatigue fracture surfaces.		
UNIT-III	STATISTICAL ASPECTS OF FATIGUE BEHAVIOUR	Classes: 12
Low cycle and high cycle fatigue - Coffin - Manson's relation - Transition life - Cyclic strain hardening and softening - Analysis of load histories - Cycle counting techniques -Cumulative damage - Miner's theory.		
UNIT-IV	OVERVIEW OF ENGINEERING FRACTURE MECHANICS	Classes: 12
Strength of cracked bodies - Potential energy and surface energy - Griffith's theory - Irwin - Orwin extension of Griffith's theory to ductile materials - Stress analysis of cracked bodies - Effect of thickness on fracture toughness.		

UNIT-V	FATIGUE DESIGN AND TESTING	Classes: 12
Safe life and fail-safe design philosophies, Importance of fracture mechanics in aerospace structure, Application to structures		
Text Books:		
1. Prashant Kumar, "Elements of Fracture Mechanics", Tata McGraw Hill, New Delhi, India, 2009. 2. Ali Fatemi, "Metal Fatigue in Engineering", 2nd edition, John Wiley and sons, inc., 2000.		
Reference Books:		
1. K. R.Y. Simha, "Fracture Mechanics for Modern Engineering Design", Universities Press (India) Limited, 2001 2. T.L. Anderson, "Fracture Mechanics - Fundamentals and Applications", 3rd Edition, Taylor and Francis Group, 2005. 3. K. Ramesh, e-Book on "Engineering Fracture Mechanics", IIT Madras, 2007. 4. Gross, Dietmar, Seelig and Thomas, "Fracture Mechanics with an Introduction to Micromechanics", ISBN 978-3-319-71090-7, Springer, 2018.		
Web References:		
https://ocw.mit.edu/courses/materials-science-and-engineering/3-35-fracture-and-fatigue-fall-2003/lecture-notes/ https://www.efatigue.com/		
E-Text Books:		
1. K. Ramesh, e-Book on "Engineering Fracture Mechanics", IIT Madras, 2007. 2. R N L Smith, e-Book on "Basic Fracture Mechanics", Butterworth-Heinemann, 1991.		
MOOC Course		
https://nptel.ac.in/courses/112106065/ https://ocw.mit.edu/courses/materials-science-and-engineering/3-35-fracture-and-fatigue-fall-2003/		

ROCKETS AND MISSILES

VII Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE45	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

This course is to develop and understand the aerodynamics of rocket and missile, major propellants used in rocket, trajectory and launching systems

1. Illustrate Fundamentals of rocket and missile systems, functions and disciplines.
2. Demonstrate full spectrum of rocket systems, application and technologies.
3. Classify technologies of guided systems used in missile.
4. Explain Fundamentals and applications of solid, liquid and hybrid rocket systems.
5. Distinguish between systems built as weapons and those built for commerce.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Interpret new technology for staging of rockets and missiles.
2. Develop methods for altitude control.
3. Identify material for nozzle of rockets and missiles.
4. Analyse the missile guidance system.
5. Determine 2-D motion rocket equation.

UNIT-I	ROCKET PROPELLANT SYSTEMS	Classes: 12
SOLID PROPELLANT ROCKET SYSTEMS: Ignition system in rockets, Types of igniters, Igniter design considerations, Combustion system of solid rockets. LIQUID PROPELLANT ROCKET SYSTEMS: Design consideration of liquid rocket combustion chamber, injector, propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems, Propellant slosh, Propellant hammer, Geysering effect in cryogenic rocket engines.		
UNIT-II	AERODYNAMICS OF ROCKETS AND MISSILES	Classes: 12
Airframe components of rockets and missiles, Forces acting on a missile while passing through atmosphere, Classification of missiles. Method of describing aerodynamic forces and moments, Lateral aerodynamic moment, Lateral damping moment and longitudinal moment of a rocket-Lift and drag forces, Drag estimation, Body upwash and downwash in missiles, Rocket dispersion.		
UNIT-III	TWO-DIMENSIONAL ROCKET MOTION IN VACUUM & MULTI-STAGE ROCKET	Classes: 12
TWO-DIMENSIONAL ROCKET MOTION IN VACUUM: Equations of motion, Rocket Motion in free space (Tsiolkovsky's equation, Rocket Parameters, Burnout range), Rocket Motion in a homogeneous gravitational field (Vertical flight, Constant Pitch angle, Gravity turns). MULTI-STAGE ROCKET: Nomenclature of the multi-stage rocket, Ideal Velocity of the multi-stage rocket, Vertical ascent in a homogeneous gravitational field and in vacuum (Burnout velocity-Culmination altitude-Vertical ascent of a two-stage rocket).		
UNIT-IV	ATTITUDE CONTROL OF ROCKETS AND MISSILES	Classes: 12
ATTITUDE CONTROL OF ROCKETS AND MISSILES: Rocket thrust vector control, Methods of thrust vector control, Thrust magnitude control, Thrust Termination. SEPARATION SYSTEMS FOR ROCKETS AND MISSILES: Stage separation dynamics, Separation		

techniques.		
UNIT-V	MATERIALS FOR ROCKETS AND MISSILES	Classes: 12
Criteria for Selection of materials for rockets and missiles, Choice of materials at cryogenic temperatures, extremely high temperatures. Requirement of materials for thermal protection and pressure vessels.		
Text Books:		
1. Martin J. L. Turner (2008), Rocket and Spacecraft Propulsion principles, practice and new developments, 3rd edition, Springer, USA. 2. Sutton G.P. (2010), Rocket Propulsion Elements, John Wiley / BSP Books, USA. 3. Cornelisse J. W. (1980), Rocket Propulsion and Space Dynamics, Pitman Publishing, London.		
Reference Books:		
1. S. S. Chin (1982), Missile Configuration Design, McGraw- Hill, New Delhi. 2. Bong Wie (2008), Space Vehicle Dynamics and Control, AIAA Educational Series, USA. 3. Earl R Parker (1998), Materials for Missiles and Spacecraft, McGraw Hill, New Delhi.		
Web References:		
http://www.nptelvideos.in/2012/12/rocket-propulsion.html https://nptel.ac.in/courses/101108054/2 https://www.drdo.gov.in/drdo/data/Guided%20Missiles.pdf		
E-Text Books:		
Guided Missiles -T V Karthikeyan & A K Kapoor		
MOOC Course		
https://online.stanford.edu/courses/aa283-aircraft-and-rocket-propulsion		

NUMERICAL TECHNIQUES

VII Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE46	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Necessary basic concepts of major numerical methods and its importance in Aeronautical domain.
2. Procedures for solving complex problems numerically occurring in engineering and technology

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Explain basic theorems and concepts in major areas of mathematics
2. Ability to consider problems that could be solved by implementing concepts from different areas in mathematics
3. Efficient use of computers, laboratories and software's to handle problems that are difficult to be solved manually
4. Understanding of professional and ethical responsibilities
5. Efficient use of the techniques, skills, and tools of modern mathematics.

UNIT-I	SOLUTION OF EQUATIONS AND EIGENVALUE PROBLEMS	Classes: 13
Solution of algebraic and transcendental equations – Fixed point iteration method – Newton Raphson method- Solution of linear system of equations – Gauss elimination method – Pivoting – Gauss Jordan method – Iterative methods of Gauss Jacobi and Gauss Seidel – Matrix Inversion by Gauss Jordan method – Eigen values of a matrix by Power method.		
UNIT-II	INTERPOLATION AND APPROXIMATION	Classes: 11
Interpolation with unequal intervals – Lagrange's interpolation – Newton's divided difference interpolation – Cubic Splines – Interpolation with equal intervals – Newton's forward and backward difference formulae.		
UNIT-III	NUMERICAL DIFFERENTIATION AND INTEGRATION	Classes: 12
Approximation of derivatives using interpolation polynomials – Numerical integration using Trapezoidal, Simpson's 1/3 rule – Romberg's method – Two point and three point Gaussian quadrature formulae – Evaluation of double integrals by Trapezoidal and Simpson's 1/3 rules		
UNIT-IV	INITIAL VALUE PROBLEMS FOR ORDINARY DIFFERENTIAL EQUATIONS	Classes: 12
Single Step methods – Taylor's series method – Euler's method – Modified Euler's method – Fourth order Runge-Kutta method for solving first order equations – Multi step methods – Milne's and Adams- Bash forth predictor corrector methods for solving first order equations.		

UNIT-V	BOUNDARY VALUE PROBLEMS IN ORDINARY AND PARTIAL DIFFERENTIAL EQUATIONS	Classes: 12
Finite difference methods for solving two-point linear boundary value problems – Finite difference techniques for the solution of two dimensional Laplace's and Poisson's equations on rectangular domain – One dimensional heat flow equation by explicit and implicit (Crank Nicholson) methods – One dimensional wave equation by explicit method.		
Text Books:		
1. Grewal. B.S., and Grewal. J.S., "Numerical methods in Engineering and Science", Khanna Publishers, 9th Edition, New Delhi, 2007. 2. Gerald. C. F., and Wheatley. P. O., "Applied Numerical Analysis", Pearson Education, Asia, 6th Edition, New Delhi, 2006.		
Reference Books:		
1. Chapra. S.C., and Canale.R.P., "Numerical Methods for Engineers, Tata McGraw Hill, 5th Edition, New Delhi, 2007. 2. Brian Bradie. "A friendly introduction to Numerical analysis", Pearson Education, Asia, New Delhi, 2007. 3. Sankara Rao. K., "Numerical methods for Scientists and Engineers", Prentice Hall of India Private, 3rd Edition, New Delhi, 2007		
Web References:		
https://www.britannica.com/science/numerical-analysis		
E-Text Books:		
1. An Introduction to Numerical Analysis by Endre Süli, David F. Mayers Cambridge University Press 2. Introduction to Numerical Analysis - J.Stoer,R.Bulirsch		
MOOC Course		
https://nptel.ac.in/courses/122102009/		

AERO ELASTICITY**VII Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE47	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

This course will address issues related to the mutual interaction of elastic, inertial, and aerodynamic forces with emphasis on aeronautical applications.

1. Understand how the aeroelastic phenomena flutter, divergence and aileron reversal arise and how they affect aircraft performance,
2. Formulate aeroelastic equations of motion and use to derive fundamental relations for aero elastic analysis,
3. Perform a preliminary aeroelastic analysis of a slender wing structure in low-speed airflow and under what circumstances an aeroelastic analysis can be expected to produce useful results.

COURSE OUTCOMES:

1. Identify the aeroelastic phenomena flutter, divergence and aileron reversal arise and how they affect aircraft performance, collar triangle
2. Demonstrate a basic understanding of modern numerical methods and the state-of-the-art in structural dynamics and aeroelasticity.
3. Differentiate between static aeroelasticity and dynamic aeroelasticity
4. Develop equation of motion for linear motion and rotary motion
5. Analyze the wing flutter, under the over damping and critical damping conditions
6. Build confidence for self learning needed for aircrafts, automobiles failures due to vibration effect

UNIT-I	INTRODUCTION	Classes: 10
Introduction to Aero elasticity COLLARS Triangle, Aerodynamics and interactions of Structural and Inertial forces Static and Dynamic Aero Elasticity Phenomena. Simple Two dimensional idealization of flow, String Theory, Fredholm Integral equations of Second Kind Exact Solutions for simple rectangular wings.		
UNIT-II	ANLAYTICAL METHODS	Classes: 11
Formulations of Structural Dynamics Equation and Coupling effects for panels and plates, generalized coordinates, Lagrange's Equations of motion Hamilton's Principle Orthogonality conditions. Static Aero elastic Studies Divergences, control reversal, Aileron reversal speed, Aileron efficiency, lift distribution, Rigid and elastic wings.		
UNIT-III	EXPERIMENTAL ANALYSIS&EQUATIONS OF AERO ELASTIC	Classes: 13
EXPERIMENTAL ANALYSIS: Non-dimensional Parameters, stiffness criteria, dynamic mass balancing, model experiments and dimensional similarity, flutter analysis. EQUATIONS OF AERO ELASTIC: Formulation of Aero elastic Equations for a Typical Section, Quasi Steady Aerodynamic derivatives, modal equations Galerkins method of analysis.		
UNIT-IV	FLUTTER	Classes: 10
Stability of motion of Continua Torsion flexure flutter, Solution of flutter determinant, method of determining the classical flutter speed, Flutter Prevention and control.		

UNIT-V	AERO ELASTICITY APPLICATIONS	Classes: 10
Application of Aero Elasticity in Engineering Problems, Galloping of transmissionlines, flow induces vibrations of tall slender structures and suspension Bidges.		
Text Books:		
1. Dewey H. Hodges, G. Alvin Pierce (2011), Introduction to Structural Dynamics and Aero Elasticity, 2nd edition, Cambridge University Press, UK. 2. Fung Y. C. (2008), An introduction to the Theory of Aero Elasticity, Dover Publications, USA. 3. Jan R. Wright (2008), Introduction to Aircraft Aero Elasticity and Loads, John Wiley, USA		
Reference Books:		
1. Raymond L. Bisplinghoff, Holt Ashely (2002), Principles of Aeroelasticity, Drovers Publications, USA. 2. Adamu Yebi (2010), Vibration Analysis of Cracked Composite Aircraft Wing Modeled as Shell, VMD Verlag, New Delhi. 3. E. H. Dwell (1995), A Modern Course in Aero elasticity, Springer Publishers, Germany.		
Web References:		
https://engineering.purdue.edu/AAECourses/aae556/2011/AAE556-text-Chapter%201-Intro		
E-Text Books:		
http://ebooksgo.org/free-aeroelasticity-ebooks-online.html		
MOOC Course		
Aero elasticity https://nptel.ac.in/courses/101104005/		

MECHANISMS AND MACHINE DESIGN**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE48	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To study the importance of mechanical engineering design theory to identify and quantify machine elements in the design of commonly used mechanical systems.
2. To apply the concepts of stress analysis, theories of failure and material science to analyze, design and/or select commonly used machine components

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Analyze the forces and motion of complex mechanisms
2. Analyze the mechanism for velocity and acceleration
3. Evaluate the gyroscopic effect of engine on various vehicles
4. Analyze the motion and design a cam based on required motion of follower
5. Evaluate and design a machine or a machine element such as gears, couplings etc. for the required constraints

UNIT-I	MECHANISMS AND MACHINES	Classes: 12
MECHANISMS: Elements of links, Classification Rigid link, flexible and fluid link. Types of kinematic pairs, sliding, turning, rolling, screw and spherical pairs, Lower and higher pairs, Closed and open pairs. Constrained motion, completely, partially or successfully constrained and incompletely constrained. MACHINES: Mechanism and machines, Classification of machines, Kinematic chain, Inversion of mechanism, Inversion of quadrilateral cycle, Chain: single and double slider crank chains.		
UNIT-II	KINEMATICS	Classes: 12
KINEMATICS: Velocity and acceleration. Determination of velocity and acceleration diagrams, Graphical method. Application of relative velocity method four bar chain. ANALYSIS OF MECHANISMS: Analysis of slider crank chain for displacement, Velocity and acceleration of sliding. Acceleration diagram for a given mechanism, Kleins construction, Coriolis acceleration, Determination of Coriolis component of acceleration.		
UNIT-III	PLANE MOTION OF BODY	Classes: 10
PLANE MOTION OF BODY: Instantaneous center of rotation. Three centers in line theorem. Graphical determination of instantaneous center, diagrams for simple mechanisms and determination of angular velocity of points and links. PRECISION: Effect of Precision on Stability of moving vehicles such as Aero planes and ships.		
UNIT-IV	CAMS	Classes: 10
CAMS: Definition of cam and followers. Types of followers and cams, Terminology, Types of follower motion. Uniform velocity, Simple harmonic motion and uniform acceleration. Maximum velocity and Maximum acceleration during out ward and return strokes in all the above three cases. ANALYSIS OF MOTION OF FOLLOWERS: Roller follower, Circular cam with straight, concave and convex flanks.		

UNIT-V	DESIGN OF MACHINE ELEMENTS	Classes: 10
DESIGN OF MACHINE ELEMENTS: Principles of mechanical design, dimensional tolerances, fits. Design of common machine elements springs, shafts, Universal coupling. GEARS AND GEAR TRAINS: Introduction to gears-types, Law of gearing, classification of Helical, Bevel and worm gears: Simple and reverted gear train, epicyclic gear trains-velocity ratio or train value		
Text Books:		
1. Dr Jagdish Lal, J. M. Shaw (2003), Theory of Mechanisms and Machines, Metropolitan Book Co-Ltd, New Delhi. 2. P. L. Ballaney (2003), Theory of Machines and Mechanisms, Khanna Publisher, New Delhi.		
Reference Books:		
1. Amithab Ghosh, Asok Kumar Malik (2001), Theory of Mechanisms and machines, East West Press Private Limited, New Delhi. 2. J. E. Shigley, Charles, R. Mischke (2009), Theory of Machines and Mechanisms, Tata McGraw Hill, New Delhi		
Web References:		
https://nptel.ac.in/downloads/112105125/		
E-Text Books:		
Theory of Machines and Mechanisms (with John J. Uicker, Jr.)		
MOOC Course		
https://nptel.ac.in/courses/112104121/		

AVIONICS AND FLIGHT CONTROL SYSTEMS

VII Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE49	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-				

COURSE OBJECTIVES

1. The course is intended to build up necessary background for understanding avionics system and its applications.
2. To understand the method of message transfer between the aircraft systems and sub-systems.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Compare and contrast avionics systems between the civil and military aircrafts
2. Illustrate the various types of navigation systems used in aircraft
3. Develop the transfer function of the feedback control systems
4. Analyze the time response and stability criteria for an aircraft
5. The modern era of flight control system

UNIT-I	BASICS & FLIGHT DECK AND DISPLAY SYSTEMS	Classes: 8
---------------	---	-------------------

BASICS: Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles.

FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, Electronic instrumentation systems.

UNIT-II	COMMUNICATION SYSTEMS	Classes: 10
----------------	------------------------------	--------------------

AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.

RANGING AND LANDING SYSTEMS: VHF Omnidirectional range, VOR receiver principles, distance measuring equipment, principles of operation, Instrument landing system, and localizer and glide slope.

POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation

UNIT-III	CONTROL SYSTEM	Classes: 12
-----------------	-----------------------	--------------------

BASICS IN CONTROL SYSTEM AND TRANSFER FUNCTION: Introduction of Control Systems, Various types of systems (Open Loop and closed loop) and their differences- Classification and Feedback Characteristics of control system-Effects of feedback. Mathematical models – Differential equations, Translational and Rotational mechanical systems. Transfer Function of DC Servo motor - AC Servo motor- Synchro transmitter and Receiver.

REPRESENTATION OF TRANSFER FUNCTION AND CONTROL DESIGN TECHNIQUES: Block diagram representation of systems considering electrical systems as examples. Block diagram algebra – Representation by Signal flow graph - Reduction using Mason's gain formula. Compensation techniques – Lag, Lead, Lead-Lag Controllers design, PID Controllers

UNIT-IV	TIME RESPONSE ANALYSIS	Classes: 12
----------------	-------------------------------	--------------------

TIME RESPONSE ANALYSIS: Standard test signals - Time response of first order systems – Characteristic Equation of Feedback control systems, Transient response of second order systems - Time domain specifications – Steady state response - Steady state errors and error constants – Effects of proportional derivative, proportional integral systems.

STABILITY ANALYSIS: The concept of stability – Routh's stability criterion – qualitative stability and conditional stability – limitations of Routh's stability. The root locus concept - construction of root loci-

effects of adding poles and zeros to $G(s)H(s)$ on the root loci.		
UNIT-V	AUTO FLIGHT SYSTEM	Classes: 8
AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight directorsystems, flight management systems. Integrated Data transfer methodology by use of MILS - STD - 1553/ ARINC - 429.		
Text Books:		
1. N. S. Nagaraja(1996),Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi. 2. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance, Sterling Book House, Mumbai. 3. Benjamin C. Kuo, Automatic Control System (Ninth Edition)		
Reference Books:		
1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India. 2. E. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi. 3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London		
Web References:		
https://www.britannica.com/technology/avionics https://www.pdc.com/aviation/		
E-Text Books:		
Advanced Avionics Handbook(FAA)		
Digital Avionics: A Computing Perspective - Elisabeth A. Strunk John C. Knight		
MOOC Course		
https://www.canvas.net/browse/erau/courses/aviation-101		

MATERIAL CHARACTERIZATION**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE50	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To provide a through introduction to the principles and practice of diffraction.
2. To provide basic descriptions of a range of common characterization methods for the determination of the structure and composition of solids.

COURSE OUTCOMES

Upon successful completion of this course, the student are able to

1. Exemplify the importance of microscope in metallographic studies.
2. Outline the crystal structure and how structures can be determined using diffraction methods.
3. Explain the operation and use of TEM in material characterization.
4. Explain the operation and use of SEM in material characterization.
5. Exemplify the importance of advanced spectroscopy in characterization of materials.

UNIT-I	INTRODUCTION TO METALLOGRAPHIC STUDIES	Classes: 12
Image formation, resolving power, numerical aperture, empty magnification, depth of focus, components of microscopes, important lens defects and their correction, principles of phase contrast, interference and polarized light microscopy, elements of quantitative metallography and image processing, sample preparation techniques.		
UNIT-II	X RAY DIFFRACTION	Classes: 12
Production and properties of X-ray, absorption of X-rays and filters, X-ray - diffraction directions, diffraction methods. X-ray - diffraction intensities, factors affecting intensity, 'structure factor' calculations for simple, body centered, face centered, diamond cubic and hexagonal crystal structures.		
UNIT-III	ELECTRON MICROSCOPES	Classes: 12
Construction and working principles of transmission electron microscopes. Image formation, resolving power, magnification, depth of focus, elementary treatment of image contrasts, important lens defects and their correction. Bright field and dark field images. Stereographic projection and their applications. Formation of selected area diffraction patterns, reciprocal lattice and Ewald sphere construction, indexing of diffraction patterns, sample preparation techniques.		
UNIT-IV	SCANNING ELECTRON MICROSCOPY (SEM)	Classes: 12
Scanning electron microscope; construction, interaction of electrons with matter, modes of operation, image formation of plane and fractured surfaces. Chemical analysis using electron beam devices like electron probe micro analysis and atomic force microscopy.		
UNIT-V	ADVANCED CHEMICAL AND THERMAL ANALYSIS	Classes: 12
Basic principles, practice and applications of X-ray photoelectron spectrometry, Augur spectroscopy, differential thermal analysis, differential scanning calorimetric and thermo gravimetric analysis.		
Text Books:		
1. Spencer, Michael, Fundamentals of Light Microscopy, Cambridge University Press, 1982. 2. Yang Leng, Wiley, Materials Characterization: Introduction to Microscopic and Spectroscopic Methods, Wiley VCH, 2nd Edition 2013. 3. Sam Zhang, Lin Li and Ashok Kumar, Materials Characterization Techniques, CRC Press, 2008.		

Reference Books:

1. David B. Williams, C. Barry Carter, " Transmission Electron Microscopy: A Textbook for Materials Science" ,Springer, pub. 2009.
2. B.D.Cullity and S.R.Stock, "Elements of X-Ray Diffraction" Third edition, Prentice Hall, NJ , 2001.
3. Joseph I Goldstein, Dale E Newbury, Patrick Echlin and David C Joy, "Scanning Electron Microscopy and X-Ray Microanalysis", 3rd Edition , 2005.
4. David B. Williams, C. Barry Carter, " Transmission Electron Microscopy: A Textbook for Materials Science" ,Springer, pub. 2009.
5. P.J. Goodhew, F.J. Humphreys, and R. Beanland, Electron Microscopy and Analysis: 3rd Edition, Taylor and Francis, 2001.

Web References:

<https://www.microscopyu.com/>

https://nptel.ac.in/courses/nptel_download.php?subjectid=113104004

E-Text Books:

Yang Leng, Wiley, Materials Characterization: Introduction to Microscopic and Spectroscopic Methods; 2009.

MOOC Course

<https://nptel.ac.in/courses/113104004/>

HEAT TRANSFER LAB**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME24	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Understand heat transfer fundamentals, and apply them to engineering problems.
2. Understanding practical heat transfer measurements in laboratories, and compare measurements with theories
3. Use Excel for data analyses and engineering plots, and use word for technical report writing

COURSE OUTCOMES:

At the end of the course the students are able to:

1. To define the various modes of heat transfer with basics laws and application of conduction heat transfer correlations in engineering field
2. To distinguish the modes of heat transfer conduction, convection, radiation, heat exchangers, different condensations.
3. To calculate the rate of heat transfer ,properties of modes of heat transfer, LMTD, NTU values
4. To analyze the concepts of LMTD, NTU, different regimes in boiling curve, different types of condensation, free convection and forced convection
5. To justify the when we use forced or free convection in different condition to justify the when we use LMTD and NTU in heat exchangers

Exp.- I	Composite Slab Apparatus – Overall heat transfer co-efficient	Classes: 3
Exp.-II	Heat transfer through lagged pipe..	Classes: 3
Exp -III	Heat Transfer through a Concentric Sphere	Classes: 3
Exp -IV	Thermal Conductivity of given metal rod.	Classes: 3
Exp -V	Heat transfer through pin-fin	Classes: 3
Exp-VI	Experiment on Transient Heat Conduction	Classes: 3
Exp-VII	Heat transfer in forced convection apparatus.	Classes: 3
Exp-VIII	Heat transfer in natural convection	Classes: 3
Exp- IX	Parallel and counter flow heat exchanger.	Classes: 3
Exp- X	Emissivity apparatus	Classes: 3
Exp-XI	Stefan Boltzmann Apparatus.	Classes: 3
Exp-XII	Critical Heat flux apparatus.	Classes: 3

Exp-XIII	Study of heat pipe and its demonstration	Classes: 3
Reference Books:		
1. D K Dixit, Heat and Mass Transfer, McGraw Hill Education, 2017. 2. C Kothandaraman, Fundamentals of Heat and Mass transfer, New Age International Private Limited, 2012. 3. Jack Holman, Heat Transfer, , McGraw Hill Education, 2009.		
Web References:		
http://home.iitk.ac.in/~gtm/turbulence/ui/TOC.htm		
http://web.mit.edu/lienhard/www/ahtt.html		

Note: Minimum twelve experiments should be performed.

COMPUTATIONAL FLUID DYNAMICS LAB**VII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE51	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

Students undergoing this course are expected:

1. To study and analyze the internal and external aerodynamic properties over major aerodynamic profiles using commercial software packages.
2. To develop a Matlab code to analyze the behavior of the governing equations of Aerodynamics, Thermodynamics, Heat transfer, Aerospace Propulsion etc., and to generate major types of grid over an aerodynamic profile.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Understanding modeling of geometries in computational tools
- 2 analyze internal and external flow using various geometries
- 3 Develop a code for solution for the one-dimensional wave equation using explicit method of lax
- 4 Generation of the algebraic grids and elliptic grids using MATLAB using ANSYS FLUENT
- 5 Evaluate supersonic flow over axi-symmetric bodies

LIST OF EXPERIMENTS

- 1 Introduction to computational process and computational software package
- 2 Modeling of basic geometries using sketcher in workbench
- 3 Laminar flow through pipe
- 4 Flow over a flat plate
- 5 Supersonic flow over a wedge
- 6 Flow past over a cylinder
- 7 Flow over an airfoil
- 8 Flow over wing
- 9 Supersonic flow over axi-symmetric bodies
- 10 Solution for the one dimensional wave equation using explicit method of lax (code development)
- 11 Solution for the one dimensional transient heat conduction equation using explicit method (code development)
- 12 Generation of the algebraic grids (code development)
- 13 Generation of the elliptic grids (code development)

Reference Books:

1. ANSYS FLUENT Tutorial Guide 18.0
2. Lab Manual
3. Stormy Attaway, 'MATLAB-A Practical Approach'

Web References:

1. <http://www.mathworks.com/matlabcentral/fileexchange/53993-quickersim-cfd-toolbox>
2. <http://www.quickersim.com/cfd-toolbox-for-matlab/index>
3. <https://www.learncax.com/class/cfd-simulation-with-ansys-fluent-detail>

FLIGHT VEHICLE DESIGN LAB**Semester: VII**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE52	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Discuss the importance of conceptual design process and studying the major phases of designing process involved in the design.
2. Understand the Integrated product development and principles of baseline design– stability & control, performance and constraint analysis
3. Analyze cost estimation, parametric analysis, optimization, refined sizing & trade studies
4. Demonstrate on direct energy conversion and their principles.
5. Discuss the trade design and trade studies of the different types of aircrafts.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Ability to distinguish and understand the design phases of an aircraft
2. Ability to analysis and estimate of different types of aircraft
3. Ability to develop layout sketch of different aircraft
4. Ability to calculate total takeoff weight and other sub categories of weights for an aircraft
5. Ability evaluate the sensitivity analysis of design parameters
6. Ability to estimate sizing of an aircraft
7. Ability to calculate performance characteristics of an aircraft

LIST OF EXPERIMENTS

1. Objectives Requirements of the vehicle
2. Conceptual Sketch and first estimate of weight
3. Initial Sizing
4. Fuselage and control surfaces
5. Configuration layout.
6. Performance and stability Estimate
7. Load estimates
8. Design of naca 4 digit aerofoil
9. Induced drag estimation
10. Fixed engine sizing
11. Trade off study on fixed engine sizing
12. Fuselage design
13. Cost analysis for anti-submarine warfare aircraft

Reference Books:

1. Raymer, Daniel P. (2006), Aircraft Design: A Conceptual Approach, 4th edition, AIAA Educational Series, USA.
2. J. F. Marchman, L. R. Jenkinson (2003), Aircraft Design Projects for Engineering students, AIAA Publishers, USA.
3. Ajoy Kumar Kunda (2010), Aircraft Design, Cambridge University Press, UK.
4. Torenbeek E. (1986), Synthesis of Subsonic Airplane Design, Delft University Press, New York.
5. Bruhn. E. H (1973), Analysis and Design of Flight Vehicles Structures, New Edition, Jacobs Publishing House, USA.
6. Scheler E. E, Dunn L.G (1963), Airplane Structural Analysis and Design, John Wiley & Sons, USA.
7. D. Howe (2005), Aircraft conceptual Design Synthesis, John Wiley & Sons Publishers, USA

Web References:

<https://nptel.ac.in/downloads/101106035/>
<https://nptel.ac.in/downloads/101104069/>

IV B.TECH VIII SEMESTER SYLLABUS

ADVANCED COMPUTATIONAL FLUID DYNAMICS**VIII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE55	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

Course aims to provide students with the necessary skills to use commercial Computational Fluid Dynamics packages and to carry out research in the area of Computational Fluid Dynamics. The Course shall impart knowledge of following aspects to the students.

1. Develop an understanding of numerical methods involved in computational fluid dynamics.
2. Apply method of characteristics for solving various problems of CFD.
3. Determine time dependent methods to solve computational fluid problems.
4. Apply time dependent methods to solve Boundary Layer Problems
5. Solve numerical methods involving transonic flows.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Develop an understanding of numerical methods involved in computational fluid dynamics.
2. Apply method of characteristics for solving various problems of CFD.
3. Determine time dependent methods to solve computational fluid problems.
4. Apply time dependent methods to solve Boundary Layer Problems
5. Solve numerical methods involving transonic flows.

UNIT-I	PANEL METHODS	Classes: 12
PANEL METHODS: Introduction to panel method, Basic aspects of uniform source and vortex flows, Source panel method, Non-lifting flows over arbitrary two-dimensional bodies. VORTEX PANEL METHOD: Vortex panel method Lifting flows over arbitrary two-dimensional bodies.		
UNIT-II	METHOD OF CHARACTERISTICS	Classes: 12
Introduction to numerical techniques for steady supersonic flows, Philosophy of method of characteristics. Determination of characteristic lines, Two-dimensional irrotational flow. Determination of the compatibility equation and unit processes. Regions of influence and Domains of dependence. APPLICATIONS OF METHOD OF CHARACTERISTICS: Supersonic nozzle design using method of characteristics, Description of Mc Cormack's predictors - Corrector techniques.		
UNIT-III	TIME DEPENDENT METHODS	Classes: 12
TIME DEPENDENT METHODS - I: Stability of Solution, Explicit time dependent methods: Euler, Backward Euler, Onestep trapezoidal, Backward differencing, methods, Leap Frog method. TIME DEPENDENT METHODS - II: Description of Lax-Wendroff Scheme and Mac Cormack's two-step predictor -Corrector method. Description of time split methods and Approximate factorization schemes..		
UNIT-IV	BOUNDARY LAYER EQUATION	Classes: 12
Introduction to boundary layer equations and their solutions. Description of the boundary layer equations. Transformation of boundary layer equations and the numerical solution method. Choice of discretization model and the generalized Crank- Nicholson Scheme. Discretization of boundary layer equations and illustration of solutions of a tridiagonal system of linear algebraic equations.		

UNIT-V	TRANSONIC RELAXATION METHOD	Classes: 12
Theoretical aspects of transonic flows, Small Perturbation flows, Transonic small perturbation equations, Central and Backward difference schemes, Shock capturing vs. shock fitting techniques: Conservation vs. non conservation forms of governing equations, Line relaxation techniques.		
Text Books:		
1. T. J. Chung (2010), Computational Fluid Dynamics, 2nd edition, Cambridge University Press, USA. 2. John D. Anderson (2010), Computational Fluid Dynamics, McGraw Hill, New Delhi. 3. John C. Tannehill, Richard H. Pletcher (1997), Computational Fluid Mechanics and Heat transfer, 2nd edition, Taylor & Francis Group, New York.		
Reference Books:		
1. Ronnie Anderson (2012), Computational Fluid Dynamics for Engineers, Cambridge University Press, USA. 2. Jean-Jacques Chattot (2010), Computational aerodynamics and fluid dynamics an introduction, Springer, Germany		
Web References:		
https://ocw.mit.edu/courses/mechanical-engineering/2-29-numerical-fluid-mechanics-spring-2015/		
E-Text Books:		
1. Ferziger J.H. & Peric M. (1999) Computational Methods for Fluid Dynamics, Springer, Berlin, Germany. 2. Hirsch C. (1988) Numerical Computation of Internal and External Flows, John Wiley & Sons, New York, USA.		
MOOC Course		
https://nptel.ac.in/course.php		

HELICOPTER ENGINEERING**VIII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE56	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To provide knowledge about aerodynamics, flight performance, stability and control of a helicopter with particular emphasis to rotor blades.
2. To provide knowledge about working principle and performance estimation of ground effect machines

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Distinguish the working characteristics of rotors and flight controllers for flights.
2. Evaluating the blade element theory and profile induced power estimation.
3. Analyze the stability and coefficient of performance of forward flight
4. Determine the performance analysis of VTOL and STOL aircraft.
5. Calculate the power requirement for jet machines and drag of hovercraft on land and water.

UNIT-I	ELEMENTS OF HELICOPTER AERODYNAMICS AND ROTOR CONTROL	Classes: 12
ELEMENTS OF HELICOPTER AERODYNAMICS: Configurations based on torque reaction, Jet rotors and compound helicopters. ROTOR CONTROL: Methods of control, Collective and cyclic pitch changes, Lead-lag and flapping hinges.		
UNIT-II	IDEAL ROTOR THEORY AND ROTOR PERFORMANCE	Classes: 12
IDEAL ROTOR THEORY: Hovering performances, Momentum and simple blade element theories. ROTOR PERFORMANCE: Figures of merit, Profile and induced power estimation, Constant chord and ideal twist rotors.		
UNIT-III	POWER ESTIMATES AND STABILITY AND TRIM	Classes: 10
POWER ESTIMATES: Induced, Profile and Parasite power requirements in forward flight, Performances curves with effects of altitude. STABILITY AND TRIM: Preliminary ideas on helicopter stability.		
UNIT-IV	LIFT AND CONTROL OF V/S TOL AIRCRAFT	Classes: 10
Various configuration, Propeller, Rotor ducted fan and jet lift, Tilt wing and vectored thrust, Performances of VTOL and STOL aircraft in hover, Transition and Forward motion.		
UNIT-V	GROUND EFFECT MACHINES	Classes: 10
Types, Hover height, Lift augmentation and power calculations for plenum chamber and peripheral jet machines. Drag of hovercraft on land and water. Applications of hovercraft		

Text Books:

1. Johnson Wayne (2011), *Helicopter Theory*, 1st edition, Sterling Publishing House, New York.
2. McCormick B. W. (2010), *Aerodynamics, Aeronautics and Flight Mechanics*, 2nd edition, Wiley India Ltd, New Delhi, India.

Reference Books:

1. Alfred Gessow, Garry C. Myers (2007), *Aerodynamics of Helicopter*, 2nd edition, F. Ungar Pub. Co, New York.
2. B.W. (1998), *Aerodynamics of V/STOL Flight*, Dover Publications, USA.
3. John M. Seddon (2011), *Basic Helicopter Aerodynamics*, John Wiley & Sons, USA.

Web References:

<http://www.khaiedu.com/airplanes-and-helicopters.html>

E-Text Books:

Helicopter Theory by Wayne Johnson (Kindle edition)

MOOC Course

<https://nptel.ac.in/courses/101104017/>

PROPELLANT TECHNOLOGY

VIII Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE57	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with the theoretical background and engineering applications. 1. Describe various types of propellants with their merits of challenges. 2. Comprehend the basic requirements of the test facilities for propellant.								
COURSE OUTCOMES: 1. Exemplify the testing procedure of liquid fuels used in aviation sector. 2. Classify the major types of solid propellant and their major parameters governing it. 3. Comprehend and illustrate the basic challenges involved in the usage and storage of cryogenic propellants. 4. Understand and examine major types of liquid propellants used in chemical rocket. 5. Relate the significance of propellant testing and their associated parameters.								
UNIT-I	LIQUID FUELS						Classes: 8	
Properties and tests for petroleum products, Motor gasoline, Aviation gasoline, Aviation turbine fuels, requirements of aviation fuels of kerosene type and high flash point type, Requirements for fuel oils.								
UNIT-II	SOLID PROPELLANTS						Classes: 12	
Single base propellants, Double base propellants, Composite propellants, CMBD propellants, metalized composite propellants. Introduction to different fuels and oxidizers of composite propellants. Brief introduction to composite theory of composite and double base propellants.								
UNIT-III	CRYOGENIC PROPELLANTS						Classes: 14	
Introduction to cryogenic propellants, Liquid hydrogen, liquid oxygen, liquid nitrogen and liquid nitrogen and liquid helium and their properties. THEORY: Behind the production of low temperature, Expansion engine, Cascade process, Joule Thompson effect, Magnetic effect, Ortho and para H2, Helium 4 and Helium 3. Ideal cycles and efficiency of cryo systems, Storing of cryogenic propellants, Cryogenic loading problems.								
UNIT-IV	LIQUID PROPELLANTS						Classes: 14	
Various liquid propellants and their properties, Monopropellants and bipropellant system, concept of ullage, Ignition studies of liquid propellants. Propellant loading tolerances, inventory, Volume versus mass loading. Loading measurement and control, Outage control.								
UNIT-V	PROPELLANT TESTING						Classes: 12	
Laboratory testing, Arc Image Furnace, Ignitability studies. Differential Thermal Analysis, Thermo-gravimetric analysis, Particle size measurement Micro-merograph, Strand burner tests impulse bomb, Performance estimation.								

Text Books:

Cornelisse J. W. (1980), Rocket Propulsion and Space Dynamics, Pitman Publishing, London.

Reference Books:

1. Sutton, G. P. (2010), Rocket Propulsion Elements, John Wiley / BSP Books, USA
2. Samir Sarkar (2009), Fuels and Combustion, 3rd edition, Universities Press /CRC Press, New York.
3. Mathur M, Sharma R. P. (2010), Gas Turbine and Jet and Rocket Propulsion, Standard Publishers, New Delhi.

Web References:

<http://www.braeunig.us/space/propuls.htm>

<https://courses.lumenlearning.com/boundless-physics/chapter/rocket-propulsion/>

E-Text Books:

<https://pdfs.semanticscholar.org/6cce/fa84a00b396c5b50b66063772f4851392b8f.pdf>

https://www.academia.edu/13290390/Sutton-Biblarz-Rocket_Propulsion_Elements

MOOC Course

<https://nptel.ac.in/courses/112106073/>

AERO ENGINE REPAIR AND MAINTENANCE**VIII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE58	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To understand the basic concepts of the maintenance and repair of both piston and jet engines
2. To acquire the knowledge of inspection and overhaul of both piston and jet engines.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Classify types of piston engine and their components
2. Explain briefly about maintenance and troubleshooting
3. Identify major checks during overhauling
4. Categorize major maintenance procedures of gas turbine engines
5. Estimate the troubleshooting procedures for rectification

UNIT-I	CLASSIFICATION OF PISTON ENGINE COMPONENTS	Classes: 8
Types of piston engines – Principles of operation – Function of components – Materials used – Details of starting the engines – Details of carburetion and injection systems for small and large engines – Ignition system components – Spark plug details – Engine operating conditions at various altitudes – Maintenance and inspection check to be carried out.		
UNIT-II	INSPECTIONS OF PISTON ENGINES	Classes: 8
Inspection, maintenance and troubleshooting – Inspection of all engine components – Daily and routine checks – Overhaul procedures – Compression testing of cylinders – Special inspection schedules – Engine fuel, control and exhaust systems – Engine mount and super charger – Checks and inspection procedures.		
UNIT-III	INSPECTIONS OF PISTON ENGINES	Classes: 10
Symptoms of failure – Fault diagnostics – Case studies of different engine systems – Tools and equipment requirements for various checks and alignment during overhauling – Tools for inspection – Tools for safety and for visual inspection – Methods and instruments for non destructive testing techniques – Equipment for replacement of part and their repair. Engine testing: Engine testing procedures and schedule preparation – Online maintenance.		
UNIT-IV	CLASSIFICATION OF JET ENGINE COMPONENTS	Classes: 11
Types of jet engines – Principles of operation – Functions of components – Materials used – Details of starting and operating procedures – Gas turbine engine inspection & checks – Use of instruments for online maintenance – Special inspection procedures : Foreign Object Damage – Blade damage – Maintenance procedures of gas turbine engines – Trouble shooting and rectification procedures – Component maintenance procedures – Systems maintenance procedures. Gas turbine testing procedures – Test schedule preparation – Storage of Engines – Preservation and de-preservation procedures.		

UNIT-V	OVERHAUL PROCEDURES	Classes:8
Engine Overhaul procedures – Inspections and cleaning of components – Repairs schedules for overhaul – Balancing of gas turbine components. Trouble Shooting - Procedures for rectification – Condition monitoring of the engine on ground and at altitude – engine health monitoring and corrective methods.		
Text Books:		
1. Kroes & Wild," Aircraft Power plants ", 7th Edition – McGraw Hill, New York, 1994.		
Reference Books:		
1. Turbomeca," Gas Turbine Engines ", The English Book Store ", New Delhi, 1993. 2. United Technologies' Pratt & Whitney, "The Aircraft Gas turbine Engine and its Operation", The English Book Store, New Delhi.		
Web References:		
https://honolulucc.lib.hawaii.edu/vwebv/holdingsInfo?bibId=3665954 https://honolulucc.lib.hawaii.edu/vwebv/holdingsInfo?bibId=3683166		
E-Text Books:		
Aero Engine Maintenance and Repair (Dover Books on Aeronautical Engineering)		
MOOC Course		
https://www.classcentral.com/course/stepik-technical-maintenance-of-aircraft-and-aircraft-engines-10489		

PRODUCT LIFE CYCLE MANAGEMENT

VIII Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME59	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The objective of this course is to enable students to learn concepts of product life cycle management such as need for PLM, PLM strategies, principles, product development approaches etc.								
COURSE OUTCOMES: At the end of the course, the student should be able to 1. Explain basic concepts of product life cycle management. 2. Demonstrate product development approaches. 3. Explain elements of product modelling. 4. Discuss in detail the concept of product data management. Discuss about integration of PLM with other applications.								
UNIT-I	INTRODUCTION TO PLM						Classes: 10	
Background, Overview, Need, Benefits, Concept of Product Life Cycle. Components / Elements of PLM, Emergence of PLM, Significance of PLM, Customer Involvement. Product Data and Product Workflow, Company's PLM vision, The PLM Strategy, Principles for PLM strategy, Preparing for the PLM strategy, Developing a PLM strategy, Strategy identification and selection, Change Management for PLM								
UNIT-II	PRODUCT DEVELOPMENT APPROACHES						Classes: 11	
Product Development Approaches: Bottom-up design, Top-down design, Front-loading design workflow, Design in context, Modular design. Concurrent engineering, partnership with supplier, collaborative and Internet based design, work structuring and team deployment, Product and process systemization, problem, identification and solving methodologies, improving product development solutions								
UNIT-III	PRODCUT MODELING						Classes: 13	
Product Modelling - Definition of concepts - Fundamental issues – Role of Process chains and product models -Types of product models – model standardization efforts-types of process chains - Industrial demands. Foundation technologies and standards (e.g. visualization, collaboration and enterprise application integration)								
UNIT-IV	PRODUCT DATA MANAGEMENT						Classes: 10	
Product Data Management (PDM) –Benefits and Terminology, PDM functions, definition and architectures of PDM systems, product data interchange, portal integration, PDM acquisition and implementation. Information authoring tools (e.g., MCAD, ECAD, and technical publishing), Core functions (e.g., data vaults, document and content management, workflow and program management), Functional applications (e.g., configuration Management).								
UNIT-V	INTEGRATION OF PLM WITH OTHER APPLICATIONS, PLM SOFTWARES						Classes: 10	
Different ways to integrate PLM systems, Transfer file, Database integration, System roles, ERP, Optimization of ERP for PLM and CAD. PLM Softwares-Basic features and modules of ENOVIA and Windchill.								

Text Books:

1. Grieves, Michael. Product Lifecycle Management, McGraw-Hill, 2006. ISBN 0071452303
2. Saaksvuori Antti / Immonen Anselmie, product Life Cycle Management Springer, Dreamtech, 3-540-25731-4

Reference Books:

1. Stark, John. Product Lifecycle Management: Paradigm for 21st Century Product Realisation, Springer-Verlag, 2004. ISBN 1852338105
2. Fabio Giudice, Guido La Rosa, Product Design for the environment-A life cycle approach, Taylor & Francis 2006

Web References:

1. <https://www.slideshare.net/anandsubramaniam/product-life-cycle-management>
2. <http://productlifecyclestages.com/>
3. <https://nxrev.com/2018/02/windchill-vs-enovia/>

E-Text Books:

1. https://books.google.co.in/books?id=q9AdtdDeuPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=CiHbLm6twJMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.cimdata.com/en/education/plm-basics-e-learning-course>
2. <https://www.cimdata.com/en/education/plm-certificate-program>

AIRPORT MANAGEMENT**VIII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE60	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Airport management and its activities studied
2. Airline management and its types
3. Role of a airport hubs
4. International and domestic airports
5. Airtraffic control methods
6. Planning and management of an airport
7. Role of a manager in a airport

COURSE OUTCOMES:

1. Analyze the various rules, ownership structures, components & facilities located on airfields at airports
2. Analyze various enhancements in air traffic control, airport terminal and ground access
3. Analyze the methods involved in airport security and ARFF and safety inspection programs
4. Compare the revenue strategies and pricing of airport facilities & services for airport financial management
5. Analyze the airport capacity and delay

UNIT-I	AIRPORTS AND AIRPORT SYSTEMS	Classes: 10
Introduction, Airport Management on an international level, Rules that govern airport management, Airport ownership and organization, Airport organization chart, Airport manager and public relations. THE AIRFIELD: Components of an airport, the airfield, Navigation aids (NAVAIDS) located on airfields, Air traffic Control and surveillance facilities located on the airfield, Weather reporting facilities located on airfields, security infrastructure on airfields.		
UNIT-II	AIRSPACE AND AIR TRAFFIC CONTROL	Classes: 11
Air traffic control management and operating infrastructure, Basics of air traffic control. Current and future enhancements to air traffic control. AIRPORT TERMINALS AND GROUND ACCESS: Historical development of airport terminals, Components of airport terminal, Airport ground access.		
UNIT-III	AIRPORT OPERATIONS MANAGEMENT	Classes: 13
Pavement management-Aircraft rescue and fire fighting (ARFF) Snow and ice control, Safety inspection programs, Bird and wildlife hazard management. AIRPORT SECURITY: Transportation Security Administration-Security at commercial service airports, Security at general aviation airports.		
UNIT-IV	AIRPORT FINANCIAL MANAGEMENT	Classes: 10
Airport financial accounting-Revenue strategies at commercial airports, Pricing of airport facilities and services, Variation in the sources of operating revenues. Rise in airport financial burdens, Air port funding, Airport financing, Private Investment, Sale of air port.		

UNIT-V	AIRPORT CAPACITY AND DELAY	Classes: 10
Defining capacity, Factors effecting capacity and delay, Estimating capacity, Simulation Models. Defining delay, Analytical estimates of delay: queuing diagram, Approaches to reducing delay, Administrative and demand management.		
Text Books:		
1. Alexander T. Wells, John G. Wensveen (2008), Air Transportation: A Management Perspective, 8th edition, Ashgate Publishing, New Delhi. 2. Alexander T. Wells and Seth B. Young (2011), Airport Planning and Management, 6th edition, McGraw-Hill, New Delhi.		
Reference Books:		
1. Amedeo Odoni, Pentder Belobaba and Tom Reynolds(2012), Airport Systems: Planning, Design AndManagement, Edition, McGraw Hill, New Delhi 2. Richard D Neufville (2012), Airport Systems: Planning, Design and Management, 2nd edition, McGraw Hill, New Delhi		
Web References:		
https://engineering.purdue.edu/AACourses/aae556/2011/AAE556-text-Chapter%201-Intro		
E-Text Books:		
http://ebooksgo.org/free-aeroelasticity-ebooks-online.html		
MOOC Course		
Introduction to Virtual Aviation https://www.udemy.com/intro-virtual-aviation/?ranMID=39197&ranEAID=SAyYsTvLiGQ&ranSiteID=SAyYsTvLiGQ-H.xgLPxAHTcFuXqRqIB.HQ&LSNPUBID=SAyYsTvLiGQ		

EXPERIMENTAL STRESS ANALYSIS**VIII Semester:**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE61	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The course covers the basic concepts of the analysis and at the end of the course students will be able to

1. Explain the basic aspects of experimental stress analysis that includes exhaustive treatment of the most versatile techniques like photo elasticity and strain gauges
2. Impart a brief introduction to the emerging techniques like digital image correlation.
3. Demonstrate the fundamental aspects of six different experimental techniques
4. Understand the Moiré, Brittle Coatings, Holography, Speckle Methods, Thermoplastic Stress with practical application

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Understand the basic concepts of experimental stress analysis
2. Demonstrate the principles of major types of extensometers
3. Apply the knowledge of Strain gauges in aeronautical domain
4. Understand the principles of Rosette analysis and fringe techniques
5. Understand NDT techniques used in the structural analysis

UNIT-I	MEASUREMENTS AND EXTENSOMETERS	Classes: 12
Principles of measurement, Accuracy, Sensitivity, Range, Types of Error, Mechanical, Optical, Acoustical and Electrical extensometers and their use, advantage and disadvantage.		
UNIT-II	STRAIN GAUGE – PRINCIPLES AND STRAIN MEASUREMENT	Classes: 12
Principles and operation of electrical strain gauge, Requirement, Type and their uses, Material for strain gauge, Calibration, Bridge sensitivity, Wheatstone bridge and potentiometer circuits for static and dynamic strain measurements, Strain indicator, Rosette Analysis.		
UNIT-III	PHOTOELASTICITY, FRINGE INTERPOLATION TECHNIQUES	Classes: 12
Concept of Light, Two dimensional Photo elasticity, Photo elastic effects, Stress optic law, Plane and circular polariscope, Interpretation of fringe pattern, Compensation and separation techniques, Photo elastic material.		
UNIT-IV	NON-DESTRUCTIVE TESTING-I	Classes: 12
Fundamentals of Non Destructive Testing, Radiography, Ultrasonic Inspection, Ultrasonic C-Scan, Magnetic particles Inspection, Fluorescent penetrant technique, Eddy current testing, Acoustic Emission Technique.		
UNIT-V	NON-DESTRUCTIVE TESTING -II	Classes: 12
Fundamentals of brittle coating methods, Analysis of brittle coatings, Industrial application of brittle coating, Introduction to Moiré Techniques, Holography, Thermography.		

Text Books:

1. Daily J. W, Riley W. F (2005), Experimental Stress Analysis, 4th edition, McGraw- Hill, New Delhi.
2. Thomas G. Beckwith, Maragoni, Lienhard (2009), Mechanical Measurements, 6th edition, Pearson Education, New Delhi.

Reference Books:

1. Sadhu Singh (2009), Experimental stress Analysis, 3rd edition, Khanna Publications, New Delhi.
2. L S Srinath (1984), Experimental Stress Analysis, 2nd edition, Tata McGraw-Hill, New Delhi.
3. K. Ramesh, "Digital Photoelasticity, Advanced Techniques and Applications", Springer, 2000.
4. Prasad (2011), Non- Destructive Test and Evaluation of Materials, 1st edition, Tata McGraw-Hill, New Delhi.
5. R. Halmshaw (1991), Non-Destructive Testing, 2nd edition, Edward Arnold, New York
6. Allesandro Freddi, Giorgio Olmi, Luca Cristofolini, "Experimental Stress Analysis for Materials and Structures", Springer, 2015.

Web References:

<https://www.slideshare.net/MAHESHHUDALI/experimental-stress-analysis>
https://www.researchgate.net/.../274374622_Experimental_Stress_Analysis_for_Mater...

E-Text Books:

<https://trove.nla.gov.au/work/11110399>
blog.twowaiters.in/notes/M&M_Beckwith.pdf
 K. Ramesh, "e-Book on Experimental Stress Analysis", IIT Madras, 2009.

MOOC Course

<https://nptel.ac.in/courses/112106068/>

Fundamentals of Combustion

VIII Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE62	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with theoretical background and engineering applications. 1. To know the composition of various types of fuels and their properties 2. To understand the pollution from combustion of fuels and controlling them. 3. To understand the thermodynamic of combustion.								
COURSE OUTCOMES: 1.Explain the thermodynamics of combustion and analyse the composition of various types of fuels and their properties. 2. Discuss the fundamental thermodynamics of combustion and basic kinetics of various rections 3. Discuss the fundamental physical and chemical principles of various combustion phenomena independent of an application 4. Address premixed flame and diffusion by extending the earlier-gained knowledge of thermodynamics. 5. Make quantitative and qualitative estimates of characteristics of various combustion processes and its impact on environment.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction to combustion, Applications of combustion, Types of fuel and oxidizers, Characterization of fuel, Various combustion mode, Scope of combustion.								
UNIT-II	THERMODYNAMICS OF COMBUSTION AND CHEMISTRY OF COMBUSTION						Classes: 12	
Thermodynamics properties, Laws of thermodynamics, Stoichiometry, Thermochemistry, adiabatic temperature, chemical equilibrium. Basic Reaction Kinetics, Elementary reactions, Chain reactions, Multistep reactions, simplification of reaction mechanism, Global kinetics.								
UNIT-III	CHEMISTRY OF COMBUSTION AND PHYSICS OF COMBUSTION						Classes: 12	
Basic Reaction Kinetics, Elementary reactions, Chain reactions, Multistep reactions, simplification of reaction mechanism, Global kinetics. Fundamental laws of transport phenomena, Conservations Equations, Transport in Turbulent Flow.								
UNIT-IV	PREMIXED FLAME AND DIFFUSION FLAME						Classes: 12	
One dimensional combustion wave, Laminar premixed flame, Burning velocity measurement methods, Effects of chemical and physical variables on Burning velocity, Flame extinction, Ignition, Flame stabilizations, Turbulent Premixed flame. Gaseous Jet diffusion flame, Liquid fuel combustion, Atomization, Spray Combustion, Solid fuel combustion.								

UNIT-V	COMBUSTION AND ENVIRONMENT	Classes: 12
Atmosphere, Chemical Emission from combustion, Quantification of emission, Emission control methods.		
Text Books:		
1. D.P. Mishra, Fundamentals of Combustion, Prentice Hall of India, New Delhi, 2008. 2. Kuo K.K. "Principles of Combustion" John Wiley and Sons, 2005. 3. Strehlow R A., "Fundamentals of combustion" McGraw Hill Book Company, 1984.		
Reference Books:		
1. Warnatz J, Combustion: Physical and Chemical Fundamentals, Modelling and Simulation, Experiments, Pollutant Formation, Springer Verlag, 1996. 2. Mishra D P, Fundamental of Combustion, PHI Learning, 2008.		
Web References:		
https://nptel.ac.in/courses/101104014 https://nptel.ac.in/downloads/101104014		
E-Text Books:		
D.P. Mishra, John Wiley and Sons, Strehlow R A		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_ae01/course		

OPEN ELECTIVES OFFERED BY AERONAUTICAL DEPARTMENT

OE1		OE2		OE3	
A4AE17	Fabrication Processes	A4AE25	Introduction to Aircraft Industry	A4AE40	Guidance and Control of Aerospace Vehicles
A4AE18	Fundamentals of Avionics	A4AE26	Non Destructive Testing Methods	A4AE41	Wind Tunnel Techniques
A4AE19	Introduction to Jets And Rockets	A4AE27	Fundamentals of Finite Element Method	A4AE42	Introduction to Aerospace Technology

FABRICATION PROCESSES**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE17	OEC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES

1. To develop competency in understanding major fabrication processes used for aircraft production.
2. To have a broad knowledge in choosing a suitable fabrication process for manufacturing aircraft components.

COURSE OUTCOMES:

The students should be able to

1. Exemplify the major metal casting processes used in aircraft industry.
2. Exemplify the major metal joining processes used in aircraft industry.
3. Exemplify hot & cold working of metals including high energy rate forming.
4. Exemplify the major surface finishing processes used in aircraft industry.
5. Exemplify the major sheet metal forming processes used in aircraft industry.
6. Suggest suitable molding process for plastics for the given application

UNIT-I	METAL CASTING PROCESSES	Classes: 12
Sand casting – Sand moulds – Types of patterns – Pattern materials – Pattern allowances – Types of Moulding sand- Properties – Core making – Moulding machines – Types of moulding machines – Melting furnaces – Working principle of Special casting processes – Shell, investment casting – Lost Wax process –Centrifugal casting –Sand Casting defects.		
UNIT-II	METAL JOINING TECHNIQUES	Classes: 12
Fusion welding processes – Types of Gas welding – Equipment's used – Flame characteristics – Filler and Flux materials – Arc welding equipment – Electrodes – Coating and specifications – Principles of Resistance welding – Spot/butt, seam welding – Percussion welding – Gas metal arc welding – Flux cored – Submerged arc welding –Electro slag welding – TIG welding – Principle and application of special welding processes – Plasma arc welding – Thermit welding – Electron beam welding – Friction welding –Weld defects – Brazing and soldering process.		
UNIT-III	BULK DEFORMATION AND SURFACE FINISHING PROCESSES	Classes: 12
Hot working and cold working of metals – Forging processes – Open and close die forging – Characteristics of the process – Types of Forging Machines – Typical forging operations – Rolling of metals –Types of Rolling mills –Tube piercing – Defects in rolled parts – Principles of Extrusion – Types of Extrusion – Hot and Cold extrusion – Principle of rod and wire drawing – Equipments used - surface roughness and measurements - grinding - honing - lapping - polishing - burnishing - thermal spray coating.		
UNIT-IV	SHEET METAL FORMING PROCESSES	Classes: 12
Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics		

UNIT-V	FORMING AND SHAPING OF PLASTICS	Classes: 12
Types of plastics – Characteristics of forming and shaping processes – Moulding of Thermoplastics – Working principles– Injection moulding – Plunger and screw machines – Blow moulding –Extrusion – Typical industrial applications – Thermoforming – Processing of Thermosets – Working principles and typical applications – Compression moulding – Transfer moulding – Bonding of Thermoplastics – Induction and Ultrasonic methods.		
Text Books:		
1. Hajra Choudhury , Elements of Workshop Technology, Vol. I and II, Media Promotors Pvt. Ltd., Mumbai, 2007 2. Serope Kalpak Jain, Steven R, Schmid, Manufacturing Engineering and Technology, Pearson Education, Inc. 4th Edition, 2009		
Reference Books:		
1. Elements of Manufacturing Processes, B.S. Magendran Parashar & R.K. Mittal, Prentice Hall of India, 2008 2. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010. 3. A text book of production technology, P.c. Sharma, S. Chand and Company, 2010 4. Manufacturing Process – Begman, John Wiley & Sons, VIII Edition, 1999		
Web References: https://www.velaction.com/fabrication/ https://www.boconline.co.uk/en/.../fabrication-processes/fabrication-processes.html		
E-Text Books: 1. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010. 2. Manufacturing Process – Begman, John Wiley & Sons, VIII Edition, 1999		
MOOC Course https://nptel.ac.in/courses/112107144/ https://nptel.ac.in/courses/112105127/ https://nptel.ac.in/courses/112104195/ https://nptel.ac.in/courses/112107078/		

FUNDAMENTALS OF AVIONICS

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE18	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with the theoretical background and engineering applications. 1. Overview on Aviation using Electronics 2. Basic understanding about major electronics systems used for communication 3. Basic understanding about major devices, display and flight controls used in aircraft								
COURSE OUTCOMES: At the end of the course the students are able to: 6 To explain the instrumentation used in avionics. 7 To classify various ranges of the communication techniques used in aircraft. 8 To distinguish between network systems, controlling parts & surfaces 9 To compare various principles of navigation systems 10 To build phenomena of auto pilot control system								
UNIT-I	BASICS & FLIGHT DECK AND DISPLAY SYSTEMS						Classes: 8	
BASICS: Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles. FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display,Electronic instrumentation systems.								
UNIT-II	COMMUNICATION SYSTEMS						Classes: 10	
AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHFcommunication system, UHF communication systems.								
UNIT-III	FREQUENCY RANGING SYSTEM						Classes: 12	
RANGING AND LANDING SYSTEMS: VHF Omnicrange, VOR receiver principles, distance maturity equipment, principles of operation, Instrument landing system, and localizer and glide slope. POSITIONG SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation								
UNIT-IV	NAVIGATION SYSTEM						Classes: 12	
INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform. SURVELLIENCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system								
UNIT-V	AUTO FLIGHT SYSTEM						Classes: 8	
AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight directorsystems, flight management systems..								

Text Books:

4. N. S. Nagaraja(1996), Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi.
5. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance,

Reference Books:

1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India.
2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi.
3. J. Powell (1999), Aircraft Radio Systems, Pitman publishers, London.

Web References:

<https://www.britannica.com/technology/avionics>
<https://www.pdc.com/aviation/>

E-Text Books:

Advanced Avionics Handbook(FAA)
 Digital Avionics: A Computing Perspective - Elisabeth A. Strunk John C. Knight

MOOC Course

<https://www.canvas.net/browse/erau/courses/aviation-101>

INTRODUCTION TO JETS AND ROCKETS**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE19	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems.

3. Illustrate an overview of aerospace propulsion system.
4. Identify the foundation in fundamentals of thermodynamics.
5. Compare the ideal components and characteristics of jet engine
6. Interpret the performance of nozzles
7. Simplify the ideal performance analysis of rocket engines.
8. Select appropriate fuel for aerospace application.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Explain the complexity in working of various engines
- 2 Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis
- 3 Analyze the process involved in individual components
- 4 Compare the nozzles with various operating conditions.
- 5 Determine Equations of motion in free space, Tsiolkovsky's equation.
- 6 Classify the types of fuel in aviation and aerospace engineering.

UNIT-I	INTRODUCTION TO AEROSPACE PROPULSION	Classes: 12
What is Propulsion system? Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.		
UNIT-II	PRINCIPLES OF JET PROPULSION	Classes: 12
Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit- significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.		
UNIT-III	RAMJET, SCRAMJET ENGINES AND NOZZLES	Classes: 12
Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.		
UNIT-IV	ROCKET THEORY	Classes: 12
Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiolkovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems		

UNIT-V	PROPELLANT ROCKETS	Classes: 12
Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.		
Text Books:		
1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2 nd Edt 2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993. 3. Fuels and Combustion, Sharma, S.P. and Mohan .C., Tata McGraw Hill Publishing Co, Ltd., 1984		
Reference Books:		
1.The Jet Engine – Rolls Royce 2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.		
Web References:		
http://engineeringvideolectures.com/course/24 http://engineeringvideolectures.com/course/28		
E-Text Books:		
Jet Propulsion (3rd ed. - Nicholas Cumpsty A Simple Guide To Understanding Jet Engines		
MOOC Course		
https://online.stanford.edu/courses/aa283-aircraft-and-rocket-propulsion		

INTRODUCTION TO AIRCRAFT INDUSTRY**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE25	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems.

- 1 Familiarize students with major types of aircraft industry.
- 2 Illustrate the role of regulatory bodies and its business context
- 3 Exemplify the organisational structure of airlines and airports
- 4 Illustrate the importance of air navigation services and its types.
- 5 Familiarize students with air safety and security as a means for ensuring passenger safety.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Discuss the types of aerospace industry and evolution of flying machine to airline alliances.
- 2 Explain standardisation and the importance of regulatory bodies.
- 3 Outline the organisational structure of airlines and airports.
- 4 Demonstrate major types of air navigation services and air traffic control services.
- 5 Illustrate the major processes involved in air safety and security.

UNIT-I	AIRCRAFT INDUSTRY OVERVIEW	Classes: 12
Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.		
UNIT-II	REGULATORY AND BUSINESS CONTEXT	Classes: 12
Regulatory bodies, Introduction to standardisation, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometre, yield etc.), Industry characteristics of passenger airlines.		
UNIT-III	AIRLINES AND AIRPORTS	Classes: 12
Introduction to airlines, Organisational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organisational culture of an airline Introduction to airports – Airport personnel, Processing passengers and freight, Security		
UNIT-IV	AIR NAVIGATION SERVICES	Classes: 12
Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers		

UNIT-V	AIR SAFETY AND SECURITY	Classes: 12
Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).		
Text Books:		
1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom. 2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York. 3. Shevell (2004), Fundamentals of Flight, 2nd edition, Pearson Education Limited, New Delhi 4. Allan Seabridge, Ian Moir (2008), Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, 3rd edition, John Willey & Sons, New York.		
Reference Books:		
1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, Apress, New York 2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modeling Applications in the Airline Industry, Ashgate Publishing Ltd, England. 3. Kermode, A.C. (1989), Flight without Formulae, 5th edition, Pearson Education Limited, New Delhi		
Web References: https://www.investopedia.com/features/industryhandbook/airline.asp http://web.mit.edu/airlines/analysis/analysis_airline_industry.html https://www.ukessays.com/essays/commerce/an-introduction-to-the-airline-industry-business-essay.php		
E-Text Books:		
Introduction to the Airline Industry (IATA Aviation Distance Learning) Kindle Edition, IATA Commercial Aviation 101, Greg Gayden.		
MOOC Course		
https://www.iata.org/training/courses/Pages/airline-industry-introduction-talg50.aspx		

NON DESTRUCTIVE TESTING METHODS

VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE26	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1.To impart knowledge about the non-destructive testing methods. 2. To provide knowledge on the selection of NDT methods for application in engineering industries.								
COURSE OUTCOMES: 1. Recognize various non-destructive techniques for engineering industries. 2. Select appropriate non-destructive technique for defects detection in manufactured/operating parts. 3. Perform inspection using major non-destructive testing methods. 4. Understand the importance and application of NDT in Aerospace structural analysis								
UNIT-I	SURFACE TECHNIQUES						Classes: 12	
Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - ASME, ASTM, AWS, BIS, SAE standard sample specifications - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - properties of liquid penetrants and developers - advantages and limitations of LPT - applications of LPT.								
UNIT-II	MAGNETIC PARTICLE TESTING						Classes: 12	
Introduction to magnetic particle testing (MPT) - magnetization methods - dry particle and wet fluorescent particle techniques - demagnetization - advantages and limitations of MPT - magnetic flux leakage testing - principle, instrumentation and applications of electromagnetic induction techniques and eddy current testing (ECT) method.								
UNIT-III	ULTRASONIC TESTING						Classes: 12	
Introduction to ultrasonic testing (UT) - characteristics of ultrasonic waves - principle of UT – UT probes - UT inspection methods (pulse echo, transmission and phased array techniques, PAUT) -types of scanning and displays - application of UT for welded parts.								
UNIT-IV	RADIOGRAPHY TESTING						Classes: 12	
Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of X-rays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) - inverse square law - exposure charts - penetrameters - safety issues.								
UNIT-V	SPECIAL TECHNIQUES						Classes: 12	
Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - pressure and leak detection - LASER shearography - acoustic holography.								

Text Books:

1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012.
2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.

Reference Books:

1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001
2. W.T. Mc Gonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.
3. Louis Cartz, "Non-Destructive Testing", ASM International, Metals Park Ohio, US, 2007.
4. Barry Hull and Vernon John, "Non Destructive Testing", ELBS/Macmillan, Hampshire, UK, 2015.

Web References:

<https://inspectioneering.com/tag/nondestructive+testing>
<https://www.element.com/materials-testing-services/non-destructive-testing-and-inspection>
<https://www.twi-global.com/technical-knowledge/faqs/what-is-non-destructive-testing>

E-Text Books:

1. Nondestructive Testing Methods and New Applications by Mohammed Omar, Intech, 2012.
2. Non-Destructive Testing by S Ramachandran, A Anderson and T Rajasanthosh Kumar, Airwalk Publications, 2017, Kindle Edition.

MOOC Course

<https://nptel.ac.in/courses/113106070/>

FUNDAMENTALS OF FINITE ELEMENT METHODS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE27	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the concept of numerical analysis of structural components.
2. To understand the basic knowledge of Isoparametric Element Formulation.
3. To impart knowledge on both ANSYS, NASTRAN.

COURSE OUTCOMES:

1. Demonstrate an understanding of the principles and theories of finite element method.
2. Understand the concepts of discretization, approximations, mathematical tools, FEM tools and error control
3. Analyze 2D problems using bar, beam and truss elements.
4. Demonstrate the basic concepts of isoparametric elements and its application in aircraft.
5. Apply Finite element methods for heat transfer and dynamic loading problems.

UNIT-I	INTRODUCTION TO FEM & 1-D PROBLEMS	Classes: 12
Basic concept, historical background, application of FEM, general description, comparison of FEM with other methods. Basic equations of elasticity, Stress, strain relations, Strain, Displacement relations. ONE DIMENSIONAL PROBLEM: Finite element modeling coordinates and shape functions. Potential Energy approach: Assembly of Global stiffness matrix and load vector. Finite element equations, Treatment of boundary conditions.		
UNIT-II	ANALYSIS OF BEAMS	Classes: 12
Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, deflection, stresses.		
UNIT-III	2-D PROBLEMS	Classes: 12
CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems.		
UNIT-IV	STEADY STATE HEAT TRANSFER ANALYSIS	Classes: 12
One dimensional analysis of a fin and two dimensional analysis of thin plate.		
UNIT-V	DYNAMIC ANALYSIS	Classes: 12
Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar.		
Text Books:		
3. R. Tirupathi Chandrapatla (2011), Introduction to Finite Elements in Engineering, 4 th edition, Pearson Education, India. 4. S. S. Rao (2012), The Finite Element Methods in Engineering, 5 th edition, Elsevier, USA. 3. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1 st edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India		

Reference Books:

5. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India.
6. J. N. Reddy (2010), An introduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
7. Kenneth H. Huebner, Donald L. Ridd, Douglas E. Smith, Ted G. Byrom (2009), The Finite Element Method for Engineers, 3rd edition, John Wiley & sons (ASIA) Pvt. Ltd., New York.

Web References:

<https://nptel.ac.in/courses/112104116/>

E-Text Books:

http://web.mit.edu/kjb/www/Books/FEP_2nd_Edition_4th_Printing.pdf

MOOC Course

<https://nptel.ac.in/courses/112104193/>

GUIDANCE AND CONTROL OF AEROSPACE VEHICLES

VII Semester: OPEN ELECTIVE -III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE40	OEC	L	T	P	C	CIA	SEE	Total
		3	1	-	3	30	70	100
COURSE OBJECTIVES								
The purpose of this subject is to provide the students with the theoretical background and engineering applications.								
6. To introduce the concepts of Navigation, guidance and control								
7. To familiarize with various ways in which aerospace vehicles are guided and controlled								
8. The dynamic objectives which students also learn to achieve by designing flight control systems.								
9. Familiarize with the control principles of rockets and missiles								
10.To give Insight into the manoeuvres of the space craft								
COURSE OUTCOMES:								
At the end of the course the students are able to:								
1 Formulate the navigational equations of the space vehicle								
2 Describe the guidance of the vehicle with state feed back								
3 Explain the automatic control and guidance of the aircraft								
4 Evaluate the control techniques of the rockets and missiles								
5 Describe major manoeuvres of the space aircraft.								
UNIT-I	NAVIGATION						Classes: 13	
Introduction, Basic Principles and Definitions; Dead reckoning and Position Fixing, Celestial, Radio, Inertial Navigation; Principle and Construction of Accelerometers, Mechanical Gyros and Ring Laser Gyros, Inertial Measurement Units, Navigation Equations, Sensor Error Models, Kalman Filter, Attitude Heading Reference System, GPS, Terrain Reference Navigation.								
UNIT-II	GUIDANCE						Classes: 11	
Optimal Terminal Guidance of Interceptors, Optimal Terminal Guidance - planar and non-planar, Robust and Adaptive Guidance, Guidance with State Feedback , Guidance with Normal Acceleration Input , Minimum Energy Orbital Transfer.								
UNIT-III	GUIDANCE AND CONTROL OF AIRCRAFT						Classes: 13	
Powered Flying Controls, Helicopter Flight Controls, Fly-by-Wire Flight Control, Control laws, Redundancy and Failure Survival, Digital Implementation, Fly-by-Light Flight Control, Auto Pilot, Flight Management Systems, Unmanned Aerial Vehicle.								
UNIT-IV	CONTROL TECHNIQUES/ CONTROL OF ROCKETS AND MISSILES						Classes: 12	
Open-loop and Closed Loop Control Systems, Multi-variable Optimization, Optimal Control of Dynamic Systems, Hamiltonian and Minimum Principle and Jacobi-Bellman Equation, Linear Time-Varying System with Quadratic Performance Index..								
UNIT-V	CONTROL OF SPACECRAFT						Classes: 12	
Launch of Satellite/ Spacecraft, Terminal Control of Spacecraft Attitude, Optimal Single-Axis Rotation of Spacecraft, Multi-axis Rotational Manoeuvres of Spacecraft, Spacecraft Control Torques, Rocket Thrusters, Reaction Wheels, Momentum Wheels and Control Moment Gyros, Torque.								
Text Books:								

4. Tewari, A.—Advanced Control of Aircraft, Spacecraft and Rocketsll, John Wiley & Sons, Ltd, Chichester, UK, 2011
5. Nelson R. C - Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York, 2007.
6. R.P.G Collinson —Introduction to Avionics Systemsll, Springer; 3rd ed. edition, 2011.

Reference Books:

1. Noton,M. —Spacecraft navigation and Guidancell, Springer-Verlag, Germany, 1998
2. Mc. Cormic 2. B. W - Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA, 2010.
3. Richard H. Battin —An Introduction to the Mathematics and Methods of Astrodynamicsll, AIAA,

Web References:

<https://nptel.ac.in/courses/101106043/6>

<https://nptel.ac.in/courses/101106042/>

E-Text Books:

<https://www.amazon.com/Guidance-Control-Aerospace-Vehicles-Cornelius/dp/1258807521>

MOOC Course

https://swayam.gov.in/nd1_noc19_ae10

<https://www.mooc-list.com/course/abc-flight-mechanics-how-do-planes-fly-futurelearn>

WIND TUNNEL TECHNIQUES

VII Semester: OPEN ELECTIVE - III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE41	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To interpret the basic concepts of measurement of forces and moments on models during the wind tunnel testing. 2. To understand the application of various types of wind tunnels. 3. To learn the basic measurement procedure involving wind tunnel testing.								
COURSE OUTCOMES: 1. Ability to understand basics of aerodynamics and to identify the type of wind tunnel 2. Ability to develop and understand flow visualization techniques over model 3. Ability to understand concepts of low speed and high speed wind tunnels 4. Ability to understand measurement and balancing of loads on model 5. Ability to understand the different types of equipment's for measuring pressure and velocity								
UNIT-I	WIND TUNNELS						Classes: 12	
Wind Tunnel, layouts and nomenclature, Types of Wind Tunnels – continuous and intermittent -closed circuit and open circuit - closed jet and open jet test section – application. Special purpose tunnels - Smoke Tunnels – Water Tunnels – Spin tunnel, automobile wind tunnel and environmental wind tunnel Important parameters of flow similarity. types of flow similarities for compressible and incompressible flows Model power consideration.								
UNIT-II	FLOW VISUALIZATION TECHNIQUES						Classes: 12	
Path – Streak – Stream and Timelines; Techniques: Smoke, Tuft, Streaks, Surface oil flow. Pressure measurements: Manometers – U-Tube, Inclined and Precession. Bourdon Gauge and Pressure Transducer – Strain Gauge, Semi conductor – Absolute and Differential. Velocity Measurements: Pivot Tube – Static and Total. Calibration of test section: Test section flow calibration and Boundary Layers.								
UNIT-III	MEASUREMENTS OF FORCES AND MOMENTS						Classes: 12	
Forces, moments and Reference Frames – Balances – Internal and External - Requirements and Specifications – Fundamentals of Model Installations. Boundary correction, types of blockages: 2-d.								
UNIT-IV	HIGH SPEED WIND TUNNELS						Classes: 12	
Supersonic Wind Tunnels and - Classification - Runtime - Compressors - Charging Times - nozzle Mass Flows - Starting Loads - Model Size – Calibration. Hypersonic Wind Tunnels: Classification – Runtime – Vacuum Tanks – Vacuum pumps – Evacuation Times. Shock Tube: Driver – driven – Vacuum Pumps – Diaphragm.								
UNIT-V	HIGH SPEED FLOW VISUALIZATIONS AND MEASUREMENTS						Classes: 12	
Schlieren and Shadow Graph – Pressure sensitive Paints – Temperature sensitive Paints – Force Measurements – Strain Gauge Balances – Pressure Measurements.								
Text Books: 1. Rae, W.H. and Pope, A. —Low Speed Wind Tunnel TestingII, John Wiley Publication, 1999 2. Pope, A., and Goin, L., —High Speed wind Tunnel TestingII, John Wiley Publication , 1999 3. Pope, J B Barlow —low speed wind tunnel testing — 3 edition j.w publication								

Reference Books:

1. John D. Anderson, Jr., "Fundamentals of Aerodynamics", Third edition, McGraw-Hill publications, 2001
2. E L Houghton and PW Carpenter, "Aerodynamics for Engineering students", Fourth edition, Edward Arnold publications, 1993.
3. L.M Milne Thomson, —Theoretical Aerodynamics II, 1996 McGraw-Hill, New Delhi.
4. R. Halmshaw (1991), Non-Destructive Testing, 2nd edition, Edward Arnold, New York

Web References:

wingsofaero.in/introduction-to-wind-tunnel-techniques
<https://www.keyshone.com/wind-tunnel-techniques-lecture-notes-for-aero/>

E-Text Books:

<https://www.scribd.com/document/324894849/wind-tunnel-technique-notes-for-aeronautical-engineers-tp-1-pdf>
https://www.academia.edu/.../john_d_anderson_jr_fundamentals_of_aerodynamics_

MOOC Course

<https://nptel.ac.in/courses/101106040>

INTRODUCTION TO AEROSPACE TECHNOLOGY

VII Semester: OPEN ELECTIVE - III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE42	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To introduce the basic concepts of Aerospace technology and the current developments in the field. 2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.								
COURSE OUTCOMES: Students should be able to 1. Compare the atmosphere conditions of different altitudes for spacecraft system 2. Analyze how lift, drag and thrust are generated and understand which components constitute them 3. Analyze the flight performance parameters with respective stability condition 4. Distinguish the different systems used in a satellite mission 5. Design lighter than air vehicle using Catia software								
UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT						Classes: 12	
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments								
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION						Classes: 12	
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.								
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY						Classes: 12	
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes								
UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION						Classes: 12	
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity								

UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION	Classes: 12
Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.		
Text Books:		
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082		
Reference Books:		
1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017. 2. Ajoy kumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.		
Web References: https://www.grc.nasa.gov/www/k-12/airplane/index.html		
E-Text Books:		
https://www.researchgate.net/publication/259576416_Fundamentals_of_Aerospace_Engineering http://www.adl.gatech.edu/extrovert/Ebooks/ebook_Intro.pdf		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_ae07/		

OPEN ELECTIVES OFFERED BY COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

OE1		OE2	
A4CS43	Fundamentals of Databases	A4CS46	Introduction to Computer Networks
A4CS44	Introduction to C++	A4CS47	Introduction to Linux Programming
A4CS45	Core Java Programming	A4CS48	Fundamentals of Artificial Intelligence
OE3		OE4	
A4CS49	Soft Computing	A4CS52	Introduction to Network Security
A4CS50	Introduction to Cloud Computing	A4CS53	Data Analytics
A4CS51	Web Programming	A4CS54	.NET & C# Programming

FUNDAMENTALS OF DATABASES

V SEMESTER - OPEN ELECTIVE – I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS43	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To understand the basic database concepts, data models including the architecture and operation of the relational data model.
2. To make a study of Structured Query Language (SQL) and construct simple and moderately advanced database queries using SQL.
3. To conceptualize relational database design and depict a database system using ERdiagram.
4. To know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.
5. To develop a database application using commercial databases

COURSE OUTCOMES:

1. Able to design effective database schema for realistic applications.
2. Design a database using ER modelling.
3. Make use the basic concepts of database systems in database design.
4. Apply SQL queries to interact with database.
5. Apply normalization on database design to eliminate anomalies.

UNIT-I	INTRODUCTION	Classes: 12
Introduction :Relational model, File System vs. Database System -- Data Models – Database System Architecture – Relational Databases - Relational Model - Keys -- Integrity Constraints - Relational Algebra		
UNIT-II	SQL FUNDAMENTALS	Classes: 10
SQL FUNDAMENTALS :SQL fundamentals - SQL Data Definition - Basic Structure of SQL Queries - Additional Basic Operations - Set Operations - Aggregate Functions - Nested Subqueries - Join Operations - Views		
UNIT-III	DATABASE DESIGN	Classes: 15
DATABASE DESIGN : Entity-Relationship Model – ER Diagrams - Mapping ER Model to Relational Model - Functional Dependencies Normalization - First Normal Form - Second Normal Form - Third Normal Form		
UNIT-IV	TRANSACTION MANAGEMENT	Classes: 13
TRANSACTION MANAGEMENT: Transaction Concepts - Transaction Recovery – ACID Properties - Concurrency Control – Need for Concurrency Control – Serializability - Locking Protocols – Two Phase Locking – Deadlock – Recovery		
UNIT-V	APPLICATION DEVELOPMENT	Classes: 10
APPLICATION DEVELOPMENT :Commercial Databases - Oracle/MySQL - PL/SQL – Stored Procedures - Functions - Triggers - Database Security using SQL commands		
TEXT BOOKS:		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, “Database System Concepts”, Sixth Edition, Tata McGraw Hill, 2010.		

REFERENCE BOOKS:

1. C.J.Date, A.Kannan, S.Swamynathan, "An Introduction to Database Systems", Eighth Edition, Pearson Education, 2006.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Sixth Edition, Pearson Education, 2010.
3. Raghu Ramakrishnan, Johannes Gehrke "Database Management Systems", Fourth Edition, Tata McGraw Hill, 2010.
4. G.K.Gupta, "Database Management Systems", Tata McGraw Hill, 2011.
5. Carlos Coronel, Steven Morris, Peter Rob, "Database Systems: Design, Implementation and Management", Ninth Edition, Cengage Learning, 2011
6. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.

WEB REFERENCES:

1. <http://iips.icci.edu.iq/images/exam/databases-ramaz.pdf>
2. https://www.w3schools.com/sql/sql_intro.asp

E-TEXT BOOKS:

1. <http://www.freebookcentre.net/Database/dbCategory.html>

MOOC Course

1. <https://swayam.gov.in/course/1322-fundamentals-of-database-system>

OUTCOMES:

The course should enable the students to:

1. List and explain the fundamental concepts of a relational database system.
2. Understand and use DML, DDL and DCL to query, update, and manage a database.
3. Convert the ER-model to relational tables, populate relational database and formulate SQL Queries on data.
4. Design database using E-R modeling and apply normalization techniques over it.
5. Manage the transactions that happens in a database.
6. Design and build a simple database system and demonstrate competence with the Fundamental tasks involved with modelling, designing, and implementing a DBMS

INTRODUCTION TO C++

V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS44	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES:								
1. To familiarize the students with the different techniques pertaining to problem solving skills. 2. To familiarize the students with basic concepts of object oriented programming. 3. To facilitate students in understanding the syntax and semantics of the “C++” language as well as basic constructs offered by the language 4. To facilitate students in solving real world problems using Object Oriented approach. 5. To impart the concepts of File handling techniques 6. To familiarize students with Generic Programming								
COURSE OUTCOMES:								
Students will be able to								
1. Understand different paradigms in solving the problem and the different concepts of object oriented programming. Analyze the importance of Procedure Oriented over Object Oriented Programming. 2. Analyze the performance of OOP system with POP in terms of the complexity. 3. Understand the key concepts of OOP. Apply OOP concepts and design a new system. Analyze various Inheritance & polymorphism techniques to design a system. 4. Understand file handling mechanism in C++ and design data base applications. 5. Understand Templates in C++ and Design C++ programs using Templates and Analyze the performance of OOP & POP program in terms of reusability.								
UNIT-I	OBJECT ORIENTED PROGRAMMING						Classes: 10	
Introduction to Object Oriented Programming, Structure of C++ program, Tokens, Keywords, Identifiers and Constants, Basic Data Types, Operators and Type Conversions. Functions: Returning values from functions. Reference arguments, Overloaded function, Inline function, Default arguments, returning by reference.								
UNIT-II	OBJECT, CLASSES, CONSTRUCTORS AND DESTRUCTORS						Classes: 12	
Object and Classes: Specifying a Class, Member function, Nesting of Member functions, Arrays in a Class, Memory Allocation for Objects, Static Members and Member Functions, Array of Objects, Friend Functions, Pointers to Members. Constructors and Destructors: Parameterized Constructor, Copy Constructor, Dynamic Constructor, Destructors.								
UNIT-III	OPERATOR OVERLOADING AND INHERITANCE						Classes: 12	
Operator Overloading: Overloading unary, binary operators, data conversion, pitfalls of operators overloading and conversion keywords, Explicit and Mutable. Inheritance: Derived class and Base class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Virtual Base Classes, Abstract Classes, Constructor in Derived Classes.								
UNIT-IV	POINTER, STREAMS AND FILES						Classes: 13	
Pointer: Introduction, Pointers to Objects, this Pointer, Pointers to Derived Class, Virtual Functions, Pure Virtual Functions. Streams and Files: C++ Streams, Stream Classes, Unformatted IO Operations, Formatted Console IO Operations, Managing Output with Manipulators.								

UNIT-V	FILES, TEMPLATES AND EXCEPTION HANDLING	Classes: 13
Files: Classes for File Handling, File Pointer, Updating a File, Error Handling in Files. Templates and Exception Handling: Class Templates, Function Templates, Exception Handling Mechanism, And Components of STL: Containers, Algorithms, Iterators.		
TEXT BOOKS:		
1. Object Oriented Programming with C++, <i>E Balaguruswamy</i> , Third Edition, and McGraw Hill Publications.		
REFERENCE BOOKS:		
1. Object Oriented Programming in C++ by <i>Robert Lafore</i> Techmedia Publication. 2. Object Oriented Programming in C++ <i>Saurav Sahay</i> Oxford University Press.		

CORE JAVA PROGRAMMING

V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS45	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES:								
1. Learning principles of object oriented programming paradigm including abstraction, encapsulation, inheritance and polymorphism. 2. Understand fundamentals of programming such as variables, conditional and iterative execution, methods etc. 3. Understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc. 4. Learning the concept of inheritance to create new classes from existing one & Design the classes needed given a problem specification; 5. Understand the concept of packages and interfaces. 6. Learning how to detect exceptions and to handle strings &Implement the designed classes using the object oriented programming language 7. Learn how to test, verify, and debug object-oriented programs; and Learning about multithreading and multitasking. 8. Learn & demonstrate the concept of event handling, JDBC Connectivity used in GUI. 9. Create Applications using object-oriented principles with concepts of JAVA								
COURSE OUTCOMES:								
1. Analyze the necessity for Object Oriented Programming paradigm over structured programming and become familiar with the fundamental concepts in OOP like encapsulation, Inheritance and Polymorphism 2. Demonstrate an ability to design and develop java programs, analyze, and interpret object oriented data and report results. 3. Demonstrate an ability to design an object oriented system, AWT components and multithreaded processes as per needs and specifications.. 4. Demonstrate an ability to visualize and work on laboratory and multidisciplinary tasks like console and windows applications both for stand alone and Applets programs. 5. Demonstrate skills to use latest object oriented programming language and software like java to analyze OOP problems 6. Develop confidence for self education and ability for life-long learning needed for advanced java technologies 7. Able to participate and succeed in competitive examinations like GATE, Engineering services, recruitment interviews etc.								
UNIT-I	OPEN CONCEPTS AND JAVA PROGRAMMING						Classes: 11	
Oops concepts- Procedural and objects oriented programming paradigms, classes and objects,Data abstraction, encapsulation, polymorphism, inheritance, benefits of inheritance. Java Programming- History of java, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expression, type conversion and casting , control flow- block scope, condition statements, loops, break, and continue statements, simple java stand alone programs, arrays, constructors, methods, parameter passing, static keyword, access control, this pointer, overloading methods and constructor, recursion, garbage collection, exploring string class								
UNIT-II	INHERITANCE AND POLYMORPHISM						Classes: 12	
Inheritance- Inheritance hierarchies super and sub classes, member access rules, super keyword, final keyword, the Object class. Polymorphism- dynamic binding, method overriding, abstract classes and methods. Interface- Interfaces vs. Abstract classes, defining an interface, implementing interfaces, implementing multiple inheritance using interfaces, extending interfaces. Inner classes- Uses of inner classes, local inner classes, anonymous inner classes, static inner classes.								

UNIT-III	PACKAGES, EXCEPTION HANDLING AND FILES	Classes: 12
Packages- Defining, Creation and Accessing Packages, Understanding CLASSPATH, importing packages. Exception handling- Types of errors, benefits of exception handling, classification of exceptions-exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception sub classes. Files- streams- byte streams, character streams, text input/output, binary input/output, File management		
UNIT-IV	MULTITHREADING AND AWT CONTROLS	Classes: 13
Multithreading- Difference between multiprocessing and multithreading, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem. AWT CONTROLS: The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, ScrollBar. Working with Frame class, Colour, Fonts and layout managers.		
UNIT-V	GUI PROGRAMMING WITH JAVA AND EVENT HANDLING	Classes: 12
GUI Programming with Java-Introduction to Swing, Hierarchy for Swing components, Swing vs.AWT, Containers- JFrame, JApplet, JDialog, JPanel, JButton, JLabel, JtextField, JtextArea Event Handling- Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model, Examples: handling a button click, handling mouse events, Adapter classes.		
TEXT BOOKS:		
1. Java Fundamentals – A comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH 2. Java A Beginner's Guide Sixth Edition , Herbert Schildt		
REFERENCE BOOKS:		
1. Java for Programmers, P.J. Deitel and H.M. Deitel, Pearson education (OR) Java: How to Program P.J.Deitel and H.M. Deitel, PHI. 2. Object Oriented Programming through Java, P. Radha Krishna, Universities Press. 3. Thinking in Java, Bruce Eckel, Pearson Education 4. Programming in Java, S. Malhotra and S. Choudary, Oxford Univ. Press.		

INTRODUCTION TO COMPUTER NETWORKS

VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS46	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To introduce the fundamentals of various types of computer networks 2. To demonstrate the TCP/IP and OSI models with merits and demerits 3. To explore the various layers of OSI model 4. To introduce UDP and TCP models 5. To explore routing and application layer protocols								
COURSE OUTCOMES: By the end of this course, student will be able to 1. Define and describe functionality and services offered at various layers of OSI and TCP/IP 2. Analyse the internal implementation of Data link layer, Network Layer, Transport Layer and 3. Application layer. 4. Implement the various routing protocols, congestion control protocols. 5. Analyse and solve problems with respect to various layers of reference model.								
UNIT-I	INTRODUCTION TO COMPUTER NETWORK AND PHYSICAL LAYER						Classes: 12	
INTRODUCTION TO COMPUTER NETWORKS:Uses of Computer Networks, Network Hardware, Network Software, Reference Models (OSI, TCP/IP), Example Networks. PHYSICAL LAYER- Basis for Data Communication, Guided Transmission Media, Wireless Transmission, Communication Satellites, Digital Modulation And Multiplexing, The Public Switched Telephone Network.								
UNIT-II	DATA LINK LAYER						Classes: 12	
DATA LINK LAYER- Data Link Layer Design Issues, Error Detection and Correction, Elementary Data Link Protocols, Sliding Window Protocols. The Channel Allocation Problem, Multiple Access Protocols.								
UNIT-III	NETWORK LAYER						Classes:12	
NETWORK LAYER- Virtual Circuit and Datagram Networks, Inside A Router, The Internet Protocol (IP): Forwarding and Addressing in The Internet, Routing Algorithms, Routing in The Internet, Broadcast And Multicast Routing.								
UNIT-IV	TRANSPORT LAYER						Classes: 12	
TRANSPORT LAYER: Transport-Layer Services, Multiplexing and Demultiplexing, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control, TCP Congestion Control.								
UNIT-V	APPLICATION LAYER						Classes: 12	
APPLICATION LAYER: Principles of Network Applications, The Web and HTTP, File Transfer: FTP, Electronic Mail In The Internet, DNS—The Internet’s Directory Service, Peer-To-Peer Application.								
TEXT BOOKS: 1. Andrew S Tanenbaum David J Wetherall, Computer Networks, 5th Edition, Pearson Education. 2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education, India.								

REFERENCE BOOKS:

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers.

INTRODUCTION TO LINUX PROGRAMMING

VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS47	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES:								
1. To teach principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linux commands Scripts and filters. 2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts. 3. To impart fundamentals of file concepts kernel support for file, File structure related system calls (file API's). 4. To facilitate students in understanding Inter process communication. 5. To facilitate students in understanding semaphore and shared memory. To facilitate students in understanding process								
COURSE OUTCOMES:								
1. Ability to use various Linux commands that are used to manipulate system operations at admin level and a prerequisite to pursue job as a Network administrator. 2. Ability to write Shell Programming using Linux commands. 3. Ability to design and write application to manipulate internal kernel level Linux File System. 4. Ability to develop IPC-API's that can be used to control various processes for synchronization. 5. Ability to develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.								
UNIT-I	INTRODUCTION TO LINUX AND LINUX UTILITIES						Classes: 12	
INTRODUCTION TO LINUX AND LINUX UTILITIES: A brief history of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor. Linux commands- PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, lp, od, tar, gzip, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp, telnet, rlogin.Text Processing utilities and backup utilities , tail, head , sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, cpio								
UNIT-II	INTRODUCTION TO SHELLS AND FILTERS						Classes: 13	
Introduction to Shells: Linux Session, Standard Streams, Redirection, Pipes, Tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization. Filters: Filters and Pipes, Concatenating files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files								
UNIT-III	GREP, SED, UNIX FILE STRUCTURE AND FILE MANAGEMENT						Classes: 12	
Grep: Operation, grep Family, Searching for File Content. Sed :Scripts, Operation, Addresses, commands, Applications, grep and sed. UNIX FILE STRUCTURE: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers. File Management :File Structures, System Calls for File Management – create, open, close, read, write, lseek, link, symlink, unlink, stat, fstat, lstat, chmod, chown, Directory API – opendir, readdir, closedir, mkdir, rmdir, umask.								

UNIT-IV	PROCESS AND SIGNALS	Classes: 12
PROCESS AND SIGNALS: Process, process identifiers, process structure: process table, viewing processes, system processes, process scheduling, starting new processes: waiting for a process, zombie processes, orphan process, fork, vfork, exit, wait, waitpid, exec, signals functions, unreliable signals, interrupted system calls, kill, raise, alarm, pause, abort, system, sleep functions, signal sets. File locking: creating lock files, locking regions, use of read and write with locking, competing locks, other lock commands, deadlocks.		
UNIT-V	INTER PROCESS COMMUNICATION AND INTRODUCTION TO SOCKET	Classes: 11
INTER PROCESS COMMUNICATION: Pipe, process pipes, the pipe call, parent and child processes, and named pipes: fifos, semaphores: semget, semop, semctl, message queues: msgget, msgsnd, msgrcv, msgctl, shared memory: shmget, shmat, shmdt, shmctl, ipc status commands. INTRODUCTION TO SOCKETS: Socket, socket connections - socket attributes, socket addresses, socket, connect, bind, listen, accept, socket communications.		
TEXT BOOKS:		
1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3 rd edition, Pearson Education, New Delhi, India. 2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg. Thomson		
REFERENCE BOOKS:		
1. Linux System Programming, Robert Love, O'Reilly, SPD. 2. Advanced Programming in the UNIX environment, 2nd Edition, W.R.Stevens, Pearson Education. 3. UNIX Network Programming, W.R. Stevens, PHI. UNIX for Programmers and Users, 3rd Edition, Graham Glass, King Ables, Pearson Education		

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

VI SEMESTER - OPEN ELECTIVE – II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS48	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: - To learn the difference between optimal reasoning vs human like reasoning - To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities - To learn different knowledge representation techniques - To understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing								
COURSE OUTCOMES: - Possess the ability to formulate an efficient problem space for a problem expressed in English. - Possess the ability to select a search algorithm for a problem and characterize its time and space complexities. - Possess the skill for representing knowledge using the appropriate technique - Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing								
UNIT-I	Introduction						Classes:10	
Introduction, History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction. Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning								
UNIT-II	Logic Concepts and Logic Programming & Knowledge Representation						Classes:12	
Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.								
UNIT-III	Expert System and Applications & Uncertainty Measure – Probability Theory						Classes:13	
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools. Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster-Shafer Theory.								
UNIT-IV	Machine-Learning Paradigms & Artificial Neural Networks						Classes:13	
Machine-Learning Paradigms: Introduction. Machine Learning Systems. Supervised and Unsupervised Learning. Inductive Learning. Learning Decision Trees, Deductive Learning. Clustering, Support Vector Machines. Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks. Recurrent Networks.								

UNIT-V	Advanced Knowledge Representation Techniques	Classes:12
Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, Universal Networking Knowledge.		
TEXT BOOKS:		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS:		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W. Patterson. PHI. 4. Artificial Intelligence by George Fluger Pearson fifth edition.		

SOFT COMPUTING

VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS49	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Basics of ANN and its learning algorithms. 2. Fuzzy principles and relations. 3. Genetic algorithms and its applications. 4. Hybrid systems and usage of MATLAB toolbox.								
COURSE OUTCOMES: 1. To understand neural network (NN) paradigms 2. To learn fuzzy logic, to have a knowledge of evolutionary computations, genetic algorithm(GA), evolutionary programming, classifier systems, genetic programming parse trees, mathematical foundation of GA variants of GA.								
UNIT-I	NEURAL NETWORKS						Classes: 14	
Fundamentals of Neural Networks–History-Architectures-Learning methods-XOR problem-Delta rule-derivation-Back propagation-applications-parameters inBPN-Associative memory–Hetero associative-BAM-energy function-problems-applications of associative memories-ART1-ART2-applications of adaptive Networks.								
UNIT-II	FUZZY LOGIC						Classes: 10	
Fuzzy set theory–crisp sets–fuzzysets–crisp relations–Fuzzy relations–Fuzzy systems-Crisp logic–predicate logic–fuzzy logic-fuzzy based systems-Defuzzification methods–applications.								
UNIT-III	GENETIC ALGORITHMS						Classes: 14	
Fundamentals of GA–creation of offsprings–encoding–fitness function-reproduction–crossover-Insertion & deletion-mutation-bitwise operators–applications.								
UNIT-IV	HYBRID SYSTEMS						Classes: 12	
Hybrid systems–Neuro Fuzzy–Neuro Genetic–fuzzy Genetic hybrids-GA based weight determination and applications-fuzzy BPN–simplified fuzzy ARTMAP.								
UNIT-V	FPROGRAMMING USING MATHLAB						Classes: 10	
Using Neural Network toolbox–Using Fuzzy Logic toolbox-Using Genetic Algorithm & directed search toolbox.								
Text Books: 1. Rajasekaran.S and VijayalakshmiPai.G.A, “Neural Networks, Fuzzy Logic and Genetic Algorithms”, PHI, 2011								
Reference Books: 1. Timothy J.Ross,”Fuzzy Logic with Engineering applications ”, John Wiley and Sons, 2010. 2. Jang.J.S.R, Sun.C.T, Mizutani.E, ”Neuro fuzzy and Soft Computing ”, PHI Learning Pvt. Ltd., 2012. 3. Davis E.Goldberg, “Genetic Algorithms: Search, Optimization and Machine Learning ”, Addison Wesley, N.Y., 1989.								

Web References:

1. http://www.myreaders.info/html/soft_computing.html
2. https://en.wikipedia.org/wiki/Soft_computing

E-Text Books:

1. <http://freecomputerbooks.com/Introduction-to-Soft-Computing.html>
2. <https://www.shroffpublishers.com/books/9789351106159/>
3. <https://bookboon.com/en/introduction-to-soft-computing-ebook>

MOOC Course

1. https://onlinecourses.nptel.ac.in/noc18_cs13/course
2. <https://swayam.gov.in/course/4574-introduction-to-soft-computing>

INTRODUCTION TO CLOUD COMPUTING

VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS50	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. Understand the Virtualization paradigms 2. Learn the Cloud Computing fundamentals and its importance to various organizations. 3. Analyze the concepts of IaaS, PaaS, SaaS, Public and Private Clouds. 4. Develop applications in cloud security.								
COURSE OUTCOMES: At the end of the course, student will be able to: 1. Understand the main concepts, key technologies, strengths, and limitations of virtualization and cloud computing and the possible applications for state-of-the-art cloud computing. 2. Describe the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc. 3. Analyze the core issues of cloud computing such as security, privacy, and interoperability. 4. Identify problems, analyze, and evaluate various cloud computing solutions.								
UNIT-I	INTRODUCTION						Classes: 12	
System Modeling, Clustering And Virtualization: Distributed system models and enabling technologies, computer clusters for scalable parallel computing, virtual machines and virtualization of clusters and data centers.								
UNIT-II	FOUNDATIONS						Classes: 15	
Foundations: Introduction to cloud computing, migrating into cloud, enriching the integration of service paradigm for cloud era, the enterprise cloud computing paradigm.								
UNIT-III	INTER PROCESS COMMUNICATION						Classes: 17	
Inter Process Communication: Infra Structure As Service (IAAS) & Platform And Software Service(PAAS/SAAS): Virtual machine provisioning and migration services, on the management of virtual machines for cloud infrastructure, enhancing cloud computing environments using a cluster as service. Secure distributed data storage in cloud computing. Aneka, comet cloud, T-systems, work flow engine for clouds, understanding scientific applications for cloud environments								
UNIT-IV	MONITORING, MANAGEMENT AND APPLICATIONS						Classes: 10	
Monitoring, Management And Applications : An Architecture for federated cloud computing ,SLA management in cloud computing, performance prediction for HPC on clouds, best practices in Architecting cloud applications in the AWS cloud, building content delivery networks using clouds, resource cloud mashups.								
UNIT-V	GOVERNANCE AND CASE STUDIES						Classes: 10	
Governance and case studies: organizational readiness and change management in cloud age, data security in cloud, legal issues in cloud computing, achieving production readiness for cloud services.								

TEXT BOOKS:

1. Cloud computing: principles and paradigms by rajkumar buyya, James Broberg and Andrzej M.Goscinski,Wiley,2011.
2. Distributed and cloud computing , kai Hwang, Geofferyu C.fox, jack J.dongarra, Elsevier, 2012 .

REFERENCE BOOKS:

1. Cloud Computing :A practical approach, Anthony T.velte, Toby J.velte, Robert Isenpeter, Tata McGraw Hill,2011.
2. Enterprise Cloud Computing, Gautam Shroff, Cambridge University press,2010.
3. Cloud computing: implementation, management and security, john W .Ritting house ,James F. Ransome ,CRC press,rp2012.
4. Cloud applications architectures: building Applications and infrastructure in the cloud,George Reese , O reilly,SPD,rp2011.
5. Cloud security and privacy: An Enterprise perspective on Risks and compliance,im Mather, Subra Kumaraswamy, Shahed Latif,Oreilly,SPD,rp2011

E-Text Books:

1. <https://www.amazon.com/Cloud-Computing-Principles-Rajkumar-Buyya/dp/0470887990>
2. <https://eniac2017.files.wordpress.com/2017/03/distributed-and-cloud-computing.pdf>

MOOC Course

1. https://onlinecourses.nptel.ac.in/noc18_cs33/preview
2. <https://www.alison.com/courses/Introduction-to-cloud-computing>
3. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-cloud-computing-january-iap-2014/index.htm>

WEB PROGRAMMING**VII SEMESTER - OPEN ELECTIVE – III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS51	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To teach students the basics of server side scripting using HTML 2. To explain web application development procedures 3. To impart servlet technology for writing business logic 4. To facilitate students to connect to databases using JDBC 5. To familiarize various concepts of application development using XML								
COURSE OUTCOMES: 1. Students will be able to use HTML, CSS to develop web pages. 2. Students will be able to apply concepts of Java. 3. Students will be able to build JDBC connectivity. 4. Students will be able to identify the basic building blocks of XML and describe the applications of web services. 5. Students will be able to write server side programming using applets.								
UNIT-I	SCRIPTING						Classes: 11	
Web page Designing using HTML, Scripting basics- Client side and server side scripting. Java Script- Object, names, literals, operators and expressions- statements and features-events - windows - documents - frames - data types - built-in functions- Browser object model - Verifying forms.-HTML5- CSS3- HTML 5 canvas - Web site creation using tools.								
UNIT-II	JAVA						Classes: 11	
Introduction to object oriented programming-Features of Java – Data types, variables and arrays – Operators – Control statements – Classes and Methods – Inheritance. Packages and Interfaces – Exception Handling – Multithreaded Programming –Input/Output – Files – Utility Classes – String Handling.								
UNIT-III	JDBC						Classes: 12	
JDBC Overview – JDBC implementation – Connection class – Statements – Catching Database Results, handling database Queries. Networking– Inet Address class – URLclass- TCP sockets - UDP sockets, Java Beans –RMI.								
UNIT-IV	APPLETS						Classes: 14	
Java applets- Life cycle of an applet – Adding images to an applet – Adding sound to an applet, Passing parameters to an applet, Event Handling, Introducing AWT: Workingwith Windows Graphics and Text. Using AWT Controls, Layout Managers and Menus.Servlet – life cycle of a servlet, The Servlet API, Handling HTTP Request and Response, using Cookies, Session Tracking. Introduction to JSP.								
UNIT-V	XML AND WEB SERVICES						Classes: 12	
Xml – Introduction-Form Navigation-XML Documents- XSL – XSLT- Web services-UDDI-WSDL-Java web services – Web resources.								

TEXT BOOKS:

1. Java™ : The Complete Reference, Seventh Edition, Herbert Schildt
2. J2EE: The complete Reference by James Keogh
3. Java EE and HTML5 Enterprise Application Development (Oracle Press) by John Brock (Author), Arun Gupta (Author), Geertjan Wielenga (Author)
4. Struts: The Complete Reference, 2nd Edition by James Holmes
5. ASP.NET 4 Unleashed by Stephen Walther (Author), Kevin Scott Hoffman• (Author), Nate Dudek (Author)
6. Microsoft Visual C# 2013 Step by Step by John Sharp

INTRODUCTION TO NETWORK SECURITY

VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS52	HS	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Able to demonstrate the basic components of security and mathematics of cryptography. 2. Able to demonstrate various traditional symmetric key ciphers 3. Analyze DES and AES its variants. 4. Able to demonstrate private and public-key cryptosystem, Digital signature and attacks on it.								
COURSE OUTCOMES: At the end of the course students able to 1. Identify and explain various types of security threats/vulnerabilities attacks, consequences and security mechanisms 2. Apply various Cryptographic and Authentication techniques for secure transmission of data. 3. Compare different security algorithms with respect feasibility, cost and security service 4. Choose the most suitable security mechanisms based on the nature of data and complexity of algorithm.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction: Security Goals, Attacks, Services and Mechanism, Techniques, Mathematics of Cryptography, Integer arithmetic, Modular Arithmetic, Matrices, Linear Congruence, Primes, Primality Testing, Factorization, Chinese Remainder Theorem. Traditional Symmetric key Ciphers: Introduction, Substitution Ciphers, Transposition Ciphers, Streams and Block Ciphers.								
UNIT-II	SYMMETRIC CIPHERS						Classes: 12	
Advanced Encryption Standards: AES Structure, AES Transformation Function, AES Key Expansion, AES Implementation. Data Encryption Standard: Introduction, DES Structure, DES Analysis, Multiple DES, Security of DES, Differential Cryptanalysis, Linear Cryptanalysis of DES, Encipherment Using Modern Symmetric Key Ciphers Use of Modern Block Ciphers, Use of Stream Ciphers								
UNIT-III	ASYMMETRIC CIPHERS, DATA INTEGRITY ALGORITHMS						Classes: 12	
Asymmetric Key Cryptography and Message Integrity: Introduction, RSA Cryptosystem, Message Integrity, Message Authentication Cryptographic Hash Functions, Introduction, SHA-512, Digital Signature Comparison, Attacks on Digital Signature. User Authentication Protocols: Remote User Authentication Principles, Remote User Authentication using Symmetric Encryption, Kerberos								
UNIT-IV	MUTUAL TRUST						Classes: 10	
Key Management And Distribution : Symmetric key Distribution Using Symmetric and Asymmetric Encryption, Distribution of Public Key, X.509 Certificate , Public Key Infrastructure. Network Access Control : Network Access Control, Extensible Authentication Protocol, IEEE802.1,								
UNIT-V	NETWORK SECURITY						Classes: 10	
Transport-Level Security: Transport-Level Security, HTTPS, Secure Shell(SSH) Wireless Network Security: Web Security ,Mobile Device Security, IEEE802.11 Wireless LAN Overview, IEEE802.11i Wireless LAN Security								
Text Books:								

1. Cryptography and Network Security”by Behrouz A. Forouzan, Special Indian Edition, 2008
2. Cryptography and Network Security”by William Stallings, 6th Edition, Pearson Education/PHI, 2014

Reference Books:

1. “Cryptography and Network Security”by Atul Kahate, Tata McGraw Hill, 2003.
2. "RSA Security's official guide to cryptography" by Steve Burnett, Stephane Paine, RSA Press, Tata McGraw Hill Edition, 2001

Web References:

1. www.open.edu/openlearn/science.../network-security/content-section---references
www.cse.iitm.ac.in/~chester/courses/16e_cns/slides/01_Introduction.pdf
williamstallings.com/Crypto/Crypto4e.html
2. pubs.sciepub.com/iteces/3/1/1/
3. <https://www.sanfoundry.com/best-reference-books-cryptography-network-security/>
4. <https://dl.acm.org/citation.cfm?id=1824151>
5. web.mit.edu/6.857/OldStuff/Fall03/references.html

E-Text Books:

1. Online resources of Stallings's Cryptography and Network Security. Principles and Practice
<http://williamstallings.com/Crypto3e.html>
2. Online resources of Menezes, van Oorschot, Vanstone's Handbook of Applied Cryptography available at <http://www.cacr.math.uwaterloo.ca/hac/>

MOOC Course

1. <https://www.mooc-list.com/tags/network-security>
2. nptel.ac.in/courses/106105031/
3. https://onlinecourses.nptel.ac.in/noc18_cs07/course

DATA ANALYTICS

VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS53	HS	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To explore the fundamental concepts of data analytics 2. Formulate and use appropriate models of data analysis to solve hidden solutions to business related challenges.								
COURSE OUTCOMES: 1. Students will demonstrate the ability to use technical skills in Descriptive and prescriptive modelling to support business decision-making 2. Apply the knowledge of classification and regression techniques for various data analysis. 3. Examine the results of Classification for further predictions. 4. Make use of regression techniques for predicting future outcomes on various data 5. Illustrate the concepts of data analytics for data visualization and bigdata analysis.								
UNIT-I	DESCRIPTIVE STATISTICS						Classes: 10	
Descriptive Statistics , Introduction to the course, Descriptive Statistics, Probability Distributions, Inferential Statistics, Inferential Statistics through hypothesis tests, Permutation & Randomization Test								
UNIT-II	REGRESSION AND ANOVA						Classes: 12	
Regression & ANOVA , Regression , ANOVA(Analysis of Variance), Machine Learning: Introduction and Concepts, Differentiating algorithmic and model based frameworks. Regression: Ordinary Least Squares, Ridge Regression, Lasso Regression, K Nearest Neighbours Regression & Classification								
UNIT-III	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - I						Classes: 13	
Supervised Learning with Regression and Classification techniques -1 , Bias-Variance Dichotomy, Model Validation Approaches, Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis Regression and Classification Trees, Support Vector Machines								
UNIT-IV	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - 2						Classes: 12	
Supervised Learning with Regression and Classification techniques -2, Ensemble Methods: Random Forest Neural Networks, Deep learning, Unsupervised Learning and Challenges for Big Data Analytics Clustering Associative Rule Mining Challenges for big data analytics.								
UNIT-V	PRESCRIPTIVE ANALYTICS						Classes: 10	
Prescriptive analytics , Creating data for analytics through designed experiments, Creating data for analytics through Active learning, Creating data for analytics through Reinforcement learning								
TEXT BOOKS: 1. Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: springer, 2009. 2. Montgomery, Douglas C., and George C. Runger. Applied statistics and probability for engineers. John Wiley & Sons, 2010								

REFERENCE BOOKS:

1. Data Analytics- Models and Algorithms for Intelligent Data Analysis, Author: Thomas Runkler
ISBN-10: 3834825883
2. Data Analytics, Anil Maheshwari, McGraw Hill Education; First edition, 9352604180.

.NET & C# PROGRAMMING

VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS54	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: This course enables to student to know: - To impart basic concepts of .NET Framework Components. - To familiarize the C# Core Concepts. - To facilitate students in understanding OOP Concepts. - To facilitate students in understanding Exception Handling. - To impart the concepts of Delegates, Events.								
COURSE OUTCOMES: At the end of the course, students will - Explain the purpose of the .net framework, and understand how to use c# and visual studio 2010 to build .net framework applications. - Understand the syntax of basic c# programming constructs. Create and use new types (enumerations, classes, and structures), and understand the differences between reference types and value types. - Use inheritance to create new reference types. control the visibility and lifetime of members in a type. - Manage the lifetime of objects and control the use of resources, catch, handle and throw exceptions. - Understanding the .net delegate type & event handling.								
UNIT-I	INTRODUCTION TO .NET						Classes: 10	
Introduction - The Philosophy of .NET, An Initial Look at the .NET Platform, Introducing the Building Blocks of the .NET Platform (the CLR, CTS, and CLS), An Overview of .NET Assemblies, Understanding the Common Type System, Understanding the Common Language Specification, Understanding the Common Language Runtime, The Assembly/Namespace/Type Distinction, The Platform-Independent Nature of .NET.								
UNIT-II	CORE C# PROGRAMMING CONSTRUCTS						Classes: 13	
Core C# Programming Constructs, Part I - The Anatomy of a Simple C# Program, An Interesting Aside: Some Additional Members of the System.Environment Class, The System.Console Class, System Data Types and Corresponding C# Keywords, Working with String Data, Narrowing and Widening Data Type Conversions, Understanding Implicitly Typed Local Variables, C# Iteration Constructs, Decision Constructs and the Relational/Equality Operators. Core C# Programming Constructs, Part II- Methods and Parameter Modifiers, Understanding C# Arrays, Understanding the enum Type, Understanding the Structure Type, Understanding Value Types and Reference Types, Understanding C# Nullable Types.								
UNIT-III	OBJECT-ORIENTED PROGRAMMING WITH C#						Classes: 15	
Object-Oriented Programming with C# Understanding Encapsulation - Introducing the C# Class Type, Understanding Constructors, The Role of the this Keyword, Understanding the static Keyword, Defining the Pillars of OOP, C# Access Modifiers, The First Pillar: C#'s Encapsulation Services, Understanding Automatic Properties, Understanding Object Initialization Syntax , Working with Constant Field Data, Understanding Partial Types. Understanding Inheritance and Polymorphism - The Basic Mechanics of Inheritance, The Second Pillar of OOP: The Details of Inheritance, Programming for Containment/Delegation, The Third Pillar of OOP: C#'s Polymorphic Support, Understanding Base Class/Derived Class Casting Rules, The Master Parent Class: System.Object.								

UNIT-IV	STRUCTURED EXCEPTION HANDLING	Classes: 12
Understanding Structured Exception Handling - Ode to Errors, Bugs, and Exceptions, The Role of .NET Exception Handling, The Simplest Possible Example, Configuring the State of an Exception, System-Level Exceptions (System.SystemException), Application-Level Exceptions (System.ApplicationException), Processing Multiple Exceptions, Who Is Throwing What?, The Result of Unhandled Exceptions.		
UNIT-V	WORKING WITH INTERFACES, DELEGATES AND EVENTS	Classes: 14
Working with Interfaces - Understanding Interface Types, Defining Custom Interfaces, Implementing an Interface, Invoking Interface Members at the Object Level, Interfaces As Parameters, Interfaces As Return Values, Arrays of Interface Types, Implementing Interfaces Using Visual Studio, Explicit Interface Implementation, Designing Interface Hierarchies, The IEnumerable and IEnumerator Interfaces, The ICloneable Interface, The IComparable Interface. Delegates, Events - Understanding the .NET Delegate Type, The Simplest Possible Delegate Example, Sending Object State Notifications Using Delegates, Understanding Generic Delegates, Understanding C# Events, Understanding C# Anonymous Methods.		
TEXT BOOKS:		
1. Andrew Troelsen, Pro C-Hash and the .NET 4 Platform, 5th edition, Springer (India) Private Limited, New Delhi, India., 2010		
REFERENCE BOOKS:		
1. E. Balagurusamy (2004), Programming in C#, 5 th edition, Tata McGraw-Hill, New Delhi, India. 2. Herbert Schildt (2004), The Complete Reference: C#, Tata McGraw-Hill, New Delhi, India. 3. Simon Robinson, Christian Nagel, Karli Watson, Jay GI (2006), Professional C#, 3rd edition, Wiley & Sons, India.		

OPEN ELECTIVES OFFERED BY ELECTRONICS AND COMMUNICATION ENGINEERING DEPARTMENT

OE1		OE2	
A4EC59	Logic Design	A4EC62	Fundamentals of Integrated Circuits
A4EC60	Principles of Communications	A4EC63	Signals Transmission Through Linear Systems
A4EC61	Measurements And Instrumentation	A4EC64	Fundamentals of VLSI Design
OE3		OE4	
A4EC65	Introduction of Microprocessors and Microcontrollers	A4EC68	Introduction to Signal Processing
A4EC66	Fundamentals of Image Processing	A4EC69	Introduction to Mobile Communication
A4EC67	TV Engineering	A4EC70	Basic Embedded Systems and Design

LOGIC DESIGN

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Explain how to design a digital circuit.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Study the characteristics and performance of digital circuits built using various MOS technologies

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain how digital circuit of large complexity can be built in a methodological way, starting from Boolean logic and applying a set of rigorous techniques.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Study the characteristics and performance of digital circuits built using various MOS technologies

UNIT-I	BOOLEAN ALGEBRA	Classes: 09
Boolean Algebra: Axiomatic Definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates, IC Digital Logic Families		
UNIT-II	SIMPLIFICATION OF BOOLEAN FUNCTIONS	Classes: 09
Simplification of Boolean Functions: The Map Method, Two & Three Variable Maps, Four Variable Map, Five & Six Variable Maps, Product of Sum Simplification, NAND & NOR Implementation, Two-Level Implementations, Don't-Care Conditions, Tabulation Method, Determination of Prime Implicants, Selection of Prime Implicants.		
UNIT-III	COMBINATIONAL LOGIC	Classes: 09
Combinational Logic: Design Procedure, Adders, Subtractors, Code Conversion, Multilevel NAND Circuits, Multilevel NOR Circuits, Exclusive-Or and Equivalence Functions. Combinational Logic with MSI And LSI: Binary Parallel Adder, Decimal Adder, Magnitude Comparator, Decoders, Multiplexers, ROM, PLA.		
UNIT-IV	SEQUENTIAL CIRCUIT DESIGN AND ANALYSIS	Classes: 09
Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics		

UNIT-V	DIGITAL INTEGRATED CIRCUITS	Classes: 09
Digital Integrated Circuits: Introduction, Bi-Polar Transistor Characteristics, Integrated Injection Logic, Transistor-Transistor Logic (TTL), Emitter-Coupled Logic(ECL), MOS, Complementary MOS.		
Text Books:		
1. M.Morris Mano, —Digital DesignII, 2nd Edition, 1997, PHI. (Unit I, II, III & IV).		
Reference Books:		
1. Zvi Kohavi, —Switching and Automata TheoryII, 2nd Edition, 1978, McGraw-Hill. 2. Thomas L. Floyd —Digital FundamentalsII, 10th Edition, Pearson Education India. 3. Stephen Brown, Zvonko Vrsanec, IIFundamentals of Digital Logic with Verilog DesignII, 2nd Edition, McGraw-Hill.		
Web References:		
1. http://www.freebookcentre.net/electronics-ebooks-download/logic design.html 2. http://coen.boisestate.edu/sml00/sml00-courses/ece-332-logic designs-fall07/lecture-notes/ 3. http://www.freebookcentre.net/electronics-ebooks-download/logic design-Lecture-Notes.html		
E-Text Books:		
1. http://gen.lib.rus.ec/book/logic design 2. http://www.faadooengineers.com/threads/logicdesign -by-RAMESH-GANOKAR-ebook-pdf-download 3. https://e.edim.co/123389964/logic design.pdf		
MOOC Course		
1. https://www.mooc-list.com/tags/logic design 2. https://www.coursera.org/courses?query=logicedesign		

PRINCIPLES OF COMMUNICATIONS

V Semester: OPEN ELECTIVE -I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC60	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OUTCOMES: The course should enable the students to 1. Describe the basic concepts of Communication technology 2. Explain the Models of Communication 3. Discourse Analog and Digital Signal transmission 4. Discourse Computer Networking 5. Explain the types of Communication Media 6. Explain the Application of computer in data transmission and Security.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Describe the basic concepts of Communication technology 2. Explain the Models of Communication 3. Discourse Analog and Digital Signal transmission 4. Discourse Computer Networking 5. Explain the types of Communication Media 6. Explain the Application of computer in data transmission and Security.								
UNIT-I	FUNDAMENTALS OF ANALOG COMMUNICATION						Classes: 10	
Principles of amplitude modulation, AM envelope, frequency spectrum and bandwidth, modulation index and percent modulation, AM Voltage distribution, AM power distribution, Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation, Frequency analysis of angle modulated waves. Bandwidth requirements for Angle modulated waves.								
UNIT-II	DIGITAL COMMUNICATION						Classes: 09	
Introduction, Shannon limit for information capacity, digital amplitude modulation, frequency shift keying, FSK bit rate and baud, FSK transmitter, BW consideration of FSK, FSK receiver, phase shift keying binary phase shift keying QPSK, Quadrature Amplitude modulation, bandwidth efficiency, carrier recovery – squaring loop, Costas loop, DPSK.								
UNIT-III	DIGITAL TRANSMISSION						Classes: 10	
Introduction, Pulse modulation, PCM,PCM sampling, sampling rate, signal to quantization noise rate, companding analog and digital percentage error, delta modulation, adaptive delta modulation, differential pulse code modulation, pulse transmission – Inter symbol interference, eye patterns.								
UNIT-IV	SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES:						Classes: 08	
Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, processing gain, FH spread spectrum, multiple access techniques wireless communication, TDMA and CDMA in wireless communication systems, source coding of speech for wireless communications.								
UNIT-V	SATELLITE AND OPTICALCOMMUNICATION						Classes: 08	
Satellite Communication Systems ,Keplers Law, LEO and GEO Orbits, footprint, Link model Optical Communication Systems-Elements of Optical Fiber Transmission link, Types, Losses, Sources and Detectors								

Text Books:

1. Wayne Tomasi, —Advanced Electronic Communication SystemsII, 6/e, Pearson Education, 2007.
2. Simon Haykin, —Communication SystemsII, 4th Edition, John Wiley & Sons., 2001.

Reference Books:

1. H.Taub, D L Schilling, G Saha ,II Principles of CommunicationII3/e, 2007.
2. B.P.Lathi, II Modern Analog And Digital Communication systemsII, 3/e, Oxford University Press, 2007
3. Blake, —Electronic Communication SystemsII, Thomson Delmar Publications, 2002.
4. Martin S.Roden, —Analog and Digital Communication SystemII, 3rd Edition, PHI, 2002.
5. B.Sklar, II Digital Communication Fundamentals and ApplicationsII2/e Pearson Education 2007.

Web References:

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

E-Text Books:

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

MOOC Course

1. https://onlinecourses.nptel.ac.in/noc18_ee26/preview

MEASUREMENTS AND INSTRUMENTATION

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students:

1. To understand the measuring methods and instruments of electrical quantities.
2. To understand, design aspects and performance criterion of measuring instruments.
3. To understand the working principle of various transducers.
4. To aware the students about the advances in instrumentation.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply knowledge of electronic instruments for measurement of electrical quantities.
2. Understand the principles and operations of instruments.
3. Select and use latest hardware for measurements.
4. Identify the various instruments for various measurements
5. Use the instruments in laboratory and real life

UNIT-I	MEASUREMENT CONCEPTS	Classes: 10
Measurement concepts : Quantities of Measurement- Performance Characteristics – Static and dynamic characteristics – units and standards of measurements – errors in measurements- accuracy and precision, statistical analysis – moving coil, moving iron types Instruments –DC Voltmeter multi-meter & Calibrations		
UNIT-II	DIGITAL INSTRUMENTS	Classes: 08
Digital instruments: Digital multimeters – Digital frequency meter-digital tachometer-digital PH meter-digital phase meter-digital capacitance meter-Microprocessor based instruments- IEEE 488 bus		
UNIT-III	SIGNAL GENERATORS CATHODE RAY OSCILLOSCOPES	Classes: 08
Signal generators: Function generators – pulse and square wave generators, RF signal generators – Sweep generators – Frequency synthesizer Cathode ray oscilloscopes: Basic block diagram – operation-applications special Oscilloscopes: delayed time base oscilloscopes, sampling oscilloscope		
UNIT-IV	MEASURING INSTRUMENTS AND BRIDGES	Classes: 09
Measuring instruments: Introduction-field strength meter –Q- meter- LCR Bridge-Transistor tester Bridge: Wheatstone's bridges for resistance measurements-Maxwell's bridge for inductance measurements-Schering's bridge for capacitance measurements measurement		
UNIT-V	TRANSDUCERS & DATA ACQUISITION SYSTEMS	Classes: 10
Transducer: Introduction-electrical transducers-selecting a transducers-active and passive transducers with examples Data acquisition systems: Elements of a digital data acquisition system – interfacing of transducers – multiplexing –data loggers –computer controlled instrumentation		

Text Books:

1. Albert D. Helfrick and William D. Cooper – Modern Electronic Instrumentation and Measurement Techniques, Pearson / Prentice Hall of India, 2007.
2. Ernest O. Doebelin, Measurement Systems- Application and Design, TMH, 2007.

Reference Books:

1. Joseph J. Carr, Elements of Electronics Instrumentation and Measurement, Pearson Education, 2003.
2. Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition, Prentice Hall of India, 2003.
3. David A. Bell, Electronic Instrumentation and measurements, Prentice Hall of India Pvt Ltd, 2003.
4. B.C. Nakra and K.K. Choudhry, Instrumentation, Measurement and Analysis, 2nd Edition, TMH, 2004.
5. James W. Dally, William F. Riley, Kenneth G. McConnell, Instrumentation for Engineering Measurements, 2nd Edition, John Wiley, 2003.

Web References:

1. <https://www.sciencedirect.com/science/book/9780123819604>
2. <https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>

E-Text Books:

1. https://books.google.co.in/books/about/Electronics_Measurements_And_Instrumenta.html?id=mxVRJRYCi-QC
2. <https://sway.com/V1Uclj1JCiLhqCcb>

MOOC Course

1. <https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>
2. <https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>
3. <https://www.coursera.org/courses?query=measurements%20and%20instrumentat>

FUNDAMENTALS OF INTEGRATED CIRCUITS

VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students: 1. To understand the Basic NMOS, CMOS & BiCMOS circuits and their processtechnology. 2. To learn the concepts of modeling of Delay techniques and MOSlayers. 3. To learn the concepts of Technology Scaling of MOStransistors. 4. To understand the concepts of testing of combinational and sequential circuits								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Analyze the basic fabrication steps of semiconductor devices. 2. Analyze silicon oxidation and photolithography. 3. Analyze various etching techniques. 4. Analyze the Ion implantation process and metallization. 5. Evaluate various IC technologies such as Bipolar, MOSFET and MESFET.								
UNIT-I	INTEGRATED CIRCUITS AND OPERATIONAL AMPLIFIER						Classes: 09	
Semiconductor materials, Semiconductor Devices, Semiconductor process technology, Basic fabrication steps. Crystal Growth: Silicon Crystal Growth from melt, Silicon Float-Zone Process, GaAs Crystal Growth Techniques, Material Characterization.								
UNIT-II	SILICON OXIDATION						Classes: 09	
Thermal oxidation, Impurity Redistribution during oxidation, masking properties of silicon dioxide, oxide quality, oxide thickness characterization, Photolithography.								
UNIT-III	ETCHING						Classes: 09	
Wet chemical etching, Dry etching. Diffusion: Basic Diffusion Process, Extrinsic Diffusion, Lateral Diffusion.								
UNIT-IV	ION IMPLANTATION						Classes: 09	
Range of Implanted ions, Implant Damage and annealing, Implantation-related Process. Film Deposition: Epitaxial growth techniques, Structures and defects in epitaxial layers, Dielectric deposition, Poly-silicon deposition, Metallization.								
UNIT-V	PROCESS INTEGRATION						Classes: 09	
Passive Components, Bipolar Technology, MOSFET technology, MESFET technology, MEMS technology. IC Manufacturing: Electrical testing, Packaging.								
Text Books:								
1. Gary S. May, Simon M. Sze, "Fundamentals of Semiconductor Fabrication", John Wiley Inc., 2004.								

Reference Books:

1. C.Y. Chang and S.M.Sze (Ed), "ULSI Technology", McGraw Hill Companies Inc, 1996.
2. S.K. Ghandhi, "VLSI Fabrication Principles", John Wiley Inc., New York, 1983.
3. S.M. Sze (Ed), "VLSI Technology", 2nd Edition, McGraw Hill, 1988
4. The Science and Engineering of Microelectronic Fabrication, Stephen Cambell, Oxford University Press, 2001

Web References:

1. <https://banavathvalyaitmadras.files.wordpress.com/2015/12/vlsi-fabrication-principles-silicon-and-gallium-arsenide-1994-ghandi-wiley-2e.pdf>

E-Text Books:

1. rku.ac.in/soe/wp-content/uploads/2015/04/VL204_CMOSDeviceFabrication.pdf
2. https://books.google.co.in/books/.../The_Science_and_Engineering_of_Microelectronics

MOOC Course

1. nptel.ac.in/courses/117103066/35
2. <https://www.edx.org/course/circuits-electronics-1-basic-circuit-mitx-6-002-1x-0>

SIGNALS TRANSMISSION THROUGH LINEAR SYSTEMS

VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC63	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students: 1. To understand various fundamental characteristics of signals and systems. 2. To study the importance of transform domain. 3. To analyze and design various systems. 4. To study the effects of sampling.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Analyze various signals and systems by applying Laplace and Z-transforms. 2. Apply Fourier series, Fourier transform and inverse Fourier transform techniques to find magnitude and phase spectrum of various signals. 3. Apply convolution and correlation operations to various signals and systems. 4. Design IIR and FIR analog and digital filters. 5. Analyze multirate Digital Signal Processing.								
UNIT-I	TIME DOMAIN SIGNALS						Classes: 09	
Introduction: Definitions of signals and systems, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sync, Signum, and Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.								
UNIT-II	SPECTRAL DOMAIN REPRESENTATION OF SIGNALS						Classes: 09	
Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transforms, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.								
UNIT-III	CONVOLUTION AND CORRELATION						Classes: 09	
Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.								
UNIT-IV	DIGITAL FILTERS						Classes: 09	
IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method. FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.								

UNIT-V	MULTI RATE DIGITAL SIGNAL PROCESSING	Classes: 09
Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.		
Text Books:		
1. John G. Proakis, Dimitris G. Manolakis , Digital Signal Processing, Pearson Prentice Hall, 2007. 2. B.P. Lathi ,Signals, Systems & Communications ,BS Publications 2013 . 3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.		
Reference Books:		
1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed., 2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008. 3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007		
Web References:		
1. http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-Haver).html 2. https://www.sciencedirect.com/science/article/pii/B9780123965028000012 3. Applied Digital Signal Processing (1st Edition)		
E-Text Books:		
1. http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59 2. http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html		
MOOC Course		
1. https://www.mooc-list.com/course/digital-signal-processing-coursera 2. http://nptel.ac.in/courses/117102060/		

FUNDAMENTALS OF VLSI DESIGN**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC64	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-		30	70	100

COURSE OBJECTIVES:

The course should enable the students:

- 1.To understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
- 2.To designing of stick diagrams and layouts for MOS transistors.
- 3.To learn the concepts of modeling of Delay techniques and MOS layers.
- 4.To learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. Designing of stick diagrams and layouts for MOS transistors.
3. Learn the concepts of modeling of Delay techniques and MOS layers.
4. Learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

UNIT-I	REVIEW OF MICROELECTRONICS AND INTRODUCTION TO MOS TECHNOLOGY	Classes: 09
Introduction MOS and related VLSI technology – NMOS-CMOS-BICMOS-GaAs Technologies – thermal aspects of processing – production of E beam masks.		
UNIT-II	MOS AND BICMOS CIRCUIT DESIGN PROCESS	Classes: 09
MOS layers – stick diagrams – design rules and layout – 2m meter – 1.2 m meter CMOS rules – Layout diagrams – Symbolic diagrams.		
UNIT-III	BASIC CIRCUIT CONCEPTS	Classes: 09
Sheet resistance – Area capacitance of layers – delay unit – wiring capacitances – choice of layers.		
UNIT-IV	SCALING OF MOS CIRCUITS	Classes: 09
Scaling of MOS circuits: Scaling modes I – Scaling function for device parameters – Limitation of Scaling Subsystem design process: Architectural issues – switch logic – examples of structural design (Combinational logic)– design of ALU subsystem – commonly used storage elements – aspects of design rules.		
UNIT-V	TEST AND TESTABILITY	Classes: 09
Test and Testability: Design for testability built in self test (BIST) – teaching combinational logic – testing sequential logic – practical design for test guide lines – scan design techniques – etc.		
Text Books:		
1. Basic VLSI design by Douglas A, Pucknell, Kamran Eshraghian, Prantice Hall, 1996 3rd edition.		

Reference Books:

1. Mead, C.A and Conway, L.A., Introduction to VLSI Systems, Wesley – Wesley.

Web References:

1. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi .html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-vlsi-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi design-Lecture-Notes.html>

E-Text Books:

1. <http://gen.lib.rus.ec/book/vlsi design>
2. <http://www.faadooengineers.com/threads/vlsi design -by-kamran eshragghan-ebook-pdf-download>
3. <https://e.edim.co/123389964/vlsi design.pdf>

MOOC Course

1. <https://www.mooc-list.com/tags/vlsi design>
2. <https://www.coursera.org/courses?query=vlsi design>

INTRODUCTION OF MICROPROCESSORS AND MICROCONTROLLERS**VII Semester: OPEN ELECTIVE -III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC65	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

UNIT-I	MICROPROCESSOR ARCHITECTURE	Classes: 09
Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization, Signal descriptions of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine.		
UNIT-II	INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086	Classes: 09
Instruction set, Assembler directives, Addressing modes, programming model, Programs involving logical, branch and call instructions, Sorting.		
UNIT-III	INTERFACING WITH 8086	Classes: 09
8255 PPI, various modes of operation and interfacing to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086, DMA controller (8257).		
UNIT-IV	INTRODUCTION TO MICROCONTROLLERS	Classes: 09
Overview of 8051 microcontroller, Architecture, I/O ports, Memory organization, addressing modes and instruction set of 8051, Simple programs.		
UNIT-V	8051 REAL TIME CONTROL	Classes: 09
Programming Timer interrupts, programming external hardware interrupts, Programming the serial communication interrupts, programming 8051 timers and Counters.		
Text Books:		
1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006. 2. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning, 2010 3. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006. 4. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.		

Reference Books:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,
3. Microcontrollers and application, Ajay. V. Deshmukh, TMGH, 2005.

Web References:

1. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
2. <http://coen.boisestate.edu/smluo/smluo-courses/ece-332-microprocessors-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

E-Text Books:

1. <http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
2. <http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
3. https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf
4. https://e.edim.co/123389964/A.K._Ray_and_K.M._Bhurchandi-Advanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC Course

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

FUNDAMENTALS OF IMAGE PROCESSING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC66	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Know the Fundamentals of Digital Image Processing.
2. Apply the concepts in real time applications
3. Develop a code for various image processing concepts

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Know the Fundamentals of Digital Image Processing.
2. Apply the concepts in real time applications
3. Develop a code for various image processing concepts

UNIT-I	INTRODUCTION	Classes: 10
Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures.		
UNIT-II	IMAGE ENHANCEMENT	Classes: 12
Image Enhancements and Filtering-Gray level transformations, histogram equalization and specifications, pixel-domain smoothing filters – linear and order-statistics, pixel-domain sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass		
UNIT-III	IMAGE SEGMENTATION	Classes: 12
Image Segmentation- Detection of discontinuities, edge linking and boundary detection, thresholding – global and adaptive, region-based segmentation.		
Wavelets and Multi-resolution image processing- Uncertainty principles of Fourier Transform, Time-frequency localization, continuous wavelet transforms, wavelet bases and multi-resolution analysis, wavelets and Subband filter banks, wavelet packets.		
UNIT-IV	IMAGE COMPRESSION	Classes: 08
Image Compression-Redundancy–inter-pixel and psycho-visual; Lossless compression – predictive, entropy; Lossy compression- predictive and transform coding; Discrete Cosine Transform; Still image compression standards – JPEG and JPEG-2000.		
UNIT-V	VIDEO PROCESSING	Classes: 12
Fundamentals of Video Coding- Inter-frame redundancy, motion estimation techniques – full-search, fast search strategies, forward and backward motion prediction, frame classification – I, P and B; Video sequence hierarchy – Group of pictures, Elements of a video encoder and decoder; Video coding standards – MPEG and H.26X.		
Video Segmentation- Temporal segmentation–shot boundary detection, hard-cuts and soft-cuts; spatial segmentation – motion-based; Video object detection and tracking.		

Text Books:

1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008
2. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India.2nd edition 2004

Reference Books:

1. Murat Tekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015

Web References:

1. <http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf>

E-Text Books:

1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC

MOOC Course

1. <https://nptel.ac.in/courses/117105079/>

TELEVISION ENGINEERING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC67	OEC	L	T	P	C	CIA	SEE	Total
		4	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Know the fundamentals of Television transmitters and receivers
2. Understand different signal processing step monochrome television,
3. Introduce color television transmitters and receivers.
4. Remember the concepts of speech recognition and synthesis
5. Know the functions of the TV

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Know the fundamentals of Television transmitters and receivers
2. Understand different signal processing step monochrome television,
3. Introduce color television transmitters and receivers.
4. Remember various transmission and receiving antennas
5. Know the function of the TV

UNIT-I	INTRODUCTION	Classes: 14
TV transmitter and receivers, synchronization. Geometric form and aspect ratio, image continuity, interlaced scanning, picture resolution Composite video signal, TV standards. Camera tubes: image Orthicon Plumbicon, vidicon, silicon Diode Array vidicon, Comparison of camera tubes Monochrome TV camera, TV Signal Transmission and Propagation: Picture Signal transmission positive and negative modulation, VSB transmission, sound signal transmission, standard channel BW, TV transmitter, TV signal propagation interference, TV broadcast channels, TV transmission Antennas.		
UNIT-II	MONOCHROME TV RECEIVER	Classes: 12
RF tuner, IF subsystem, video amplifier, sound section, sync separation and processing, deflection circuits, scanning circuits, AGC, noise cancellation, video and inter carrier sound signal detection, vision IF subsystem of Black and White receivers, Receiver sound system: FM detection, FM Sound detectors, and typical applications.		
UNIT-III	SYNC SEPARATION AND DETECTION	Classes: 12
TV Receiver Tuners, Tuner operation, VHF and UHF tuners, digital tuning techniques, remote control of receiver functions. Sync Separation, AFC and Deflection Oscillators: Synchronous separation, k noise in sync pulses, separation of frame and line sync pulses. AFC, single ended AFC circuit, Deflection Oscillators, deflection drive IOs, Receiver Antennas, Picture Tubes.		
UNIT-IV	SPEECH RECOGNITION	Classes: 10

Large Vocabulary Continuous Speech Recognition: Architecture of large vocabulary continuous speech recognition system – acoustics and language models – n-grams, context dependent sub-word units; Applications and present status.

UNIT-V**SPEECH SYNTHESIS****Classes: 12**

Color Receiver: Introduction to colour receiver, Electron tuners, IF subsystem, Y-signal channel, Chroma decoder, Separation of U & V Color, Phasors, synchronous demodulators, Sub carrier generation, raster circuits.

Digital TV: Introduction to Digital TV, Digital Satellite TV, Direct to Home Satellite TV, Digital TV Transmitter, Digital TV Receiver, Digital Terrestrial TV, LCD TV, LED TV, CCD Image Sensors, HDTV.

Text Books:

1. Television and Video Engineering- A.M.Dhake, 2nd Edition.
2. Modern Television Practice Principles, Technology and Service R .R.Gallatin, New Age International Publication, 2002.
3. Monochrome and Colour TV- R.R. Gulati, New Age International Publication, 2002.

Reference Books:

1. Colour Television Theory and Practice-S.P.Bal, TMH, 1994.
2. Basic Television and Video Systems-B.Grob and C.E.Herndon, McGraw Hill, 1999.

Web References:

1. <http://www.sasurieengg.com/e-course-material/ECE/IV-Year%20Sem%207/EC2034%20Television%20and%20Video%20Engineering.pdf>

E-Text Books:

1. https://books.google.co.in/books?id=AvQAa5Zfuj0C&printsec=frontcover&dq=tv+by+rr+gulati&source=gbs_similarbooks_s&cad=1#v=onepage&q=tv%20by%20rr%20gulati&f=false

MOOC Course

1. <https://nptel.ac.in/courses/109103018/34>

INTRODUCTION TO SIGNAL PROCESSING

VIII Semester: OPEN ELECTIVE -IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC68	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1. Identify various signals and systems. 2. Understand various transformation techniques and apply them to find spectral domain representation of various signals. 3. Apply convolution and correlation operations between various signals and systems. 4. Develop the digital filter designs. 5. Understand Multirate digital signal processing concepts.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Identify various signals and systems. 2. Understand various transformation techniques and apply them to find spectral domain representation of various signals. 3. Apply convolution and correlation operations between various signals and systems. 4. Develop the digital filter designs. 5. Understand Multirate digital signal processing concepts.								
UNIT-I	TIME DOMAIN SIGNALS						Classes: 09	
Introduction, Definitions of signal and system, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sinc, Signum, Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.								
UNIT-II	SPECTRAL DOMAIN REPRESENTATION OF SIGNALS						Classes: 09	
Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transform, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.								
UNIT-III	CONVOLUTION AND CORRELATION						Classes: 09	
Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.								
UNIT-IV	DIGITAL FILTERS						Classes: 09	
IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method. FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.								
UNIT-V	MULTIRATE DIGITAL SIGNAL PROCESSING						Classes: 09	

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

Text Books:

1. John G. Proakis, Dimitris G. Manolakis, Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi , Signals, Systems & Communications ,BS Publications 2013 .
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

Reference Books:

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

Web References:

1. [http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-\(T.-OHaver\).html](http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-OHaver).html)
2. <https://www.sciencedirect.com/science/article/pii/B9780123965028000012>
3. [Applied Digital Signal Processing \(1st Edition\)](#)

E-Text Books:

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. [http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-\(Digital-Signal-Processing\).html](http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html)

MOOC Course

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

INTRODUCTION TO MOBILE COMMUNICATION

VIII Semester: OPEN ELECTIVE -IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC69	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to: 1.Understand impairments due to multi path fading channel. 2.Understand the fundamental techniques to overcome the different fading effects. 3.Understand Co-channel and Non Co channel interference 4. Familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Understand impairments due to multi path fading channel. 2. Understand the fundamental techniques to overcome the different fading effects. 3. Understand Co-channel and Non Co channel interference 4. Familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.								
UNIT-I	INTRODUCTION TO CELLULAR MOBILE RADIO SYSTEMS						Classes: 11	
Limitations of Conventional Mobile Telephone Systems, Basic Cellular Mobile System, First, Second, Third and Fourth Generation Cellular Wireless Systems, Uniqueness of Mobile Radio Environment- Fading -Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time. Fundamentals of Cellular Radio System Design: Concept of Frequency Reuse, Co-Channel Interference, Co-Channel Interference Reduction Factor, Desired C/I From a Normal Case in a Omni Directional Antenna System, System Capacity, Trunking and Grade of Service, Improving Coverage and Capacity in Cellular Systems- Cell Splitting, Sectoring, Microcell Zone Concept.								
UNIT-II	CO-CHANNEL INTERFERENCE						Classes: 09	
Measurement Of Real Time Co-Channel Interference, Design of Antenna System, Antenna Parameters and Their Effects, Diversity Techniques-Space Diversity, Polarization Diversity, Frequency Diversity, Time Diversity. Non-Co-Channel Interference: Adjacent Channel Interference, Near End Far End Interference, Cross Talk, Effects on Coverage and Interference by Power Decrease, Antenna Height Decrease, Effects of Cell Site Components.								
UNIT-III	MULTIPLE ACCESS & MODULATION SCHEMES						Classes: 10	
Signal Reflections in Flat And Hilly Terrain, Effect of Human Made Structures, Phase Difference Between Direct and Reflected Paths, Constant Standard Deviation, Straight Line Path Loss Slope, General Formula for Mobile Propagation Over V'later and Flat Open Area, Near and Long Distance Propagation, Path Loss From a Point to Point Prediction Model in Different Conditions, Merits of Lee Model. Cell Site and Mobile Antennas: Space Diversity Antennas, Umbrella Pattern Antennas, Minimum Separation of Cell Site Antennas, Mobile Antennas.								
UNIT-IV	FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT						Classes: 07	
Numbering And Grouping, Setup Access And Paging Channels, Channel Assignments to Cell Sites and Mobile Units, Channel Sharing and Borrowing, Sectorization, Overlaid Cells, Non Fixed Channel Assignment.								

UNIT-V	HANDOFFS AND DROPPED CALLS	Classes: 08
Handoff Initiation, Types of Handoff, Delaying Handoff, Advantages of Handoff, Power Difference Handoff, Forced Handoff, Mobile Assisted and Soft Handoff, inter system Handoff, Introduction to Dropped Call Rates and their Evaluation.		
Text Books:		
1. Mobile Cellular Telecommunications — W.C.Y. Lee, Mc Graw Hill, 2nd Edn., 1989. 2. Wireless Communications – Theodore. S. Rapport, Pearson Education, 2nd Edn., 2002. 3. Mobile Cellular Communication – Gottapu sashibhushana Rao, Pearson, 2012.		
Reference Books:		
1. Principles of Mobile Communications — Gordon L. Stuber, Springer International, 2nd Edn., 2001 2. Modern Wireless Communications-Simon Haykin, Michael Moher, Pearson Education, 2005 3. Wireless Communications Theory and Techniques, Asrar U. H .Sheikh, Springer, 2004.		
Web References:		
1. https://web.fe.up.pt/~mricardo/08_09/cmouv-mieic/slides/introduction.pdf		
E-Text Books:		
1. http://ee.sharif.edu/~pr.wireless.comm/references/Schwartz.pdf		
MOOC Course		
1. http://nptel.ac.in/courses/117102062/		

BASIC EMBEDDED SYSTEMS DESIGN**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC70	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-		30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

- 1.Explains about the Basic functions, Structure, Concepts and Applications of Embedded systems.
- 2.Build simple programs to solve well-defined problems on an Embedded platform
- 3.Understanding various communication interfaces available.
- 4.Identify different processors for various applications.
- 5.Explains about the tools used to develop in an embedded environment and Implement Real time applications on embedded platform

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explains about the Basic functions, Structure, Concepts and Applications of Embedded systems.
2. Build simple programs to solve well-defined problems on an Embedded platform
3. Understanding various communication interfaces available.
4. Identify different processors for various applications.
5. Explains about the tools used to develop in an Embedded environment and Implement Real time applications on Embedded platform

UNIT-I	INTRODUCTION TO EMBEDDED SYSTEMS	Classes: 12
Definition and Classification – Overview of Processors and hardware units in an embedded system – Software embedded into the system – Exemplary Embedded Systems – Embedded Systems on a Chip (SoC) and the use of VLSI designed circuits.		
UNIT-II	INTRODUCING THE 8051 MICROCONTROLLER FAMILY	Classes: 10
Introduction, What's in a name, The external interface of the Standard 8051, Reset requirements ,Clock frequency and performance, Memory issues, I/O pins, Timers, Interrupts, Serial interface, Power consumption ,Conclusions		
UNIT-III	DEVICES AND BUSES FOR DEVICES NETWORK	Classes: 12
I/O Devices - Device I/O Types and Examples – Synchronous - Iso-synchronous and Asynchronous Communications from Serial Devices - Examples of Internal Serial-Communication Devices - UART and HDLC - Parallel Port Devices - Sophisticated interfacing features in Devices/Ports- Timer and Counting Devices - '12C', 'USB', 'CAN' and advanced I/O Serial high speed buses- ISA, PCI, PCI-X, cPCI and advanced buses.		
UNIT-IV	PROGRAMMING CONCEPTS AND EMBEDDED PROGRAMMING IN C, C++	Classes: 14
Programming in assembly language (ALP) vs. High Level Language - C Program Elements, Macros and functions -Use of Pointers - NULL Pointers - Use of Function Calls – Multiple function calls in a Cyclic Order in the Main Function Pointers – Function Queues and Interrupt Service Routines Queues Pointers – Concepts of EMBEDDED PROGRAMMING in C++ - Objected Oriented Programming – Embedded Programming in C++, 'C' Program compilers – Cross compiler – Optimization of memory codes.		

UNIT-V	REAL TIME OPERATING SYSTEMS – PART - 1	Classes: 12
Tasks and task states, tasks and data, semaphores, and shared data; message queues, mailboxes and pipes, timer functions, events, memory management, interrupt routines in an RTOS environment. Semaphores, queues, hard real-time scheduling considerations, saving memory and power, an example RTOS like uC-OS (open source).		
Text Books:		
1. Rajkamal, Embedded Systems Architecture, Programming and Design, TATA McGraw-Hill, First reprint Oct. 2003 2. Shibu K.V, Introduction to Embedded Systems, Mc Graw Hill.		
Reference Books:		
1. Steve Heath, Embedded Systems Design, Second Edition-2003, Newnes, 2. David E.Simon, An Embedded Software Primer, Pearson Education Asia, First Indian Reprint 2000.		
Web References: 1. https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/notes/BEI605-%20Embedded-System.pdf		
E-Text Books: 1. http://read.pudn.com/downloads158/ebook/707037/Embedded%20Systems%20Design%20-%202ed%20-%200750655461.pdf		
MOOC Course 1. https://nptel.ac.in/courses/108102045/		

OPEN ELECTIVES OFFERED BY ELECTRICAL AND ELECTRONICS ENGINEERING DEPARTMENT

OE1		OE2	
A4EE52	Electrical Wiring and Safety Measures	A4EE55	Power Plant Engineering
A4EE53	Electrical Materials	A4EE56	Analysis of Linear Systems
A4EE54	New Trends Electrical Energy	A4EE57	Neural Networks and Fuzzy Logic
OE3		OE4	
A4EE58	Electrical Engineering Materials	A4EE61	Instrumentation and Control
A4EE59	Non-Conventional Power Generation	A4EE62	Energy Audit and Management Systems
A4EE60	Solar Energy and Applications	A4EE63	Energy Storage Systems

ELECTRICAL WIRING AND SAFETY MEASURES**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE52	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring

COURSE OUTCOMES:

The students should be able to

1. Know safety measures & state safety precautions.
2. Test single phase, three phase transformer, DC & AC machine as per IS.
3. Ascertain the condition of insulation & varnishing if necessary
4. Identify faults & measures to repair faults.

UNIT-I**BASICS OF ELECTRICAL INSTALLATIONS****Classes: 12**

Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.

UNIT-II**EARTHING****Classes: 08**

Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.

UNIT-III**SAFETY & PREVENTION OF ACCIDENTS****Classes: 08**

Definition of terminology used in safety , I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation . Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.

UNIT-IV**RESIDENTIAL BUILDING ELECTRIFICATION****Classes: 10**

General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub- circuits. Method of drawing single line diagram & wiring diagram.

UNIT-V**R ENDS& SUBSTATION AND METERS****Classes: 12**

Rule 28 : Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises.
 Rule 31: Cut-out on consumer's premises.
 Rule 46: Periodical inspection & testing of consumer's installation.
 Rule 47: Testing of consumer's installation.
 Rule 54: Declared voltage of supply to consumer.

Rule 55: Declared frequency of supply to consumer.
 Rule 56: Sealing of meters & cut-outs.
 Rule 77: Clearances above ground of the lowest conductor.
 Rule 79: Clearances between conductors & trolley wires.
 Rule 87: Lines crossing or approaching each other.
 Rule 88: Guarding.

Text Books:

1. K.B. Raina, S.K.Bhattacharya Electrical Design;Estimating and costing New Age International (p) Limited,
New Delhi Surjit Singh.
2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta
- 3.A course in Electrical Installation, Estimating & costing S.K.Kataria&sons,S.L. Uappal .
4. Electrical wiringEstimating and costing Khanna Publication. ,A.K.Sawhney

Reference Books:

1. Electrical Machine Design Danpat Rai & co.
2. The Electricity Rule 2005Universal Law Publishing Co. Pvt. Ltd. N. AlagapanS. Ekambaram
3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi ,Surjit Singh
4. TarlokSibgh Installation, Commissioning & Maintenance of Electrical Equipment S.K.Kataria& Sons
5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd Mumbai

Web References:

1. <https://electrical-engineering-portal.com> › Technical Articles
2. <https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/>
3. <https://www.cpwd.gov.in/Publication/Internal2013.pdf>

E-Text Books:

1. <https://books.google.co.in/books?isbn=0323170064>
2. <https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment>

MOOC Course

<https://nptel.ac.in/courses/103106071/5>
<https://nptel.ac.in/courses/108108099/28>
<https://nptel.ac.in/courses/124107001/>

ELECTRICAL MATERIALS

V Semester: OPEN ELECTIVE -I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE53	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES To understand about various electrical engineering material								
COURSE OUTCOMES: The students should be able to 1. Analyze the various engineering materials. 2. Application of various engineering materials.								
UNIT-I	CONDUCTORS						Classes: 10	
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.								
UNIT-II	SEMICONDUCTORS						Classes: 08	
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.								
UNIT-III	DIELECTRICS AND INSULATORS						Classes: 12	
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.								
UNIT-IV	MAGNETIC MATERIALS						Classes: 10	
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.								
UNIT-V	OPTICAL PROPERTIES OF SOLIDS						Classes: 10	
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).								

Text Books:

1. "Electrical Engineering Materials", Dekker, PHI Pbs.
2. "Electrical Engineering Materials", Indulkar, S.Chand

Reference Books:

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

Web References:

1. <https://physics.info/dielectrics/>
 2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>.
- web.mit.edu/course/6/6.732/www/6.732-pt2.pdf

E-Text Books:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

MOOC Course

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
3. https://onlinecourses.nptel.ac.in/noc18_ee14/

NEW TRENDS IN ELECTRICAL ENERGY**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE54	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. The fundamental principles of wireless power transfer (WPT) for cable-free transfer of power.
2. Theories for inductive power transfer (IPT) based on the coupled inductor model and low-order circuit compensation.
3. Specific converter topologies for lighting and battery charging applications.
4. Technology trends in the adoption of WPT for key consumer applications.

COURSE OUTCOMES:

The students should be able to

1. Understand the characteristics of power transfer through coupled inductors and the significance of leakage inductance.
2. Analyze and design appropriate compensation circuits and efficient power converters for WPT applications.
3. Understand technical requirements for applications involving solid-state loads and battery loads using WPT technologies.
4. Appreciate the factors affecting adoption of WPT in consumer applications including lightings, charging of smartphones and electric vehicles.

UNIT-I	BASIC CIRCUIT THEORY	Classes: 10
Review of transformers. Leakage inductance. Circuit compensation principles. Low-order compensations; series and parallel compensations. Resonance and operating frequency. Efficiency equation.		
UNIT-II	POWER CONVERTERS FUNDAMENTALS	Classes: 10
DC-DC converters. AC-DC converters and inverters. PWM and soft switching principles. Basic topologies with transformers. Input, output and transfer characteristics of power converters. Incorporation of leaky transformer. Control methods.		
UNIT-III	COMPENSATION CONFIGURATIONS	Classes: 10
Types of compensation for inductor power transfer. Characteristics for various termination requirements. Design for load-independence output voltage and output current. Efficiency optimization.		
UNIT-IV	APPLICATIONS	Classes: 10
Circuit requirements for various loading conditions. Characteristics of LED loads, resistors and battery loads. Appropriate compensation design. Lighting systems. Battery charging profiles. Electric vehicle charging. Energy efficiency metric for charging.		
UNIT-V	TECHNOLOGY TRENDS	Classes: 10
Demand for safe power transfer and durable operation. Portable and smart devices. Mobile communication devices. IoT devices and systems. Sensors. Solid-state lighting development. Battery technologies. Electric vehicle development. Renewable source integration trends. Future trends and demand for wireless power transfer.		

Text Books:

1. C. T. Rim and C. Mi, Wireless Power Transfer for Electric Vehicles and Mobile Devices, New York: IEEE Press-Wiley, 2017.
2. J. I. Agbinya, Wireless Power Transfer, River Publishers, 2015

Reference Books:

1. Z. Huang, S. C. Wong, and C. K. Tse, "Design of a single-stage inductive power-transfer converter for efficient EV battery charging," IEEE Transactions on Vehicular Technology, vol. 66, no. 7, pp. 5808-5821, July 2017.
2. L. Xu, Q. Chen, X. Ren, S. C. Wong, and C. K. Tse, "Self-oscillating resonant converter with contactless power transfer and integrated current sensing transformer," IEEE Transactions on Power Electronics, vol. 32, no. 6, pp. 4839-4851, June 2017.
3. W. Zhang, S. C. Wong, C. K. Tse, and Q. Chen, "Load-independent duality of current and voltage outputs of a series or parallel compensated inductive power transfer converter with optimized efficiency," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, pp. 137-146, March 2015.
4. J. Hou, Q. Chen, X. Ren, X. Ruan, S. C. Wong, and C. K. Tse, "Precise characteristics analysis of series/series-parallel compensated contactless resonant converter," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, 101-110, March 2015.

Web References:

1. https://www.researchgate.net/.../267512972_Wireless_Power_Transmission_Trends
2. <https://ieeexplore.ieee.org/document/6403386>
3. <https://www.cse.wustl.edu/~jain/cse574-14/ftp/power/>

E-Text Books:

1. https://books.google.co.in/books/about/Wireless_Power_Transfer.html?id...
2. <https://www.intechopen.com/books/wireless-power-transfer-principles-and-engineering-explorations>

MOOC Course:

1. <https://ocw.mit.edu/.../6-452-principles-of-wireless-communications-spring-2006/>
2. www.iitg.ac.in/e_mobility/WPT.html
3. <https://nptel.ac.in/courses/117102062/>
4. <https://nptel.ac.in/courses/117106108/150>
5. <https://nptel.ac.in/courses/117104099/2>

POWER PLANT ENGINEERING**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring

COURSE OUTCOMES:

The students should be able to

1. Know safety measures & state safety precautions.
2. Test single phase, three phase transformer, DC & AC machine as per IS.
3. Ascertain the condition of insulation & varnishing if necessary
4. Identify faults & measures to repair faults.

UNIT-I	INTRODUCTION TO THE SOURCES OF ENERGY	Classes: 10
Resources and Development of Power in India. Steam Power Plant: Plant Layout, Working of different Circuits, Fuel and handling equipment, types of coals, coal handling, choice of handling equipment, coal storage, Ash handling systems. Combustion Process: Properties of coal — overfeed and underfeed fuel beds, traveling grate stokers, spreader stokers, retort stokers, pulverized fuel burning system and its components, combustion needs and drought system, cyclone furnace, design and construction, Dust collectors, cooling towers and heat rejection. Corrosion and feed water treatment.		
UNIT-II	INTERNAL COMBUSTION ENGINE PLANT	Classes: 10
DIESEL POWER PLANT: Introduction — IC Engines, types, construction— Plant layout with auxiliaries fuel supply system, air starting equipment, lubrication and cooling system—super charging. Gas Turbine Plant: Introduction — classification — construction — Layout with auxiliaries — Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison. Direct Energy Conversion: Solar energy, Fuel cells, Thermo electric and Thermo ionic, MHD generation.		
UNIT-III	HYDRO ELECTRIC POWER PLANT	Classes: 08
Water power — Hydro logical cycle / flow measurement — drainage area characteristics — Hydro graphs — storage and Poundage — classification of dams and spill ways. Hydro Projects And Plant: Classification — Typical layouts — plant auxiliaries — plant operation pumped storage plants. Power From Non-Conventional Sources: Utilization of Solar- Collectors- Principle of Working, Wind Energy — types — HAWT, VAWT Tidal Energy.		
UNIT-IV	NUCLEAR POWER STATION	Classes: 10
Nuclear fuel — breeding and fertile materials — Nuclear reactor — reactor operation. Types of Reactors: Pressurized water reactor, Boiling water reactor, sodium-graphite reactor, fast Breeder Reactor, homogeneous Reactor, Gas cooled Reactor, Radiation hazards and shielding radioactive waste disposal.		

UNIT-V	POWER PLANT ECONOMICS AND ENVIRONMENTAL CONSIDERATIONS	Classes: 12
Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration curve. Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor — related exercises. Effluents from power plants and Impact on environment — pollutants and pollution standards — Methods of Pollution control.		
Text Books:		
1. Power Plant Engineering! P.C.Sharma / S.K.Kataria Pub. 2. A Course in Power Plant Engineering: I Arora and S. Domkundwar.		
Reference Books:		
1. A Text Book of Power Plant Engineering I Rajput I Laxmi Publications. 2. Power Plant Engineering: P.K.Nag/ II Edition /TMH. 3. An Introduction to Power Plant Technology I G.D. Rai/Khanna Publishers. 4. Power plant Engg I Elanchezhian/ I.K. International Pub. 5. Power plant Engineering! Ramalingam / Scietech Publishers.		
Web References:		
1. www.cbspd.co.in/power-plant-engineering-9788123919461-black-veatch.html 2. https://www.semcomaritime.com/en-en/references/power-generation		
E-Text Books:		
1. https://www.intechopen.com/books/subject/power-plant-engineering 2. https://books.google.co.in/books?isbn=1482221519		
MOOC Course		
1. https://nptel.ac.in/courses/108105058/8 2. www.nptelvideos.in/2012/11/energy-resources-and-technology.html 3. www.ignou.ac.in/upload/Unit-2-58.pdf		

ANALYSIS OF LINEAR SYSTEMS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To develop ability to analyze linear systems and signals 2. To develop critical understanding of mathematical methods to analyze linear systems and signals								
COURSE OUTCOMES: The students should be able to 1. Use mathematical modeling tools to represent linear systems 2. Use mathematical modeling tools to analyze linear systems								
UNIT-I	STATE VARIABLE ANALYSIS						Classes: 08	
Choice of state variables in Electrical networks-Formulation of state equations for Electrical networks Equivalent source method. Network topological method – Solution of state equations-Analysis of simple networks with state variable approach.								
UNIT-II	FOURIER SERIES AND FOURIER TRANSFORM REPRESENTATION						Classes: 12	
Introduction, Trigonometric form of Fourier series, Exponential form of Fourier series, Wave symmetry, Fourier integrals and transforms, Fourier transform of a periodic function , Properties of Fourier Transform , Parseval's theorem , Fourier transform of some common signals, Fourier transform relationship with Laplace Transform. Applications of Fourier series and Fourier Transform Representation: Introduction, Effective value, and average values of non sinusoidal periodic waves, currents, Power Factor, Effects of harmonics, Application in Circuit Analysis, Circuit Analysis using Fourier Series.								
UNIT-III	LAPLACE TRANSFORM APPLICATIONS						Classes: 10	
Application of Laplace transform Methods of Ananalysis – Response of RL, RC, RLC Networks to Step, Ramp, and impulse functions, Shifting Theorem – Convolution Integral – Applications Testing of Polynomials: Elements of realisability – Hurwitz polynomials-positive real functions-Properties-Testing-Sturm's Test, examples. Network Synthesis: Network synthesis: Synthesis of one port LC networks-Foster and Cauer methods-Synthesis of RL and RC one port networks-Foster and Cauer methods								
UNIT-IV	SAMPLING						Classes: 10	
Sampling theorem – Graphical and Analytical proof for Band Limited Signal impulse sampling, natural and Flat top Sampling, Reconstruction of signal from its samples, effect of under sampling – Aliasing, introduction to Band Pass sampling, Cross correlation and auto correlation of functions, properties of correlation function, Energy density spectrum, Power density spectrum, Relation between auto correlation function and Energy / Power spectral density function.								
UNIT-V	Z-TRANSFORMS						Classes: 10	
Fundamental difference between continuous and discrete time signals, discrete time complex, exponential and sinusoidal signals, periodicity of discrete time complex exponential, concept of Z Transform of a discrete sequence. Distinction between Laplace, Fourier, and Z-Transforms. Region of convergence in Z-Transforms, constraints on ROC for various classes of signals, Inverse Z-Transform properties of Z-Transforms.								

Text Books:

1. B. P. Lathi, "Signals, Systems and Communications", BS Publications 2003.
2. "Umesh Sinha" "Network Analysis and Synthesis", Satya Prakashan Publications, 2013.

Reference Books:

1. "A. N. Tripathi", "Linear System Analysis", New Age International, 2nd Edition 1987.
2. "D. Roy Chowdhary", "Network and Systems", New Age International, 2005.
3. "Gopal G Bhise, Prem R. Chadha", Engineering Network Analysis and Filter Design, Umesh Publications 2009.
4. "A. Cheng", linear system analysis, Oxford publishers, 1999

Web References:

1. <https://archive.org/details/introductiontoli00brow>
2. <https://ieeexplore.ieee.org/iel5/9/24171/01101971.pdf>

E-Text Books:

1. www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_04Mar10.pdf
2. <https://www.springer.com/gp/book/9780387975733>

MOOC Course:

1. www.nptelvideos.in/2012/11/estimation-of-signals-and-systems.html
2. <https://nptel.ac.in/courses/108104100/6>

NEURAL NETWORKS AND FUZZY LOGIC**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

COURSE OUTCOMES:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback network.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	Classes: 12
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	Classes: 10
Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training. Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	Classes: 08
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		
UNIT-IV	ASSOCIATIVE MEMORIES-II	Classes: 10
Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall		

Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield		
Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		
UNIT-V	FUZZY LOGIC	Classes: 10
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.		
Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.		
Text Books:		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.		
Reference Books:		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Haykin, Pearson Education. 8. Neural Engineering, C. Elasmith and CH. Anderson, PHI.		
Web References:		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		
E-Text Books:		
1. https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book 2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf 3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf		
MOOC Course:		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07 2. https://nptel.ac.in/courses/108104049/16 3. https://nptel.ac.in/courses/117105084/ 4. https://nptel.ac.in/syllabus/127105006		

ELECTRICAL ENGINEERING MATERIALS

VII Semester: OPEN ELECTIVE -III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES								
To understand about various electrical engineering material								
COURSE OUTCOMES:								
The students should be able to								
1. Analyze the various engineering materials.								
2. Application of various engineering materials.								
UNIT-I	CONDUCTORS						Classes: 10	
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.								
UNIT-II	SEMICONDUCTORS						Classes: 08	
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.								
UNIT-III	DIELECTRICS AND INSULATORS						Classes: 12	
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.								
UNIT-IV	MAGNETIC MATERIALS						Classes: 10	
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.								
UNIT-V	OPTICAL PROPERTIES OF SOLIDS						Classes: 10	
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).								

Text Books:

1. "Electrical Engineering Materials", Dekker, PHI Pbs.
2. "Electrical Engineering Materials", Indulkar, S.Chand

Reference Books:

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

Web References:

1. <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf

E-Text Books:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

MOOC Course

3. 1. <https://nptel.ac.in/courses/108108076/>
4. 2. <https://nptel.ac.in/courses/112104203/3>
3. https://onlinecourses.nptel.ac.in/noc18_ee14/

NON-CONVENTIONAL POWER GENERATION**VII Semester: OPEN ELECTIVE -III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. Demonstrate power generation systems including major subsystems.
2. Understand basic working principles of nuclear power generation systems.
3. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

COURSE OUTCOMES:

The students should be able to

1. Understand basic working principles of nuclear power generation systems.
2. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

UNIT-I**PRINCIPLES OF SOLAR RADIATION****Classes: 08**

Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, the solar constant, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and Sun shine, solar radiation data.

UNIT-II**SOLAR ENERGY COLLECTORS****Classes: 10**

Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

STORAGE AND APPLICATIONS: Different methods of solar energy storage, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating /cooling technique, solar distillation and drying.

UNIT-III**PHOTO VOLTAICS (PV)****Classes: 12**

Fundamentals of solar cells, types of solar cells, semiconducting materials, band gap theory, absorption of photons, excitations and photoemission of electrons, band engineering.

PV CELL PROPERTIES: Solar cell properties and design, p-n junction photodiodes, depletion region, electrostatic field across the depletion layer, electron and holes transports, device physics, charge carrier generation, recombination and other losses, I-V characteristics, output power.

UNIT-IV**SOLAR CELL APPLICATIONS****Classes: 10**

PV cell interconnection, module structure and module fabrication, Equivalent circuits, load matching, efficiency, fill factor and optimization for maximum power, Design of stand-alone PV systems, system sizing, device structures, device construction, DC to AC conversion, inverters, on-site storage and grid connections.

UNIT-V	COST ANALYSIS AND ENVIRONMENTAL ISSUES	Classes: 10
Cost analysis and pay back calculations for different types of solar panels and collectors, installation and operating costs, Environmental and safety issues, protection systems, performance monitoring. ALTERNATIVE ENERGY SOURCES: Solar Energy: Types of devices for Solar Energy Collection, Thermal Storage System, Control Systems, Wind Energy, Availability, Wind Devices, Wind Characteristics, Performance of Turbines and systems.		
Text Books:		
1. G. D. Rai (2009), Non-Conventional Energy Sources, 4th edition, Khanna Publishers, New Delhi. 2. Martin A. Green (2008), Solar Cells: Operating Principles, Technology and system Applications, 1st edition, Prentice Hall, New Delhi.		
Reference Books:		
1. Sukatme (2008), Solar Energy, 3rd Edition, McGraw Hill Companies, New Delhi. 2. D. Yogi goswami, Frank Kreith, Jan F. Kreider (2000), Principles of Solar Engineering, 2nd edition, Taylor & Francis, USA.		
Web References:		
1. https://nptel.ac.in/courses/Webcourse-contents/IISc-BANG/notused/Non-Conventional%20Energy%20Systems-/Learning%20Material%20-%20NCES.pdf 2. https://www.ukessays.com/essays/environmental-sciences/promoting-non-conventional-energy-resource		
E-Text Books:		
1. bie.telangana.gov.in/files/Nonconventionalenergysources.pdf 2. https://www.researchgate.net/.../236857095_Non_Conventional_Energy_Resources		
MOOC Course		
1. https://nptel.ac.in/courses/108102047/ 2. https://onlinecourses.nptel.ac.in/noc19_ge11/course 3. https://nptel.ac.in/downloads/108108078/		

SOLAR ENERGY AND APPLICATIONS

VII Semester: OPEN ELECTIVE -III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE60	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To Study about solar modules and PV system design and their applications 2. To Deal with grid connected PV systems 3. To Discuss about different energy storage systems								
COURSE OUTCOMES: The students should be able to 1. Understanding on solar energy storage systems 2. Basic knowledge on standalone PV system 3. Understand the issues in grid connected PV systems 4. Study about the modelling of different energy storage systems and their performances 5. Attain more on different applications of solar energy								
UNIT-I	INTRODUCTION						Classes: 10	
Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection								
UNIT-II	STAND ALONE PV SYSTEM						Classes: 10	
Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing								
UNIT-III	GRID CONNECTED PV SYSTEMS						Classes: 10	
PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs								
UNIT-IV	ENERGY STORAGE SYSTEMS						Classes: 10	
Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage								
UNIT-V	APPLICATIONS						Classes: 10	
Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.								
Text Books: 1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015. 2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994								
Reference Books: 1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011. 2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 S.P. Sukhatme , "Solar Energy", Tata McGraw Hill,1987.								

Web References:

1. <https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798>
3. <https://link.springer.com/journal/11949>

E-Text Books:

1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf
2. bookstore.teri.res.in/books/9788179935736

MOOC Course:

1. <https://nptel.ac.in/courses/112105051/>
2. <https://nptel.ac.in/courses/121106014/18>
3. <https://nptel.ac.in/courses/112105050/>
4. <https://nptel.ac.in/syllabus/112105051/>

INSTRUMENTATION AND CONTROL**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

COURSE OUTCOMES:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	Classes: 10
Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.		
UNIT-II	POTENTIOMETERS & INSTRUMENT TRANSFORMERS	Classes: 08
Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications. CT and PT — Ratio and phase angle errors.		
UNIT-III	TRANSDUCERS & OSCILLOSCOPES	Classes: 12
Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes. CRO:Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.		
UNIT-IV	CLASSIFICATION OF CONTROL SYSTEMS	Classes: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences-Different examples of control systems- Classification of control systems Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse		

Response and transfer functions . Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.

UNIT-V**TIME RESPONSE ANALYSIS****Classes: 10**

Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.

Text Books:

1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd.
2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications.
3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009
4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.

Reference Books:

1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications.
2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd.
3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications.
4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rd Edition, 1998.
5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011.
6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.

Web References:

1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf
2. https://www.mccdaq.com/handbook/chapt_4.aspx

E-Text Books:

1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf
2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf
3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf

MOOC Course:

1. <https://nptel.ac.in/syllabus/108106070/>
2. <https://nptel.ac.in/courses/108105064/>
3. <https://nptel.ac.in/courses/108101037/>
4. <https://nptel.ac.in/courses/107106081/>

ENERGY AUDIT AND MANAGEMENT SYSTEMS**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To understand energy efficiency, scope, conservation and technologies.
2. To design energy efficient lighting systems.
3. To estimate/calculate power factor of systems and propose suitable compensation techniques.
4. To understand energy conservation in HVAC systems.
5. To calculate life cycle costing analysis and return on investment on energy efficient technologies.

COURSE OUTCOMES:

The students should be able to

1. Explain energy efficiency, conservation and various technologies.
2. Design energy efficient lighting systems.
3. Calculate power factor of systems and propose suitable compensation techniques.
4. Explain energy conservation in HVAC systems.
5. Calculate life cycle costing analysis and return on investment on energy efficient technologies.

UNIT-I	ENERGY AUDIT AND MANAGEMENT ENERGY AUDIT	Classes: 12
Basic Principles of Energy Audit and management Energy audit – Definitions – Concept – Types of audit – Energy index – Cost index – Pie charts – Sankey diagrams – Load profiles – Energy conservation schemes and energy saving potential – Numerical problems – Principles of energy management – Initiating, planning, controlling, promoting, monitoring, reporting – Energy manager – Qualities and functions – Language – Questionnaire – Check list for top management.		
UNIT-II	LIGHTING	Classes: 10
Modification of existing systems – Replacement of existing systems – Priorities: Definition of terms and units – Luminous efficiency – Polar curve – Calculation of illumination level – Illumination of inclined surface to beam – Luminance or brightness – Types of lamps – Types of lighting – Electric lighting fittings (luminaries) – Flood lighting – White light LED and conducting Polymers – Energy conservation measures.		
UNIT-III	POWER FACTOR AND ENERGY INSTRUMENTS	Classes: 08
Power factor – Methods of improvement – Location of capacitors – Power factor with non linear loads – Effect of harmonics on Power factor – Numerical problems. Energy Instruments – Watt-hour meter – Data loggers – Thermocouples – Pyrometers – Lux meters – Tong testers – Power analyzer.		

UNIT-IV	SPACE HEATING AND VENTILATION	Classes: 08
Ventilation – Air–Conditioning (HVAC) and Water Heating: Introduction – Heating of buildings – Transfer of Heat–Space heating methods – Ventilation and air–conditioning – Insulation–Cooling load – Electric water heating systems – Energy conservation methods.		
UNIT-V	ECONOMIC ASPECTS AND ANALYSIS	Classes: 12
Economics Analysis – Depreciation Methods – Time value of money – Rate of return – Present worth method – Replacement analysis – Life cycle costing analysis – Energy efficient motors (basic concepts) Computation of Economic Aspects : Calculation of simple payback method – Net present worth method – Power factor correction – Lighting – Applications of life cycle costing analysis – Return on investment		
Text Books:		
1. Energy management by W.R. Murphy & G. McKay Butter worth, Elsevier publications. 2012 2. Energy efficient electric motors by John .C. Andreas, Marcel Dekker Inc Ltd–2nd edition, 1995		
Reference Books:		
1. Electric Energy Utilization and Conservation by S C Tripathy, Tata McGraw hill publishing company Ltd. New Delhi. 2. Energy management by Paul o' Callaghan, Mc–Graw Hill Book company–1st edition, 1998. 3. Energy management hand book by W.C.Turner, John wiley and sons. 4. Energy management and conservation –k v Sharma and pvenkateshaiah-I K International Publishing House pvt.ltd,2011.		
Web References:		
1. https://beeindia.gov.in/sites/default/files/1Ch3.pdf 2. https://www.researchgate.net/publication/309463130_Energy_Audit_Management		
E-Text Books:		
1. lab.fs.uni-lj.si/kes/erasmus/Energy%20Management%20Handbook.pdf 2. https://fenix.tecnico.ulisboa.pt/downloadFile/.../Energy%20Audit%202013.pdf 3. enersupply.euke.sk/wp-content/uploads/Introduction-to-EM-and-EA_P1_Fotini.pdf		
MOOC Course:		
1. https://nptel.ac.in/courses/108106022 2. https://nptel.ac.in/courses/105107156/12 3. shodhganga.inflibnet.ac.in/bitstream/10603/46067/11/11_chapter%201.pdf		

ENERGY STORAGE SYSTEMS

VIII Semester: OPEN ELECTIVE -IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE63	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES 1. To enable the student to understand the need for energy storage, devices and technologies available and their applications								
COURSE OUTCOMES: The students should be able to 1. Analyze the characteristics of energy from various sources and need for storage 2. Classify various types of energy storage and various devices used for the purpose 3. Identify various real time applications.								
UNIT-I	ELECTRICAL ENERGY STORAGE TECHNOLOGIES						Classes: 12	
Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.								
UNIT-II	NEEDS FOR ELECTRICAL ENERGY STORAGE						Classes: 08	
Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.								
UNIT-III	FEATURES OF ENERGY STORAGE SYSTEMS						Classes: 08	
Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES),Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H2),Synthetic natural gas (SNG).								
UNIT-IV	TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS						Classes: 10	
Electrical storage systems, Double-layer capacitors (DLC) ,Superconducting magnetic energy storage (SMES),Thermal storage systems, Standards for EES, Technical comparison of EES technologies.								
UNIT-V	APPLICATIONS						Classes: 12	
Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications ,Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems , Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA– aggregation of many dispersed batteries.								
Text Books: 1. “James M. Eyer, Joseph J. Iannucci and Garth P. Corey “, “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004. 2. The Electrical Energy Storage by IEC Market Strategy Board.								

Reference Books:

1. "Jim Eyer, Garth Corey", Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.

Web References:

1. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212>

E-Text Books:

1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf
2. <https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf>
3. <https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf>

MOOC Course:

1. nptel.ac.in/courses/112105221/56
2. nptel.ac.in/courses/108108036/9
3. <https://nptel.ac.in/courses/108102047/7>
4. <https://nptel.ac.in/courses/108105058/37>

OPEN ELECTIVES OFFERED BY INFORMATION TECHNOLOGY DEPARTMENT

OE1		OE2	
A4IT11	Fundamentals of Data Structures	A4IT14	Basics of Python Programming
A4IT12	Software Engineering Principles	A4IT15	Human Computer Interaction
A4IT13	Computer Organization and Operating Systems	A4IT16	Software Testing Fundamentals
OE3		OE4	
A4IT17	E- Commerce	A4IT20	Introduction to Mobile Application Development
A4IT18	Cyber Forensics	A4IT21	Distributed Databases
A4IT19	Discrete Structures	A4IT22	Big Data

FUNDAMENTALS OF DATA STRUCTURES

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT11	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Impart the basic concepts of data structures and algorithms.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching and hashing.
6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate algorithms in terms of time and memory complexity.
2. Formulate new solutions for problems or improve existing code using data structures and algorithms.
3. Implement basic data structures such as arrays, linked lists, stacks and queues.
4. Solve problem involving graphs, trees and heaps
5. Apply Algorithms for solving problems like sorting, searching, and hashing.
6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

UNIT-I	INTRODUCTION TO DATA STRUCTURES	Classes: 12
Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations Introduction to Linear and Non Linear data structures-Singly Linked Lists- Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion.		
UNIT-II	STACKS AND QUEUES	Classes: 10
Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation. Queues-Queue ADT, definition and operations, array and linked Implementations in C, Circular queues-Insertion and deletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.		
UNIT-III	TREESANDGRAPHS	Classes: 14
Trees – Terminology, Representation of Trees, binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals. Max Priority Queue-ADT- implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from		

a Max Heap. Graphs , Introduction, Definition, Terminology, Graph ADT, Graph Representations- Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.

UNIT-IV**SEARCHING AND SORTING****Classes: 12**

Searching- Linear Search, Binary Search, Comparison of search techniques. **Sorting**-Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.

UNIT-V**BINARY SEARCH TREES****Classes: 12**

Search Trees-Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees- Definition and Examples, Insertion into an AVL Tree, B-Trees, Definition, B-Tree of order m, operations-Insertion and Searching, Comparison of Search Trees.

Pattern matching algorithm- The Knuth-Morris-Pratt algorithm, Tries (examples only)

Text Books:

1. Fundamentals of Data Structures II, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and Susan Anderson-Freed, Universities Press

Reference Books:

1. Algorithms, Data Structures, and Problem Solving with C++, Illustrated Edition by Mark Allen Weiss, Addison- Wesley Publishing Company
2. How to Solve it by Computer II, 2nd Impression by R.G.Dromey, Pearson Education

SOFTWARE ENGINEERING PRINCIPLES

V Semester: OPEN ELECTIVE - I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT12	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES To learn 1. Be familiar with basic Software engineering methods and practices, and its applications. 2. Apply the implementation of software engineering layered technology and Process framework. 3. Understand the concepts of software measurement and software risks. 4. Be familiar with software requirements and the SRS documents. 5. Be familiar with role of project management including planning, scheduling, risk management.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Apply basic knowledge and understanding of the analysis, synthesis and design of complex systems. 2. Apply software engineering principles and techniques. 3. Design and evaluate large-scale software systems. 4.. Explain the principles, tools and practices of IT project management. 5. Apply the fundamental knowledge of science in emerging technologies.								
UNIT-I	INTRODUCTION TO SOFTWARE ENGINEERING						Classes: 11	
The evolving role of software, Changing Nature of Software, Software myths. A Generic view of process: Software engineering- A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models.								
UNIT-II	PROCESS MODELS						Classes: 17	
The waterfall model, Incremental process models, Evolutionary process models, The Unified process. Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document. Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.								
UNIT-III	SYSTEM MODELS						Classes: 15	
Context Models, Behavioral models, Data models, Object models, structured methods. Design Engineering: Design process and Design quality, Design concepts, the design model. Creating an architectural design: Software architecture, Data design, Architectural styles and patterns, Architectural Design.								
UNIT-IV	OBJECT-ORIENTED DESIGN						Classes: 17	
Objects and object classes, An Object-Oriented design process, Design evolution. Performing User interface design: Golden rules, User interface analysis and design, interface								

analysis, interface design steps, Design evaluation.

Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging.

UNIT-V

RISK MANAGEMENT

Classes: 16

Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan.

Quality Management: Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

Text Books:

1. Software Engineering a practitioner's Approach, Roger S Pressman, 6th edition. Mc. Graw Hill International Edition.
2. Software Engineering, Ian Somerville, 7th edition, Pearson education.

Reference Books:

1. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010.
2. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw-Hill, 2008
3. Fundamentals of Software Engineering, Rajib Mall, PHI, 2005

COMPUTER ORGANIZATION AND OPERATING SYSTEMS

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT13	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To understand the basic structure and operation of a digital computer.
2. To discuss in detail the operation of the arithmetic unit including the algorithms & implementation of fixed-point and floating-point addition, subtraction, multiplication & division.
3. To study different ways of communicating with I/O devices and standard I/O interfaces.
4. To study hierarchical memory system including cache memories and virtual memory.
5. To demonstrate the knowledge of functions of operating system memory management scheduling, file system and interface, distributed systems, security and dead locks.
6. To implement a significant portion of an Operating System

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain different types of Addressing modes
2. Explain different synchronous and asynchronous data transfer techniques.
3. Explain different I/O data transfer techniques with performance comparison.
4. Outline various concepts and features of Operating systems.

UNIT-I	BASIC STRUCTURE OF COMPUTERS	Classes: 24
Computer Types, Functional UNIT, Basic Operational Concepts, Bus Structures, Software, Performance, Multiprocessors and Multi Computers, Data Representation, Fixed Point Representation, Floating - Point Representation. Register Transfer Language and Micro Operations: Register Transfer Language, Register Transfer Bus and Memory Transfers, Arithmetic Micro Operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit, Instruction Codes, Computer Registers Computer Instructions - Instruction Cycle. Memory - Reference Instructions, Input - Output and Interrupt, STACK Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation.		
UNIT-II	MICRO PROGRAMMED CONTROL	Classes: 13
Control Memory, Address Sequencing, Micro Program Examples, Design of Control Unit, Hard Wired Control, Micro Programmed Control. Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer Modes, Priority Interrupt, Direct Memory Access, Input-Output Processor (IOP).		
UNIT-III	OPERATING SYSTEMS OVERVIEW	Classes: 09
Overview of Computer Operating Systems Functions, Protection and Security, Distributed Systems, Special Purpose Systems, Operating Systems Structures-Operating System Services and Systems		

Calls, System Programs, Operating System Generation.

Principles of Deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

UNIT-IV**MEMORY MANAGEMENT****Classes: 13**

Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation, Virtual Memory, Demand Paging, Page-Replacement Algorithms, Allocation of Frames, Thrashing Case Studies -UNIX, Linux, Windows.

UNIT-V**PRINCIPLES OF DEADLOCK****Classes: 15**

System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

File System Interface: The Concept of a File, Access Methods, Directory Structure, File System Mounting, File Sharing, Protection. Allocation Methods, Free-Space Management.

Text Books:

1. Computer Organization - Carl Hamacher, Zvonks Vranesic, SafeaZaky, 5th Edition, McGraw Hill.
2. Computer System Architecture - M. moris mano, 3rd edition, Pearson
3. Operating System Concepts - Abreham Silberchatz, Peter B. Galvin, Greg Gagne, 8th Edition, John Wiley.

Reference Books:

1. Computer Organization and Architecture - William Stallings 6th Edition, Pearson
2. Structured Computer Organization - Andrew S. Tanenbaum, 4th Edition, PHI
3. Fundamentals of Computer Organization and Design - Sivaraama Dandamudi, Springer Int. Edition
4. Operating Systems - Internals and Design Principles, Stallings, 6th Edition - 2009, Pearson Education.
5. Modern Operating Systems, Andrew S Tanenbaum 2nd Edition, PHI
6. Principles of Operating System, B. L. Stuart, Cengage Learning, India Edition

BASICS OF PYTHON PROGRAMMING**VI Semester: OPEN ELECTIVE - II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT14	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To Describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To Describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methods through use of these in the context of a Python application.

COURSE OUTCOMES:

Up on successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

UNIT-I	INTRODUCTION TO PYTHON	Classes: 08
Overview, Basic. Python Installation, Comments in Python, Concept of Indentation in python.		
UNIT-II	DATA TYPES	Classes: 08
Tuples, Lists More advanced data types (dictionary, string), Python operators, control flows, Loops, Functions.		
UNIT-III	MODULES AND PACKAGES	Classes: 08
File handling in python, Module and Packages, Object oriented programming.		
UNIT-IV	ADVANCED FUNCTIONS	Classes: 08
Exceptions, sorting, advanced function: map, filter, and reduce.		
UNIT-V	INTRODUCTION TO STANDARD LIBRARIES	Classes: 08
Multi-Processing And Multi-Threading, Introduction To Standard Libraries (pandas,Turtle, numpy, os).		

Text Books:

1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013.

Reference Books:

1. Python Programming for Beginners: Python Programming Languageby Joseph Joyner

HUMAN COMPUTER INTERACTION**VI Semester: OPEN ELECTIVE - II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT15	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain Computer components functions regarding interaction with human
3. Demonstrate Understanding of Interaction between the human and computer components.
4. Use Paradigms
5. Implement Interaction design basics
6. Use HCI in the software process

UNIT-I	INTRODUCTION	Classes: 12
Importance of user Interface – definition, importance of good design, Benefits of good design. A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics-Principles of user interface.		
UNIT-II	DESIGN PROCESS	Classes: 15
Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.		
UNIT-III	WINDOWS	Classes: 12
New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.		

UNIT-IV	SOFTWARE TOOLS	Classes: 08
Specification methods, interface – Building Tools.		
UNIT-V	INTERACTION DEVICES	Classes: 08
Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.		
Text Books:		
1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech. 2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.		
Reference Books:		
1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEALG, PEARSON. 2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech, 3. User Interface Design, Soren Lauesen , Pearson Education.		

SOFTWARE TESTING FUNDAMENTALS

VI Semester: OPEN ELECTIVE - II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT16	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES: To learn Software Testing has different goals and objectives. 1. Finding defects which may get created by the programmer while developing the software. 2. Gaining confidence in and providing information about the level of quality. 3. To prevent defects.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Understand software testing methods and apply various testing techniques 2. Design and conduct a software test process for a software testing project 3. Designing solutions for various software testing problems by selecting appropriate software test model 4. Implement various practice oriented software testing projects								
UNIT-I	INTRODUCTION						Classes: 08	
Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.								
UNIT-II	TESTING METHODOLOGIES						Classes: 12	
White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.								
UNIT-III	INTEGRATION TESTING						Classes: 10	
Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing								
UNIT-IV	TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING						Classes: 10	
Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.								

UNIT-V	TEST MANAGEMENT AND AUTOMATION TEST PLANNING	Classes: 10
Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool framework, Test tool selection.		
Text Books:		
1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education. 2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.		
Reference Books:		
1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley 2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.		

E - COMMERCE

VII Semester: OPEN ELECTIVE - III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT17	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES To learn 1. To become familiar with state of the art Electronic Model, Payment Mechanisms. 2. To understand the basic principal of E-Government, Securities, Supply Chain Mechanisms. 3. To Evaluate and observe various online businesses Management.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Understand how the E-Commerce is done in the IT environment when the data is huge and in a variety of formats (e.g. multi media) like on the Internet and in digital libraries. 2. To enable them to use the search and retrieval techniques taught in any of the IT applications that they may develop in future. 3. Understand the concept of electronic data interchange. 4. To learn different types of payment modes.								
UNIT-I	ELECTRONIC COMMERCE						Classes: 08	
Electronic Commerce - Frame work, anatomy of E-Commerce applications, E-Commerce Consumer applications, and E-Commerce organization applications, Consumer Oriented Electronic commerce - Mercantile Process models.								
UNIT-II	ELECTRONIC PAYMENT SYSTEMS						Classes: 08	
Electronic payment systems – Digital Token-Based, Smart Cards, Credit Cards, Risks in Electronic Payment systems. Inter-Organizational Commerce – EDI, EDI Implementation, Value added networks.								
UNIT-III	INTRA ORGANIZATIONAL COMMERCE						Classes: 08	
Intra Organizational Commerce – work Flow, Automation Customization and internal Commerce, Supply chain Management.								
UNIT-IV	CORPORATE DIGITAL LIBRARY						Classes: 11	
Corporate Digital Library - Document Library, digital Document types, corporate Data Warehouses. Advertising and Marketing - Information based marketing, Advertising on Internet, on-line marketing process, market research.								
UNIT-V	CONSUMER SEARCH AND RESOURCE DISCOVERY & MULTIMEDIA						Classes: 12	
Consumer Search And Resource Discovery: Information search and Retrieval, Commerce Catalogues, Information Filtering Multimedia - key multimedia concepts, Digital Video and electronic Commerce, Desktop video processing, Desktop video conferencing.								

Text Books:

1. Frontiers of electronic commerce – Kalakata, Whinston, Pearson.
2. E-Commerce fundamentals and applications Hendry Chan, Raymond Lee, Tharam Dillon, Elizabeth Chang, John Wiley.

Reference Books:

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turbon, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson.
4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

CYBER FORENSICS**VII Semester: OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT18	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. A brief explanation of the objective is to provide digital evidences which are obtained from digital media.
2. In order to understand the objectives of computer forensics, first of all, people have to recognize the different roles computer plays in a certain crime.
3. According to a snippet from the United States Security Service, the functions computer has in different kinds of crimes.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Students will understand the usage of computers in forensic, and how to use various forensic tools for a wide variety of investigations.
2. It gives an opportunity to students to continue their zeal in research in computer forensics.

UNIT-I	COMPUTER FORENSICS FUNDAMENTALS	Classes: 15
What is Computer Forensics?, Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps taken by Computer Forensics Specialists Types of Computer Forensics Technology: Types of Military Computer Forensic Technology, Types of Law Enforcement — Computer Forensic Technology — Types of Business Computer Forensic Technology Computer Forensics Evidence and Capture: Data Recovery Defined — Data Back-up and Recovery — The Role of Back-up in Data Recovery — The Data-Recovery Solution.		
UNIT-II	EVIDENCE COLLECTION AND DATA SEIZURE	Classes: 15
Why Collect Evidence? Collection Options — Obstacles — Types of Evidence — The Rules of Evidence — Volatile Evidence — General Procedure — Collection and Archiving — Methods of Collection — Artifacts — Collection Steps — Controlling Contamination: The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene — Computer Evidence Processing Steps — Legal Aspects of Collecting and Preserving Computer Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication — Practical Consideration — Practical Implementation.		
UNIT-III	COMPUTER FORENSICS ANALYSIS AND VALIDATION	Classes: 12
Determining what data to collect and analyze, validating forensic data, addressing data-hiding techniques, performing remote acquisitions Network Forensics: Network forensics overview, performing live acquisitions, developing standard procedures for network forensics, using network tools, examining the honeynet project. Processing Crime and Incident Scenes: Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash, reviewing a case.		

UNIT-IV	CURRENT COMPUTER FORENSIC TOOLS	Classes: 15
Evaluating computer forensic tool needs, computer forensics software tools, computer forensics hardware tools, validating and testing forensics software E-Mail Investigations: Exploring the role of e-mail in investigation, exploring the roles of the client and server in e-mail, investigating e-mail crimes and violations, understanding e-mail servers, using specialized e-mail forensic tools. Cell phone and mobile device forensics: Understanding mobile device forensics, understanding acquisition procedures for cell phones and mobile devices.		
UNIT-V	WORKING WITH WINDOWS AND DOS SYSTEMS	Classes: 08
Understanding file systems, exploring Microsoft File Structures, Examining NTFS disks, Understanding whole disk encryption, windows registry, Microsoft startup tasks, MS-DOS startup tasks, virtual machines.		
Text Books:		
1. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi. 2. Computer Forensics and Investigations by Nelson, Phillips Enfinger, Steuart, CENGAGE Learning		
Reference Books:		
1. Real Digital Forensics by Keith J. Jones, Richard Bejtich, Curtis W. Rose, Addison Wesley Pearson Education 2. Forensic Compiling, A Tractitioneris Guide by Tony Sammes and Brian Jenkinson, Springer International edition. 3. Computer Evidence Collection & Presentation by Christopher L.T. Brown, Firewall Media. 4. Homeland Security, Techniques & Technologies by Jesus Mena, Firewall Media. 5. Software Forensics Collecting Evidence from the Scene of a Digital Crime by Robert M. Slade, TMH 2005 6. Windows Forensics by Chad Steel, Wiley India Edition		

DISCRETE STRUCTURES**VII Semester: OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT19	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Reason mathematically about basic data types and structures (such as numbers, sets, graphs, and trees) used in computer algorithms and systems;
2. Distinguish rigorous definitions and conclusions from merely plausible ones; synthesize elementary proofs, especially proofs by induction.
3. Model and analyze computational processes using analytic and combinatorial methods.
4. Apply principles of discrete probability to calculate probabilities and expectations of simple random processes.
5. Work in small teams to accomplish all the objectives above.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Use logical notation to define and reason about fundamental mathematical concepts such as sets, relations, functions, and integers.
2. Synthesize induction hypotheses and simple induction proofs.
3. Prove elementary properties of modular arithmetic and explain their applications in Computer Science, for example, in cryptography and hashing algorithms.
4. Apply graph theory models of data structures and state machines to solve problems of connectivity and constraint satisfaction, for example, scheduling.
5. Calculate numbers of possible outcomes of elementary combinatorial processes such as permutations and combinations.

UNIT-I	MATHEMATICAL LOGIC	Classes: 15
Statements and Notations, Connectives, Statement Formulas and Truth Tables, well formed formulas, Tautologies, Equivalence of Formulas, Normal Forms: Disjunctive Normal Forms (DNF), Conjunctive Normal Forms (CNF), Principle Disjunctive Normal Forms (PDNF), Principle Conjunctive Normal Forms (PCNF).		
UNIT-II	PREDICATES	Classes: 15
The Predicate calculus, Free and Bound Variables, Rules of Inference, Consistency Of Premises and Indirect Method of Proof, Automatic Theorem Proving.		
UNIT-III	RELATIONS AND ORDERING	Classes: 15
Relations, Properties of Binary Relations in a Set, Equivalence Relations, Compatibility Relations, and Partial Ordering, Partial Ordered Set – Representation and Associated Terminology.		
FUNCTIONS: Definition and Introduction, Composition of Functions, Inverse Functions, Recursive Functions.		

UNIT-IV	TREES	Classes: 11
Trees, Properties of trees, Representation of trees, Binary trees, Tree traversals, spanning trees, DFS, BFS, Minimal spanning trees, Prim's algorithm and Kruskal's algorithm.		
UNIT-V	GRAPHS	Classes: 10
Representation of Graphs, Planar Graphs, Graph theory and applications. Basic concepts of Isomorphism and sub graphs, Multi graph and Euler circuits, Hamiltonian Graphs, chromatic number.		
Text Books:		
1. J. P. Trembly, R. Manohar (1997), Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, India. 2. Joe L. Mott, Abraham Kandel, Theodore P. Baker (2011), Discrete Mathematics for Computer Scientists and Mathematicians, 2nd edition, Prentice Hall of India Learning Private Limited, New Delhi, India.		
Reference Books:		
1. Kenneth H. Rosen (2007), Discrete Mathematics and its Applications, 6th edition, Tata McGraw Hill, India. 2. C.L. Liu, D.P. Mohapatra (2008), Elements of Discrete Mathematics, 3rd edition, McGraw Hill, India. 3. Ralph P. Grimaldi, B.V. Ramana (2006), Discrete and Combinatorial Mathematics - An Applied Introduction, 5th Edition, Pearson Education, India.		

INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT20	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. Produce apps for Android platform devices
2. Gain a basic understanding of computer architecture and object-oriented programming
3. Develop a working knowledge of Apple's Xcode app development tool
4. Understand mobile design principles
5. Identify need and opportunity in app markets

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe those aspects of mobile programming that make it unique from programming for other platforms,
2. Critique mobile applications on their design pros and cons,
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces,
4. Program mobile applications for the Android operating system that use basic and advanced phone features, and
5. Deploy applications to the Android marketplace for distribution.

UNIT-I	INTRODUCTION TO ANDROID	Classes: 10
The Android Platform, Android SDK, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file.		
UNIT-II	ANDROID APPLICATION DESIGN ESSENTIALS	Classes: 08
Android terminologies, Application Context, Activities.		
UNIT-III	ANDROID USER INTERFACE DESIGN ESSENTIALS	Classes: 08
User Interface Screen elements, Designing User Interfaces with Layouts.		
UNIT-IV	TESTING ANDROID APPLICATIONS	Classes: 08
Testing Android applications, Publishing Android application, Using Android preferences, managing Application resources in a hierarchy, working with different types of resources.		
UNIT-V	USING COMMON ANDROID APIS	Classes: 10
Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers.		

Text Books:

1. Lauren Darcey and Shane Conder, —Android Wireless Application DevelopmentII, Pearson Education, 2nd ed. (2011)

Reference Books:

1. R1. Reto Meier, —Professional Android 2 Application DevelopmentII, Wiley India Pvt Ltd
2. R2. Mark L Murphy, —Beginning AndroidII, Wiley India Pvt Ltd.
3. R3. Android Application Development All in one for Dummies by Barry Burd, Edition.

DISTRIBUTED DATABASES

VIII Semester: OPEN ELECTIVE - IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT21	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES								
To learn								
1. Introducing Distributed Database Management System and its Design issues 2. Exploring several algorithms for processing queries and be able to use them 3. Describe the methods to translate complex conceptual data models into logical and Physical database designs 4. Demonstrating query optimization and its algorithms 5. Enumerating the concepts behind distributed transaction processing								
COURSE OUTCOMES:								
Upon successful completion of the course, the student is able to								
1. Identify the introductory distributed database concepts and its structures. 2. Describe terms related to distributed object database design and management. 3. Produce the transaction management and query processing techniques in DDBMS. 4. Relate the importance and application of emerging database technology.								
Pre-requisites: Basic concept of DBMS and RDBMS.								
UNIT-I	INTRODUCTORY CONCEPTS AND DESIGN OF (DDBMS)						Classes: 12	
Data Fragmentation; Replication, and allocation techniques for DDBMS; Methods for designing and implementing DDBMS, designing a distributed relational database; Architectures for DDBMS: cluster federated, parallel databases and client server architecture.								
UNIT-II	QUERY PROCESSING & TRANSACTION MANAGEMENT						Classes: 12	
Overview Of Query Processing: Query processing problem; Objectives of Query Processing; Complexity of Relational Algebra operations; characterization of Query processors; Layers of Query Processing Introduction To Transaction Management: Definition of Transaction, Properties of Transaction, types of transaction ; Distributed Concurrency Control: Serializability theory; Taxonomy of concurrency control mechanisms; locking bases concurrency control algorithms.								
UNIT-III	DISTRIBUTED OBJECT DATABASE MANAGEMENT SYSTEMS						Classes: 08	
Fundamental Object concepts and Object models, Object distribution design, Architectural issues, Object management, Distributed object storage, object query processing.								
UNIT-IV	CURRENT TRENDS & DEVELOPMENTS RELATED TO DISTRIBUTED DATABASE APPLICATIONS TECHNOLOGIES						Classes: 10	
Distributed Object/component-based DBMS; Database Interoperability including CORBA; DCOM and Java RMI; Distributed document-based systems; XML and Workflow management.								

UNIT-V	EMERGING RELATED DATABASE TECHNOLOGIES	Classes: 08
Parallel Database Mobile database; Multimedia Database; Spatial Database and Web Databases.		
Text Books:		
1. Distributed Databases - Principles and Systems; Stefano Ceri; Giuseppe Pelagatti; Tata McGraw Hill; 1985. 2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia. 3. Database System Concepts; Korth & Sudarshan; TMH. 4. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall.		
Reference Books:		
1. Data Base Management System; Leon & Leon; Vikas Publications 2. Introduction to Database Systems; Bipin C Desai; Galgotia.		

BIG DATA**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT22	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Compare various file systems and use an appropriate file system for storing different types of data.
2. Demonstrate the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. Apply the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. Connect to web data sources for data gathering, Integrate data sources with hadoop components to process streaming data.
5. Tabulate and examine the results generated using hadoop components.

UNIT-I	INTRODUCTION TO BIG DATA	Classes: 12
---------------	---------------------------------	--------------------

Data and its importance, Big Data- Definition, V's of Big Data, Hadoop Ecosystem

HADOOP ARCHITECTURE

Hadoop Storage : HDFS, Hadoop

Processing : Map Reduce Framework

Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, Task Tracker

HDFS-HADOOP DISTRIBUTED FILE SYSTEM

Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE- File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

UNIT-II	MAPREDUCE PROGRAMMING MODEL	Classes: 12
----------------	------------------------------------	--------------------

Introduction to Map Reduce Programming model to process BigData, key features of MapReduce, MapReduce Job skeleton, Introduction to MapReduce API, Hadoop Data Types, Develop MapReduce Job using Eclipse, build a MapReduce Job export it as a java archive(.jar file).

MAPREDUCE JOB LIFE CYCLE: How Mapreduce Works? Understanding Mapper ,Combiner, Partitioner ,Shuffle & Sort and Reduce phases of MapReduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

MAPREDUCE API: Understanding new MapReduce API from org.apache.hadoop.mapreduce and its sub packages to develop MapReduce applications ,key difference between old MapReduce API and the new MapReduce API.

UNIT-III	INTRODUCTION TO PIG	Classes: 12
Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage. PIG LATIN LANGUAGE–DATA TYPES IN PIG Pig Latin Basics, Key words ,Pig Data types ,Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell ,expressions, Data processing operators, using Built in functions. WRITING PIG SCRIPTS USING PIG LATIN: Writing pig scripts and saving them in text editor, running pig scripts from command line.		
UNIT-IV	INTRODUCATION TO HIVE	Classes: 12
Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write. HIVE QL DATA TYPES, SEMANTICS: Introduction to Hive QL (Query Language), Language semantics, Hive Data Types. HIVE DDL, DML AND HIVE SCRIPTS: Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and commands line.		
UNIT-V	SQOOP, FLUME, OOZIE	Classes: 12
SQOOP: Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS. FLUME: Introduction to Flume agent, understanding Flume components Source, Channel and Sink, Writing flume configuration file, running flume configuration file to ingest the data into HDFS. OOZIE: Introduction to Oozie, Understanding work flow and how to create Work flow using Work Flow definition language in XML, running a basic Oozie workflow to run a Map Reduce job.		
Text Books:		
1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media 2. Big Data and Hadoop- Learn by Example, Mayank Bhushan		
Reference Books:		
1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007. 2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.		

OPEN ELECTIVES OFFERED BY MECHANICAL ENGINEERING DEPARTMENT

OE1		OE2	
A4ME60	Elements of Mechanical Engineering	A4ME63	Fundamentals of Mechatronics
A4ME61	Fundamentals of Engineering Materials	A4ME64	Basics of Thermodynamics
A4ME62	Fundamentals of Theory of Machines	A4ME65	Fundamentals of Smart Materials
OE3		OE4	
A4ME66	Fundamentals of Operation Research	A4ME69	Introduction to Material Handling
A4ME67	Fundamentals of Energy Sources and Applications	A4ME70	Non Conventional Energy Sources
A4ME68	Basics of Robotics	A4ME71	Aspects of Heat Transfer

ELEMENTS OF MECHANICAL ENGINEERING

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME60	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSEOBJECTIVES:

The course aims to enable the students learn basic concepts of mechanics, materials, thermodynamics, manufacturing and CAD/CAM.

COURSEOUTCOMES:

After completing this course the student must demonstrate the knowledge and ability:

1. To demonstrate basic concepts of mechanics of rigid and deformable bodies.

2. To explain fundamentals of metallurgy and material science.

3. To discuss basic concepts and principles of thermo dynamics.

4. To demonstrate the elements of fabrication processes and equipment.

5. To explain basic concepts of CAD/CAM.

UNIT-I	MECHANICS	Classes: 14
--------	-----------	-------------

MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law-derivation and problems, polygon law, resolution of forces in coplanar-concurrent force system-problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, definition of centroid and centre of gravity, pappus and guldinus theorems, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia.

MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strain, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.

UNIT-II	METALLURGY & MATERIAL SCIENCE	Classes: 12
---------	-------------------------------	-------------

Bonds in Solids, crystallization of metals, Classification of steels, , Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide diagram, Heat treatment-Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.

UNIT-III	THERMO DYNAMICS	Classes: 12
----------	-----------------	-------------

System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry , First law of Thermodynamics – Corollaries, First law applied to a Closed System ,applied to a flow system , Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.

UNIT-IV	MANUFACTURING	Classes: 11
Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies; Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping ; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.		
UNIT-V	CAD/CAM	Classes: 11
Fundamentals of CAD/CAM- Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, dis-advantages, wire frame entities-analytic entities and synthetic entities; Surface modeling: Definition, advantages, dis-advantages, surface entities-analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.		
Text Books:		
1.Singer's Engineering Mechanics by K.Vijay Kumar Reddy and J.Suresh Kumar, B.S Publications 2. Strength of Materials By Dr.Sadhu Singh, Khanna Publishers. 3.Introduction to physical metallurgy by Avner, TATA McGrawHill Edition. 4.Thermodynamics by P.K Nag, McGrawHill Education. 5.Production Technology by R.K Jain, Khanna Publishers. 6.CAD/CAM Theory & Practice by Ibrahim Zeid & R.Sivasubramanian, McGrawHill Education.		
Reference Books:		
1. Engineering Mechanics By R.K.Bansal, Laxmi Publications 2. Mechanics of Solids By GVV Satyanarayana & Pakirappa, Durga Publishing House 3. Material science and metallurgy by OP Khanna, Dhanpat Ray Publications 4.Engineering Thermodynamics By Pakirappa, V.Naveen Kumar, V.Naresh, Durga Publishing House 5. Production Technology by Dr. P.C Sharma, S.CHAND 6. CAD/CAM By P.Radha Krishnan, S.Subramanyan and V.Raju, New Age International Publishers.		
Web References:		
1. https://www.iitg.ac.in/rkbc/me101/Presentation/L01-03.pdf 2. https://nptel.ac.in/courses/113106032/ 3. https://nptel.ac.in/downloads/112108148/ 4. https://lecturenotes.in/subject/195/cad-and-cam-cac		
E-Text Books:		
1. https://books.google.co.in/books?id=8kA8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=XYorDAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://ocw.mit.edu/courses/mechanical-engineering/		

FUNDAMENTALS OF ENGINEERING MATERIALS**V Semester: OPEN ELECTIVE-I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME61	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

COURSE OUTCOMES:

At the end of course students are able to

- 1.Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
- 2.Demonstrate the concept of alloying and formation of different types of phases in alloys.
- 3.Differentiate ferrous and non ferrous alloys.
- 4.Explain various heat treatment processes.
- 5.Classify and explain polymers, ceramics and composites.

UNIT-I	CRYSTAL STRUCTURE	Classes: 10
Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.		
UNIT-II	ALLOYS & PHASE DIAGRAMS	Classes: 10
Alloys- substitutional and interstitial solid solutions. Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.		
UNIT-III	FERROUS AND NON FERROUS ALLOYS	Classes: 12
Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons; copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.		
UNIT-IV	HEAT TREATMENT OF STEEL	Classes: 12
Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.		
UNIT-V	POYMERS, CERAMICS AND COMPOSITES	Classes: 8
Classification, properties and applications of polymers, ceramics, composites and nano materials.		

Text Books:

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener (2007,) *Introduction to Physical Metallurgy*, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.

Reference Books:

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. D. Kodgire (2006), *Material Science and Metallurgy for engineers*, 1st Edition, Everest, Pune, India.

Web References:

1. <https://nptel.ac.in/courses/113106032/>
2. <http://web.pdx.edu/~pmoeck/phy381/lecture1.pdf>

E-Text Books:

1. https://books.google.co.in/books?id=XXE8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=kBM8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://ocw.mit.edu/courses/materials-science-and-engineering/>
2. <https://www.mooc-list.com/tags/materials-science>

FUNDAMENTALS OF THEORY OF MACHINES**V Semester: OPEN ELECTIVE-I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME62	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

This course aims to enable students to learn basic concepts of kinematics and dynamics of machinery.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Demonstrate various kinematic links, pairs, chains, mechanisms and their inversions.
2. Illustrate and analyze exact and approximate straight line motion mechanisms.
3. Explain concepts of higher pairs.
4. Demonstrate and analyze the effect of gyroscopic couples on stability of motor cycle, motor car and aeroplane.
5. Illustrate and analyze various governors.

UNIT-I	MECHANISMS	Classes: 14
Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains.		
UNIT-II	STRAIGHT LINE MOTION MECHANISMS	Classes: 12
Exact, approximately copied and generated types – Peaucellier - Hart and Scott Russel – Grasshopper – Watt – Tchebichef's and Robert Mechanisms - Pantograph.		
UNIT-III	GEARS & GEAR TRAINS	Classes: 12
Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion, Form of teeth-cycloid and involute profiles, phenomena of interference.		
Gear trains: Introduction – Types – Simple, compound and reverted gear trains-simple problems.		
UNIT-IV	GYROSCOPES	Classes: 11
Gyroscopes – effect of precession – motion on the stability of moving vehicles such as motorcycle motorcar and aeroplane.		
UNIT-V	GOVERNORS	Classes: 11
Governors: Types of governors - Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. (No derivations)		

Text Books:

1. Theory of Machines by S.S.Rattan.
2. Theory of Machines by R.S.Khurmi

Reference Books:

1. Theory of Machines by Shigley, Mc Graw Hill Publishers
2. Theory of Machines by Thomas Bevan, CBS Publishers
3. Theory of Machines by R.K.Bansal (Lakshmi publications)
4. Mechanism and Machine Theory by JS Rao and RV Duggipati, New Age International

Web References:

1. <http://nptel.ac.in/courses/112104121/>
2. <https://freevidelectures.com/course/2359/kinematics-of-machines>
3. <https://nptel.ac.in/courses/112104114/>
4. <https://freevidelectures.com/course/2364/dynamics-of-machines>

E-Text Books:

1. https://books.google.co.in/books?id=T_WlquM4EI4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://cosmolearning.org/courses/kinematics-of-machines>
2. <https://nptel.ac.in/courses/112101096/>
3. <https://www.mooc-list.com/course/mechanics-kinematics-and-dynamics-edx>

FUNDAMENTALS OF MECHATRONICS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME63	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the basic principles of mechatronics, Differentiating between conventional and mechatronics design process and product.
2. Differentiate various sensors, actuators, electromechanical systems.
3. To understand and apply interfacing of electromechanical products to microcontrollers and microprocessor.
4. To understand the working of various programmable logic controllers (PLCs), Architecture of PLCs.
5. Evaluate different mechatronic systems.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various elements underlying mechatronic systems, electronics, control systems and able to differentiate their purpose in the system.
2. Analyze and select sensors, actuators, electromechanical components needed for an application.
3. To evaluate microprocessor and microcontroller interfacing to mechanical application.
4. To choose PLC to a mechanical application.
5. To design mechatronic systems for mechanical applications.

UNIT-I	INTRODUCTION	Classes: 12
Introduction to Mechatronics —Mechatronics systems - Mechatronics design process - Mechatronics in Manufacturing –Adaptive and distributed control systems – Modelling and simulation of mechatronics systems.		
UNIT-II	SENSORS AND ACTUATORS	Classes: 12
Sensors and actuators: Overview of sensors and transducers – Microsensors - Signal conditioning – Operational amplifiers – Protection – Filtering - Analog and Digital converters. Electro – pneumatics and Electro – hydraulics - Solenoids – Direct Current motors – Servomotors – Stepper motors - Micro actuators; Drives selection and application.		
UNIT-III	INTERFACING	Classes: 12
Interfacing: Microprocessor based Controllers Architecture of microprocessor and microcontroller – System interfacing for a sensor, keyboard, display and motors - Application cases for temperature control, warning and process control systems		
UNIT-IV	PLCs	Classes: 12
PLCs: Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/Output modules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting.		

UNIT-V	ARTIFICIAL INTELLIGENCE	Classes: 12
AI: Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer – based instrumentation - Real-time Data Acquisition and Control – Software integration – Man Machine interface -Vision system – Mechatronics system case studies.		
Text Books:		
1. Introduction to Mechatronics and Measurement Systems, Tata McGraw Hill		
Reference Books:		
1. Designing Intelligent Machines, Michel B. Histan and David G. Alciatore, Open University London		
2. Control Sensors and Actuators, ICW. Desiha, Prentice Hall		
Web References:		
1. https://lecturenotes.in/subject/137/mechatronics-mech		
2. https://sites.google.com/site/profnarendralakal/Home/presentations/mtrx		
E-Text Books:		
1. https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
2. https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://ocw.mit.edu/courses/mechanical-engineering/2-737-mechatronics-fall-2014/		
2. https://nptel.ac.in/courses/112103174/		
3. https://www.igi-global.com/chapter/massive-online-open-course-assisted-mechatronics-learning/137327		

BASICS OF THERMODYNAMICS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME64	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To understand the treatment of classical Thermodynamics and to apply the First and Second laws of Thermodynamics for the analysis of thermal equipment

COURSE OUTCOMES:

At the end of the course, the student should be able to

1. Demonstrate basic concepts of thermodynamics.
2. Explain laws of thermodynamics.
3. Explain pure substances and power cycles.
4. Explain perfect gas laws and concepts of mixture of perfect gases.
5. Discuss fundamental concepts of psychrometry.

UNIT-I	INTRODUCTION	Classes: 12
Introduction: System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Exact & Inexact Differentials, Cycle – Reversibility – Quasi – static Process, Irreversible Process, Causes of Irreversibility – Energy in State and in Transition, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry		
UNIT-II	LAWS OF THERMO DYNAMICS	Classes: 12
First law of Thermodynamics – Corollaries – First law applied to a Closed System – applied to a flow system – Steady Flow Energy Equation. Limitations of the First Law Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements and their Equivalence / Corollaries, Entropy, Principle of Entropy Increase – Energy Equation, Availability and Irreversibility – Elementary Treatment of the Third Law of Thermodynamics		
UNIT-III	PURE SUBSTANCES & POWER CYCLES	Classes: 12
Pure Substances, p-V-T- surfaces, T-S and h-s diagrams, Mollier Charts, Phase Transformations – Triple point at critical state properties during change of phase, Dryness Fraction – Clausius – Clapeyron Equation Property tables. Power Cycles: Otto, Diesel, Dual Combustion cycles, Sterling Cycle – Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis – comparison of Cycles.		
UNIT-IV	PERFECT GAS LAWS & MIXTURE OF PERFECT GASES	Classes: 12
Perfect Gas Laws – Equation of State, specific and Universal Gas constants – various Non-flow processes, properties, end states Mixtures of perfect Gases – Mole Fraction, Mass fraction Gravimetric and volumetric Analysis – Dalton's Law of partial pressure, Avogadro's Laws of additive volumes – Mole fraction, Volume fraction and partial pressure.		

UNIT-V	PSYCHROMETRY	Classes: 12
Psychrometric Properties – Dry bulb Temperature, Wet Bulb Temperature, Dew point Temperature, Thermodynamic Wet Bulb Temperature, Specific Humidity, Relative Humidity, saturated Air, Vapour pressure, Degree of saturation – Adiabatic Saturation, Carrier's Equation – Psychrometric chart.		
Text Books:		
1. Engineering Thermodynamics, P.K. Nag, 6th Edition, Mc Graw Hill Education. 2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8th Edition, Mc Graw Hill Companies. 3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7th Edition, Wiley.		
Reference Books:		
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi. 3. Applied Thermodynamics, Onkar Singh, 3rd Edition, New Age, India		
Web References:		
1. https://www.askiitians.com/revision-notes/physics/thermodynamics/ 2. http://www.ncert.nic.in/ncerts//kech106.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=91MZ2ZOb3n8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.coursera.org/learn/thermodynamics-intro 2. https://www.edx.org/course/thermodynamics-2		

FUNDAMENTALS OF SMART MATERIALS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME65	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The objective of this course is to enable the students to learn principles of smart materials such as electronic materials, ionic polymer matrix composites etc.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various smart materials and their principles.
2. Explain different types of electro-active materials.
3. Explain principles of piezoelectricity.
4. Discuss principles of Magnetostriction.
5. Explain about various shape memory alloys.

UNIT-I	INTRODUCTION	Classes: 12
Overview of Smart Materials: Introduction and definition of Smart Materials, principles of smart materials, input – output decision ability, devices based on conductivity changes, devices based on changes in optical response, biological systems smart materials. Devices based on magnetization, artificial structures, surfaces, hetero structures, polycrystalline, amorphous, liquid crystalline materials		
UNIT-II	ELECTRO ACTIVE MATERIALS & PIEZO ELECTRICITY	Classes: 12
Electro-active Materials: Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composites (IPMC)		
UNIT-III	PIEZOELECTRICITY	Classes: 12
Piezoelectricity: Principles of Piezoelectricity, Perovskite Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers		
UNIT-IV	MAGNETOSTRICTION	Classes: 12
Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magnetoresistance Effect, magnetorheological fluids, electrorheological materials and their applications		
UNIT-V	SHAPE MEMORY ALLOYS	CLASSES: 12
Shape Memory Alloys: Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Self- healing materials.		

Text Books:

1. B. Culshaw, Smart Structures and Materials, Artech House, 2000
2. P. Gauenzi, Smart Structures: Physical Behaviour, Mathematical Modelling and Applications, Wiley Publishers, 2009
3. W. G. Cady, Piezoelectricity, Dover Publication, 1964
4. M. Addington, Schodek, L. Daniel.: Smart materials and new technologies, Architectural Press, 2005
5. W.D Callister, Materials Science and Engineering an Introduction, Willey 1999

Reference Books:

1. M. Schwartz, Encyclopedia of Smart Materials, Volumes 1-2 ,Willey, 2002
2. Smith, C.: Smart material systems, Ralph, SIAM, 2005
- 3.K Vijay, K.Varadan, J. Vinoy, S.Gopalakrisham, Smart Material Systems and MEMS: Design and Development Methodologies , Willey 2006

Web References:

1. <https://www.sciencedirect.com/topics/chemistry/smart-material>
2. <https://www.mide.com/collections/smart-materials>

E-Text Books:

1. https://books.google.co.in/books?id=KdBr38J-bxoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=CQxwDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. https://nptel.ac.in/noc/individual_course.php?id=noc18-me60
2. <https://nptel.ac.in/courses/112104251/>

FUNDAMENTALS OF OPERATIONS RESEARCH

VII Semester: OPEN ELECTIVE-III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME66	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSEOBJECTIVES: The objectives of this course are to learn quantitative methods and techniques for effective decisions-making; model formulation and applications that are used in solving business decision problems.								
COURSEOUTCOMES: At the end of course students will be able to 1. Describe types of models and solve linear programming problem. 2. Solve transportation and assignment problems. 3. Analyze sequencing and replacement models and apply them for optimization. 4. Apply gaming theory for optimal decision making. 5. Analyze inventory models to optimize the cost								
UNIT-I	INTRODUCTION & ALLOCATION						Classes: 12	
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem - Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.								
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS						Classes: 12	
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.								
UNIT-III	SEQUENCING & REPLACEMENT						Classes: 12	
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.								
UNIT-IV	THEORY OF GAMES						Classes: 12	
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games –m x 2 & 2 x n games - graphical method – m x n games - dominance principle.								
UNIT-V	INVENTORY						Classes: 12	
INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.								

Text Books:

1. Operation Research by J.K.Sharma, MacMilan.
2. Operations Research by ACS Kumar, Yesdee

Reference Books:

1. Operations Research: Methods and Problems by Maurice Saseini, Arhur Yaspan and Lawrence Friedman
2. Operations Research by A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Pearson Education.
3. Operations Research by Wagner, PHI Publications.
4. Introduction to O.R by Hillier & Libermann, TMH.

Web References:

1. <http://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf>
2. http://www.mathcity.org/msc/notes/operation_research
3. http://biobharati.com/admin/img/gall/1467763196_BCA-504.pdf

E-Text Books:

1. https://books.google.co.in/books?id=6khDDAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
3. https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <http://www.orcomplete.com/internet/enesbilgin/open-courses-on-operations-research>
2. <https://www.coursera.org/learn/wharton-operations>

FUNDAMENTALS OF ENERGY SOURCES AND APPLICATIONS

VII Semester: OPEN ELECTIVE-III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME67	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The objectives of this course are to study classifications and comparison of various renewable energy sources, flat plate and concentrating collectors, wind energy, principles of bio mass, geothermal and ocean energy sources and principles of direct energy conversion.								
COURSE OUTCOMES: At the end of the course, the student shall be able to 1. Classify renewable energy sources and explain environmental impact of solar energy, solar radiation, their measurement and instruments. 2. Classify concentrating collectors and explain in detail about various solar gadgets. 3. Explain about wind mill, types of wind mills, its construction, wind energy conversion systems and principles of bio mass. 4. Explain in detail geothermal and ocean energies. 5. Demonstrate the concepts of direct energy conversion.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction – Renewable energy sources – Limitations – Classifications – Comparison – Global and Indian energy scenario. Solar Energy: Introduction – Environmental impact of solar power – extraterrestrial and terrestrial solar radiation – solar radiation on tilted surface – Solar Radiation Measurement and Instruments								
UNIT-II	SOLAR ENERGY						Classes: 12	
Flat plate and concentrating collectors – classification of concentrating collectors – orientation and thermal analysis – advanced collectors. Solar Gadgets: Solar furnace, Solar air heater, Solar water heater, Soar cooker, solar still, solar pond, solar pump, solar dryer, solar cell.								
UNIT-III	WIND ENERGY & BIO MASS						Classes: 12	
Wind energy utilization: Introduction – Origin and nature of winds – Types of wind mills and its construction – Wind energy conversion systems – performance characteristics. Bio-Mass: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, combustion characteristics of bio-gas, applications and economic aspects.								
UNIT-IV	GEOTHERMAL ENERGY & OCEAN ENERGY						Classes: 12	
Geothermal Energy: Resources, types of wells, methods of harnessing the energy, potential in India. Ocean Energy: OTEC, Principles utilization, setting of OTEC plants. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.								
UNIT-V	DIRECT ENERGY CONVERSION						Classes: 12	
Direct Energy Conversion: Need, limitations, principles of DEC. Thermo-electric generators, seebeck, peltier and Joule-Thomson effects – applications, MHD generator principle, Fuel cells: Classification of fuel cells; Operating principles. Bio-fuels, Urban waste to energy conversion								

Text Books:

1. Rai G.D, Non-Conventional Energy Sources, Khanna Publishers, New Delhi, 2011.
2. Twidell, J.W. & Weir, A., Renewable Energy Sources, EFN Spon Ltd., UK, 2006

Reference Books:

1. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003.
2. Mittal K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003.
3. Khan B H, Non-Conventional Energy Resources, Tata McGraw Hill, 2nd Edn. New Delhi, 2009.
4. Ramesh R & Kumar K U, Renewable Energy Technologies, Narosa Publishing House, New Delhi, 2004
5. Tiwari. G.N., Solar Energy – Fundamentals Design, Modelling & Applications, Narosa Publishing House, New Delhi, 2002.

Web References:

1. http://www.afdc.energy.gov/fuels/natural_gas.html
2. <https://cleancities.energy.gov/>
3. <http://www.firmgreen.com/>
4. <https://www.renewableresourcescoalition.org/alternative-energy-sources/>
5. <https://www.edx.org/learn/renewable-energy>

E-Text Books:

1. <https://www.sciencedirect.com/book/9780126561531/renewable-energy>
2. <https://www.pdfdrive.com/renewable-energy-resources-e14705840.html>
3. <https://www.pdfdrive.com/search?q=renewable&pagecount=&pubyear=&searchin=>

MOOC Course:

1. <https://www.coursera.org/learn/renewable-energy-entrepreneurship>
2. <https://www.coursera.org/learn/wind-for-renewable-energies>
3. <https://online-learning.tudelft.nl/courses/sustainable-energy-design-a-renewable-future/>

BASICS OF ROBOTICS

VII Semester: OPEN ELECTIVE-III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME68	OEC	L	T	P	C	CIE	SEE	Total
		3		-	3	30	70	100
COURSE OBJECTIVES: The objectives of this course are to 1. Study about basic components, architecture, anatomy and specifications of robot. 2. Study about various end effectors and actuators. 3. Study homogeneous transformation , D-H notation and direct kinematics 4. Study Inverse kinematics and Trajectory planning techniques 5. Study feedback components and applications of robots.								
COURSE OUTCOMES: At the end of the course, the student will be able to 1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots. 2. Explain various types of end effectors 3. Evaluate rotation matrices, forward kinematics of RR, RP and 3R Manipulators. 4. Explain inverse kinematics of RR manipulator, RP manipulator and trajectory planning techniques. 5. Explain feedback components used in robots and industrial applications.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction: Automation and Robotics, Asimov's laws, Robot Architecture, Components, , Anatomy of robot, Factors to be considered in the selection of robot, present and future applications, Specifications-Degree of freedom, Pay load, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems								
UNIT-II	END EFFECTORS & ACTUATORS						Classes: 12	
End effectors: Mechanical and Non-mechanical grippers, requirements for the design of grippers, considerations for the selection of grippers, Types of actuation mechanisms. Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators								
UNIT-III	MOTION ANALYSIS & DIRECT KINEMATICS						Classes: 12	
Motion Analysis: Basic Rotation Matrices, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems. Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP and 3R manipulators								
UNIT-IV	INVERSE KINEMATICS & TRAJECTORY PLANNING						Classes: 12	
Inverse kinematics : Inverse kinematics of 2R and RP manipulators. Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic polynomial trajectory								

UNIT-V	FEEDBACK COMPONENTS & APPLICATIONS	Classes: 12
Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor. Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.		
Text Books:		
1. Industrial Robotics by Groover M P, Pearson Edu. 2. Robotics by Fu K S, McGraw Hill. 3. Theory of Applied Robotics (kinematics,Dynamics and Control-Jazar, Springer.		
Reference Books:		
1. Robotics and Control by Mittal R K & Nagrath I J, TMH. 2. Robot Dynamics and Controls by Spony and Vidyasagar, John Wiley 3. Robot Analysis and control by Asada and Slotine, Wiley Inter-Science 4. Introduction to Robotics by John J Craig, Pearson Education		
Web References:		
1. https://www.sciencedaily.com/terms/industrial_robot.htm 2. https://www.robotics.org/robotics/industrial-robot-industry-and-all-it-entails		
E-Text Books:		
1. https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.classcentral.com/tag/robotics 2. https://www.mooc-list.com/tags/robotics		

INTRODUCTION TO MATERIAL HANDLING**VIII Semester: OPEN ELECTIVE-IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME69	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The aim of this course is to enable the students to learn basic concepts of material handling, consideration in material handling system, materials transport and storage system, various inventory models and forecasting techniques.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts and principles of material handling.
2. Discuss about various material transport systems.
3. Explain conventional and automated storage systems.
4. Explain various inventory models.
5. Demonstrate different forecasting techniques.

UNIT-I	INTRODUCTION	Classes: 12
---------------	---------------------	--------------------

Introduction: Overview of material handling equipment, considerations in material handling system design, The 10 principles of material handling.

UNIT-II	MATERIAL TRANSPORT SYSTEMS	Classes: 12
----------------	-----------------------------------	--------------------

Material Transport Systems: Industrial trucks, Automated guided vehicle systems, Monorails and other rail guided vehicles, conveyor systems, cranes and hoists, Analysis of material transport systems.

UNIT-III	STORAGE SYSTEMS	Classes: 12
-----------------	------------------------	--------------------

Storage Systems: Storage system performance, storage location strategies, conventional storage methods and equipment, automated storage systems and analysis of automated storage systems.

UNIT-IV	INVENTORY	Classes: 12
----------------	------------------	--------------------

Inventory: Introduction, Inventory management, Static Inventory models, ABC Analysis, VED analysis, Just in Time, Aggregate inventory analysis, Inventory models with quantity discounts.

UNIT-V	FORECASTING	Classes: 12
---------------	--------------------	--------------------

Forecasting: Forecasting techniques, Make or Buy decision, Acceptance sampling, Materials requirement planning, objectives of master schedule.

Text Books:

1. Mikell P. Groover, Automation Production Systems and Computer Integrated Manufacturing, Pearson International Edition.
2. A.K Singh, Materials Management, Laxmi Publications (P) Ltd.

Reference Books:

1. Dr.K.C Arora, Vikas V.Shinde, Aspects of Material Handling, Laxmi Publications (P) Ltd.
2. Charles Reese, Material Handling Systems, Taylor & Francis Network

Web References:
1. https://www.scribd.com/doc/222647028/Material-Handling-Full-Notes 2. http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20(MH).pdf
E-Text Books:
1. https://books.google.co.in/books?id=AN8dzzlp1wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=d0mhPIDEm0gC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
MOOC Course
1. https://www.mooc-list.com/course/materials-and-materials-processing-saylororg 2. https://www.mooc-list.com/tags/material-processing

NON CONVENTIONAL ENERGY SOURCES

VIII Semester: OPEN ELECTIVE-IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME70	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES: To impart the knowledge of basics of different non conventional types of power generation & power plants in detail so that it helps them in understanding the need and role of Non-Conventional Energy sources particularly when the conventional sources are scarce in nature.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate various energy sources and their availability. 2. Explain concepts and applications of solar radiation 3. Discuss elaboratively the principles of wind energy conversion 4. Explain Biomass conversion techniques and Geothermal Sources and resources 5. Explain about tidal energy and Ocean Thermal Energy.								
UNIT- I	ENERGY SOURCES & THEIR AVAILABILITY						Classes: 12	
Energy Sources & their Availability - Importance of Non Conventional Energy Resources - Classification of NCES - Solar, Wind, Geothermal, Bio-mass, Nuclear Energy & Fuel Cells, Ocean Energy Sources, Comparison of these Energy Sources, Prospects of Renewable Energy Sources - Criteria for Assessing the Potential of NCES, Statistics on Conventional Energy Sources and supply in Developing Countries								
UNIT- II	INTRODUCTION TO SOLAR RADIATION & APPLICATIONS OF SOLAR ENERGY						Classes: 15	
Introduction to Solar radiation - beam and diffuse radiation, solar constant, earth sun angles, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, collector efficiency, Focusing and non-focusing type Applications of Solar Energy - solar water heater- Solar Cooker-Box type- Solar dryer - solar greenhouse— Summer and winter greenhouse-solar electric power generation- Solar Photo- Voltaic, Solar Cell Principle, Conversion efficiency and power output, Basic Photo Voltaic System for Power Generation.								
UNIT- III	PRINCIPLE OF WIND ENERGY CONVERSION						Classes: 12	
Principle of Wind Energy Conversion - Basic components of Wind Energy Conversion Systems : Wind data and energy estimation, Site Selection Considerations - Wind mill Components, Various Types and their Constructional Features - Effect of Density, Frequency Variances, Angle of attack, and Wind Speed - Design Considerations of Horizontal and Vertical Axis Wind Machines - Analysis of Aerodynamic Forces Acting on Wind Mill Blades and Estimation of Power Output.								
UNIT- IV	BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES						Classes: 15	
Biomass conversion techniques - Biogas Generation - Factors affecting biogas Generation-Types of biogas plants - Advantages and disadvantages of biogas plants - site selection, digester design consideration, filling a digester for starting, maintaining biogas production, Fuel properties of bio gas, utilization of biogas - urban waste to energy conversion. Geothermal Sources and resources like hydrothermal, geo-pressured hot dry rock, magma. advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India - exhaust types of conventional steam turbines.								

UNIT- V	TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION	Classes: 12
Tidal Energy - Principle of working, performance and limitations. Wave Energy - Principle of working, performance and limitations. Ocean Thermal Energy Conversion - Availability, theory and working principle, performance and limitations. OTEC power plants, Operational of small cycle experimental facility, Economics of OTEC, Environmental impacts of OTEC.		
Text Books:		
1. G. D. Rai, "Non-Conventional Energy Sources", 4th Edition, Khanna Publishers, 2000 2. B H Khan, " Non-Conventional Energy Resources", 2nd Edition, Tata Mc Graw Hill Education Pvt Ltd, 2011		
Reference Books:		
1. S.Hasan Saeed and D.K.Sharma , "Non-Conventional Energy Resources", 3rd Edition, S.K.Kataria & Sons, 2012 2. Renewable energy sources and conversion technology by N.K. Bansal, M. Kleemann, M. Heliss, Tata McGraw Hill 1990. 3. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003 4. K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003 5. S. P. Sukhatme, "Solar Energy Principles and Application", TMH, 2009		
Web References:		
1. https://lecturenotes.in/subject/57/non-conventional-energy-systems-nces 2. https://www.pdfdrive.com/non-conventional-energy-sources-e10086374.html 3. http://www.vssut.ac.in/lecture_notes/lecture1428910296.pdf		
E-Text Books:		
1. http://nptel.ac.in/courses/121106014 2. http://nptel.ac.in/courses/121106014/13 3. http://nptel.ac.in/courses/12110601427		
MOOC Course:		
1. https://nptel.ac.in/courses/121106014/ 2. https://nptel.ac.in/syllabus/121106014/ 3. https://nptel.ac.in/downloads/121106014/		

ASPECTS OF HEAT TRANSFER

VIII Semester: OPEN ELECTIVE-IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME71	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100
COURSE OBJECTIVES:								
1. To understand the mechanisms of heat transfer under steady and transient conditions. 2. To understand the concepts of heat transfer through free and forced convection. 3. To learn the thermal analysis and sizing of heat exchangers. 4. To understand the principles of radiation heat transfer.								
COURSE OUTCOMES:								
1 Demonstrate basic modes of heat transfer and compute temperature distribution in steady state and unsteady state heat conduction. 2 Analyze the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces. 3 Interpret and analyze the free and forced convection heat transfer. 4 Comprehend the concepts of free convection and heat exchangers. 5 Explain the principles of heat transfer with phase change.								
UNIT-I	INTRODUCTION						Classes: 09	
Introduction: Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier rate equation , general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.								
UNIT-II	ONE DIMENSIONAL STEADY STATE HEAT CONDUCTION						Classes: 15	
One dimensional steady state heat conduction: Homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy, Critical radius of insulation. Variable thermal conductivity and Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips.								
UNIT-III	CONVECTIVE HEAT TRANSFER						Classes: 12	
Convective Heat Transfer: Classification of systems based on causation of flow, condition of flow, configuration of flow and medium of flow – Dimensional analysis as a tool for experimental investigation – Buckingham Theorem and method, application for developing semi – empirical non- dimensional correlation for convection heat transfer – Significance of non-dimensional numbers – Concepts of Continuity, Momentum and Energy Equations.								
UNIT-IV	FREE CONVECTION & HEAT EXCHANGERS						Classes: 12	
Free Convection: Development of Hydrodynamic and thermal boundary layer along a vertical plate - Use of empirical relations for Vertical plates and pipes. Heat Exchangers: Classification of heat exchangers – overall heat transfer Coefficient and fouling factor .								

UNIT-V	RADIATION HEAT TRANSFER	Classes: 12
Radiation Heat Transfer : Emission characteristics and laws of black-body radiation – Irradiation – total and monochromatic quantities – laws of Planck, Wien, Kirchoff, Lambert, Stefan and Boltzmann– heat exchange between two black bodies – concepts of shape factor – Emissivity – heat exchange between grey bodies – radiation shields – electrical analogy for radiation networks.		
Text Books:		
1. Yunus A. Cengel (2012), Heat Transfer a Practical Approach, 4th edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 2. R. C. Sachdeva (2012), Fundamentals of Engineering, Heat and Man Transfer, 3rd edition, New Age, New Delhi, India.		
Reference Books:		
1. Holman (2012), Heat Transfer (SI Units), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 2. P. S. Ghoshdastidar (2012), Heat Transfer, 2nd edition, Oxford University Press, New Delhi, India. 3. Incropera, Dewitt (2012), Fundamentals of Heat Transfer, 6th edition, John Wiley, UK.		
Web References:		
1. http://nptel.ac.in/courses/112104113/ 2. http://nptel.ac.in/courses/112105123/ 3. https://www.udemy.com/fundamentals-of-engineering-heat transfer		
E-Text Books:		
1. https://www.pdfdrive.net/heat_transfer-e18770201.html 2. https://www.pdfdrive.net/fundamentals-of-heat transfer-e18762062.html 3. https://www.pdfdrive.net/heat transfer-e15057935.html		
MOOC Course		
1. https://onlinecourses.nptel.ac.in/ 2. https://www.coursera.org/learn/heat transfer-intro 3. https://swayam.gov.in/courses/4642-july-2018-conduction and convection		

OPEN ELECTIVES OFFERED BY HS DEPARTMENT

OE1		OE2	
A4HS10	Business Economics and Financial Analysis	A4HS13	Advanced Entrepreneurship
A4HS11	Basics of Entrepreneurship		
A4HS12	Human Values and Professional Ethics		
OE3		OE4	
A4HS14	Indian Ethos and Business Ethics	A4HS15	Management Science
		A4HS16	Intellectual Property Rights
		A4HS17	Number Theory
		A4HS18	Physics and Technology of Thin Films
		A4HS19	Polymer Chemistry

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS10	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:
To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSE OUTCOMES:
At the end of the course, the student will

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Develop an understanding of Analyse how capital budgeting decisions are carried out.
4. Understanding the framework for both manual and computerised accounting process
5. Know how to analyse and interpret the financial statements through ratio analysis.

UNIT-I	INTRODUCTION & DEMAND ANALYSIS	Classes: 12
Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.		
UNIT-II	PRODUCTION & COST ANALYSIS	Classes: 12
Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.		
UNIT-III	MARKETS & NEW ECONOMIC ENVIRONMENT	Classes: 12
Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.		
UNIT-IV	CAPITAL BUDGETING	Classes: 12
Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).		

UNIT-V	INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS	Classes: 12
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Trasing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
Text Books:		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
Reference Books:		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		

BASIC OF ENTREPRENEURSHIP

V Semester: OPEN ELECTIVE-I										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A4HS11		Foundation		L	T	P	C	CIA	SEE	Total
				3	-	-	3	30	70	100
Contact 50	Classes:	Tutorial Nil	Classes:	Practical Classes: Nil				Total Classes: 50		
COURSE OBJECTIVES: The course should enable the students to: The curriculum helps students 1. Understand and discover entrepreneurship 2. Build a strong foundation for students to start, build, and grow a viable and sustainable venture 3. Develop an entrepreneurial outlook and mindset , critical skills and knowledge 4. Mitigate three types of risks : Customer, Business Model, and Product/Technical COURSE OUTCOMES: 1. Entrepreneurship and Innovation minors will be able to sell themselves and their ideas. Students master oral and visual presentation skills and establish a foundation of confidence in the skills necessary to cause others to act. 2. The ability to imagine, recognize, and seize opportunities for innovation and new venture creation in the arts and culture sector 3. The know-how to bring capital to ideas in both commercial and nonprofit domains.										
UNIT-I		DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING							Classes: 10	
Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style, Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples, Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"										
UNIT-II		CUSTOMER, BUSINESS MODEL, VALIDATION							Classes: 10	
CUSTOMER :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and "Jobs-To- Be-Done" (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch . BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.										
UNIT-III		MONEY AND TEAM							Classes: 12	
MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each. What investors expect from										

you, Practice Pitching to Investors and Corporates.

TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

UNIT-IV**MARKETING & SALES****Classes: 10**

MARKETING & SALES: Positioning - Understand the difference between product and brand and the link between them, Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

Channels & Strategy : • Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-V**PLANNING & TRACKING****Classes: 08**

Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale

Planning & Tracking : Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

Business Regulation : Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

Text Books:

1. S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007.
2. Steven Fisher, Ja-nae' Duane, The Startup Equation -A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016.
3. D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)

Reference Books:

1. Vasant Desai —Small Scale industries and entrepreneurship|| Himalaya publishing 2012.
2. Rajeev Roy —Entrepreneurship|| 2e, Oxford, 2012.
3. B. Janakiram and M. Rizwan|| Entrepreneurship Development: Text & Cases, Excel Books, 2011.
4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013.
5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013

Web References:

1. <http://freevideolectures.com/Course/3641/Entrepreneurship-Through-the-Lens-of-Venture-Capital>
2. <http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship>
3. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf
4. <https://www.scribd.com/doc/21516826/Entrepreneurship-Notes>
5. <http://freevideolectures.com/Course/3514/Economics-/Management-/Entrepreneurship/50>
6. Journal of Entrepreneurship & Organization Management, Vikalpa, IIMA, IIMB Review, Decision, IIMC, Vision, HBR.

HUMAN VALUES AND PROFESSIONAL ETHICS

V Semester: OPEN ELECTIVE-I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS12	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behaviour and mutually enriching interaction with Nature.

COURSE OUTCOMES:

It ensures students sustained happiness through identifying the essentials of human values and skills.

1. It facilitates a correct understanding between profession and happiness
2. It helps students understand practically the importance of trust, mutually satisfying human behaviour and enriching interaction with nature.
3. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

UNIT-I	Introduction	Classes: 12
Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.		
UNIT-II	Understanding Harmony in the Human Being - Harmony in Myself!	Classes: 12
Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.		
UNIT-III	Understanding Harmony in the Family and Society - Harmony in Human - Human Relationship	Classes: 12
Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha) - from family to world family!		
UNIT-IV	Understanding Harmony in the nature and Existence - Whole existence as Co-existence	Classes: 12
Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four		

orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.

UNIT-V**Implications of the above Holistic Understanding of Harmony on Professional Ethics****Classes: 12**

Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics:

- A. Ability to utilize the professional competence for augmenting universal human order,
- B. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
- C. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- D. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order.
- E. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
- F. At the level of society: as mutually enriching institutions and organizations.

TEXT BOOKS:

1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.

REFERENCE BOOKS:

1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
2. E. F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
3. A Nagraj, 1998 Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
4. Susan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991.
5. P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers.
6. A. N. Tripathy, 2003, Human Values, New Age International Publishers.
7. Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
8. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books.
9. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press.
10. M Govindrajan, S Natrajan & V. S Senthil kumar, Engineering Ethics (including Human Values), Eastern Economy Edition, Prentice Hall of India Ltd.

Web References:

1. Value Education website, <http://www.uptu.ac.in>
2. Story of Stuff, <http://www.storyofstuff.com>
3. Al Gore, An Inconvenient Truth, Paramount Classics, USA
4. Charle Chaplin, Modern Times, United Artists, USA
5. IIT Delhi, Modern Technology - the Untold Story

ADVANCED ENTREPRENEURSHIP

VI Semester: OPEN ELECTIVE-II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS13	Foundation	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			
COURSE OBJECTIVES: The course should enable the students to: The curriculum helps students 1. Understand and discover entrepreneurship 2. Build a strong foundation for students to start, build, and grow a viable and sustainable venture 3. Develop an entrepreneurial outlook and mindset , critical skills and knowledge 4. Mitigate three types of risks : Customer, Business Model, and Product/Technical								
COURSE OUTCOMES: 1. The knowledge and mastery of organizational and management techniques in the arts, including leadership, organizational design, human resources, legal issues, and finance 2.The knowledge of the legal and ethical environment impacting business organizations and exhibit an understanding and appreciation of the ethical implications of decisions. 3. An ability to engage in critical thinking by analyzing situations and constructing and selecting viable solutions to solve problems.								
UNIT-I	ORIENTATION TO GROWTH						Classes: 10	
ORIENTATION TO GROWTH : Getting Ready for Growth-Why growth stage is different compared to startup phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth,To chart a growth path.								
UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION						Classes: 10	
CUSTOMERS : Expanding Customer Base-Revisit your business model and develop fewvariants (more business modeltypes),Identify additional customersegmentsthat yoursolution can address,Evaluate business modelsforthe new customer segments,Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize. BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.								
UNIT-III	TRACTION AND TEAM						Classes: 12	
TRACTION : Scaling - How to gain traction beyond earlycustomers,Defining traction (in quantifiable terms) and identifying themost important metricstomeasure traction,Calculate cost of new customeracquisition,Estimate your customer lifetime value(LTV),Identifying waste in your operations and focusing your team on what is important fortraction. Channels and Strategy - The Bullseye framework,Identify Channels using Bulls EyeFramework,Measuring the effectiveness ofselected channels,Budgeting and planning.								
UNIT-IV	MONEY & SALES						Classes: 10	
MONEY : Growing Revenues - Stabilizing key revenue streams,Developing additionalrevenue streams(licensing,franchising),Exploring new channels and partnerships,Sales Planning - Understanding why customers buy and how buying decisions are made; Listenerskills,Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features,Follow-up and closing a sale; Asking for the sale. Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional salesteam,Sales compensation and incentives,Sales planning, setting targets. Improving Margins - Testing price elasticity,Optimizing costs and operational expenses, Advanced concepts of unit costing.Financial Modeling - Financial modeling of your								

venture's growth, Analyzing competitor and peer's financial models.		
UNIT-V	SUPPORT	Classes: 08
SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks. Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan		
Text Books:		
1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley. 2. Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D. B., Acs, Z. J. (eds). Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.		
Reference Books:		
1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth 2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism" Technovation 3. Audretsch, D. B., Grilo, I., Thurik, A. R. (eds). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.		
Web References:		
1. https://www.ediindia.org/EDIILibrary.aspx 2. https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd		

INDIAN ETHOS AND BUSINESS ETHICS

VII Semester: OPEN ELECTIVE-III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS14	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To understand the importance of ethics in business 2. To acquire knowledge and capability to develop ethical practices for effective management 3. To understand the Business Ethics and to provide best practices of business ethics. 4. To learn the values and implement in their careers to become a good managers. 5. To develop various ethical Responsibilities and practice in their professional life 6. To imbibe the ethical issues and to adhere to the ethical codes.								
COURSE OUTCOMES: 1. Understand the dynamics of morality 2. Identify the constant in morality 3. Recognize the variable values in morality 4. Students will be able to understand the business ethics. 5. The student will be able to analyze various ethical codes in corporate governance 6. Student will be able to analyze the Employees conditions and Business Ethics								
UNIT-I	INTRODUCTION TO INDIAN ETHOS						Classes: 12	
History & Relevance, Principles Practiced by Indian Companies, Role of Indian Ethos in Managerial Practices, Management Lessons from Vedas, Mahabharata, Bible and Quran.								
UNIT-II	UNDERSTANDING VALUES IN BUSINESS						Classes: 12	
Kautilya's Arthashastra, Indian Heritage in Business, Management-Production and Consumption. Ethics v/s Ethos , Indian v/s Western Management, Work Ethos and Values for Indian Managers- Relevance of Value Based Management in Global Change- Impact of Values on Stakeholders, Trans-Cultural Human Values, Secular v/s Spiritual Values , Value System in Work Culture, Stress Management-Meditation for mental health, Yoga.								
UNIT-III	CONTEMPORARY APPROACHES TO INDIAN ETHOS						Classes: 12	
Contemporary Approaches to Leadership- Joint Hindu Family Business–Leadership Qualities of Karta, Indian Systems of Learning-Gurukul System of Learning, Advantages- Disadvantages of Karma, importance of Karma to Managers-Nishkama Karma-Laws of Karma, Law of Creation- Law of Humility- Law of Growth- Law of Responsibility- Law of Connection-Corporate Karma Leadership.								
UNIT-IV	UNDERSTANDING THE BUSINESS ETHICS						Classes: 12	
Understanding the need for ethics, Ethical values, myths and ambiguity, ethical codes, Ethical Principles in Business; Theories of Ethics, Absolutism verses Relativism, Teleological approach, the Deontological approach, Kohlberg's six stages of moral development (CMD).								
UNIT-V	ETHICAL CULTURE IN ORGANIZATION						Classes: 12	
Ethical Culture in Organization, Developing codes of Ethics and conduct, Ethical and value based leadership. Role of scriptures in understanding ethics, Indian wisdom & Indian approaches towards business ethics.								
TEXT BOOKS: 1.M.G. Velasquez, Business Ethics, Prentice Hall India Limited, New Delhi, 2. R.C. Sekhar, Ethical Choices in Business, Response Books, New Delhi, 2007								

REFERENCE BOOKS:

1. Chakraborty S.K., —Management Transformation by Valuesll, New Delhi, Sage Publication, 1990.
2. Chakraborty, S.K., Ethics in Management-Vedantic Approach, New Delhi, Oxford India Ltd. 1995.
3. Fernando A.C., Business Ethics: An Indian Perspective, Pearson, 2009.
4. Kautilya's Arthasastra, King, Governance, and Law in Ancient India, Oxford University Press, 2016.
5. Murthy, C.S.R. Business Ethics, Himalaya Publishing House, Mumbai, 2009.
6. Narayana G., —The Responsible Leader: A Journey through Gitall, Ahmedabad, AMA 2000.

Web References:

1. <http://lsib.co.uk/lms/wp-content/uploads/2015/02/Indian-Ethos-and-Management.pdf>
2. www.vikaspublishing.com/books/business-economics/management/ethics

MANAGEMENT SCIENCE

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS15	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. Familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices. 2. Understand the structure & Designing of an Organization. 3. Knowledge on the aspects of Production. 4. Analyse the market and the strategies involved in Marketing. 5. Knowledge on concepts related to Human Resources. 6. Understand the techniques used in Project Management. 7. Familiarize with strategies used for analysis of an Organization. 8. Understand the Contemporary Management Issues. 9. Familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.								
COURSE OUTCOMES: 1. Knowledge on management theories and practices. 2. Understanding designing organizational structure. 3. Understanding on the methods & charts used in operations management. 4. Ability to understand the market and its environment. 5. Understand the processes, functions etc in Human Resources Management. 6. Ability to solve problems in managing the Project. 7. Knowledge on Strategic alternatives. 8. Familiar with the practices implemented in management. 9. Understand the social responsibilities of Management. 10. Understand the basic concepts of Management.								
UNIT-I	INTRODUCTION TO MANAGEMENT AND ORGANIZATION						Classes: 12	
Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management-Taylor's Scientific Management Theory- Fayal's Principles of Management- Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Herzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.								
UNIT-II	OPERATIONS AND MARKETING MANAGEMENT						Classes: 12	
Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.								
UNIT-III	HUMAN RESOURCES MANAGEMENT(HRM)						Classes: 12	
Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.								

UNIT-IV	PROJECT MANAGEMENT (PERT/ CPM)	Classes: 12
Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems)		
UNIT-V	STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES	Classes: 12
Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.		
TEXT BOOKS:		
1.A.R.Aryasri : Management Science, TMH, (Latest edition) 2.Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.		
REFERENCE BOOKS:		
1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012. 2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012. 3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012. 5. Samuel C. Certo: Modern Management, 2012. 6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012. 7. Parnell: Strategic Management, Cengage, 2012. 8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.		
Web References:		
1. https://www.smartworld.com/notes/management-science-pdf-notes-ms-notes-pdf 2. https://www.alljntuworld.in/download/management-science-notes/		

INTELLECTUAL PROPERTY RIGHTS**VIII Semester: OPEN ELECTIVE-IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS16	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To introduce fundamental aspects of Intellectual property Rights to students who are going to play a major role in development and management of innovative projects in industries. 2.To disseminate knowledge on patents, patent regime in India and abroad and registration aspects 3. To disseminate knowledge on copyrights and its related rights and registration aspects 4. To aware about current trends in IPR and Govt. steps in fostering IPR 5. To familiarize them with the kind of rights, remedies and licensing regime associated with each kind of intellectual property so that students can have a basic understanding of Intellectual Property laws.								
COURSE OUTCOMES: 1 .Skill to understand the concept of intellectual property rights. 2. Develops procedural knowledge to Legal System and solving the problem relating to intellectual property rights. 3. Skill to pursue the professional programs in Company Secretaryship, Law, Business(MBA), International Affairs, Public Administration and Other fields. 4. Employability as the Compliance Officer, Public Relation Officer and Liaison Officer. 5. Establishment of Legal Consultancy and service provider.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction, types of intellectual property, international organizations, agencies and treaties, importance of intellectual property rights.								
UNIT-II	TRADE MARKS						Classes: 12	
Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.								
UNIT-III	LAW OF COPY RIGHTS AND PATENTS						Classes: 12	
Law of copy rights: Fundamental of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law. Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer.								
UNIT-IV	TRADE SECRETS AND UNFAIR COMPETITION						Classes: 12	
Trade Secrets: Trade secrete law, determination of trade secrete status, liability for misappropriations of trade secrets, protection for submission, trade secrete litigation. Unfair competition: Misappropriation right of publicity, False advertising.								
UNIT-V	NEW DEVELOPMENT OF INTELLECTUAL PROPERTY						Classes: 12	
new developments in trade mark law; copy right law, patent law, intellectual property audits. International overview on intellectual property, international - trade mark law, copy right law, international patent law, international development in trade secrets law.								
TEXT BOOKS: 1. Intellectual property right, Deborah, E. Bouchoux, cengage learning. 2.Intellectual property right - Unleashing the knowledge economy, prabuddha ganguli, Tata Mc Graw Hill Publishing Company Ltd.								
REFERENCE BOOKS:								

1. Deborah. E. Bouchoux (2009), Intellectual property, Cengage learning, India.
2. Deborah. E. Bouchoux (2001), Protecting your companies intellectual property, AMACOM, USA.
3. Prabuddaganguli (2003), Intellectual property right, Tata McGraw Hill Publishing company Ltd., India.
4. Robert Hisrich, Michael P. PETER, Dean A. Shepherd (201), Entrepreneurship, Tata McGraw Hill., India.

Web References:

- 1.Subramanian, N., & Sundararaman, M. (2018). Intellectual Property Rights – An Overview. Retrieved from <http://www.bdu.ac.in/cells/ipr/docs/ipr-eng-ebook.pdf>
- 2.World Intellectual Property Organisation. (2004). WIPO Intellectual property Handbook. Retrieved from https://www.wipo.int/edocs/pubdocs/en/intproperty/489/wipo_pub_489.pdf
- 3.Journal of Intellectual Property Rights (JIPR): NISCAIR

NUMBER THEORY

VIII Semester: OPEN ELECTIVE - IV								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4HS17	HS	L	T	P	C	CIA	SEE	Total
		3	0	-	3	30	70	100
Contact Classes: 42	Tutorial Classes: 00	Practical Classes:--			Total Classes: 42			
COURSE OBJECTIVES: To learn 1. Divisibility test and Euclidian Algorithms. 2. The Arithmetic function and Dirichlet multiplication. 3. The properties of congruences. 4. Application of Lagrange's theorem. 5. Quadratic Residues and Reciprocity Law.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Apply the Fundamental theorem of Arithmetic.. 2. Identify the Liouville's function and divisor function. 3. Apply the Euler Fermat theorem in polynomial congruence modulo P 4. Apply Chinese remainder theorem. 5. Find Quadratic residues.								
UNIT-I						Classes: 10		
Divisibility, GCD, Prime Numbers, Fundamental theorem of Arithmetic, the series of reciprocal of the Primes, The Euclidean Algorithm.								
UNIT-II						Classes: 08		
A Arithmetic function and Dirichlet Multiplication, The functions $\varphi(n)$, $\mu(n)$ and a relation connecting them, Product formulae for $\varphi(n)$, Dirichlet Product, Dirichlet inverse and Mobius inversion formula and Mangoldt function $\Lambda(n)$, multiplication function, multiplication function and Dirichlet multiplication, Inverse of a completely multiplication function, Liouville's function $\lambda(n)$, the divisor function is $\sigma\alpha(n)$								
UNIT-III						Classes: 08		
Congruences, Properties of congruences, Residue Classes and complete residue system, linear congruences conversion, reduced residue system and Euler Fermat theorem, polynomial congruence modulo P								
UNIT-IV						Classes: 08		
Lagrange's theorem, Application of Lagrange's theorem, Chinese remainder theorem and its application, polynomial congruences with prime power module.								
UNIT-V						Classes: 08		
Quadratic residue and quadratic reciprocity law, Quadratic residues, Legendre's symbol and its properties, evaluation of $(-1/p)$ and $(2/p)$, Gauss Lemma, the quadratic reciprocity law and its applications								
TEXT BOOKS:								
1. " Introduction to analytic Number Theory "by Tom M. Apostol. Chapters 1, 2, 5, 9.								

REFERENCE BOOKS:

1. Number Theory by Joseph H. Silverman.
2. Theory of Numbers by K.Ramchandra.
3. Elementary Number Theory by James K Strayer.
4. Elementary Number Theory by James Tattusall.

WEB REFERENCES:

1. <http://14.139.172.204/nptel/CSE/Web/111103020-old/NumberTheory-A.Saikia.pdf>
2. <https://www.btechguru.com/courses--nptel--mathematics-video-lecture--maths.html>

E-TEXT BOOKS:

<https://wstein.org/ent>
<https://www.pdfdrive.com/number-theory-books.html>

MOOC COURSE

<https://nptel.ac.in/courses/111103020/>
<http://nptel.ac.in/courses/111106086/5>

PHYSICS AND TECHNOLOGY OF THIN FILMS

VIII Semester: OPEN ELECTIVE - IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS18	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
Contact Classes: 42	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 42			
COURSE OBJECTIVES: The course should enable the students to: 1. Understand the bonding, nucleation process in materials. 2. Gain the knowledge on fabrication of thin films. 3. Learn the kinetics of different thin films. 4. Understand the properties of thin films by using characterization methods. 5. Learn the applications of thin films in different fields.								
COURSE OUTCOMES: The student will able to: 1. Analyze the defects in solids and kinetics. 2. Evaluate the epitaxy of thin films and growth process. 3. Justify how the growth kinetics is more efficient than nucleation kinetics in thin film technology. 4. Recommend appropriate synthesis method and explain the characterization techniques. 5. Analyze which thin films are used in different applications.								
UNIT-I	A REVIEW OF MATERIALS SCIENCE					CLASSES: 06		
Introduction, Structure Defects in Solids Bonding of Materials, Thermodynamics of Materials, Kinetics, Nucleation and Conclusion.								
UNIT-II	PREPARATION OF THIN-FILMS					CLASSES: 10		
Classifications of vacuum ranges – Vacuum pumps - Rotary, Diffusion, Turbo molecular and Ion Pumps –Thin film (epitaxy) – definition & advantages – Types of epitaxy. Different Growth Techniques: Liquid Phase Epitaxy, Molecular Beam Epitaxy, Metal Organic Vapour Phase Epitaxy, Sputtering (RF & DC), Pulsed Laser Deposition. Thickness Measurement: Microbalance technique, Photometry, Interferometry (MBI, FECO)								
UNIT-III	KINETICS OF THIN FILMS					CLASSES: 08		
Nucleation Kinetics: types of nucleation – kinetic theory of nucleation – energy formation of a nucleus – critical nucleation parameters; spherical and non-spherical nucleus (cap, disc and cubic shaped)on the substrates. Growth Kinetics: Kinetics of binary (GaAs, InP, etc.), ternary (Al _{1-x} Ga _x As, Ga _{1-x} In _x P, etc.) and quaternary (Ga _{1-x} In _x As _{1-y} P _y , etc.) semiconductors – derivation of growth rate and composition expressions.								
UNIT-IV	PROPERTIES &CHARACTERIZATION OF THIN FILMS					CLASSES: 10		
Properties of Thin Films: Dielectric properties – Important parameters, Measurement of dielectric properties- Effect of annealing and film thickness. Optical properties – Optical constants, determination of optical constants by Brewster angle method, Normal incidence method and graphical method. Mechanical								

properties – Concept and origin of stress and strain, Lattice misfit, Thermal misfit, Hardness test and Bulge test.

Characterization

X-ray diffraction –Photoluminescence –UV-Vis-IR spectrophotometer – Atomic Force Microscope – Scanning Electron Microscope – Hall Effect – Vibrational Sample Magnetometer – Secondary Ion Mass Spectrometry.

UNIT-V

APPLICATIONS OF THIN FILMS

CLASSES: 08

Optoelectronic devices: LED, LASER and Solar cell – Micro Electromechanical Systems (MEMS) – Fabrication of thin film capacitor – application of ferromagnetic thin films; Data storage, Giant Magneto resistance (GMR).

TEXT BOOKS:

1.A. Goswami, Thin Film Fundamentals, New Age international (P) Ltd. Publishers, New Delhi(1996).

REFERENCE BOOKS:

- 1.K. L. Chopra, Thin Film Phenomena, McGraw- Hill book company New York,(1969).
- 2.L. Eckertova, Physics of Thin Films, Plenum press, New York(1977).
- 3.Milton Ohring, Material Science of Thin films, 2nd Edition, Academic Press(2002).

E-TEXT BOOKS:

- 1.<https://www.safaribooksonline.com/library/view/materials-science-of/9780125249751/?ar&orpq>
- 2.<http://doi-ds.org/doi/10.1016/B978-0-12-375189-7>
- 3.<https://ieeexplore.ieee.org/document/5488896>

MOOC COURSE:

- 1.<https://nptel.ac.in/courses/113106034/44>
2. www.facweb.iitkgp.ac.in/~jdas/mfm/Teaching.html
3. https://swayam.gov.in/nd1_noc19_mm13

POLYMER CHEMISTRY

VIII Semester: OPEN ELECTIVE - IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS19	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100
Contact Classes:	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 48			
COURSE OBJECTIVES: The course should enable the students to: 1. Understand the structure of the polymers, identify and explain types of polymerization. 2. Gain the knowledge to use techniques of polymerization. 3. Understand the properties, processing and applications of natural rubber. 4. Gain the knowledge to design polymers for industrial applications. 5. Learn the structure properties of polymers ,inorganic and its characterization techniques								
COURSE OUTCOMES: The student will able to: 1. Describe the structures of the polymers, identify and explain types of polymerization and account for the concept of the molecular weight. 2. Explain the various properties of polymers and use techniques of polymerization. 3. Account the properties, processing and applications of natural rubber. 4. Design polymers for industrial application using suitable additives. 5. Correlate structure-properties of polymer composite, inorganic, conducting polymers and its characterization techniques.								
UNIT-I	BASICS OF POLYMERS						CLASSES: 10	
Introduction: Polymer Definition, Classifications of Polymers: Natural & Synthetic, Biopolymers, Thermoplastic, Thermosets, Elastomers, Fibers etc. Types of Polymerization: Addition & Condensation Polymerization. Molecular Weight : No. Average and Weight Average Molecular Weight of Polymers, Distribution & Poly dispersity Index, Structure of Polymers: Amorphous, Semi crystalline and Crystalline states								
UNIT-II	PROPERTIES & POLYMERIZATION TECHNIQUES						CLASSES: 10	
Polymerization Techniques: Mass Polymerization, Bulk Polymerization, Solution Polymerization, Emulsion Polymerization, Suspension Polymerization, Mechanisms with explanation. Structure Property Relationship in Polymers: glass transition, melting and crystallization temperature. Effect of structure on the chemical, mechanical, electrical and optical properties of polymers.								
UNIT-III	NATURAL RUBBER : PROPERTIES, PROCESSING AND APPLICATIONS						CLASSES: 10	
Natural Rubber (NR): Origin – Natural Rubber Latex, Tapping, Processing- Vulcanization, Properties and Applications – Conversion of Latex into Dry Rubber – Properties of Dry Rubber. Synthetic Polyisoprene (IR) Rubbers: Preparation, Properties and Applications, Thermoplastic Elastomers Based on Plastics.								

UNIT-IV	NON RUBBER ADDITIVES	CLASSES: 10
Non Rubber Additives: Part A: Vulcanizing ingredients & other additives: Vulcanizing ingredients & their sequence of mixing: Activators and Accelerators: mechanisms of action Part B: Fillers Carbon black-Its preparation, structure, properties and their effect on rubber properties Silica fillers & coupling agents, Other fillers: Clay, Calcium carbonate, titania etc. Nano-fillers: Reinforcement by filler: Reinforcement, Part C: Processing aids & other additives: Processing aids, plasticizers, process additives, release agents, Prevention of Ozone Attack with the use of waxes & saturated polymer for Ozone Protection.		
UNIT-V	INORGANIC, COMPOSITE AND CONDUCTING POLYMER, CHARACTERIZATION TECHNIQUES OF POLYMERS	CLASSES: 08
Inorganic Polymers: Synthesis, Structure- property Relationship and Applications of Polyphosphates, Polysiloxanes, Polymer Composite- Properties and its Applications. Conducting Polymers- Poly Aniline (PaNi)- Mechanism of Conduction and Applications. Characterization Techniques of Polymers: Thermo gravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC).		
Text Books:		
1. F. W. Billmeyer Jr., Text Book of Polymer Science, Ed. Wiley-Interscience, 1984. 2. V. T. Gowariker, N. V. Viswanathan, and J. Sreedar, Polymer Science, 1988.		
Reference Books:		
1. M. Morton, Rubber Technology, Chapman Hall, 1995. 2. Robert J. Young and Peter A. Lovell, Introduction to Polymers, 3 rd Edition, 2011.		